P- ISSN : 0973 - 7057

BIOSPECTRA

(An International Biannual Refereed Journal of Life Sciences)

March, 2023

Vol. 18(1), pp 01-290

BIOSPECTRA

(An International Biannual Refereed Journal of Life Sciences)

(www.mset-biospectra.org)

SJIF 2023- 8.546

www.sjifactor.com

Indexed in Clarivate Analytics as Zoological Records
(https://mjl.clarivate.com/search-results)
Listed in UGC-CARE (Group II)

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Published by :
Madhawi Shyam Educational Trust (MSET)
Ranchi, Jharkhand (India)
Reg. No. 20560/IV-1815/2005

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BIOSPECTRA:P- ISSN.0973-7057 is an innovative, refereed, biannual **(every March and September)** multicolumn research journal in the field of Life Sciences & Bioinformatics published by the **Madhawi-Shyam Educational Trust, Ranchi, Jharkhand, India (Reg. No. 20560/IV-1815/2005)**. It is dedicated to cover wide spectrum original research papers under following categories:-

- **Review Articles:** On any pre- published idea or topic with latest information and with more than 50 references.
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- **Full Length Papers:** Covering standard write-up span of 6-8 *Pagemaker* pages (10-12 MS Word pages) 1.5 space typed material with tables and graphs/figs. Photographs up to 4-6, may also be accommodated in black and white **(colour photograph to be printed on extra tariff)** strictly in the standard format given below.
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Biospectra : Vol. 18(1), March, 2023

An International Biannual Refereed Journal of Life Sciences

Published every **March & September** by
Madhawi Shyam Educational Trust, Ranchi

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An International Biannual Refereed Journal of Life Sciences

BIOSPECTRA

ISSN : 0973-7057

Reg No. 20560/IV - 1815/2005

An International Biannual Refereed Journal of Life Sciences
(Every **March & September**)

ISSN : 0973-7057

(An official publication of Madhawi-Shyam Educational Trust & ICCB, Ranchi, Jharkhand, India)
Reg. No. 20560/IV 1815/2005

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ISSN : 0973-7057

An International Biannual Refereed Journal of Life Sciences

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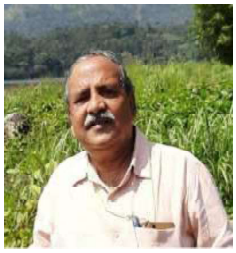
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From the Editor Desk

Biospectra is an official publication of MSET-ICCB, Ranchi. On behalf of the Editorial Board of the Journal Biospectra and my Co-Editors, I am glad to present the "Volume 18, Issue 1, March, 2023" of the Journal. It is with immense pleasure that we welcome you to the latest edition of the Journal BIOSPECTRA. As we delve into the captivating realms of life sciences, we find ourselves at a critical juncture in the advancement of our understanding of living organisms and their intricate mechanisms. This issue, filled with a diverse array of research articles, reviews, and perspectives, reflects the relentless pursuit of knowledge that characterizes our scientific community.

In this edition, we are delighted to present studies that illuminate various facets of life sciences, from molecular intricacies to ecosystem dynamics. Our authors have not only expanded our comprehension of existing theories but have also unveiled novel insights that challenge conventional wisdom. The interdisciplinary nature of modern life sciences is vividly illustrated by the collaborative efforts that have yielded discoveries at the intersections of biology, chemistry and computer science.

As we celebrate the achievements showcased in this issue, we also recognize the collaborative nature of scientific progress. The exchange of ideas and the fusion of diverse perspectives fuel the engine of innovation. We extend our heartfelt gratitude to the authors, reviewers, and editorial board members who have devoted their expertise and time to curate a collection that embodies the spirit of scientific inquiry.

In conclusion, we invite you to immerse yourselves in the pages of this edition, to explore the intricate tapestry of life sciences and to be inspired by the relentless pursuit of knowledge that defines our community. The voyage of discovery is unending, and it is our privilege to share in this journey with you, our esteemed readers.

Wishing you an enlightening and enriching reading experience.

31.03.2023

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From the Secretary Desk

Dear Members,
Greetings

As you know Late Prof. (Dr.) Mahendra Prasad had established a trust in the sweet memories of his parents in the year of 2005. **MADHAWI SHYAM EDUCATIONAL TRUST (MSET)** has been registered in December, 2005. After registration of the trust, he had constituted an **INTERNATIONAL CONSORTIUM OF CONTEMPORARY BIOLOGIST (ICCB)** wing with the consent of his colleagues and family members. Under the **MSET-ICCB** he had launched a Life Science Journal named **"BIOSPECTRA"** in the year of 2006 which is a biannual journal publishing every March and September. The first issue of Journal was published in September, 2006. Since then, the same journal has been published regularly. Gradually **"BIOSPECTRA"** has improved its quality with co-operation and valuable suggestions of all over the world wide members.

He had organised several National/ International Conferences with collaboration of the different prestigious organisations. His last commitment was with Department of Zoology, Patna University, Patna **"ICEBRUD-2021"** but unfortunately he couldn't attend the same conference due to his sudden demise on the 20th April, 2021.

He was the whole and sole Settler, President cum Director of the MSET-ICCB and always will be remain in our hearts. After his death the trustees of MSET-ICCB, have selected **Ms. Astha Kiran** (daughter of Late President), as the new President of MSET-ICCB with consent of all family members. Our former Executive Editor, Prof. (Dr.) Jyoti Kumar has been nominated as Editor-in-Chief of Biospectra for a period of three years.

Further, I would like to inform all the esteemed members that MSET-ICCB is an integral body and its headquarter is in Ranchi only. The trust bodies do not authorise anybody else to take any descion in behalf of ICCB separately without prior permission of the trust.

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Astha Kiran emerges as the ideal candidate for the presidency of MSET-ICCB, given her rich tapestry of experience, unwavering dedication, and an illustrious track record. Her qualifications, which span from an M.Sc. in Fashion Technology from Himalayan University to specialized fashion design credentials from IIFT New Delhi, showcase her multidisciplinary approach. Astha's proficiency extends to boutique management, creative pattern making, and an innate understanding of luxury fabrics. These qualifications have not only established her as a prominent figure in the fashion world but also equipped her with the versatility required for dynamic leadership.

However, what truly sets Astha apart is her exceptional array of awards and recognitions. She has been honored with prestigious accolades including the coveted "Best Achievement Award" from INIFD, the distinguished "Gold medal" from the "Grih Shobha" group, and the esteemed "Best Female Entrepreneur of the Year Award." These awards are not mere tokens of recognition; they underscore her profound impact in the realms of fashion and entrepreneurship.

Astha's vision for MSET-ICCB is marked by a commitment to growth and innovation. She envisions a trust that continues to elevate education, art, and societal betterment. Her mission is to foster a collaborative environment that enables knowledge sharing and propels interdisciplinary projects. Under her guidance, MSET-ICCB is poised to extend its reach globally, becoming a beacon for progressive initiatives.

Among her numerous achievements, Astha's role in furthering and sustaining "BIOSPECTRA," a biannual Life Science Journal launched by her father late Professor (Dr.) Mahendra Prasad. This reflects her commitment to collaboration and knowledge dissemination. She has been working relentlessly in order to honour her father's glorious legacy. Beyond fashion, her contributions span across reviving endangered Indian art forms, championing Indian textiles and handloom fabrics, and chairing the Women's Sub-Committee at the Federation of Jharkhand Chamber of Commerce and Industries (FJCCI). Her position as a fashion design faculty at Ranchi University adds another layer to her illustrious career. Astha is also a distinguished member of the World Designing Forum. These achievements underscore her holistic approach to leadership and her ability to drive change in various spheres.

In sum, Astha Kiran's exceptional qualifications, forward-thinking vision, and remarkable achievements, notably her distinguished awards, make her not only suitable but also an inspiring choice for the presidency of MSET-ICCB.

Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 1-6

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Vehicular impact on *Varanus bengalensis* (Daudin, 1803) around the roads of Girnar Eco-Sensitive Zones

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Received : 13th November, 2022 ; Revised : 14th December, 2022

Abstract- *Varanus bengalensis* a monitor lizard found all over India. The study focused on the threats faced by this species due to vehicle traffic. The study was conducted in Girnar Eco-sensitive zones and utilized two survey methods: Visual Encounter Survey (VES) to study habitat utilization and Nocturnal Road Cruising to collect road kill data for a year (Oct-21 to Sept-22). The study reveals that crossing road pose a significant risk to the species, with 18 individuals found dead on the road. Furthermore, 31 individuals were seen basking or wandering near the roadside, emphasizing the importance of preserving such habitats for their survival. The findings of this study suggests that further conservation efforts are necessary to protect this species from road mortality. This study emphasizes the importance of protecting this species and its habitats to ensure its long-term survival.

Key words: *Varanus bengalensis*, Eco-sensitive zones, Girnar wildlife sanctuary, Road kill.

INTRODUCTION

The Bengal monitor lizard, *Varanus bengalensis*, has exhibited remarkable ecological resilience by thriving in various habitats. The species has adapted well to anthropogenically modified environments and demonstrated adaptability in different natural landscapes. Its ability to persist and reproduce in diverse ecosystems, such as dense forests, sparsely wooded regions, open grasslands, and thorny scrublands, is particularly noteworthy. This diversity in *V. bengalensis* characteristics reflects the species adaptations to the varying environmental conditions, such as differences in temperature, precipitation, vegetation, and prey availability, which exist across its range.[1-3] The species is well adapted to live in various terrestrial habitats,[2,4] and thrives in anthropogenic situations such as home gardens and homesteads, even occupying households and densely populated urban environments.[5] *V. bengalensis* is a culturally and ecologically important top predator. It requires conservation efforts against habitat destruction, fragmentation, and roadkill mortality.

Roads are a significant threat to wildlife populations due to vehicle collisions, with approximately 25 million kilometers of new roads projected to be constructed by 2050, 90% of which will be in developing countries.[6,7] Roads can also harm wildlife through oil leaks and noise-related stress,[8] but the most severe impact is animal injury and mortality from vehicle collisions,[9] which can be influenced by driver fatigue.[10] Animal-vehicle collisions are linked to traffic flow, barriers, and speed,[11] causing alterations in the distribution of age, sex, and fecundity in animal populations.[12] The impact of the collision can cause serious injury or death to the animal, and can result in significant ecological consequences.[11] However, any animal that crosses the road can become a victim of a collision

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with a vehicle. Herpetofauna were found to be the most affected group.[13]

Reptiles, particularly monitor lizards, are no exception to this threat. In recent years, there have been growing concerns about the impact of vehicular traffic on the population of *Varanus bengalensis*, particularly in areas where their habitats intersect with roads and highways. In this study, we investigate the occurrence and extent of road kill incidents involving the *Varanus bengalensis* species in the Eco-sensitive zones of Girnar Wildlife Sanctuary. We examine the response of monitor lizards to vehicular disturbance, their behaviour in seeking shelter, and the effectiveness of existing protective measures. This research aims to provide valuable insights into the impact of road traffic on the population of *V. bengalensis* and to suggest ways to minimize the damage to these ecologically significant creatures.

METHODS

Study area

The peripheral region encompassing the roads that surround the Girnar Eco-sensitive Zone, located in Junagadh, is estimated to cover an area of approximately 9,317.58 hectares. (Fig. 1) This region spans across 27 villages that fall under the jurisdiction of the Junagadh and Bhesan Talukas. The Eco-sensitive Zone itself has a radius of 0 to 5 kilometres, which is an area designed to preserve and protect the unique flora and fauna within it. The area is designated as an Eco-sensitive Zone due to its ecological significance and importance in maintaining the environmental balance of the surrounding regions. The protective measures within the zone help to conserve the natural resources and biodiversity of the area, ensuring their long-term sustainability.

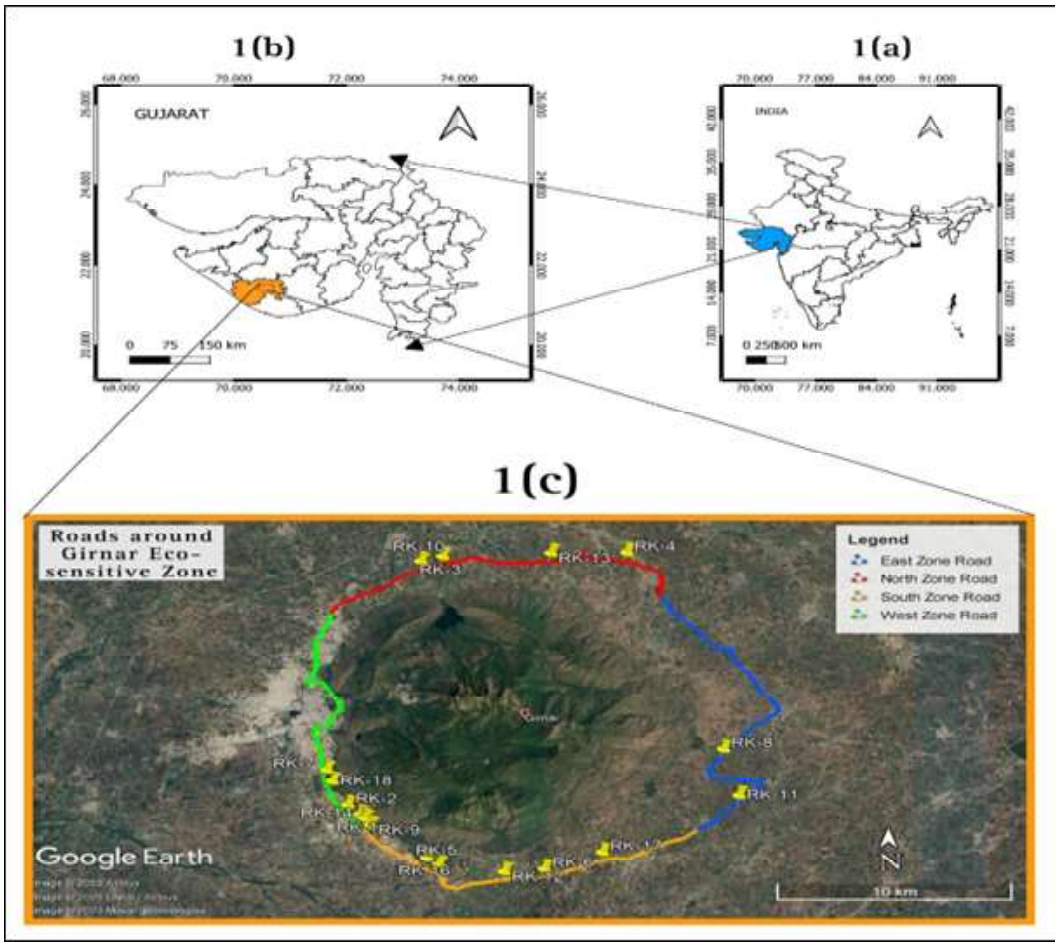


Fig. 1, 1(a): Map of India; 1(b): Map of Gujarat; 1(c): Road around Girnar Eco-sensitive zone divided into four different zones denoted my different colours, (North zone road – Red; East zone road – Blue; South zone road – Orange; West zone road – Green). RK denotes geo-location of road kill specimen found during the study period. (Image Source: Google Earth)

Data collection

During the period of October 2021 to September 2022, we conducted a comprehensive study to collect data on the occurrence and distribution of *V. bengalensis* around the roads in the Eco-sensitive zones of Girnar Wildlife Sanctuary. For the road kill survey, a Nocturnal Road cruising approach was employed to gather data. In addition, the Visual Encounter Survey (VES) method,[14] was utilized to perform a field survey of the lizard populations in the study area. We also surveyed an area of 5-10 meters around the roads to detect any basking lizards. The use of both data collection methods increases the accuracy and reliability of the obtained results in our study.

A field study was undertaken from October 2021 to September 2022, to investigate roadkill incidents on a designated road segment. A total of 48 systematic road trips were conducted during the study period. Data collection was conducted at specific times, with 07:00 h to 10:00 h during winter and 06:00 h to 09:00 h in summer being the standard times, selected based on visibility. The motor vehicle travelled at a constant speed of 20-40 kmph. All carcasses or individuals encountered were identified to the species level, and the geographic coordinates of road-killed specimens were recorded using Google Maps to enable accurate geo-referencing.

RESULT & DISCUSSION

The current study unveils the *V. bengalensis* is highly vulnerable to road kill incidents during the summer (March to June), with eight recorded cases, followed by the monsoon (July to October), which saw seven incidents. This pattern can be attributed to the species higher levels of foraging activity during these periods. However, the impact is most pronounced in the west and south zones road, where high vehicle transportation and the presence of the main highway from Junagadh to Bilkha have resulted in elevated road kill rates of 34% and 33%, respectively (Fig. 2). Although some of incidents may have missed out due to eaten by other animals or collected by humans.

Table 1. Number of individuals that were found dead on the roads throughout the study period and its geo-location. Each specimen is denoted by "RK = Road Kill"

Road kill data of *Varanus bengalensis*				
Month	**No. of Kills**	**Specimen no.**	**Latitude**	**Longitude**
Oct-21	1	RK-1	21°27'35.85"N	70°28'50.89"E
Nov-21	1	RK-2	21°27'53.87"N	70°28'33.50"E
Dec-21	0	-	-	-
Jan-22	1	RK-3	21°35'55.62"N	70°30'54.04"E
Feb-22	1	RK-4	21°36'2.70"N	70°35'27.69"E
Mar-22	1	RK-5	21°25'58.82"N	70°30'48.31"E
Apr-22	2	RK-6	21°25'50.42"N	70°33'25.33"E
		RK-7	21°29'1.82"N	70°28'1.89"E
May-22	2	RK-8	21°29'42.27"N	70°37'51.21"E
		RK-9	21°27'14.27"N	70°29'7.63"E
Jun-22	3	RK-10	21°35'46.48"N	70°30'21.90"E
		RK-11	21°28'12.01"N	70°38'14.41"E
		RK-12	21°25'45.02"N	70°32'25.50"E
Jul-22	2	RK-13	21°36'1.84"N	70°33'35.16"E
		RK-14	21°27'27.17"N	70°28'58.75"E
Aug-22	2	RK-15	21°27'29.92"N	70°28'56.40"E
		RK-16	21°26'11.76"N	70°30'30.38"E
Sep-22	2	RK-17	21°26'22.86"N	70°34'51.40"E
		RK-18	21°28'39.18"N	70°28'10.51"E
Total = 18				

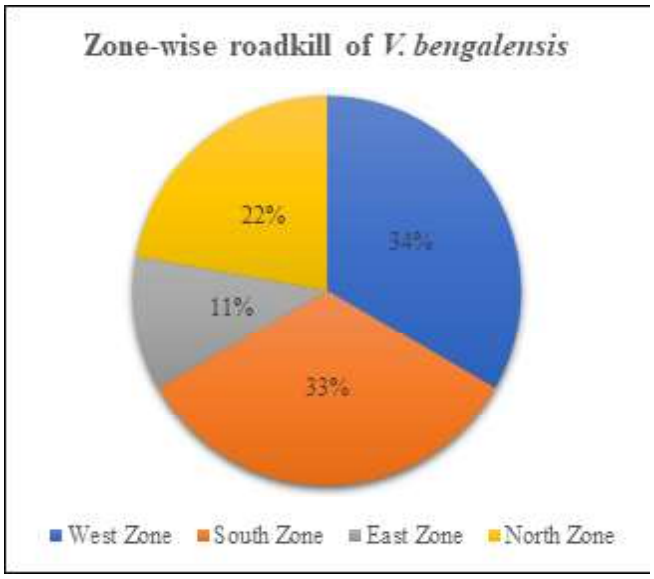


Fig. 2. Zone-wise roadkill of *Varanus bengalensis* around the Girnar Eco-sensitive Zone

According to the findings of the survey conducted in the study area, a total of 49 Bengal monitor lizards were recorded. Out of these, a concerning number of 18 individuals (Table 1) were found dead on the road, indicating the devastating impact of vehicular traffic on the species. On the other hand, 31 individuals were observed basking or roaming near the roadside, suggesting their tendency to utilize this area for foraging, thermoregulation, or other activities. (Fig. 3, 4)



Fig.3. Individuals that are observed basking and roaming near the roadsides



Fig. 4. Road-killed *V. bengalensis* sighted during the study period.

The loss of a *Varanus bengalensis* due to roadkill is a blunt reminder of human impact on wildlife. These lizards maintain ecological balance by controlling populations of smaller animals. Even a single loss can have far-reaching consequences for the local environment, emphasizing the need for action to reduce negative human impact. To mitigate roadkill, measures like creating wildlife crossings and reducing speed limits are necessary. Promoting awareness and encouraging conservation actions is crucial for harmonious coexistence with nature. Citizen science databases have potential benefits for improving roadkill data.[15]

These findings highlight the need for urgent conservation measures to mitigate the impact of human activities, particularly road infrastructure development and vehicular traffic, on the Bengal monitor lizard population. Such measures may include creating wildlife crossings or underpasses, reducing speed limits in areas known to be frequented by the species.

CONCLUSION

Based on the present study, vehicular impact on *V. bengalensis* involves direct effects such as injury or mortality, as well as indirect effects such as habitat loss, fragmentation and alteration. Such changes may influence the behavior, survival, growth and reproductive success of individual animals. The findings of these research highlight the importance of taking action to protect the Bengal monitor lizard from the threat of habitat loss and road mortality. We need to develop and implement strategies to manage human-wildlife conflict and mitigate the impact of urbanization on these animals. This study can serve as a basis for further research and conservation efforts to safeguard the Bengal monitor lizard and other species facing similar threats.

ACKNOWLEDGEMENT

The authors would like to thank Department of Life Sciences, BKNM University for support. The first author JKC is thankful to Mr. Vipul V. Bamaniya, Ms. Usha J, Zala and Mr. Malay A. Vyas for their continuous support.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 7-12

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

A systemic review on mupirocin resistance in *Staphylococcus aureus*: Current status and future prospects

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Received : 15th November, 2022 ; Revised : 20th December, 2022

Abstract- Gram positive *Staphylococcus aureus* bacteria have been identified as prevalent pathogen causing mild tissue infections. In the mid of the 20th century, penicillin and methicillin antibiotics had been widely applied to treat *S. aureus* infection. This widely and ruthless use of these antibiotics resulted into the development of these antibiotic resistant in the *Staphylococcus aureus.* With the increasing demand to treat MRSA, Mupirocin has become the first choice of treatment for the heath service providers worldwide. This is because it is most effective against MRSA and its decolonization. However, large-scale use of Mupirocin antibiotic has resulted in Mupirocin resistant *Staphylococcus aureus* (MuRSA). Conjugative plasmid *mup A gene* which harbors the antimicrobial resistance determinants, mediates higher side Mupirocin resistance in *Staphylococcus aureus* (HL-MuRSA). Mutation in native ileS gene mediates lower side Mupirocin resistance in *Staphylococcus aureus*, which has no medically significant. Reduction in the effectiveness of mupirocin against MRSA has resulted in high risk for invasive infection of MRSA, therefore monitoring of MuRSA is critically needed in present scenario. Keeping this in view, we have discussed mupirocin resistance mechanism, pathological and clinical significance and future perspectives in the current review.

Key words: *Staphylococcus aureus*, Mupirocin, antimicrobial, clinical significance, penicillin

INTRODUCTION

Gram positive *Staphylococcus aureus bacteria* have been identified as the commonest prevalent pathogen causing soft tissue infection, endocarditis, bacteramia, osteomyeliti and nosocomial infections across the globe by the medical professionals Acute pneumonia, septicemia, are some of the acquired infections which prolong the hospital stay and usually are not present at the time of admission.[1-6] In 1940s, penicillin was extensively used to treat of various infectious diseases, which resulted as β lactams resistance in *Staphylococcus aureus.* The gene mec*A,* synthesizes mutated penicillin binding protein PBP2a which shows low affinity with β lactams and allocates methicillin resistance in *Staphylococcus aureus* (MRSA) was first time noted in 1960.[7-9] Later, it was explored that *Staphylococcus aureus* has become resistant to penicillin, Pseudomonic acid A which is also known as Mupirocin, was the only approved antibiotic to treat the methicillin resistance and methicillin sensitive *Staphylococcus aureus* infections.[1,6,10] Mupirocin antibiotics impede protein formation via binding to the Isoleucine residue of tRNA aminoacyl synthetase. The effective clinical data of this antibiotic against MRSA made this antibiotic best bet of clinicians especially to treat the decolonization, bedsore and soft tissue skin lesions. Mupirocin is one of the components of the topical

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antibacterial ointment and nasal formulation used to treat primary and secondary infection caused by *Staphylococcus* and *Streptococcal* species.[11,12] Food and drug administrative (FDA) department of USA has provided the approval for the use of mupirocin in nasal formulations applied to treat *Staphylococcus aureus* in patients and health service providers who are susceptible to MRSA. MRSA infection has become a serious health threat across the globe and associated with higher infection and death rate along with causing high economic burden on the healthcare system.[3,11-13] Multiple drug resistance (MDR) (resistant to ≥ 3 non beta lactum antibiotics) was also observed in methicillin resistance in *Staphylococcus aureus* (MRSA).[14,15] Topical use of mupirocin as ointment and nasal formulation has been found to reduce *Staphylococcus aureus* pathogen related infections in haemodialysis patients and in peritoneal dialysis patients by 80% and 63% respectively. Application of mupirocin and chlorhexidine has been found to reduce 37% MRSA infections and 44% blood infections which were result of secondary infections in Intensive Care Unit admitted patients.[16]

Mupirocin: Resistance and its mechanism

Pseudomonic acid A i.e. Mupirocin antibiotic was derived from *Pseudomonas fluorescens at* first time, and has been found to inhibit translational step at molecular level in bacteria through bacterial *Isoleuyl RNA synthetase* (IleRS).[16] It is widely used antibiotic for the infections causing microbes viz., *Staphylococci*, *Streptococci* and various gram negative bacteria such as Haemophilus influenza, Neisseria gonorrhea.[17] Minimum Inhibitory Concentration (MIC) for higher level resistance for mupirocin is defined as ≥ 512 mg/L and lower level resistance for mupirocin is defined as 8-256mg/L respectively.[18-21] Higher side Mupirocin antibiotic resistance in *Staphylococcus aureus* bacteria (HL-MuRSA) have been associated with conjugative plasmid *mup A gene* which harbours antimicrobial resistance determinants and lower side Mupirocin antibiotic resistance in *Staphylococcus aureus* bacteria due to mis-sense point mutations in *Isoleucyl RNA synthetase* (IleRS) gene.[10,12,22,23] Few studies have shown that plasmid mediated conjugate mupA gene which is present on mobile element of genetic material facilitates in the resistance mechanism. Mupirocin resistant *Staphylococcus* bacterial isolates showed various strains of mutated *Staphylococcus aureus* mediated replicate mupA plasmids in various generations.[24,25] Conjugative plasmid A was identified as the most common factor of higher-level mupirocin resistance during US epidemic caused by *Staphylococcus aureus* strain USA300.[26,27]

Epidemiology:

Mupirocin resistance has been globally reported with different levels of resistance in different countries. In 2009-10, a survey was conducted in 23 hospitals of USA to find out mupirocin resistance in MRSA. 3% bacteria isolated from nasal carriage and 5% from blood cultures were identified as mupirocin resistant MRSA. Between 1995 to 2004, from the 32 hospitals of Canada, out of 4980 MRSA strain, 200 mupirocin resistance *Staphylococcus aureus* were identified.[28,29] The high level of mupirocin antibiotic resistance in *Staphylococcus aureus* bacteria reported in various countries in the various time span has been shown in table 1.

Laboratory detection of Mupirocin antibiotic resistance:

Conventional microbiological studies based on Agar dilution, broth micro dilution and E-test[30-32] are most commonly used laboratory techniques to detect minimum inhibitory concentration (MIC) threshold of mupirocin antibiotic.[30,33] Disc diffusion test for the assessment of high side mupirocin antibiotic resistance in *Staphylococcus*

Table 1: High level mupirocin resistance in *Staphylococcus aureus* prevalence rate reported among various countries

Sl. No.	Time duration of the survey	Geographical area of the survey	Mupirocin resistance among identified MRSA in specific survey	References
1	2009-10	23 hospitals in USA	3% from nasal Carriage and 5% from blood cultures	(Dadashi *et al.* 2020) (Eum *et al.* 2021).
2	2009-10	32 Canadian Hospitals	4%	
3	2006-2007	Ireland	3%	
4	2011-12	France	1%	
5	2011-12	Spain	13%	
6	2011-12	Turkey	47%	
7	2006-2009	England	5%	
8	2008-2011	USA	12%	

aureus bacteria was recommended by the international organizations namely European Committee for antimicrobial Susceptibility Testing (EUCAST), Clinical and Laboratory Standards Institute (CLSI), and British Society for Antimicrobial Chemotherapy (BSAC). The CLSI and EUCAST recommended 200µg discs to detect mupirocin resistance whereas 20 µg discs were recommended by BSAC. Thresholds and interpretation of mupirocin sensitivity test were recommended by EUCAST, CLSI and BSAC have been shown in table 2.

Table 2: Thresholds and interpretation of mupirocin susceptibility testing

Organization	Laboratory Method	Thresholds and interpretation		
		Susceptible	Intermediate	Resistant
European Committee for antimicrobial Susceptibility Testing (EUCAST)	Disc diffusion tablet: 200µg MIC	≥ 30 mm	18-29 mm	< 18mm
Clinical and Laboratory Standards Institute (CLSI)	Disc diffusion tablet: 200µg MIC	≥ 27 mm	7-26 mm	< 7mm
British Society for Antimicrobial Chemotherapy (BSAC)	Disc diffusion tablet: 20µg MIC	≤ 1 mg/L	2-256mg/L	> 256mg/L

Clinical implication and relevance of mupirocin antibiotic resistance:

Increased and inadequate use of mupirocin antibiotic to cure the various infections caused by *Staphylococcus aureus* bacteria have been resulted into the development of mupirocin antibiotic resistance. It has been well established that higher side mupirocin antibiotic resistant in *Staphylococcus aureus* bacteria resulted in the failure of mupirocin decolonization. However, lower level of mupirocin resistance was not demonstrated to inhibit mupirocin decolonization.[11,12,34] In Swiss, Brazilian and Dutch hospitals, mupirocin was broadly used for the eradication of MRSA due to hospital wide policies.[33,35-37] In New Zealand, during 1991 to 2000, Mupirocin was extensively used for the treatment of tissue infection at the end of this decade, 28% mupirocin resistant isolates of *Staphylococcus aureus* had been identified.[38] As a result of higher level mupirocin resistance in *Staphylococcus aureus*, Australian government's health department issued a guideline to regulate the use of mupirocin to minimum, which resulted in 0.3% reduction in resistance rate during the period of 4 years.[39] In some studies, it was found that lower level mupirocin resistant was not found insignificant with decolonization of MRSA carriage.[15,40-42] Overall, both higher and lower level mupirocin resistance in *Staphylococcus aureus* directly connected to fail with decolonization.[12]

Future perspectives:

In many cases, only the Real Time PCR method was effectively found to confirm mupirocin exposed staphylococcus aureus. It is cost effective to screen all the patients with this PCR based technique instead of culture-based technique.[43,44] Decolonization of all patients undergoing surgery would be another cost-effective approach better than PCR screening.[12] Noticeably, universal decontamination of all the patients is also unnecessary application of mupirocin, where approximately 80% of the patients are not carrying *Staphylococcus aureus*. This could again direct towards the resistance for *Staphylococcus aureus*. Therefore, mupirocin prophylaxis should be carefully considered before its application and mupirocin resistance statistics for *Staphylococcus aureus* should be considered. The recommendation of the international organization "European Committee for antimicrobial Susceptibility Testing" (EUCAST), intermediate threshold > 1 and ≤ 4 mg/L should be undertaken for the determination of resistance mechanism and clinical relevance in mupirocin exposed *Staphylococcus aureus* bacteria.

CONCLUSION

Large-scale use of mupirocin application to cure soft tissue infections, nasal decolonization and bed sores due to *Staphylococcus aureus* resulted in increased mupirocin resistance. Higher and lower level mupirocin antibiotic resistant strain is the basis of epidemiology of mupirocin resistance in *Staphylococcus aureus* which has enormously affected the health sector. Therefore, it should be focused on the use of mupirocin, closely monitored to avoid development of mupirocin antibiotic resistance in

Staphylococcus aureus bacteria in health care setup of the country. It is also suggested that, to treat soft skin tissue infection and bed sores, use of mupirocin should be avoided as much as possible and alternatives of mupirocin should be used.

DECLARATION OF INTEREST

Authors declared that they have no conflict of interest.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 13-18 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

A new nematode from Loktak Lake, Manipur

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Received : 13th November, 2022 ; Revised : 14th December, 2022

Abstract- During a taxonomic study of parasitic insect nematode, five mature medium-sized gravid female nematodes were recovered from the intestine of a host insect *Gryllotalpa africana* which was found inhabited on the floating phumdi (a thick mass of floating vegetation) of Loktak Lake (24° 32' 24" N and 93° 47' 37" E), Manipur, India. On microscopic examination the specimen exhibited many features which were not encountered in other species of the family. The present specimen possesses only two annules at the cephalic end unlike other species. Body is covered with thick, transparent and smooth cuticle. Oesophagus long consisting three parts-corpus, isthmus and rounded valvular bulb. Numerous oval eggs are present in the body. Tail conoidal with short caudal appendage. Basing on these unique characteristics, the present specimen is proposed as a new genus of the family Thelatomatidae Travassos, 1929.

Key words: Insect, *Gryllotalpa* sp., Nematode, *Gryllotum* n.gen., Loktak, Manipur

INTRODUCTION

Manipur belongs in Indo-Burma hotspot region having high diversity of species richness. The present host *Gryllotalpa* was collected from a floating vegetation called phumdi found at the middle of Loktak lake. The present Nematode is recovered from the gut of host insect. This specimen-nematode is very different from other nematodes reported from the terrestrial area. However, it shows some resemblance to two other genera of the family Thelastomatidae viz., *Periplaneticola* Basir, (1940,1942)[1,2] and *Isobinema* Rao, (1958)[3]. But, when studied in detail the present specimen is quite different from any other genera, not only from the two mentioned genera. Thus, on the basis of the notable characters exhibited by the present specimen, such as presence of two distinct cephalic annules surrounding the mouth, a distinct excretory pore having a protruded conical appearance, a thick and smooth cuticle, etc. make this specimen differentiated from other similar species. Therefore, in order to keep it in a distinct status, it is proposed to be recognised as a new genus.

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MATERIALS & METHODS

Many insects were collected from the different localities of the Manipur including Loktak lake. The insect hosts were brought to the laboratory in a ventilated plastic vessel in live condition and dissected in a petridish with normal saline solution. The nematode specimen recovered from intestine of the host insect *Gryllotalpa africana* were fixed in A.F.A. (Alcohol Formalin Acetate) for few minutes. The fixed nematodes were transferred to G.A. (Glycerine Alcohol) in a cavity block which are ultimately transferred in the desiccator for complete dehydration. After few days the dehydrated specimen were mounted on a drop of pure dehydrated glycerine on a slide, then covered with a coverslip and sealed with nail polish and then observed under the microscope. Diagram was drawn

with the help of camera lucida and measurements were taken with the help of ocular micrometer (figure 1). The methodology followed is the Seinhorst's (1959)[4] rapid glycerin method. All measurement values are given in mm, if otherwise not mentioned.

OBSERVATION

Generic Characters:

Body is long, slender, cylindrical. Mouth is surrounded by six elevated labial-papillae. Only the first and second annules are distinctly demarcated at the cephalic end. Body covered with a thick and smooth cuticle. Oesophagus has three parts-corpus, short isthmus and rounded valvular bulb. Excretory pore is distinctly protruded from body wall like a pyramidal model, open at a little below the oesophagus. Beginning of intestine is wider than the valvular bulb. Vagina directed anteriorly, opens just below the mid body. Gonad mono-prodelphic. Eggs oval shaped and densely arrange inside the body with a capsulated mass. Tail very short, constricted in the middle and symmetrically chiselled tip having a very short caudal appendage lies in the mid axis of posterior body.

DESCRIPTION

Female: Body is elongated, cylindrical, 3.57-3.81 long and 0.31-0.32 wide. Mouth surrounded by six elevated structures with labial-papillae. Cuticle is very thick and smooth, covering whole body. The body has only two distinct annules at the cephalic region, the first annule surrounds the mouth opening and followed by the second annule. Oesophagus, 0.38-0.43 long, having three parts-corpus, isthmus and valvular bulb. Corpus is tubular structure, 0.28-0.32 long and 0.03-0.04 wide. Isthmus is very short connects both corpus and isthmus, 0.02-0.03 long and 0.02-0.03 wide. Valvular bulb is rounded, thick, muscular and roughly distillation flask shaped, 0.09-0.10 long and 0.09-0.10 wide. Cardia is not seen clearly. Nerve ring encircles the corpus at 0.16-0.18 from the anterior end. Excretory pore is protruded and opens at 0.59-0.71 from the anterior end. Intestinal wall is thick at the beginning near the oesophago-intestinal junction but soon continued as very thin wall giving an appearance of expanded lumen. Then after the mid body region the intestine become narrower and continued as tubular

Table 1. Showing the comparatives description of present specimen with the other two species *Periplaneticola* Basir, (1940)[1] and *Isobinema* Rao, (1958)[3].

Characters	*Periplaneticola* Basir, (1940)[1]	*Isobinema* Rao, (1958)[3]	Present Specimen
Labial Papillae	Eight labial-papillae	Eight labial-papillae	Six labial-papillae
Annules	Nine small dense annules present at the cephalic end	Annulation completely is absent	Only two distinct annules are present
Excretory Pore	Excretory pore opens as a simple hole	Excretory pore opens as a simple hole	Excretory pore protruded distinctly like a pyramidal structure
Intestine	Intestine begins as broader and gradually narrower posteriorly	Intestine begins as similar wide with the valvular bulb	Intestine begins as similar wide of valvular bulb with a very thin wall and gradually broader posteriorly
Ovary	Two long chain of ovary present	Two long chain of ovary present	Only a long chain of ovary present
Polar cap	Present	Absent	Absent
Eggs	Eggs are arranged in capsulated mass and do not possess polar filament	Eggs are not arranged in beaded structure without capsulated mass, held together by polar filament	Eggs are arranged in capsulated mass without polar filament
Tail	Tail very short with small blunt caudal appendage which closer the dorsal side of tail	Tail elongated and conical structure with a long filiform caudal appendage	Tail is very short with small caudal appendage which comes downward just in the line of mid axis

structure up to the anus. Vulva is simple, opens below the midbody at 2.05-2.23 from the anterior end. Vagina directed anteriorly. Eggs are oval shaped surrounded by thin smooth shell, 0.05-0.06 long and 0.03-0.04 wide and arranged densely in the body. A long chain of ovary present. Anus is much closer to the posterior end of the body and opens at 3.45-3.66 from the anterior end. Tail is short 0.10-0.19 long, constricted in the middle, having a short caudal appendage, lies in the mid axis of posterior body.

Male**:** Male not found.

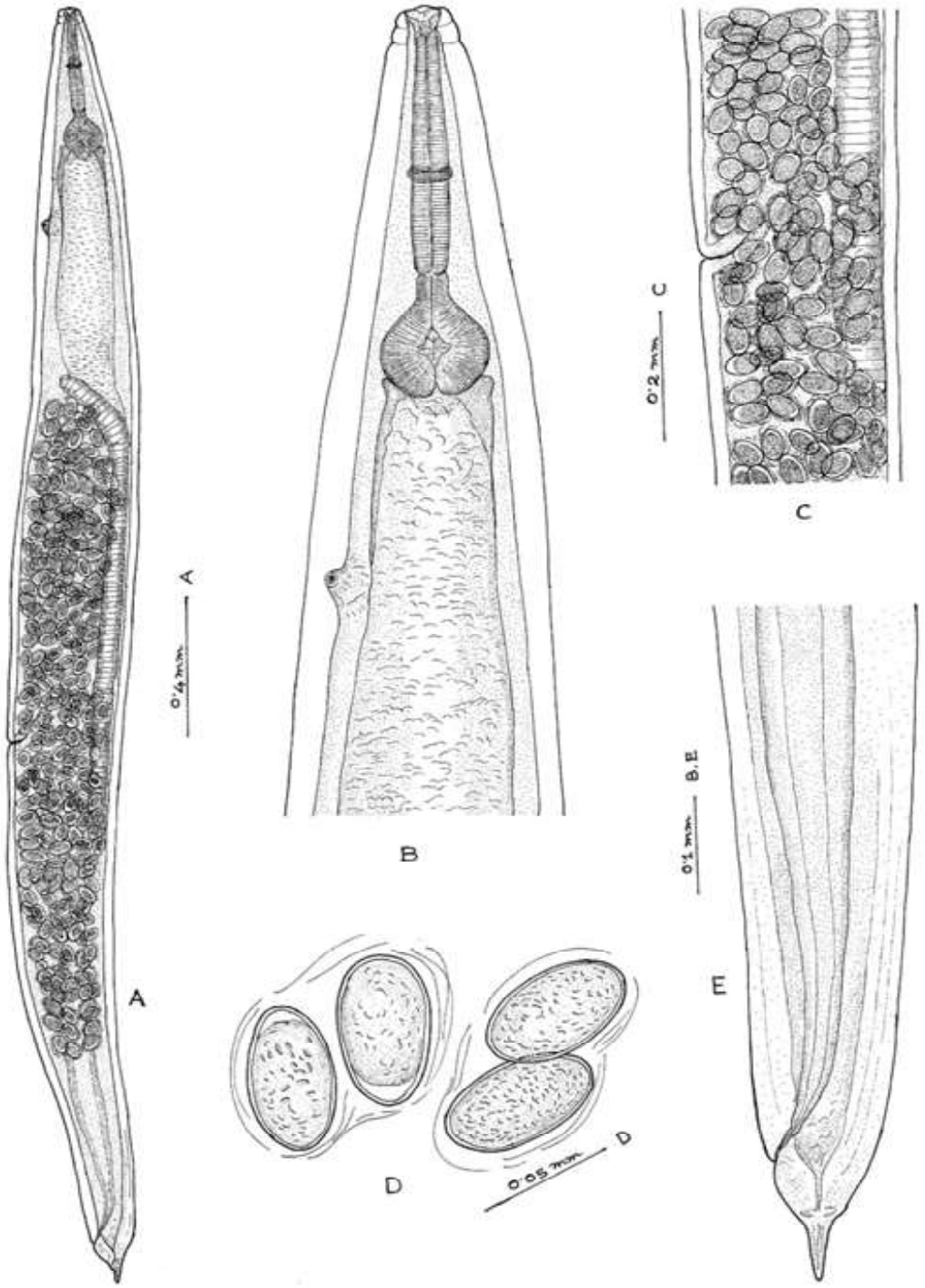
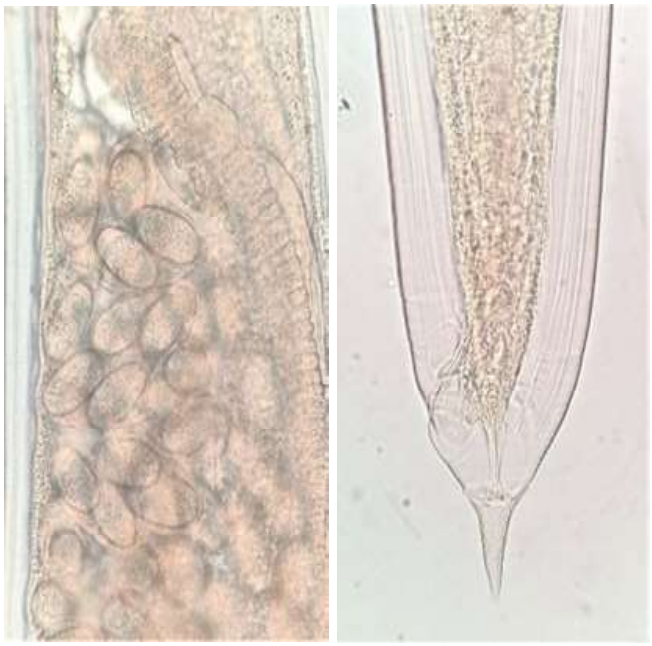


Figure-1: Camera lucida drawing of *Gryllotum loktaklica* n.gen. & n.sp. A. Entire Female; B. Anterior end; C. Vulval region; D. Eggs; E. Female posterior end.



A. B.



C. D.

Figure-2. Photomicrograph of *Gryllotum loktaklica* n.gen. and n.sp. A. Chephalic end showing prominent two annules, mouth part and entire oesophagus, B. Vulval region showing vaginal opening, C. Body showing ovary and numerous eggs, D. Posterior end showing short caudal appendage.

RESULTS & DISCUSSION

The present specimen belongs to the order Oxyurida and the largest family Thelastomatidae Travassos, (1929)[5] by the presence of cuticular wall, annulated cephalic end and posterior tail region, etc. Some more commonly occurring genera that belongs to the family Thelastomatidae, which are slightly resembling the present specimen are *Binema* Trvassos (1925)[6], *Blatticola* Schwenck (1926)[7], *Cameronia* Basir (1948)[8], *Cephalobellus* Cobb (1920)[9], *Gryllophila* Basir (1942)[2], *Hammerchmidtiella* Chitwood (1932)[10], *Isobinema* Rao (1958)[3]. *Leidynema* Schwenck (in Travassos, 1929)[5], *Linstowiella* Basir (1956)[11], *Periplaneticola* Basir, (1940,1942)[1,2], *Pseudonymus* Diesing (1857)[12], *Schwenkiella* Basir (1956)[11], *Thelastoma* Leidy (1849)[13], *Zonothrix* Todd (1942)[14]. However, when it is studied in detail many distinctive characters are come out, differentiating the present specimen from any known genera. Out of the fourteen genera mentioned the present specimen have some more closer resemblance with only two genera viz., *Periplaneticola* and *Isobinema* but in detail studies with the two genera and the present specimen have many distinguishing characters which are given below:

Comparison of present specimen with the genus *Periplaneticola* Basir, (1940,1942)[1,2] and *Isobinema* Rao, (1958)[3].

The mouth opening of the present specimen is surrounded by six labial-papillae (eight labial-papillae in *Periplaneticola* and *Isobinema*).

Body has only two large annules at the cephalic end. The first annule enclosed the mouth openning followed by second annule (body has nine small dense annules at the cephalic region of *Periplaneticola*, annulation is completely absent in *Isobinema*).

Excretory pore is protruded as pyramidal like outgrowth in the present specimen (it is opened as a simple hole in *Periplaneticola* and *Isobinema*).

Intestine begins with similar wide of valvular bulb having a very thin wall and become gradually broader posteriorly (it begins similar size with valvular bulb and gradually narrower posteriorly in *Periplaneticola* and *Isobinema*).

A long chain of ovary present in the present specimen (two long chain of ovary in *Periplaneticola* and *Isobinema*).

Eggs do not bear polar cap (polar cap present in *Periplaneticola*).

Eggs are arranged in capsulated mass as cluster in the present specimen and do not possess any polar filament (eggs are single but arranged in beaded like structure which are held together by polar filament in *Isobinema*).

Tail very short with a small caudal appendage which is beautifully chiselled just in the mid axis of the posterior body (tail elongated and conical structure with a long filiform caudal appendage in *Isobinema* whereas caudal appendage is very short which closer to dorsal side from mid axis of the tail region in *Periplaneticola*).

TAXONOMIC SUMMARY

Genus : *Gryllotum* n.gen.
Species : *loktaklica* n.sp.
Type Host : *Gryllotalpa africana*
Type Habitat : Posterior part of alimentary canal.
Locality of Host: Loktak Lake (24° 32' 24'' N & 93° 47' 37'' E), Manipur, India.
Etymology : The genus and species name are based on the name of host and place collection of the host, respectively.

CONCLUSION

Due to the above points of differences, the present specimen cannot be accommodated under any known genera of the family Thelastomatidae. Therefore, it becomes necessary to accommodate the present specimen and proposing a new genus and giving a name *Gryllotum* n.gen with *loktaklica* as its type species. The generic name *Gryllotum* n.gen. is derived from the name of the host. The species name *loktaklica* n.sp. is given after the name of Loktak Lake from where the host was recovered.

ACKNOWLEDGEMENT

The authors wish to thank the Head of the Department of Life Sciences (Zoology), Manipur University, for providing necessary laboratory facilities. The first author shows gratefulness to the UGC, Government of India, for extending some financial assistance under Rajiv Gandhi National Fellowship, during the research work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 19-24

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Aquatic bird diversity in the Badwai Pond of Chittorgarh District of Rajasthan

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Received : 13th November, 2022 ; Revised : 14th December, 2022

Abstract- Birds are the best indicator of environmental changes. The richness of bird diversity indicates the richness of habitat. Our survey was carried out to monitor aquatic bird diversity during the winter season when various migratory bird species arrive in Badwai pond. In this study, around 50 species were identified. Among these 3 species reported as threatened category, 5 species at the verge of threatened category and the rest come in a less concerned category. Identified species namely Bar-headed goose, Common Coot, Greater Flamingo, Common Pochard, Sarus crane, Eurasian Spoonbill, Black Ibis, etc. Some factors like climatic variability, the productivity of the environment, food availability, geographical status, habitat, invasive species, anthropogenic activity, etc. play an important role in species diversity and richness. The presence of different types of bird species depicts the richness of pond productivity. By taking some conservative steps this place can be developed as a better wetland.

Key words: Badwai, threatened, less concerned, richness, habitat, conservation

INTRODUCTION

Birds that are ecologically dependent on wetlands are broadly defined as water birds.[1] These include various categories like seabirds, diving birds, shorebirds, waders, aquatic raptors, waterfowls, migratory birds, etc. Changes in their population, behavior, and reproductive ability are mostly used to study the long-term effects of habitat changes.[2] Birds that migrate from one habitat to another to breed and feed are called migratory birds. Some birds migrate during the breeding season; some migrate on the availability of food resources and some on the change of season.

Approximately 1300 species of bird are found in the Indian subcontinent, among these around 310 species are wetland-dependent birds found in India, and half of these are migratory birds.[3] Wetland birds contain about 10% of the globally threatened species and 20% of Asian threatened species.[4] Many species are close to extinction through disturbance or loss of their habitats, as well as through intensive searching pressure.[5] The majority of the wetland species within the Indian subcontinent are not unusual and a quarter (24%) is rare. Of which, 34 are globally threatened species, 34 are critically endangered and one is conservation structured.[6]

Chittorgarh is well known as a city of pride and honor and the best tourist destination in the country. There are many water bodies in the Chittorgarh district, out of which Badwai and Kishan Kareri ponds are very famous for their enriched habitat and biodiversity as many water bird species migrate here in different seasons. This is the first scientific study of the bird diversity at Badwai pond to understand the present status and the conservation measures that need to be done in future years.

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Study Area



Figure 1: GIS map of study area created by using google earth application



Figure 2: Camera picture of studied water body

The studied water body Badwai pond is situated near Badwai village which comes in the Chittorgarh district. This pond is located on the boundary line of Udaipur and Chittorgarh district, which is about 65 km from Udaipur and about 70km from Chittorgarh at 24º29'34" N and 74º15" 06" E at the elevation of 481 m with a perimeter of around 2.41 km and surface area of around 0.36 km². Rainwater is the main source of water for such vegetation and avian diversity. There is a man-made island in the middle of the pond, on which various types of vegetation are there, which are used as food and as a residence by different bird species.

MATERIALS & METHODS

The study was carried out for a period of 4 months from December 2021 to March 2022. The data was collected by continuous observation and bird watching by walking on fixed routes. Almost all possible places were thoroughly searched for birds and birds were identified with the help of a colored field guide. The maximum number of aquatic birds were observed during the morning (7:00 am to 10:00 am) and evening (4:00 pm to 6:00 pm) hours. On most occasions, the pond and surrounding area were surveyed during the day hours. The birds were identified through naked eyes and with the help of binoculars (10 x 50) and digital cameras (Sony alpha 58) were used to capture the pictures of birds. The information like bird species, their behaviors, their numbers, and other activities was written down in a notebook. The status of birds as residential or migratory has been denoted by R and M respectively. Similarly, according to the red list category of the International Union for Conservation of Nature[7] (IUCN), the status of birds has also been denoted as vulnerable (VU), near threatened (NT), and least concerned (LC) category.

RESULTS & DISCUSSION

In the study total number of 52 aquatic bird species were identified which belong to 16 families. The species of family Phasianidae, Anatidae, Podicipididae, Pheonicopteridae, Ardeidae, Ciconiidae, Threskiornithidae, Pelecanidae, Cruidae, Burhinidae, Recurvirostridiae, Charadriidae, Scolopacidae, Laridae, Acledinidae, Rallidae were found at the study site.

Nearly 250 species of birds are known to be highly dependent on freshwater habitats out of which a large proportion (60%) belongs to a single family Anatidae, which comprises ducks, geese, and swans.[8] Wading birds such as sandpipers (Scolopacidae), plovers (Charadridae), herons and bitterns (Ardeidae), and another diverse assemblage of families are also associated with inland freshwater habitats.[9]

There are many trees around this pond and a man-made island right in the middle, on which these birds get shelter and utilize it as a perch site. Moreover, the factors such as food source availability, an abundance of

zooplankton, and phytoplankton in the water, amount of vegetation, natural habitat, predators, seasonal changes, less anthropogenic activities, etc. affect adequate bird diversity in any waterbody, therefore this pond and the surrounding area fulfill all of these, hence the diversity of all these types of bird species is seen here.

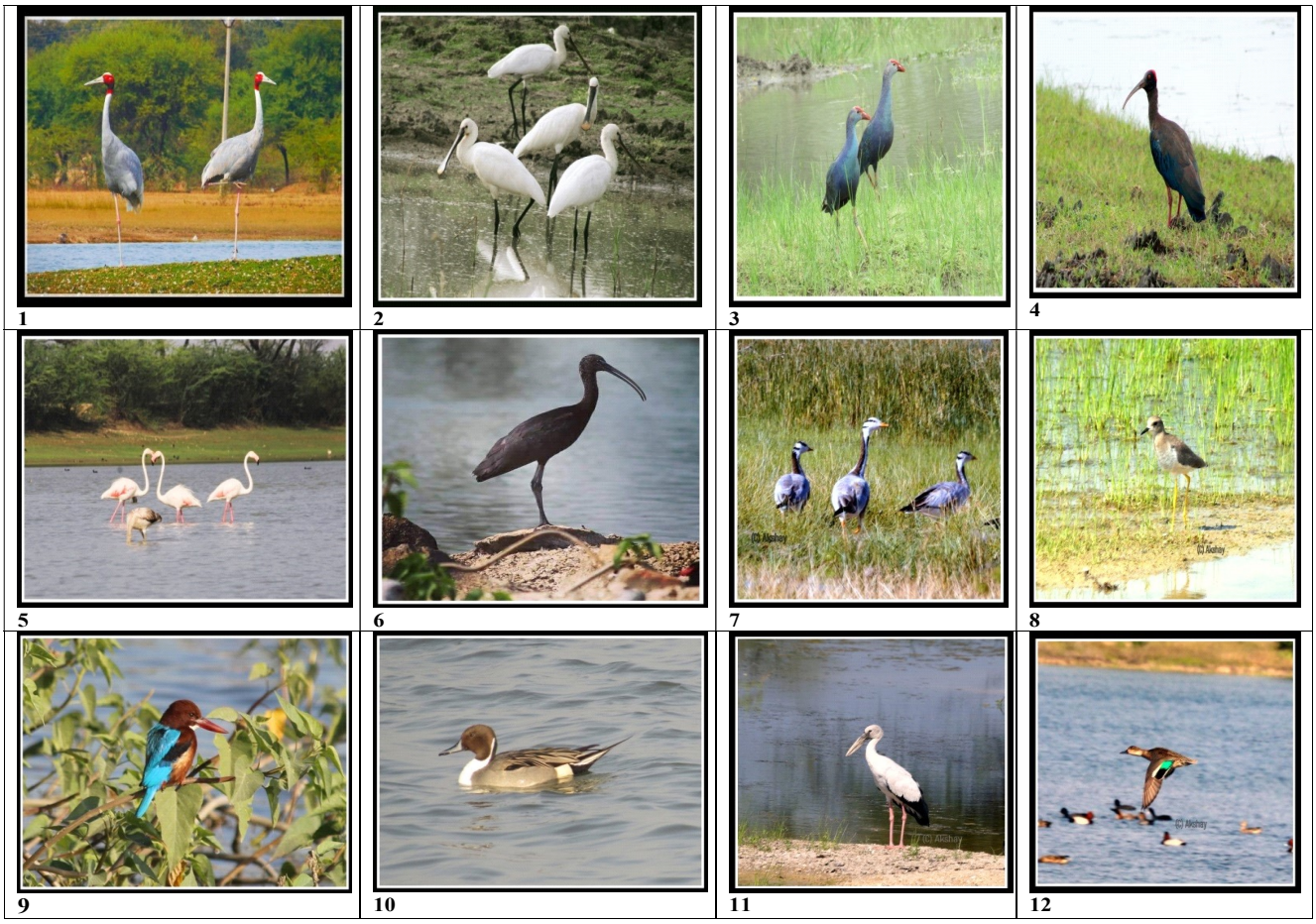


Figure 3:- 1. Sarus Crane; 2.Eurasian spoonbill; 3.Western Swamphen; 4. Red Naped Ibis; 5. Greater Flamingo; 6. Glossy Ibis; 7. Bar-headed Goose; 8. White Tailed Lapwing; 9. White Throated Kingfisher; 10.Pintail; 11. Asian Open bill; 12.Green Winged Hen

Table 1: Status of Identified Aquatic Bird at Badwai Pond

S.No.	Common Name	Scientific Name	Status (R/M)	IUCN Status (VU/NT/LC)
Family 1:Phasianidae				
1	Grey Francolin	*Francolinus pondicerianus*	R	LC
Family 2:Anatidae				
2	Bar-headed goose	*Anser indicus*	M	LC
3	Ruddy shelduck	*Tadorna ferruginea*	M	LC
4	Gadwall	*Anasstrepera*	M	LC
5	Northern shoveler	*Anas clypeata*	M	LC
6	Northern Pintail	*Anas acuta*	M	LC
7	Common Teal	*Anas crecca*	M	LC
8	Eurasian Wigeon	*Anas penelope*	M	LC
9	Common Pochard	*Aythya ferina*	M	**VU**
10	Tufted Duck	*Aythya fuligula*	M	LC

Family 3:Podicipididae				
11	Little Grebe	*Tachybaptus ruficollis*	R	LC
Family 4: Pheonicopteridae				
12	Greater Flamingo	*Phoenicopterus roseus*	M	LC
Family 5:Ardeidae				
13	Little Egret	*Egretta garzetta*	R	LC
14	Cattle Egret	*Bubulcus ibis*	M	LC
15	Intermediate Egret	*Mesophoyx intermedia*	R	LC
16	Great Egret	*Casmerodius albus*	M	LC
17	Indian Pond Heron	*Ardeola greyii*	R	LC
18	Grey Heron	*Ardea cineria*	R	LC
19	Purple Heron	*Ardea purpurea*	M	LC
Family 6: Ciconiidae				
20	Painted Stork	*Mycteria leucocephala*	M	NT
21	Asian Openbill	*Anastomus oscitans*	M	LC
22	Wooly Necked Stork	*Ciconia episcopus*	M	NT
Family 7: Threskiornithidae				
23	Glossy Ibis	*Plegadis falcinellus*	M	LC
24	Black Ibis	*Pseudibis papillosa*	M	LC
25	Black Headed Ibis	*Threskiornis melanocephalus*	M	NT
26	Eurasian Spoonbill	*Platalea leucorodia*	M	LC
Family 8 : Pelecanidae				
27	Great White Pelican	*Pelecanus onocrotalus*	M	LC
Family 9: Gruidae				
28	Sarus Crane	*Grus antigone*	R	VU
Family 10: Burhinidae				
29	Great Thicknee	*Esacus recurvirostris*	R	NT
Family 11: Recurvirostridiae				
30	Black Winged Stilt	*Himantopus himantopus*	M	LC
Family 12: Charadriidae				
31	Little Ringed Plover	*Charadrius dubius*	M	LC
32	Kentish Plover	*Charadrius alexandrinus*	M	LC
33	Red Wattled Lapwing	*Vanellus indicus*	R	LC
34	Yellow Wattled Lapwing	*Vanellus malabaricus*	R	LC
35	White Tailed Lapwing	*Vanellus leucursus*	M	LC
Family 13 : Scolopacidae				
36	Common Snipe	*Gallinago gallinago*	M	LC
37	Black Tailed Godwit	*Limosa limosa*	M	NT
38	Spotted Redshank	*Tringa erythropus*	M	LC
39	Common Sandpiper	*Actitis hypoleucos*	M	LC
40	Marsh Sandpiper	*Tringa stagnatilis*	M	LC
41	Green Sandpiper	*Tringa ochropus*	M	LC
42	Wood Sandpiper	*Tringa glareola*	M	LC
43	Ruff	*Philomachus pugnax*	M	LC
44	Little Stint	*Calidris minuta*	M	LC
Family 14: Laridae				
45	River Tern	*Sterna aurantia*	R	**VU**
Family 15: Acledinidae				
46	Common Kingfisher	*Alcedo atthis*	M	LC
47	White Breasted Kingfisher	*Halcyon smyrnensis*	R	LC
48	Pied Kingfisher	*Ceryle rudis*	R	LC
Family 16: Rallidae				
49	Common Moorhen	*Gallinula chloropus*	R	LC
50	White Breasted Waterhen	*Amaurornis peoenicurus*	R	LC
51	Common Coot	*Fulica atra*	R	LC
52	Grey Headed Swamphen	*Porphyrio poliocephalus*	R	LC

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 25-34

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Socio-economic condition of fishermen community in Patna District, Bihar: Present status and probable strategies for improvement

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Received : 18th January, 2023 ; Revised : 18th February, 2023

Abstract- The present study was conducted along the 39 Ghats from Digha to Didarganj in the Patna District of Bihar, with an objective to explore the demographic status of local fishermen communities, their academic, hygienic and health status, availability of the basic infrastructural facilities for livelihood, fish culture practices, marketing facilities and role of fisher in fisheries development. First of all, a list of various Ghats located along the bank of river Ganga between Digha to Didarganj in Patna was procured from the district fisheries department and concerned Thana, and then fish Ghats were surveyed for the demographic details and the socio-economic condition of the fishermen community residing there. For this the questionnaire was designed to get information on the demography, socioeconomic profile of fisher, the major fisheries activities performed by them, and to find out the problems encountered by the fish farmers for adoption of scientific fish farming practices. The primary data were collected directly as of the sampled respondents (randomly selected 101 fishermen and women) by visiting their location a number of times to get necessary relevant information like their family structure, their assets, occupation, investment, expenditure incurred in fish farming etc. with the help of precise and pre-tested schedule-cum questionnaire. The response sheets were systematically studied and data were generated separately for family structure, demography, academic status, health & hygiene, socio-economic status, availability and implementation of government schemes for the fishermen community, any positive intervention of non-government organization and cooperative societies like COFFED for the upliftment of their socio economic status etc. Majority of the fisher were middle aged, poorly educated, living in medium size joint family with low income and poor economic status. Most of the fisherwomen had poor extension contact and low level of mass media exposure and majority of the fishermen had a low level of knowledge, less social participation and they were traditional in their outlook. The present study revealed that large family size, non-availability of fishing gear/net, scarcity of opportunities for alternate employment etc. are the main cause of poverty of fisher families in the area. The findings of the present study may be critically analysed by the technocrats, government officials and state fishery department to decipher an amicable way for the benefit of the fishermen community of the state and to develop strategies for upliftment of their socio economic status in the state. It will definitely help in planning, development and execution of fishery policy and programme by the planners and administrators, policy makers and fishery extension educationists in a more meaningful and scientific manner.

Key words: Socioeconomic status, demography, fishermen community, Patna, strategies, upliftment.

INTRODUCTION

Ponds, tanks, small reservoirs, rivers and water logged areas like ox-bow lakes and chaurs etc. principally constitute the fisheries resources in the state of Bihar.[1,2] Around 65,000 ha of water areas are covered by ponds and tanks and nearly 35,000 ha of water areas consist of ox-bow lakes and chaurs.[1] At present fish production in Bihar doubled in last 10 years due to sustained efforts made

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by the state government of Bihar. As per the state fisheries department, Govt. of Bihar, the total fresh water fish production in Bihar in 2004-05 was 2.68 lakh metric tons which has gone up to 5.7 lakh metric tons in 2016, 6.83 and 7.62 lakh metric ton in 2021 and 2022 respectively. The data clearly reveals that an annual dividend of average 3.09 lakh metric ton is being added yearly since Mar 1999 to 2022. Fisheries sector plays a predominant role not only in terms of contributing to our food basket, but also acts as a cheap source of animal protein supporting our rural health, resource potential and employment opportunities. Most importantly, fishery is a source of livelihood for a large section of the economically backward population.

Fish farmers are the backbone of fish culture and they play a very important role in fish culture. The livestock production potential is based on the socio economic factors of the farmer,[3] which play a vital role in the adoption of improved technologies.[4] To a great extent, the socio-economic background of people determines their habits and attitudes, which shape their perceptions on entrepreneur activities.

The investigations on socio-economic status of fish community and fish farmers have been carried out in different states of India by different workers.[5-14] Somashekar and Shashikanth (2020)[15] surveyed the socio economic profile of fish farmers in Shivamogga districts of Karnataka. The most important impediments of fisheries progress in Bihar is need of knowledge about scientific fish culture, illiteracy and disorganized extension linkages amongst fishermen community. On account of these, bulk of fish farmers is still involved fish culture in traditional method. There are around 1.5 crore fishermen in Bihar. The economically backward community does not own cultivable land, and fishing is the only means of livelihood for them There is need for creating the knowledge about the emerging development in aquaculture technology in the fishermen society. It is essential to systematize short term training for rural youth with importance on learning for creation of self-employment opportunities and socio-economic progress of fishermen community. Mechanization needs to be enhanced substantially in order to meet the recommended level and to enhance the productivity.[16] Mechanization has been defined as the use of improved hand and animal operated tools.[16] Mechanization of system gives better output, as well as serves as a cost recovery option to minimize or recover operational investments, and is mostly eco-friendly.[17] Study depicted the pitiable situation of fish marketing practices in Bihar, which are based on exceedingly traditional patterns, unaffected and unfound elsewhere.[18] The review of literature clearly revealed that in recent years, the systemic detailed study on the socio- economic status of the fishermen community of Patna from Digha to Didarganj has not been carried out particularly in context of their demography, educational status, health and hygiene along with the assessment of government and NGO's involvement for the up-liftment of their economic status. Hence, the present study has been designed to assess and analyze the actual socio-economic status of the fishermen community of Patna (Bihar) and to design and recommend some of the strategies to improve the quality of their life.

MATERIALS & METHODS

The study area is located in between 250.63'83" N and 850.10'050 E at Digha Ghat, Patna to 250.36'53.6" N and 850.09'10.2" E at Didarganj Ghat, Patna. A survey was conducted for the period from September 2022 to January 2023. A total of 110 fish farmers from 39 Ghats between Digha to Didarganj locality of Patna district in Bihar were selected randomly and they were interviewed at their respective riverine sites (Ghats) to collect the first hand information with a pre-tested structural questionnaire. The selected sites were-Nasariganj Ghat; Shiva Ghat; Patipul Ghat; Minar Ghat; Bind Toli Ghat; Digha Ghat; Kurji Ghat; LCT Ghat; Baans Ghat; Collectorate Ghat; Club Ghat; Anta Ghat; Mahendru Ghat; Adalat Ghat; Mishri Ghat; T.N Banerjee Ghat; Kali Ghat; Bansi Ghat; Kadam Ghat; Krishna Ghat; Gandhi Ghat; Baharwa Ghat; Law College Ghat; Rani Ghat; Gulbi Ghat; Ghagha Ghat; Roushan Ghat, Sultanganj; Pathari Ghat; Chaudhary Tola Ghat; Koyala Ghat; Alamganj Ghat; Loharwa Ghat; Hanuman Ghat; Gosai Ghat; Raja Ghat; Gai Ghat; Bhadar Ghat; Mittan Ghat; and Kangan Ghat, Patna City. A well-read questionnaire having all detailed information related to demography, age, family size, health and hygiene status, academic background, basic amenities, occupation, socio-economic status, government and NGO's facilities, procurement of fishing crafts, nets, gears, package and practices of fish culture, operational economics were studied in terms of personal interview as well as response sheet made by them in the questionnaire. The data were also collected based on local availability of different species

of fishes under capture as well as culture fishery, agricultural practices associated with aquaculture, mechanization gap and available mechanization status. The response sheets were systematically studied and data were generated separately for family structure, demography, academic status, health & hygiene, socio-economic status, availability and implementation of government schemes for the fishermen community, any positive intervention of non-government organization and cooperative societies like COFFED for the upliftment of their socio economic status etc.

QUESTIONNAIRE RELATED TO A STUDY ON SOCIO- ECONOMIC STATUS OF FISHERMEN COMMUNITY IN PATNA (BIHAR)

Demography:

Name- Age- Sex-

Address-

Adhaar no.- Wife - Age-

Total No. of Family Member:

Male- Female-

No. of Children -

Male- Name- Age-

Female- Name- Age-

Academic Qualification:

Fisherman - Wife -

Academic Evaluation of Children-

Nearby Government School-

Facilities at Government School

Books - Copies -

Academic Facilities -

Ward Commissioner -

Government Facilities-

Tap Water - Electricity -

Public Health Center (PHC) - Ration Card -

Types of Ration Provided -

L.P.G- Subsidiary-

Eradication of Mosquitoes -

Drainage Cleaning -

Spraying of Bleaching Powder -

(How Many Days) -

Municipal Disposal of Waste -

No. of Toilets -

Gents -

Ladies -

Medical Status -

Health Status -

Male - Female - Children -

Facilities made available by the Government -

Public Health Center -

Free Medicine -

Help from Fisheries Department -

Concept of Cooperative Societies -

Socioeconomic Status -

Self- Boat -

Hired Boat -

Fishing Work as Labour Fisherman -

Fishing Without Boat -

Major Fish Catch -

Catch Statistics -

Fish Marketing -

Earning (Monthly) -

Any Idea of Government Plan of Fisheries Department -

Supply of Net -

Supply of Boat -

Cooperative Societies -

Any idea of NGO like

COFFED / COMFED -

Thumb impression / Signature

Statistical Analysis

The percentage estimation and statistical calculation have been done as per standard methods.

RESULTS & DISCUSSIONS

Traditionally, Bihar's economy is dominated by rural sector. Around 88% of the population live in the rural areas. Ganga is the main river which is joined by tributaries such as Ghaghra, Gandak, Burhi Gandak, Bagmati, Kamla-Balan, Kosi and Mahananda. The species of major and minor carps found in the Ganges river system are extensively used for culture in ponds, lakes, reservoirs and enclosures throughout the country.

The state is endowed with adequate freshwater resources. The state's average rainfall of 1091 mm is considered to be adequate; however, owing to change in climate over the years, droughts and floods are adversely affecting the agriculture, fisheries output and GSDP. Capture and culture of fish are traditional activities in Bihar. Both traditional communities and new entrepreneurs are engaged in such activities.

Capture fisheries resources are evident in vast span of aquatic ecosystems like rivers, reservoirs and flood prone

wetlands in the form of ox-bow lakes and chaurs. Ox-bow lakes are cut off segments of rivers and popularly known as Mauns. These water bodies usually get connected with the main course of the river during floods or rainy season. However, some ox-bow lakes have permanently lost their connection with the river of their origin. The state has also extensive network of irrigation canals which retains water for considerable period and offer opportunities for aquaculture through cage and other enclosure based aquacultural practices. However, this potential is still lying untapped even at national level. Besides, vast area of lands adjoining the canals remains often water logged for several months which could also be developed for aquaculture and fish seed production. Fisheries resources of the state such as reservoirs are common public property which provide productive source of livelihoods for a large number of resource poor landless and marginal farming communities, especially traditional fishing communities. Chaurs, on the other hand, are largely multi-ownership based resources which offer an additional crop of fish during the period of water availability. So far, this resource is lying unutilized.

Aquaculture is one of the hopeful, fast growing foods farming area of Bihar. It positions 12th biggest fisheries resources in India and engaged 3rd place in inland fish production and 6th in fresh water seeds production within India.[19]

Demographic Status:

Age and Family size: During study, it was found that the age of fishermen surveyed varies from 20 to 60 years or more. The family size of fisher family was categorized into three classes like small, medium and large size family. Based on study, it was found that maximum family, nearly 44.48% were considered to be large family, which was composed of 7 or more members, followed by 37.29% Medium sized family, composed of 5 to 6 members. About 19% of the surveyed families were small family i.e., composed of 2 to 4 members.

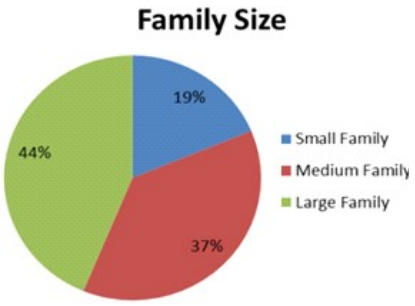


Fig. 1: Family size of the fisherman of Selected Area

Occupational status: Based on the data collected, it has been observed that capture fishing practice is the more common activity of the people residing in the adjoining area of Gangetic Deltaic region between Digha to Didarganj in Patna. The main region behind it is the presence of maximum families belonging to fisher community. However, besides fishing some of them were also engaged in agricultural activities such as cereal crops and seasonal vegetable cultivation apart from fishing practices. Some of them were working as a labour under private as well as governmental project besides fishing. They performed labour work either privately from some social construction work or engaged themselves in any project running under government like Manrega etc.

Data shows 34.44% people of these Ghats were only concern towards fishing practice either capture or culture. About 20.66% of people were working as vegetable seller including fishing practices. Fishing and agriculture practice is done by 29.50% people of this region. Rest remaining about 15.50% people of this region were fisher as well as working as labour. From the data collected from the fisher families, it is revealed that the highest production of fish and benefit from the same was found from captured fisheries practice. It was also observed that production increased from last decade. During study, it was found that the age of fishermen surveyed varied from 20 to 60 years or more.

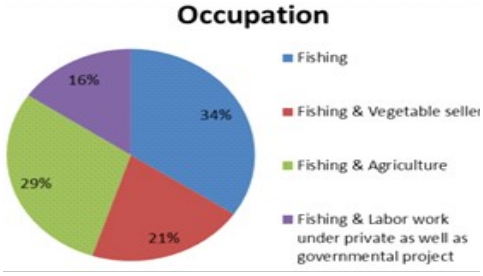


Fig. 2: Occupation of people (Generally fisher families were joint families).

Literacy status: Data collected from survey showed that the maximum number of individuals (around 43%) were illiterate and amongst them maximum belong from old generation), even some fisher has never gone to school but somehow they had learned to sign only (around 22.50%). Some of them have studied up to primary level (around 24.25%) and hardly a few individual had gone up to 6th standard (around 10%). The maximum number associated to the categories going for study above 6th standard was of young generation. Young generation came

forward for study. This happened only due to the several attempts made by Government of Bihar.

However, a sharp increase in the number of higher secondary pass fishers were noticed in contrast to the previous census. In nearly ten fishermen families a dozens of graduates were also reported. But nearly 80 of them were engaged in some other professions than the fishery practices. Only three post graduate individuals were spotted in the entire fishermen community of the surveyed area. Out of 39 Ghats surveyed only ten Government primary and six middle schools were located. Even the books, copies and other amenities made available by the Government to the fishermen children were not satisfactory. Although there was provision of mid-day meal (MDM).

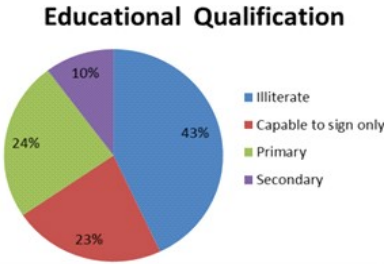


Fig 3: Educational Qualification details

More than 70% of fisher families had their own fishing gear, the rest of same had to depend upon their neighbour, retailer and wholesaler for fishing gears resulting low income.

Basic amenities/ facilities available:

Survey was also conducted for the enquiry of living condition of fisher family. The condition of housing was categorized as:

1. House with straw roof (nearly 10.20%)
2. House with tin roofing (nearly 24.95%)
3. Semi pacca house with tin roof upto base of the window (nearly 35.43%) and
4. Pacca house with concrete floor (nearly 30%).

The survey states that the most of the people had semi-pacca house and pacca house.

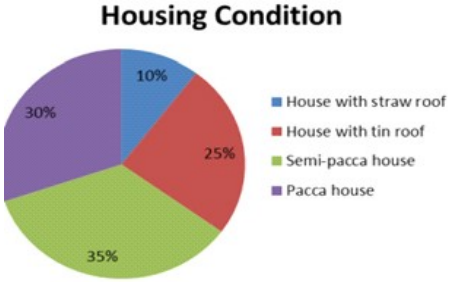


Fig. 4: Housing condition of Fishermen community

Health and hygienic Status:

Besides, some basic amenities like municipal tap water facility, common Boring hand pump, Sulabh Shauchalaya, community hall etc. were found to be satisfactory. Majority of the fish farmers had their own ration card and they were consistently receiving the allotted quota of their ration regularly (approx. 16 kg to 40kg).

Although, improper drainage system, sewage waste disposal, unorganized crematorium, abundance of pigs and dead body eating dogs, disposal of carcases at majority of Ghats have made them unhygienic for the fishermen communities. Sanitary condition of the fishermen locality was not satisfactory. Nearly 40% of the members of the fishermen community reported about their own toilets, while 40% were utilizing paid public toilets, rest 20% of the community including women defecate in the open field, predominantly on the impediments of river Ganga.

A detailed survey on the health condition of the fishermen, women and the children including adolescent boys and girls were considered. It was found that nearly 40% of the fishermen and women of age group 50-60 years were anaemic. Although the haemoglobin content of individuals of age group 12-45 years was quite satisfactory. Not a single case of Malaria was reported by the individuals. However nearly 17.21% aged fishermen as well as few women showed the filarial symptoms. Majority of the fishermen and women of age group 45-65 years were quite unknown about the diabetes. Although, nearly 28% of them reported the symptom of pre diabetic and diabetic condition. The disorders related to prostate gland and kidney were mostly reported in elderly fishermen. Any report of CKD was not declared by the fishermen community. Although, a clinical diagnosis was suggested for a few of them. Nearly 35% of the fishermen and women were diagnosed with hypertension. Surprisingly, 60-70% of the patients had never visited a physician nor taken any hypertensive drug. Nearly 5% of the fishermen were quite aware of their cardiac problems and nearly 80% of them were under proper medications. Respiratory diseases like asthma etc. were found to be more common among the aged fishermen. However, among females it was prevalent in those families were still foods are cook on wood and coal. Nearly 1.05% known cases of tuberculosis in the fishermen community were also reported. ENT disorders were reported from elderly fishermen. Some fishermen also reported the symptoms of vertigo and insomnia. Besides, fragmentary

complains of acute gastritis, liver cirrhosis and even undetected hepatitis were also made by the fishermen. Nearly 0.68% fishermen were known cancer patients and they were under chemotherapy. Nearly 32% of the children were reported to be suffered from viral fever and cough and cold at one or more times every year. Data clearly reveals that the overall hygienic condition was found to be poor and the health status was also not good. The medical facilities available in the local public health centres was found to be unsatisfactory. It seemed to be far away from the reach of the local fish farmers. Provisions of free medicines supply to the fishermen were not executed properly. Surprisingly, various schemes of the state fisheries department, meant for the welfare of the fishermen community were not known to the local fish farmers, which is evident from their response sheet. Even the fishermen community were not at all acquainted with the various schemes and policies of the fish cooperative societies like COFFED and COMFED etc.

Daily working hour of fishermen was also collected through survey. The working hour of fisherman was divided in three different categories as 2-3 hr per day, 4-6 hr per day and almost whole day based on majority of people.

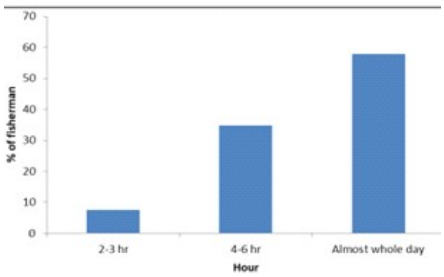


Fig. 5: Daily working hour percentage of Fishermen Community

Result obtained from data collected clearly showed the working hour percentage of fisherman as 7.50%, 34.75% and 57.75% for 2- 3 hr per day, 4-6 hr per day and almost whole day respectively. The income of fisherman was calculated by averaging the whole month income through all sources.

Economic status of the fishermen community:

Based upon the personal interview and the data submitted by the fishermen in their response sheet, it was evident that nearly 50-60% fishermen's monthly income was in between Rs 5000-6000, with an average daily income of Rs 150-200 only. 15% of the community reported their monthly income about Rs 9000-10000 (with a daily average of Rs 300-350), 10% of the fishermen have shown their monthly income of about Rs 12000-18000 (with an average of Rs 500-600 daily). 10% of the community members have reported their monthly income of Rs 20000-30000 (with an average of Rs 1000 daily), while 5% of the fishermen reported their monthly income about Rs 40000-50000 (with an average of Rs 1500-2000 daily). There was found to be direct correlation between the possession of nets and crafts and the catch statistics of the fishermen. It was found that only 20% of the fishermen had their own boat and nets. Rest 40% of them used to hire the boat and paid the monthly rent to the owners and the money lenders, while 30% used to work as daily labourers on the boat and worked as professional fish catchers on paid basis and 10% of the fishermen are found to be engaged in the profession other than fishing. The fish catch statistics at various Ghats varied considerably and it ranges in between a catch of 3-4Kg/day/fisherman to 15-20Kg/day/fisherman. The major fishes caught by the fishermen along the 39 Ghats were *Labeo rohita* (rohu), *Labeo catla* (Catla), *Cirrhina mrigala* (naini), *Cyprinus carpio* (common carp), *Hypothalmichthys molitrix* (silver carp), *Etenopharyngodon idella* (grass carp), *Pangasius bocourti* (bachwa), *Puntius chola* (pothiya), *Mystus* spp. (tengara), *Heteropneustes fossilis* (singhi), *Clarias batrachus* (mangur), *Mastacembalus armatus* (gaichi), *Macrognathus aculeatus* (pataya), *Fenneropenaeus indicus* (Indian prawn), *Channa* spp.(garai), *Pangasius pangasius* (piyas fish) etc. This result shows that the economic condition of the people in the adjoining region of Digha to Didarganj is moderate and their only source of livelihood is capture fishery in Ganga-river and they did not have any additional water body for their fishing activity. Secondly, educational progress is not satisfactory, which is considered as the most necessary requirement for the overall development to fight against social injustice, because education play a crucial role regarding economical as well as technical information about their livelihood.[20] Being uneducated the fisher families are demoralized by the middlemen.[21,22] Thirdly, the studied areas are located in remote area and due to lack of nearby market; they are fully depending on the wholesalers and middlemen for their fishes to be sold. Due to this they did not obtain the authentic price that they should get for the fishes. This difference of selling fishes at a low price against purchasing household requisite at elevated price contributes to the poor condition of the

people in terms of economy.[23] However large family size, non-availability of fishing gear/net, scarcity of opportunities for alternate employment etc. are the main cause of poverty of fisher families.

Suggestive measures for upliftment of present socio-economic status of fishermen in Bihar:

Based on the primary data obtained through rigorous survey of socio economic status of fishermen community at 39 different Ghats along the River Ganga from Digha to Didarganj, Patna city and the findings of the present study, the subsequent suggestive measures are urged for adopting and implementing measures to eradicate the drawbacks and the causes of the poor socio-economic conditions of the fishermen in Patna for proper development of the fisheries sector in Bihar.

1. The bulk of the fish farmers are illiterate. It is highly desirable that they must be made literate, so that they can fight out the adverse scenario judiciously.
2. Fishermen from scheduled castes and scheduled tribes ought to tend special coaching and incentives to require to the profession of fish catching. Now a day the number of scheduled castes and scheduled tribes within the sphere of fish catching is nearly negligible. However, in particular area committees have been formed by sahani people, known as Matshay Jeevi Committee, which is also registered under Bihar government. In terms of fishing, sahani community as well as schedule caste and schedule tribe, get subsidy upto 90% and other OBC get upto 50% of subsidy from state government. These all benefits may reach to fisher family through Matshay Jeevi Committee with the survey under Animal & Fish Resource Department, Bihar.
3. The cooperative societies of fishermen ought to be strong and created free from corruption and factionalism. COFFED and COMFED are two most potential cooperative fisheries societies in Patna, Bihar. Various schemes may be launched by such NGOs and needs to be implemented and expedite at grass route level.
4. Arrangements ought to be created for correct selling of the fish that are perishable commodities. Refrigeration of transport vehicles ought to be done on an oversized scale.
5. An organisation ought to be originated by the Govt. to relinquish technical help to the fishermen. The organisation ought to include persons, who have a huge experience within the sphere of fishing.
6. The government officers of the fisheries department ought to be equipped with sufficient information and invested the required power to unravel the day to day issues featured by the fishermen.
7. The Government of Bihar ought to establish a minimum of one 'fish market centre' in every region with correct facilities as well as cold storage. It will ease them to curtail their loss, which would otherwise maximum due to lack of preservation techniques and lack of standard fish market nearby.
8. There is a need to strengthen the 'extension service system and human resource development' to enhance the information, skill, angle and observe of individuals engaged in fishing.
9. For assortment of field data and investigation into the issues and prospects of cultivation within the region, stress ought to be arranged on the 'Participatory Approach' among the researchers, scientists, and extension market and fish farmers. Such an approach can produce sturdy and effective linkage between the extension personnel and fish farmers to create their capability.
10. Stress ought to tend to determine regional analysis and coaching centres by the govt., so as to unravel the sector issues of fishermen, it is essential to prepare coaching to create them up, thus far with the advancement in cultivation analysis. 'Long term vocational education program ought to be unionized for rural youth with stress on 'learning by doing' for generation of self-employment.
11. Stress ought to tend to strengthen the prevailing workplace, cooperative societies and to determine a lot of primary cooperative societies and assistance teams at the gram panchayat level for strengthening the service delivery system.
12. A comprehensive cultivation policy is to be developed by the govt. for providing economical extension program, planning, research, training, education, input offer, and credit and selling systems.

13. Fishermen and their families are a very important part of Bihar. The social matrix is the strength of the fishing trade. Therefore, special thought must be given to the current cluster to sustain the long way forward for the trade and therefore the life and welfare of the communities.
14. Government may also channelize the fishermen to export the yield to foreign countries at higher value by its motivation and steering.
15. Government will organize for common storage facility which might be availed by fishermen on a rental basis. This may facilitate them to earn a lot of throughout favourable reasons.
16. Throughout strike amount, the Govt. will organize for different other ways of employment and earnings
17. The Govt. ought to produce awareness among the fishermen community concerning tiny savings and self-facilitate teams, loan facilities from the banks in order that they might get warning facilitate, rather than borrowing at a high rate of interest.
18. The fishermen society ought to volunteer to create homes for them within the space far away from the near bank of river Ganga and its tributaries so as to shield them from natural calamities.
19. Most countries have policies for the event of cultivation however lack the strategic development plans and allocation of adequate resources (human and financial) to confirm that these policies really profit the poor, it is necessary that these policies be properly enforced.
20. A lot of assistance teams are often fashioned particularly for girls to affect fishing connected activities.

Scope for future studies:

For development of socio-economic assessment plan adapted to the specific context of fisheries, a ten step process is highly recommended:

1. Consider the level of participation and the constitution of the team.
2. Define the purpose of doing socio-economic monitoring and evaluation.
3. Describe and delineate the scope and scale of the project.
4. Clarify the policy, management or programmatic objectives.
5. Choose attributes and develop good indicators.
6. Develop methods and identify data sources.
7. Map the logistics and timeline for implementation.
8. Articulate a plan for analysing and presenting the results.
9. Discuss how results will be evaluated.
10. Develop a plan for communicating results.
11. Formulate a process for deliberation and deciding future actions.

A monitoring and evaluation plan is like a roadmap to guide the entire process.

Besides in context of Patna (Bihar) where the socio economic status of the fishermen is highly miserable, it depicts the apathetic attitude of the Government towards the welfare of the community. So there is an emergent need to cadre the basic needs of the community. A study on the role of various state fisheries co-operative societies for improvement of fishing community in Bihar will be highly solicited. Besides, a study on the role of state policies and schemes for fishing community upliftment in Bihar along with a comprehensive approach on the role of self help group (SHG) for girl fish staff in Patna and Aquaculture dominating areas of Bihar should be given priority.

CONCLUSION

In spite of being lamented by vast aquatic resources, Bihar is still lying far behind among the aquaculture sector of different states in India. The fishermen are generally rather poor, semi-literate or illiterate and hardly have the other different job chance to modify over. The limited quantity of capital that they may presumably save, gets drained away in medication or for mating of the nets that for the foremost a part of the year stays idle. A number of these options like poor family financial gain, low level of accomplishment and family size not to mention poor monetary inputs have left several of the fishermen in an impoverished financial condition. Instability in earning potential from fishing has in several cases compelled the offspring of the fishermen to show to different employment prospects. This trend has to be countered by bettering steps.

Besides, the correct selling facility for the harvested fish, storage facility at intervals, simple reach of the fishermen covering variety of cooperatives or teams and even provision of dissemination of time-to-time data concerning market costs and demand to the fishermen, offering viable fishing methods, causative for rising their

business, are essential. it would be troublesome to implement all of these at the same time, however the implementation of site-specific combos of many of those strategic choices at totally different degrees is very necessary to resuscitate the condition of fishermen and rejuvenate the prospects of a viable future for the fishermen.

Fishing community follows the Indian culture in their regular life. Their occupation is seasonal. They're not turned in to different field throughout off season they did not earn any financial gain from their activities and face issues financially. They are indebted, rely on the cash lenders and are exploited by agents and middlemen, whereas mercantilism their product within the market in Patna. The findings of the present study may be critically analysed by the technocrats, Government officials and state fishery department to decipher an amicable way for the benefit of the fishermen community of the state and to develop strategies for upliftment of their socio economic status in the state.

ACKNOWLEDGEMENT

The authors are thankful to Head, Department of Zoology, Patna University, Patna, Bihar for providing infrastructural facilities.

CONFLICT OF INTEREST

Authors declare no conflict of interest.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 35-38

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Efficacy of *Citrus limonum* seeds on reproductive parameters of male albino rat

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Received : 15th November, 2022 ; Revised : 20th December, 2022

Abstract- Healing with medicinal plants is as old as mankind itself. Now days, plant products catch the attention of many scientists as a primary source of naturally occurring fertility regulating agents because of their little and no side effects. Compounds found in plants could prevent fertilization without the side effects of hormone-based contraceptives. The present study was carried out to evaluate the effect of *Citrus limonum* seed on body weight as well as reproductive organ weight for 60 days. A considerable decrease in weight has been observed at ($p<0.001$). Along with this the sperm motility and sperm viability were also drastically reduced.

Key words: *Citrus limonum*, reproductive parameters, pharmacological treatments

INTRODUCTION

Fertility control is a significant issue of global and national public health concern. This increasing human population throughout the world particularly in developing and underdeveloped countries has inevitable effects on the life supporting resources on the earth.[1] Various studies have confirmed both men and women's desire for novel male contraceptives.[2] Majority of men believe that both partners share an equal responsibility of family planning.[3] Barriers to development include concerns regarding side effects and spermatogenic rebound with hormonal methods.[2] Herbal medicines have become a popular form of health care. Even though several differences exist between herbal and conventional pharmacological treatments, herbal medicine can be tested for efficacy using conventional trial methodology. Several specific herbal extracts have been demonstrated to be efficacious for specific conditions.[4] Some *Citrus* species have been reported to have antifertility effect.[5] *Citrus limonum* seeds have shown significant antifertility activity in female albino mice. The seeds have shown phytoestrogenic activities and the alcoholic extract reveals the most promising antifertility effect in female albino mice.[6] *Citrus limonum* commonly known as lemon, belongs to the *citrus-Linn* (Ruteaceae) family.[7] Therefore in this investigation ethanolic extract of *Citrus limonum* seed was studied to find out its potential antifertility effect on male albino rats.

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MATERIALS & METHOD

Citrus limonum seeds were obtained from Birsa agriculture college and was authenticated in the Department of Botany, Ranchi University, Ranchi. Ethanolic extract was prepared in soxhlet in (400) mL of ethanol (95%) solvent for (24-48) hr. then evaporated at 45°C by rotary evaporator to form a paste, and further transferred into

sterile bottles and refrigerated at 4°C until used. Adult albino mice of Wistar strain weighing between 150–200 gms housed in clean environment under 12-hour light and 12-hour dark cycle, having free access to food pellet and water *ad-libitum,* were used after approval of the protocol by the Institutional Animal Ethical Committee. 30 male albino rats were divided into three groups (n=10). Each group was kept in a separate cage. Group I served as control. Group II and group III were treated with lemon seed extract at a dose level of 250 mg and 500mg/kg b.w. respectively for 60 days.

Determination of body weight and reproductive organ weight:

After 24 hours of last treatment, weight of the animals was recorded and then sacrificed. After scarification, testes, epididymis, seminal vesicle and ventral prostrate were dissected out, trimmed off of extraneous tissues and weighed accurately on balance. The organs weights were expressed in term of mg/100g body weight.

Statistical analysis:

Data were expressed as mean ± S.D. Significance difference between means was determined by student t-test and one way (ANOVA). P <0.05 was considered as significant.

RESULTS & DISCUSSION:

The effect of oral administration of *Citrus limonum* seed extract on body weight and reproductive organ weight indices is represented in table 1. The extract significantly reduced ($p<0.001$) body weight as well as reproductive organs weight. At the same time sperm motility and viability was significantly altered upon its administration as depicted in table 2. Previous studies have revealed that lemon seeds are rich in pectin, a fibre that minimizes fat absorption into the body.[8] Fibre supplementation potentially reduces energy intake and contributes to weight loss.[9] The testes perform two highly organized and intricate events, called spermatogenesis and steroidogenesis, which are vital for the perpetuation of life.[10] Our study for the very first time revealed that *C. limonum* seed reduced the weight of the testis. This indicates that there have been disturbances in the germinal component which has led to its weight depletion. In our study the weight of epididymis was reduced, which explains that level of sperm production has declined in the testis which in turn leads to the reduction in epididymal weight.

Table 1: Effect of *Citrus limonum* seed extract on body weight and reproductive organ weight of adult male albino rats.

Parameters	Group I		Group II		Group III	
Body weight	162.53 ± 1.75		160.65± 1.24		156.02 ± 1.11	
Testes	L	R	L	R	L	R
	370.10±1.82	371.98±1.26	366.88±0.56	367.98±1.12	362.47±0.86	361.98±1.68
Epididymis	73.00±2.57	74.41± 1.23	70.94±0.78	70.92± 1.01	67.55±0.97	68.26±1.13
Seminal vesicle	208.16±1.19		205.81± 0.49		202.61± 0.84	
Prostate	51.42±3.01		50.15 ± 2.97		49.45 ± 2.35	

Table 2: Effect of *Citrus limonum* seed extract on sperm viability and sperm motility of adult male albino rats.

Parameters	Group I	Group II	Group III
Sperm motility %	74.91 ± 2.37	53.89 ± 1.56	47.64 ± 1.08
Sperm viability %	72.93± 1.14	50.34± 1.97	44.81± 1.92

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 39-42 **Animal Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Catla modifies its feeding behaviour as it develops: an approach from feeding ecology

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Received : 20th November, 2022 ; Revised : 22nd December, 2022

Abstract- A common major carp species in carp polyculture is *Labeo catla*, which is native to India. It is recognised as a plankton-eating water surface and midwater feeder. Adult feeds on zooplankton using large gill rakers, but its juvenile feeds on both zooplankton and phytoplankton. It prefers zooplankton in its natural environment at the fingerling stage, with phytoplankton serving as a supplemental diet. At this stage, all zooplanktonic species and smaller phytoplankton showed a strong positive selection. All zooplanktonic organisms exhibit severe negative selection in adults, but most phytoplanktonic organisms exhibit positive selection. Yet, in periphytic environments, it consumes nearby sub-periphytic species that are present. It is well acknowledged that *L. catla* demonstrates a variety of eating strategies as it matures from a fingerling to an adult and transitions from a planktonic to periphytic environment. The conversion of naturally occurring resources to fish biomass would be maximised by adopting a fish culture strategy in accordance with its differential feeding strategy, particularly in polyculture practices.

Key words: *L. catla,* Socioeconomic status, demography, fishermen community, strategies, upliftment.

INTRODUCTION

An important part of fish culture research is figuring out how aquatic resources are used. The eating and feeding behaviours of fish provide crucial information in choosing which species to produce in poly-species fish culture, which considers resource use as a key criterion. One of the most significant Indian Mazor Carps (IMC) in carp polyculture is *Labeo catla* (catla). Across South Asia, South-East Asia, Sri Lanka, Japan, China, the Philippines, Malaysia, Nepal, and some African nations, this IndoGangetic riverine species is found. Little data from the early 20th century is available regarding its culture. It was introduced into practically all riverine systems in India and currently plays a key role in pond fish polyculture.

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It was a prime option for carp polyculture due to its resource tolerance with other freshwater carps, particularly rohu (*Labeo rohita*) and mrigal (*Cirrhinus mrigala*).[1] Catla is one of the most well-known and delectable freshwater fish rose in India, Bangladesh, and other nearby nations in the region because to its great growth potential, compatibility, and consumer choice. It provides as a significant dietary source of n-3 PUFAs and protein. The species that supplies the most protein overall is catla. It serves as a source of calcium and vitamin A in small amounts.[2,3]

Unfortunately, there is no conclusive research on the diet and eating habits of catla in relation to natural resources. According to the earlier accounts that are now available, its primary food is plankton i.e. both zooplankton and phytoplankton. A few studies that looked at its growth

and behaviour identified it as a periphyton feeder. According to research on the diet and feeding habits of *L.catla* in polycultures with *L.rohita* and *C. mrigala*, it feeds on zooplankton as an adult and phytoplankton as a fingerling.

From the point of feed selectivity by catla, the ontogenic shift is fairly noticeable. For zooplankton such crustaceans (*Cyclops, Daphnia*), rotifers (*Keratella*), and other smaller phytoplankton like phytoflagellates (*Euglena, Volvox*), desmids (*Cosmarium, Closterium*), and fingerlings of catla exhibit substantial positive selection. Reversibly, adults were found to have significant phytoplanktonic negative selection and most zooplanktonic positive selection. Most of the Chlorophyceae, such as *Pediastrum, Selenastrum, Ankistrodesmus, Scenedesmus, Zygnema, Ulothrix,* and *Tetraspora,* were preferred over giant filamentous algae like *Oedogonium. Selenastrum, Pediastrum,* and *Scenedesmus* were the only tiny unicellular or colonial Chlorophyceae that were chosen above other species. Chlorophyceae was favoured to Bacillariophyceae. The favoured phytoflagellates during the fingerling stage were completely avoided. Compared to their abundances in the environment, zooplankton species such as protozoa, rotifers, and crustaceans only appeared in trace amounts in the gut contents.[4-6]

Periphytic organisms, which are creatures that are permanently attached to submerged surfaces in aquatic environments, quickly build up large amounts of biomass and make upto 80% of the aquatic primary production. In a natural and managed setting, it plays a crucial role in producing food for fish and other wildlife. Periphyton can provide fish with up to 75% of their metabolic energy when used as food in aquaculture, which helps to boost fish productivity. Periphyton is traditionally used as a rich source of readily accessible aquatic food for fish in nations including Bangladesh, Sri Lanka, India, and Cambodia.

MATERIALS & METHODS

Several research paper were consulted

RESULT & DISCUSSION

To comprehend and assess the level of synergizing production, a number of tests have been conducted in a periphyton-based polyculture setting using the species *Labeo catla, Labeo rohita, Cirrhinus mrigala, Oreochromis niloticus,* and *Labeo calbasu. L. catla* displayed the best growth and output in these experiments. Via periphyton-based aquaculture trials in ponds, it was attempted to assess its capacity to obtain periphytic food supplies. The biomass-based results of these investigations suggest that the fish intelligently searches naturally accessible resources in periphytic than planktonic environments, despite the fact that they did not attempt to study the feeding ecology of catla in periphyton-based tests.[7]

Despite the fact that these studies were conducted in periphytic environments, the feeding ecology of the fish species that were stocked has been disregarded. Only *Cyprinus carpio* has received more attention among the carp species as a viable option for periphyton-based aquaculture. Recently, it was shown that catla of all sizes consume a lot of algae in periphytic pond environments. The investigation focused on catla in both planktonic and periphytic habitats. They found that catla demonstrates enhanced feeding (with a wider Smith's diet breadth) in periphyton-based areas in their study, which was based on the gut content of fish from these settings. They also looked at resource selection, which revealed that catla investigates both periphyton and plankton in substrate-based areas at a later stage of growth. [8,9]

Since the fish's mouth shape does not support a grazing behaviour, they hypothesised that catla may not actually be feeding on periphyton, though it does so in a highly lucid manner. They went a step further and suggested that the fish cleverly makes use of a sub-periphytic zone that is close to the substrate and is rich in algae populations. Through the use of algal feed resources, catla can be directly enriched with micronutrients thanks to such improved feed accessibility in the natural environment. [10]

The studies that are currently accessible on the diet and feeding habits of catla in substrate-free or substrate-based habitats illustrate how catla allocates a kind of niche to various food kinds along with its ontogenetic development.

These food types includes:-

(1) extensive use of algal communities regardless of age class under periphytic conditions,
(2) selective phytoplanktonic communities at an early stage of growth,
(3) zooplanktonic communities at a later stage of growth, and
(4) phytoplanktonic communities at a later stage of growth.

The final choice may result from a synergistic interaction between resource availability and catla's active planktivorous habit. According to Ross (2003)[3], an animal's propensity to choose one food over another when eating is influenced by its innate characteristics. Alam (2011)[10] emphasised that although fish feeding behaviour plays a significant role in the selection mechanism, it is also influenced by the production of any food item by the environment and its availability.

In addition, despite the fact that the fish is thought to be a surface and midwater feeder, the presence of rotting organic debris, sand, and mud in its gut points to a partial preference for bottom feeding. The fish can feed on soft aquatic invertebrates without having to seize and crush them thanks to its nibbling mouth type, which has soft lips with soft fringes, sharp cutting edges, and no teeth in the bucco-pharyngeal area. The fish's modified thin, hair-like gill rakers further imply that it feeds on tiny planktonic creatures that filter food from the water. The primary reason why fry and fingerlings school in ponds is to eat, while adults do not display this behaviour.

CONCLUSION

It is evident that little research has been done on *L. catla's* diet and feeding ecology with regard to planktonic organisms in general and periphyton in particular, despite the species' immense potential in polyculture. A conclusive finding on the portion of its food and feeding ecology is required to ascertain the trophic position of catla and to support it as a possible candidate in converting the maximum amount of natural food resources to micronutrient rich nutritious fish biomass. Periphyton is unquestionably a solution in that situation, particularly for herbivorous fish species to adopt a "as and when need" type of feeding approach in polyculture ponds. In the Indian subcontinent, catla is one of the most widely cultivated freshwater fishes. In periphytic polyculture, it exhibits enhanced feeding and significant growth. When stocked with other column feeders, catla-based polyculture in periphytic environments must be standardised with appropriate data on its feed sharing, resource partitioning, and diet selection. Even though it grows satisfactorily when stocked in a polyculture pond with a periphytic environment, such growth results do not confirm its resource utilisation in the specified environment; as a result, more research is necessary before developing any efficient strategies for maximising periphytic resource utilisation. A report on this feeding behaviour in a periphytic polyculture environment would be more suitable for its polyculture. The feeding ecology that demonstrated its "increased feeding in periphytic environment" was examined in monoculture practise. Its ontogenic shift must be demonstrated under the presence of periphytic resources, regardless of whether it depends on size, monoculture, or polyculture combinations.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 43-48 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Seasonal variation in enzyme activities from agro ecosystem site at Ranchi, Jharkhand

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Received : 14th January, 2023 ; Revised : 26th February, 2023

Abstract- Activities of dehydrogenase, urease, cellulase and phosphatase were studied from agroecosystem site at Ranchi, Jharkhand. Phosphatase activity was higher in comparison to other enzymes. Significant positive correlation was observed between organic carbon content and the enzymes except dehydrogenase. Cellulase was significantly positively correlated with the activities of dehydrogenase, urease and phosphatase.

Key words: dehydrogenase, urease, cellulase, phosphatase

INTRODUCTION

Soil enzymes are involved in various decomposition and chemical transformation in the soil and measurement of enzyme activities can give indications of the extent of specific processes in the soil and can sometimes be used an indicators of soil fertility.[1] Soil is a complex system with the Physical, chemical and biological properties of it being in dynamic equilibrium. Studies on soil enzymatic activities provide valuable information regarding the biochemical processes in soil. Soil microorganisms enzymatic reactions forms a part of respiratory pathways and are related to the variety of soil and its temperature and moisture conditions.[2,3] Enzymes released during decomposition process play a significant role in different biogeochemical cycles.[4]

Microbial status and physico chemical conditions of the soil are indicated by the enzyme activities hence they are also used are sensors of soil degradation.[5,6] A number of reports are available dealing with various aspects of enzyme activities in soils around the world and its relation to other factors.[7-9] The present study was undertaken to determine the seasonal variation in enzyme activities and the relationship of enzymes with soil properties.

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MATERIALS & METHODS

The surface soil samples were collected from agro ecosystem site adjoining the Ranchi University campus, Morhabadi. The samples collected were mixed thoroughly, brought to the laboratory in polythene bags and screened through a 2mm sieve. The samples were kept at 4°C until used. Estimation of physico chemical parameters and enzyme activities were done within fortnight. Soil temperature was measured by the help of soil thermometer whereas moisture content of the soil was determined by Oven drying method. Organic carbon and nitrogen content of soil was estimated according to Walkley and Black (1934)[10] and Kjeldahl method of Jackson (1973)[11]. Soil respiration was estimated following the alkali absorption method of Witkamp (1966)[12].

The procedure of Casida *et al.,* (1964)[13] was followed for the assay of dehydrogenase activity in soil and urease activity was estimated following Tabatabai and Bremner (1972)[8] with modification of Kaplan (1969)[14]. Phosphatase and Cellulase activity in soil were measured following Kramer and Yardei (1959)[15] and Ross (1965)[16] respectively.

The study area experiences three well-defined seasons. The winter season starts from October to February, the summer season starts from March to mid-June and the rainy season starts from mid-June to September.

OBSERVATION

The range of variation of different physico chemical parameters during different seasons has been presented in table 1.

Table 1- Variation in physico chemical parameters of soil during the study period.

Parameters	Range
Soil Temperature	15.6 ± 0.98 - 24.73 ± 1.23 °C
Soil moisture	9.15 ± 1.02 - 22.36 ± 1.21 %
Organic carbon content	5.98 ± 0.06 – 7.493 ± 0.14 mg C g^{-1}
Nitrogen content	0.53 ± 0.01- 0.617 ± 0.03 mg N g^{-1}
Soil respiration	343.3 ± 6.45 - 472.7 ± 6.83 mg CO_2 kg^{-1} hr^{-1}

The seasonal variation in dehydrogenase activity is shown in table 2 with a peak value in rainy season (10.2 ± 1.87 µg formazan g^{-1} dry soil h^{-1}) and least in winter (6.5 ± 0.61). Dehydrogenase activity had a significant positive correlation with soil respiration ($r=0.678$, $p< 0.02$) whereas, a positive correlation was obtained with soil organic carbon matter, total nitrogen content, soil temperature and soil moisture (Table 3). Dehydrogenase activity had a positive significant correlation with enzyme activities like cellulase ($r= 0.877$, $p < 0.001$) and urease ($r= 0.62$, $p < 0.05$) whereas; a positive correlation with phosphatase was obtained (Table 4). Two way analysis of variance showed a significant difference in enzyme activity between season but it was not significant between the study sites ($F = 10.818$; $df = 2, 4$; $p < 0.025$ and $F = 3.92$; $df = 2, 4$) (Table 5).

Variation in urease activity in different season is shown in Table 2. A maximum of 47.89 ± 4.13 µg NH_3 g^{-1} soil h^{-1} was obtained in rainy season with a minimum of 25.47 ± 2.32 µg NH_3 g^{-1} soil h^{-1} in summer. The correlation between soil moisture and urease activity revealed a significant positive correlation ($r = 0.714$, $p < 0.01$). Urease activity had a significant positive correlation with organic matter, total nitrogen, soil moisture and soil respiration (Table 3). Urease activity had a significant positive correlation with phosphatase activity ($r= 0.817$, $p < 0.01$). Correlation between urease and other enzyme activity was positive and significant (Table 4). A significant difference in enzyme activity between season ($F = 35.635$; $df = 2, 4$; $p < 0.005$) but there was no significant difference in enzyme activity between the sites ($F = 0.470$; $df = 2,4$) (Table 6) was revealed by two way ANOVA.

Phosphatase activity increased from a minimum (38.96 ± 2.74 µg phenol g^{-1} soil h^{-1}) in summer to a maximum (55.71 ± 4.89 g phenol g^{-1} soil h^{-1}) in rainy season (Table 2). Phosphatase activity has a significant positive correlation with soil moisture content ($r = 0.611$, $p < 0.05$) (Table 3). Phosphatase activity had a positive significant correlation with organic matter ($r= 0.705$, $p < 0.02$). Phosphatase activity had a significant positive correlation with enzyme activity like urease and cellulase ($r= 0.817$, $p < 0.01$ and $r= 0.809$, $p < 0.01$) respectively (Table 4). Two way analysis of variance showed a significant difference in enzyme activity between season but it was not significant between the study sites ($F = 12.60$; $df = 2, 4$; $p < 0.025$ and $F = 0.383$; $df = 2, 4$) (Table 7).

Seasonal variation in cellulase activity is shown in Table 2 having a peak value of 53.12 ± 4.12 µg glucose g^{-1} soil h^{-1} in rainy season and least in winter 27.39 ± 2.53 µg glucose g^{-1} soil h^{-1}. Cellulase activity had positive significant correlation with soil organic matter and soil respiration (Table 3). It also had a significant positive correlation with other enzyme activity (Table 4). A significant difference in enzyme activity between seasons ($F = 31.135$; $df = 2; 4$, $p < 0.005$), whereas, no significant difference was obtained between different sites ($F= 0.141$; $df = 2, 4$) (Table 8) by two-way analysis of variance.

DISCUSSION

Pati and Sahu (1998)[17] observed a significant positive correlation between soil respiration and dehydrogenase activity in both normal and fluoride treated soil. A similar relationship between the two parameters was also reported by Stevenson (1959)[18] whereas, Mishra and Sahoo (1997)[19] did not observed a significant positive correlation between soil respiration and dehydrogenase activity. In the present study a significant positive correlation ($r = 0.678$, $p < 0.02$) was obtained between the two parameters showing agreement with previous reports of Pati and Sahu (1998)[17] and Stevenson (1959)[18].

Table 2- Enzyme activity in soil sample (Mean ± SEM) in different seasons.

Season	Dehydrogenase[1]	Urease[2]	Phosphatase[3]	Cellulase[4]
Winter	6.5 ± 0.61	33.38 ± 2.11	44.42 ± 3.16	27.39 ± 2.53
Summer	8.5 ± 1.38	25.47 ± 2.32	38.96 ± 2.74	31.81 ± 3.9
Rainy	10.2 ± 1.87	47.89 ± 4.13	55.71 ± 4.89	53.12 ± 4.12

1. μ g formazan g^{-1} soil h^{-1} 2. μ g NH_3 g^{-1} soil h^{-1}
3. μ g phenol g^{-1} soil h^{-1} 4. μ g glucose g^{-1} soil h^{-1}

Table 3- Correlation coefficient among different enzyme activity and edaphic parameters.

	Organic carbon	Nitrogen	Soil respiration	Soil moisture	Soil temperature
Dehydrogenase	0.433^{NS}	0.421 ^{NS}	0.678**	0.292 ^{NS}	0.428 ^{NS}
Urease	0.666**	0.650***	0.725*	0.714*	0.294 ^{NS}
Phosphatase	0.705**	0.382 ^{NS}	0.450 ^{NS}	0.611***	0.284 ^{NS}
Cellulase	0.586***	0.555****	0.705**	0.498****	0.509****

*p < 0.01, **p < 0.02, ***p < 0.05, ****p < 0.10, NS Non significant

Table 4- Correlation matrix among different soil enzyme activity.

	Dehydrogenase	Urease	Phosphatase	Cellulase
Dehydrogenase	-	0.620***	0.538****	0.877*
Urease	-	-	0.817**	0.876*
Phosphatase	-	-	-	0.809**
Cellulase	-	-	-	-

* p < 0.001, ** p < 0.01, *** p < 0.05, ****p < 0.10

Table 5- ANOVA test among dehydrogenase activity in soil at different sampling sites and in different seasons.

Source of variation	Sum of Square	Degree of freedom	Mean square	Variation ratio F	Significance
Between season	20.76056	2	10.38028	10.81811	p < 0.025
Between sites	7.523356	2	3.761678	3.920343	NS
Residual	3.838111	4	0.959528		

Table 6- ANOVA test among urease activity in soil at different sampling sites and in different seasons.

Source of variation	Sum of Square	Degree of freedom	Mean square	Variation ratio F	Significance
Between season	775.7646	2	387.8823	35.635	p < 0.005
Between sites	10.24427	2	5.122133	0.470	NS
Residual	43.53853	4	10.88463		

Table 7- ANOVA test among phosphatase activity in soil at different sampling sites and in different seasons.

Source of variation	Sum of Square	Degree of freedom	Mean square	Variation ratio F	Significance
Between season	437.8382	2	218.9191	12.60	p < 0.025
Between sites	13.3206	2	6.6603	0.383	NS
Residual	69.49	4	17.3725		

Table 8- ANOVA test among cellulase activity in soil at different sampling sites and in different seasons.

Source of variation	Sum of Square	Degree of freedom	Mean square	Variation ratio F	Significance
Between season	1124.896	2	562.4481	31.135	p < 0.005
Between sites	5.106467	2	2.553233	0.141	NS
Residual	72.25853	4	18.06463		

Enzyme activity can differ in soil under the same climatic conditions and even in soil with the same amount of organic matter. Besides the vegetation, temperature and soil moisture content play significant roles in regulating the soil enzyme activities. Synthesis, accumulation and activity of enzymes in heterogenous and dynamic system like soil are influenced to a great extent by various physical, physico-chemical and biological constitution of the soil.

Mishra *et al.*, (1979)[1] found a significant correlation between soil moisture, organic matter content and enzyme activity (cellulase). They also found a positive correlation of cellulase activity with total nitrogen content of the soil. In the present investigation a significant positive correlation between total nitrogen and cellulase activity ($r= 0.555$, $p < 0.10$) was found whereas, cellulase activity had a positive correlation with soil moisture ($r= 0.498$, $p < 0.10$) and a significant positive correlation with organic matter content ($r = 0.586$, $p < 0.05$) was obtained and the findings are in conformity with those of Mishra *et al.*, (1979)[1].

Level of enzyme activity in soil and total soil respiration (carbon dioxide evolution) are considered to give some guide to the microbiological activity prevailing in the soil system. The breakdown of macromolecular constituent of plant litter involves the extra cellular enzymes secreted by microorganisms and root to the soil and the assessment of soil enzyme activity gives the fertility status of the soil. Urolytic microorganisms and root exudates secretes urease enzyme. Urea added to the soil as fertilizer is hydrolysed by the enzyme urease resulting in the release of ammonia and causing damage to germinating seedlings and young plants, nitrite toxicity and other pollution problem.[20] Thus research concerning to the distribution and factor affecting urease activity in soil has helped the use of urea as nitrogen fertiliser and in the management of agricultural land.

A higher organic carbon content and microbial population supplemented with congenial soil reaction (pH 6.1 to 8.0) may have contributed significantly in production of urease at elevated levels in these soils of the study area. In contrast, soils having low organic content and microbial population with strongly to moderately acidic in reaction have low urease activity.[21] Dash *et al.*, (1981)[22] observed a positive correlation of total nitrogen, carbon and specific conductance with urease activity. Organic carbon and total nitrogen content are indices of organic matter content in soil. Positive correlation of urease activity with organic carbon, total nitrogen and specific conductance suggest that organic matter content accounted for most of the variations in soil urease activity.[22,23] Mishra (1989)[24] found that urease activity was negatively correlated with moisture of the soil supporting the suggestions that the organic matter content and specific conductance were the chief determinants of urease activity. In the present investigation urease activity was positively correlated with organic matter ($r= 0.666$, $p < 0.02$) and total nitrogen content ($r = 0.650$, $p < 0.05$) and is in conformity with the findings of Dash *et al.,* (1981)[22].

As regards levels of phosphatase in soils belonging to the three land situations, lowland exhibited a higher activity of 43.80 to 114.77 µg phenol released g^{-1} soil h^{-1}, whereas, the uplands contained the least activity of 9.56 to 32.41 µg phenol released g^{-1} soil h^{-1}. Soils of lowlands being rich in organic and inorganic adsorptive surfaces are likely to immobilize and protect the enzymes from proteolytic degradation more efficiently than upland soils.[25] A moderate phosphatase activity (38.93 ± 2.74 to 55.71 ± 4.89 µg phenol released g^{-1} soil h^{-1}) was recorded at the sampling sites.

Mishra and Sahoo (1997)[19] recorded a positive insignificant correlation between organic carbon content and dehydrogenase activity, a similar positive correlation ($r=0.433$) between organic matter content and dehydrogenase activity in the present study is in conformity with the findings of Mishra and Sahoo (1997)[19].

All the four enzymes examined excepting dehydrogenase established a highly significant and positive correlations with organic carbon of soils.[25] Urease activity with correlation coefficient of $r = 0.62388$ has been shown to vary from 41.35 µg urea-N hydrolyzed g^{-1} h^{-1} in soils having low organic carbon (0.5%) to 45.94 and 59.82 µg urea-N hydrolyzed g^{-1} h^{-1} (0.75%) organic carbon, respectively. Likewise, phosphatase has been found to increase gradually from the lowest level of 38.09 (in low organic matter soil) to 39.59 (in medium organic matter soil) and finally to the highest activity of 58.77 µg phenol g^{-1} h^{-1} by Sharma (1993)[25]. Cellulase, too, responded positively with increasing organic carbon resulting its highest activity of 31.212 n mole glucose g^{-1} m^{-1} with high organic carbon and the lowest of 0.587 n mole glucose g^{-1} m^{-1} in presence of low organic content.[25] A positive effect of organic matter on enzyme levels may be attributed partly to the better survival and growth of microorganisms due to

enhanced availability of carbon and energy source in presence of organic matter and partly to the protection of enzymes from the microbial degradation by complexing with humic substances.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 49-52 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Hematological indices of control water fish *Channa gachua*: a biochemical study

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Received : 22nd November, 2022 ; Revised : 23rd December, 2022

Abstract- Live freshwater fishes *Channa gachua* collected from Taj Baj Khan Pond, Hajipur. These fishes were acclimatized in the Zoology Laboratory of Science College, Patna and their haematological parameters and biochemical parameter were thoroughly studied in the control stage viz RBC and WBC count, haemoglobin content, viz. protein and cholesterol. The haematological parameter of *Channa gachua* showed the values as- RBC (4.1x106) WBC (3.4x103/cumm) haemoglobin (10.5 g/dl) PVC (32.0 %/dl) MCV (78.0µm) MCH (25.6 pg/dl) and MCHC (32.8 g/dl) etc. The biochemical finding included Glucose (87.5mg/dl) and Cholestrol (149.4mg/dl). The finding of the present study highlights the normal range of haematological and biochemical parameter in freshwater Snakehead *Channa gachua*.

Key words: *Channa gachua*, control, hematology, cholesterol.

INTRODUCTION

Blood parameter in *Channa gachua* is affected by many factors including water quality, temperature, food availability and physical condition of the fish. This physico chemical characteristic directly and indirectly affects the blood components of fishes. The impact blood parameter is also varied according to the sex, size, season and age of the fish. The blood parameter may be considered as indicator of the status of aquatic bodies.

Review of literature have shown a great degree of variation in protein, cholesterol and glucose level depending upon sex, size and age of the fishes. Some researcher reported that the percentage of protein varied from species to species.[1-3] The value of haematological parameter depends on the season and the slow or active movement of fish. It was reported that haematological parameterare affected by microbial infection of fish and toxins.[4-6] Although many research paper deals with the hematology of freshwater fish. This paper deals with important blood parameter of fish *Channa gachua*.

However, the systematic report of the haematological characteristics of *Channa gachua* is very scanty. Hence, the present research paper gives an insight of haematological profile of control freshwater fish *Channa gachua*.

MATERIALS & METHODS

Samples were collected from fresh water of Hajipur Bihar, Taj Baz Khan Pokhara, Hajipur and Patna fish market. The fish were brought to the Zoology Laboratory of Patna Science College, Patna.

Blood samples were collected by cardiac puncture using a 21 gauge hypodermic needle. They were collected in vials, one containing the anticoagulant EDTA, for blood cell studies and the other without EDTA to separate the clot and serum to study certain biochemical components.

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Standard hematological procedures for laboratory studies were adopted for experimental analysis. Red and white blood cells (RBC and WBC) were counted using a Spencer's haemocytometer. Whole blood parameters such as haemoglobin content (Hb), packed cell volume (PCV) mean corpuscular volume (MCV) mean corpuscular haemoglobin (MCH) mean, corpuscular haemoglobin concentration (MCHC) were determined by the standard protocol given by Welchselbam. The standard methods employed for the determination of blood chemistry were referred to serum proteins[7], glucose[8], cholesterol[9]. The data obtained were subjected to statistical analysis and explained in Results & Discussion section.

RESULTS & DISSCUSION

The haematological parameters of *Channa gachua* are shown Data obtained various haematological parameter the shown as-RBC (4.1x106) WBC (3.4x103/cumm) haemoglobin (10.5 g/dl) PVC (32.0 %/dl) MCV (78.0µm) MCH (25.6 pg/dl) and MCHC (25.6 pg/dl). Besides the value of the content of Protein was (6.6 g/dl), Glucose was (87.5mg/dl) and Cholesterol was (149.4mg/dl.)

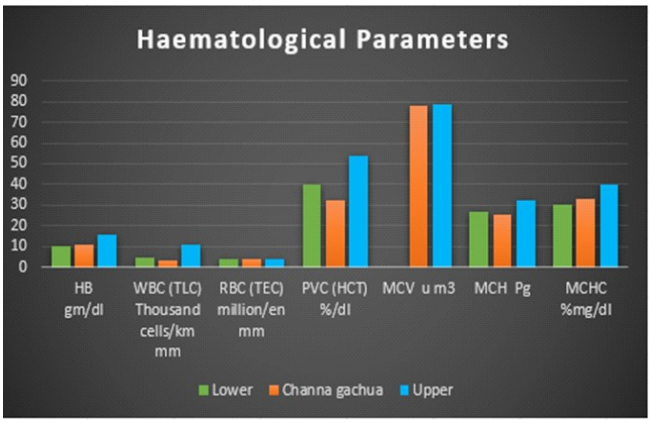
The results of heamato biochemical analysis showed that the maximum recorded protein content among the freshwater in the blood of *Channa gachua* order - Channiformes have been found and Biochemical profile such as protein & cholesterol were found to be lower sides among the order channiforme.

The levels of glucose were found very high Glucose showed higher value than cholesterol and protein. Haematological parameters in a fish are also affected by the physico-chemical conditions and its habitat. Lower values of haematological parameters were observed in sedentary and benthic condition. The female fish had more esnophill cells than male in *Channa gachua*.

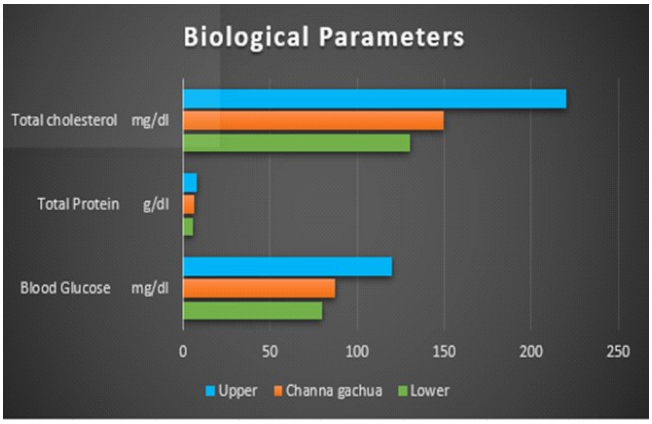
Haematological values like haemoglobin, PVC, MCV, MCH and MCHC in RBC as well as WBC count obtained in the present study are almost in agreement with the parameters observed in the blood of experimental fishes *Channa gachua* earlier workers. The number of total erythrocytes is positively correlated with body length. Seasonal changes in RBC count and haemoglobin content were observed in freshwater fish.

Table 1- The Haematological parameter of control water fish *Channa gachua*

Haematological Parameters	*Channa gachua*	Normal value in teleost
HB gm/dl	10.5	10-16
WBC (TLC) Thousand cells/km mm	3.4	4.5-11.0
RBC (TEC) million/en mm	4.1	3.9-4.2
PVC (HCT) %/dl	32.0	40-54
MCV u m3	78.0	78.94
MCH Pg	25.6	27-32
MCHC %mg/dl	32.8	30-40
Biochemical parameters		
Blood Glucose mg/dl	87.5	80-120
Total Protein g/dl	6.6	6.0-8.0
Total cholesterol mg/dl	149.4	130-220



Graphical Representation of The Haematological Parameters of control water fish *Channa gachua*



ACKNOWLEDGEMENT

The authors are thankful to the Head, Department of Zoology, Patna Science College for providing infrastructural facilities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 53-56 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Spectrophotometric estimation of total Lipid of hosts along with propagating larval instars gut content of *Callosobruchus chinensis* Linn.

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Received : 02nd December, 2022 ; Revised : 03rd January, 2023

Abstract- The estimation of total amount of Lipid of uninfected and infected hosts and larval instars gut content of *Callosobruchus chinensis* Linn. propagating on legumes like stored grains Gram (*Cicer arietinum*) and Moong (*Vigna radiata*). *Callosobruchus chinensis* Linn. (Coleoptera : Bruchidae) is a major stored grain pest of pulses of India. The pattern of variation in the different larval instars (I-IV) growing on these two hosts provide a unique gradual depletion of the total lipid content of the instar wise infected host as well as the larval gut content lipids.

Key words: Spectrophotometry, *Callosobruchus chinensis,* Lipid, Moong, Gram, Larval instars

INTRODUCTION

Callosobruchus chinensis Linn. is the pulse beetle. The pest (Coleoptera : Bruchidae) and its growth, development and metabolism mainly depends on its nutritional requirement and environmental condition. *Callosobruchus chinensis* Linn. is a pest of stored grains and are responsible for causing significant damage. The adult and grubs eat up the gram kennel and make a cavity. The concentration of biomolecules like protein, carbohydrate, lipids, amino acids, enzymes may vary during the life cycle of all organisms.

In order to strengthen the biochemical basis of host specificity in the pest has further been ascertained through quantitative estimation of total lipid through spectrophotometry or electrophoretic techniques which have been specially attempted in the present investigation.

Lipids serve as a source of metabolic energy and essential for structural components of cells. The lipid in the fat body is an energy reserve which can be mobilized rapidly during starvation, oogenesis, embryogenesis and moulting and is used to sustain continuous muscles actively.[1] Lipids are important constituents of cuticle and help in acylation of glucose-6- phosphate during chitin synthesis.[2]

Lipids are considered as an important group of compounds providing several biological functions as energy storage, cell membrane structure and signaling.[3]

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MATERIALS & METHODS

I- Preparation of Buffers for Spectrophotometric Estimation of biomolecules (Triglycerides):

(a). General Phosphate Buffer (0.1M)

It has following set of solutions:-

A- 0.2 M solution of monobasic sodium phosphate (27.6 g $NaH_2PO_4H_2O$ or 31.2g $NaH_2PO_42H_2O$ in 1000 ml)

B- 0.2M solution of dibasic solution phosphate (35.6 g $Na_2HPO_42H_2O$ or 53.60g $Na_2HPO_47H_2O$ or 71.6g of $Na_2HPO_412H_2O$ in 1000ml)

II - Estimation of Triglycerides by spectrophotometer:

The standard protocol of GPO-PAP method was used and the quantitative triglyceride in mg/dl was calculated at par with the standard kit of triglyceride through following equation. Simultaneously the blank was also used for standardizing the spectrophotometer instrument.

The GPO-PAP (Glycerol-3phosphate oxidase - phenol/4 chlorophenol aminophenazone) method for estimation of total triglycerides involves following 4 steps of reactions:-

1. Triglycerides + water – (enzyme esterase) → glycerol + carboxylic acid
2. Glycerol + ATP – (enzyme glycerol kinase) → glycerol-3-phosphate+ ADP
3. Glycerol – 3 – phosphate + oxygen - (enzyme glycerol-3-phosphate oxidase) → dihydroxyacetone phosphate + hydrogen peroxide
4. Hydrogen peroxide + 4 - aminophenazone + chloropheno – (enzyme peroxidase) → colored complex.

III- Reagent preparation for Triglycerides:

The contents of 1 bottle of L2 (Enzyme Reagent 2) was poured into 1 bottle of L1 (Enzyme Reagent). This working reagent is stable for at least 8 weeks when stored at 2-8°C. Upon storage the working reagent may develop a slight pink colour however this does not affect the performance of the reagent.

As much of working reagent may be made as and when desired by mixing together 4 parts of L1 (Enzyme reagent 1) & 1 part of L2 (Enzyme Reagent 2). Alternatively, 0.8ml of L1 and 0.2ml of L2 may also be used instead of 1 ml of the working reagent directly during the essay.

Test Sample Material:

The mixture of host samples (Gram & Moong) and the gut contents of different larval instar (I-IV) treated with phosphate buffer were used as test sample, pipetted into clean dry test tubes labeled as Blank (B), Standard (S) and Test (T).

The operational procedure was followed as per kit and the optical density of blank, standard and test were taken at 505nm wavelength at 37°C (RT).

Calculation formula:

$$Triglycerides\ in\ mg/\ dl = \frac{\text{Abs. T}}{\text{Abs. S}} \text{x} 200$$

OBSERVATION

In the present study significantly, lipids were identified during larvae growth. The body weights and diameter of larvae increased linearly with days, became larger during the course of development. The larvae may have gained fat to increase their weight during this period. Larval stage would deposit a large amount of fat to accumulate energy for later development in adults.

Estimation of Total Triglycerides

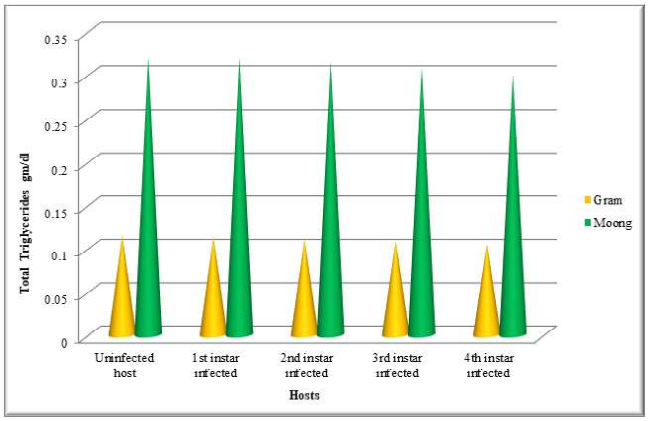
The results of estimation of total lipid by GPO/PAP method as reflected in Table 1 and Graph I A & B clearly establishes that Moong host contained more quantity of total triglyceride in mg/dl in uninfected state (0. 321) whereas, the value for Gram was 0.115 mg/dl.

The pattern of variation in the different larval instars (I-IV) growing on these two hosts provides a unique gradual depletion of the total lipid content of the instar wise infected host as well as the larval gut content lipids. For the host Moong the gradual depletion value of the total triglyceride ranged between 0.321 - 0.290 whereas for the larval instar gut content (I-IV), the value also reflected a declining range from 0.320 to 0.285 mg/dl. Similarly for the host-Gram, the gradual depletion values ranged between 0.115 - 0.104 whereas the gradual depletion value in the larval instar gut content ranged from 0.113 to 0.104.

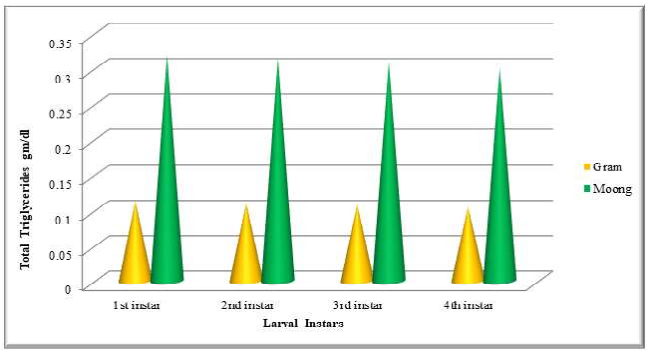
Although the difference range of depletion of host lipid was greater than that of the depletion range of gut content lipid of different larval instars thriving on the host, yet the growth of larval body did not show a significant dependency for the lipid on the host resource as per ANOVA result.

Table 1: Spectrophotometric estimation of Total Triglycerides (by GPO/PAP method) in mg/dl of uninfected & infected hosts - Gram & Moong along with propagating larval instars gut contents (I-IV) of beetle *C.chinensis L.*

Hosts	Total Triglycerides in gm/dl								
	Uninfected host	1st instar infected		2nd instar infected		3rd instar infected		4th instar infected	
		Host	Larva	Host	Larva	Host	Larva	Host	Larva
Gram	0.115	0.114	0.113	0.112	0.110	0.109	0.108	0.106	0.105
Moong	0.321	0.320	0.319	0.317	0.315	0.311	0.309	0.305	0.300



Graph I A : Spectrophotometric estimation of Total Triglycerides (GPO & PAP method) uninfected & infected hosts - Gram & Moong of Gram beetle *C.chinensis* L.



Graph I B : Spectrophotometric estimation of Total Triglycerides (GPO & PAP method) of larval instar gut content (I-IV) of Gram beetle *C.chinensis* L. from Gram & Moong

RESULT & DISCUSSION

The investigation covers the studies of spectrophotometric techniques for the quantitative estimation of lipids. The biochemical basis of host specificity in the pest has further been ascertained through quantitative estimation of the lipids through either spectrophotometric or electrophoretic techniques which have been specially attempted in the present investigation. The results of these quantitative biochemical assays furnished in the (Table 1 & Graph No. 1A & 1B) extend further strength to the findings of the biochemical investigation of host specificity in Gram beetle, *C.chinensis* L.

Stryer (Stryer's *Biochemistry*)[4] have extended the hypothesis that the quantity of the total lipids estimated from the gut content of the larval stages of the pest insects is directly proportional to the types of biomolecules consumed from the host source and their positive role in the metabolic activities. The Moong host in this investigation has been found to be rich in the biomolecular profile of monomeric nature which could have been easily displaced by the feeding larvae from the host source. The choiced by the molecules displaced from the host source played a catalytic role in the protein biosynthesis helpful in the overall morphogenetic processes and the lipids and cholesterol helpful in the fat body production, vitelo genesis and Gunaidol maturation as has been observed in the *Manduca* bugs (hemiptera). The gradual rise in the total lipid in the successive larval instar as estimated through the UV-spectrophotometer directly conveys the information that the larvae almost metabolically empowered in the Moong host over the Gram leading to better population growth. These findings have also been statistically examined and found to be significant through the applications of SD and ANOVA (Analysis of variance for multifactor experimental observations. Synthesising the concepts of the biochemical & biostatistical findings pertaining to this research work, it appears that the Gram beetle- *C.chinensis* L. has better preference to Moong over the Gram in terms of its host specificity. The detail information of spectrophotometric estimation of total of both host (uninfected & infected) and the gut content of the pest larval instar I-IV. It reflects the trend of gradual fall in the total Lipid. Almost trend has been obtained for the quantitative stimulation of total lipid (by GPO/PAP method) but the values of total lipids (triglycerides) in Gram host & the pest larvae I-IV growing on this host are substantially very low (Graphs I A). This might be considered as the constraints in the multiplication of the pest & its larval grown on this host while lipid contents of Moong & its dependent larval instars display a better range of estimated value which is a positive support for the pest multiplication (Graph I B).

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 57-60 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Toxicity of organophosphate and synthetic pyrethroid pesticides on rohu fish (*Labeo rohita*) fingerlings

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Received : 08th December, 2022 ; Revised : 09th January, 2023

Abstract- The aim of the experiment was to evaluate the toxicity of organophosphate pesticides (chlorpyrifos) and synthetic pyrethroid pesticides (fenpropathrin) to Rohu fish (*Labeo rohita*) fingerlings. The LC_{50} for Rohu fish (*Labeo rohita*) fingerlings was 5.70, 0.5, 0.015 and 0.26 ppm for chlorpyrifos, methyl parathion, fenvalerate and fenpropathrin, respectively. These pesticides are highly sensitive to fingerlings and is alarming as it is commonly used in the agricultural fields and may affect the fisheries sector.

Key words: Pesticides, Methyl parathion, Fenvalerate, Rohu fish (*Labeo rohita*) fingerlings, ecosystem.

INTRODUCTION

The indiscriminate use of pesticides has increased to protect crops from insect pests, and it contaminating land and water from production sites and storage tanks, run-offs from fields. Pesticides affects the non-target organism also e.g. invertebrates, mammals, birds and fishes, Organophosphate (OP), carbamate and synthetic pyrethroid (SP) pesticides are mostly used in agriculture to control pests due to their non-persistent nature in the environment.[1] These pesticides drastically affects and their high toxicity to some non-target species has been observed by many workers.[2-4] Laboratory studies revealed that pesticides can be acutely toxic to freshwater and estuarine organisms also.[5,6] The Rohu fish (*Labeo rohita*) is one of the important fish in composite fish culture and are very common in inland water.

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Acute toxicity is a convenient tool used extensively to evaluate the toxicity of physiologically active substances and also to evaluate the potential of chemical contamination on commercially and ecologically important species.[7] The objective of the present study is to evaluate the toxicity of selected pesticides to Rohu fingerlings, so that the farmers may be aware about the effect of pesticides on fish culture.

MATERIALS & METHODS

Methyl parathion 5% EC, fenpropathrin 20% EC, fenvalerate 20% EC, chlorpyrifos 40% EC were procured from local market. Stock solution of 100 ppm and appropriate working concentrations were prepared in filtered water. The fingerlings of Rohu fish (*Labeo rohita*) average weight 105 g were procured from local hatchery.

Standard bioassay methods were followed to evaluate toxicity of pesticide using static bioassay system.[8,9] LC_{50} were carried out in glass jars of 2 litres capacity (20.5 x 13.5cm). All glassware was acid cleaned prior to the tests. Initially all test organisms were treated with wide range of

pesticide concentration in filtered water to evaluate the at which mortality around 50% occurs. For each concentration of test pesticide ten fingerlings were exposed in groups. The experiment was repeated with five or more concentrations of different pesticides for test organism. The different concentrations of pesticides ranged between 0.1-8 ppm. The tests and controls for each experiment were in triplicate and the controls had only water. The other experimental conditions, such as, temperature (28±1°C), pH 7.57, was maintained. Acute toxicity measured as mortality of organisms exposed to each pesticides was estimated by determination of the 24 h LC_{50} (the concentration of the pesticides which kills fifty percent of the test animals after 24 h exposure). Organisms were considered dead if they did not exhibit movement and it laid immobile.

RESULTS & DISCUSSION

The results obtained in experiments, where Rohu fingerlings were tested against organophosphates and synthetic pyrethroid pesticides, show that fingerlings were sensitive to all pesticides tested. it was found higher the concentration more was mortality. The variability in the degree of sensitivity is reflected by the lethal concentration values of pesticides, at which 50% mortality occurs. The 24 h LC_{50} was 5.7, 0.5, 0.015 and 0.26 ppm for chlorpyrifos, methyl parathion, fenvalerate and fenpropathrin respectively. Among rohu fingerlings are considered as more sensitive to fenvalerate followed by fenpropathrin, methyl parathion and then chlorpyrifos.

Fingerlings appear to be highly sensitive to pesticides and have low LC_{50} values, which may be due to the fact that both OP and SPs are particularly produced to target insects which also include crustaceans.[10] As OP and SP pesticides are non-persistent in nature and readily degradable, therefore acute toxicity test of 24h LC_{50} were considered in the present study.

Fingerlings used in the present study, were highly sensitive as expected in line with findings of some previous studies, in other organisms[11] for example, copepods, brine shrimp, fish. Generally, in bioassay of fingerlings are employed for toxicity test to predict environmental risk. Juveniles are more sensitive to environmental impacts than the adult.[12] The reason is that fingerlings have higher surface area to volume ratios than adults and have faster uptake kinetics of the chemical.[12] The metabolic capacity in fingerlings may be different from that in adults.[13]

Table 1- Toxicity of organophosphate pesticides and synthetic pyrethroid pesticides on *Labeo rohita* fingerlings after 24 h of treatment showing LC_{50}.

Pesticides	No. of fingerlings	Concentration tested (ppm)	LC_{50} (ppm) level
Methyl parathion	150	1.0- 8	0.5±0.06
Chlorpyrifos	150	0.1 - 0.8	5.70±0.07
Fenvalerate	150	0.1- 2	0.015±0.05
Fenpropathrin	150	0.1-0.8	0.26±0.10

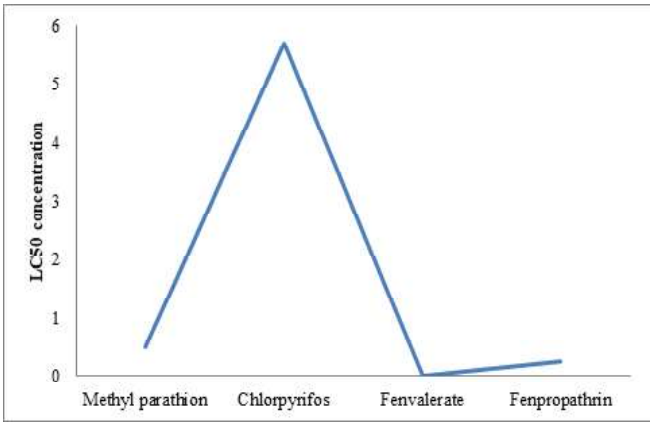


Fig 1: LC_{50} of the selected pesticides on *Labeo rohita* fingerlings

The LC_{50} values found in the present work can be compared with those reported earlier by other authors for effects of different pesticides on various fish species. The LC_{50} values for different organisms provides only a rough indication of differences in specific tolerance as a number of factors influence the bioassay results, such as, temperature[14] and degree of susceptibility of test organisms.[15] It has been reported that pesticides toxicity increases with temperature.[14] For any species, sensitivity to a given pesticide varies with age, sex, nutritional background, health, stress and the environment. Variability due to differences in sensitivity between sexual and asexual species, as well as among intra- strains and clones of the same species has been reported for aquatic invertebrates used in ecotoxicological studies. It is also reported that the toxicity differs from species to species[6] and in some cases from place to place, which may be due to differences in bioassay techniques and purity of pesticides used, the differences found could also result from differences in tolerance to the exposure between species. Different

pesticides or even different salts of same pesticide have variable effect on same organisms.[6] This is also true for the toxicity of same group of pesticide to the same organism e.g., fingerlings when exposed to OP pesticide; phorate and methyl parathion, when exposed to organ chlorine (OC) pesticide; DDT and heptachlor[16], when exposed to OC pesticide; BHC and lindane.[17]

The 24 h LC_{50} of fingerlings was very low for both OP (chlorpyrifos, methyl parathion) and SP (fenvalerate and fenpropathrin) pesticides showing sensitivity of these organisms. Since India is an agricultural country and these pesticides are used in agricultural land. There is a paucity of data of these pesticides. Deterioration in the quality of aquatic environment affects the fish industry as well as the aquaculture in the inland waters.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 61-64

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

In vitro antidiabetic property of chitosan extracted from freshwater edible crab *Sartoriana spinigera* (Wood-Mason, 1871)

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Received : 12th December, 2022 ; Revised : 13th January, 2023

Abstract- Alpha (α) amylase is the key hydrolysing enzyme involved in the digestion of carbohydrates. Retardation of starch digestion by inhibition of enzymes such as α-amylase plays a key role in the control of diabetes. Purpose of the study was to investigate *in vitro* antidiabetic property of chitosan extracted from carapace and chelate legs of freshwater edible crab *Sartoriana spinigera* using Amylase inhibition assay. The amylase inhibition assay was performed using the chromogenic DNSA method where chitosan was screened for its amylase inhibitory activity in a range of 10µg/mL-100µg/mL. IC_{50} value was calculated using Liner Regression method. The carapace chitosan and chelate legs chitosan was found to exhibit an inhibitory action on amylase enzyme up to 18.87% and 82.00% respectively at maximum concentration (100µg/ml). Acarbose was used as reference drug. Thus, it can be concluded that the chitosan extracted from carapace and chelate legs of *Sartoriana spinigera* has *in vitro* antidiabetic potential. Further studies can be carried out on molecular mechanism of amylase inhibition of chitosan extracted from *Sartoriana spinigera*, to yield an ecofriendly, cost-effective and easily accessible remedy for diabetes.

Key words: Amylase, IC_{50}, *Sartoriana spinigera,* Diabetes

INTRODUCTION

Diabetes mellitus is a major endocrine disorder affecting nearly 10% of the population all over the World.[1] In India, there are estimated 77 million people above the age of 18 years are suffering from diabetes (type 2) and nearly 25 million are prediabetics (at a higher risk of developing diabetes in near future).[2] The estimates in 2019 showed that 77 million individuals had diabetes in India, which is expected to rise to over 134 million by 2045. Approximately 57% of these individuals remain undiagnosed.[3] One of the effective methods to control diabetes is to inhibit the activity of alpha amylase enzyme which is responsible for the breakdown of starch to simpler sugars.[4] This is contributed by alpha amylase inhibitors, which delays the glucose absorption rate thereby maintaining the serum blood glucose in hyperglycaemic individuals.[5] Some inhibitors in clinical use such as acarbose, miglitol, and voglibose produce serious side effects such as bloating, and abdominal discomfort.[6]

Chitin is a polysaccharide made up of β-(1→4) linked N-acetyl-D-glucosamine that is widely found in nature forming exoskeletons of crustaceans such as crabs, lobsters, krill and shrimps and insects as well as components of bacterial cell wall.[7] Chitin the precursor to chitosan was first discovered in mushroom by French Professor Henri Braconnot[8] in 1811. The application potential of chitosan, a deacetylated derivative of chitin, is multidimensional, such as in food and nutrition, biotechnology, material

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science, drugs and pharmaceuticals, agriculture and environmental protection, and recently in gene therapy too.[9]

Sartoriana spinigera a freshwater edible crab is abundantly found in water bodies of Jharkhand.[10] Although different body parts of crabs are used in ethnomedicine for the treatment of inflammatory and related disorders by tribal folk in Jharkhand.[11] Chitosan produced by marine sources have exhibited hypoglycemic effects but the use of exoskeleton of freshwater edible crab *Sartoriana spinigera* and its *in vitro* antidiabetic property have not been investigated. Hence, here an attempt made to know the *in vitro* anti diabetic activity of chitosan extracted from carapace and chelate legs of *Sartoriana spinigera* by using α-amylase enzyme inhibition assay models.

MATERIALS & METHODS

Extraction of chitosan: Chitosan was extracted following the method of Takiguchi (1991 a,b)[12,13] exoskeleton (carapace and chelate legs) powder was demineralized, deproteinized and deacetylated and finally extracted chitosan was confirmed by FTIR studies. Degree of deacetylation was calculated by graph obtained from FTIR-spectroscopy following the formula of Brugnerotto, J. *et al*, (2001)[14].

***In vitro* evaluation of antidiabetic property of chitosan (carapace and chelate legs) using α-Amylase inhibition assay[15]:**

The inhibition assay was performed using the chromogenic Dinitrosalicylic acid (DNS) method. The total assay mixture composed of 1440 mL, 1400 mL and 1350 mL of 0.05 M sodium phosphate buffer (pH 6.9), 50 mL of amylase and carapace chitosan and chelate legs chitosan samples at concentrations of 10 µg/mL, 50 µg/mL and 100 µg/mL were incubated at 37°C for 10 minutes. After pre-incubation, 500 mL of 1% (w/v) starch solution in the above buffer was added to each tube and incubated at 37°C for another 15 minutes. The reaction was terminated with 1.0 mL DNS reagent, placed in boiling water bath for 5 minutes, cooled to room temperature and the absorbance was measured at 540 nm. The control amylase represented 100% enzyme activity and did not contain any sample of analysis. To eliminate the absorbance produced by sample, appropriate extract controls with the extract in the reaction mixture in which the enzyme was added after adding DNS.

The liberated sugar was determined by the help of standard maltose curve and activities were calculated according to the following formula:

$$\underset{(\mu mol/ml/min)}{Enzyme\ activity} = \frac{conc.of\ maltose\ liberated}{mol.wt.of\ maltose} \times \frac{ml\ of\ enzymes\ used}{incubation\ time(min)} \times dilution\ factor$$

One unit of enzyme activity is defined as the amount of enzyme required to release one micromole of maltose from starch per min under the assay conditions.

The inhibitory property shown by the sample was compared with that of control and expressed as percent inhibition. This was calculated according to the following formula:

$$\%\text{Inhibition} = \frac{activity\ in\ presence\ of\ compound}{control\ activity} \times 100$$

Analysis of Acarbose as standard inhibitor:

Acarbose was used as a standard inhibitor and it was assayed as above-mentioned test sample concentrations. The assay method was similar to the above-mentioned procedure, instead of test samples, acarbose was added. The results were compared to that of test sample.

RESULTS

Extracted chitosan was confirmed by FTIR analysis. The FTIR spectrum of chitosan samples in the present study was obtained and effective peaks were compared with that of standard chitosan. Effective peaks of carapace chitosan were observed at 709.80, 875.68, 1026.13, 1489.76, 1658.78, 2885.51 and 3267.41 cm^{-1} and that of chelate legs chitosan were observed at 894.97, 1033.85, 1423.47, 1573.91, 1662.64 and 2881.65 cm^{-1}. The peaks observed in case of chelate legs chitosan and carapace chitosan out of 9 bands, 6 and 7 bands respectively were very similar to that of standard chitosan. The degree of deacetylation (DD%) of chitosan plays a significant role for determining the specific applications of chitosan. The degree of deacetylation of carapace chitosan and chelate legs chitosan obtained from *Sartoriana spinigera* was 98.190% and 80.156% respectively.

In the present study *in vitro* α-amylase inhibitory assay, carapace chitosan exhibited potent α-amylase inhibitory activity in a less dose dependent manner**.** % inhibitory activity of carapace chitosan at 10 µg/mL, 50 µg/mL and 100 µg/mL was found to be 4.63%, 10.75% and 18.87 % respectively (Table-2) with an IC_{50} value of 296.91 µg/mL. Chelate leg's chitosan exhibited potent α-amylase inhibitory activity in a dose dependent manner. % inhibitory activity of chelate legs chitosan at 10 µg/mL, 50 µg/mL and 100 µg/mL was found to be 34.14%, 65.49%

and 82.00% respectively (Table-1) with an IC_{50} value of 33.1957 µg/mL (Table-2). Acarbose was used as a standard drug. Acarbose at a concentration of 10-100 µg/mL showed α-amylase inhibitory activity from 43.41% to 100% respectively (Table-1) with an IC_{50} value of 1.69 µg/mL (Table 2). Chelate legs chitosan exhibited higher % inhibition against Amylase enzyme than carapace chitosan. % amylase inhibition for carapace chitosan was found to be 4.63% at 10 µg/mL and only 18.87% at 100 µg/mL whereas for chelate legs chitosan 34.14% at 10 µg/mL and 82.00% at 100 µg/mL.

Table 1- Estimated value of % Amylase inhibition of control, Acarbose and carapace chitosan and chelate legs chitosan.

Groups	Concentrations (µg/mL)	OD (540 nm)	Maltose liberated (µg)	Activity of Amylase (µmol/mL/min)	Amylase Activity (%)	Amylase Inhibition (%)
Control	10	0.570	44.92	0.012466	100.00	0.00
Acarbose	10	0.336	25.42	0.007054	56.59	43.41
	50	0.015	-1.33	-0.000370	0.00	100.00
	100	0.00	-2.58	-0.000717	0.00	100.00
Carapace chitosan	10	1.699	139.00	0.038578	95.37	4.63
	50	1.592	130.08	0.036103	89.25	10.75
	100	1.450	118.25	0.032819	81.13	18.87
Chelate leg's chitosan	10	0.386	29.58	0.008211	65.86	34.14
	50	0.217	15.50	0.004302	34.51	65.49
	100	0.128	8.08	0.002243	18.00	82.00

Table 2- LC_{50} values for *in vitro* amylase inhibition activity of Acarbose, carapace chitosan and chelate legs chitosan

Groups	IC_{50} value (µg/mL)
Acarbose	1.69
Carapace chitosan	296.91
Chelate leg's chitosan	33.19

DISCUSSION

In the present study, chelate leg's chitosan exhibited higher % inhibition against α-amylase enzyme than carapace chitosan. % amylase inhibition for carapace chitosan was found to be 4.63% at 10 µg/mL and only 18.87 % at 100 µg/mL whereas for chelate legs chitosan 34.14% at 10 µg/mL and 82.00% at 100 µg/mL. Ravi *et al.* (2012)[16] reported two mollusc species which had *α*-amylase inhibiting activity. Maximum inhibition observed by acetone extract of *Hemifusus pugilinus* was 72.23±0.44% at 50 µL concentration and highest inhibitory effect exhibited by acetone extract of *Natica didyma* was 51.23±0.44% at 50 µL concentration. But, the control drug (Acarbose) had significantly higher inhibitory action. According to Sudha *et al.* (2011)[17], 3 isopropanol plant extracts exhibited concentration dependent inhibition with IC_{50} values, viz., seeds of *Linum usitatissimum* (540 µg/mL), leaves of *Morus alba* (1440 µg/mL) and *Ocimum tenuiflorum* (8.9 µg/mL) against porcine pancreatic amylase. Acarbose as the standard inhibitor exhibited an IC_{50} (half maximal inhibitory concentration) value of 10.2 µg/mL. But in the present study IC_{50} value of chelate legs chitosan (33.1957 µg/mL) of α-amylase inhibition is higher than plant extract of *Linum usitatissimum* (540 µg/mL) and leaves extract of *Morus alba* (1440 µg/mL). Sindhu *et al.* (2013)[18] reported that the α-amylase inhibition assay of methanolic extracts of *Cinnamomum zeylanicum* (130.55 µg/mL), *Artocarpus altilis* (118.88 µg/mL), *Piper betel* (84.63 µg/mL) and *Artocarpus heterophyllus* (70.58 µg/mL) exhibited 50% α-amylase inhibition activity. But in the present study the IC_{50} value of α-amylase inhibition of chelate legs chitosan (33.1957 µg/mL) was higher than the plants *Cinnamomum zeylanicum* (130.55 µg/mL), *Artocarpus altilis* (118.88 µg/mL), *Piper betel* (84.63 µg/mL) and *Artocarpus heterophyllus* (70.58 µg/mL).

CONCLUSION

The above results indicated that chitosan extracted from exoskeleton of *Sartoriana spinigera* possess the inhibitory effect on α-amylase enzyme. This enzyme activities in the body are responsible for postprandial hyperglycemia by break down of dietary carbohydrates into glucose. It may lead to reduction in post prandial hyperglycemia in diabetic condition.

ACKNOWLEDGEMENT

The researchers are grateful to faculty of Central Instrumentation Facility, BIT, Mesra, Ranchi for sharing their knowledge, equipment and precious time. The researchers are also thankful to the Head of Department of University Department of Zoology, Ranchi University, Ranchi for providing all essentials required during the research work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 65-70 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Microbial population and diversity in earthworm midden of *Ocnerodrilus occidentalis* Eisen

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Received : 11th January, 2023 ; Revised : 11th February, 2023

Abstract- The effect of earthworm (*Ocnerodrilus occidentalis* Eisen) middens on microbial population density, biomass and bacterial diversity has been studied in laboratory. Initially bacterial population was high in midden (57.2 ± 2.804 X10^9) than soil (45.2 ± 0964 X 10^9) there after a sharp decline in bacterial population was observed both in soil and earthworm midden. The bacterial species reported in the midden of *O. occidentalis* belong to the genus *Aeromonas, Bacillus, Pseudomonas, Sphingomonas, Kocuria, Acinetobacter* and *Methylobacterium*. The objective of the present study was to analyse the state of art on the microbial population and bacterial diversity in the midden of *O. occidentalis*.

Key words: Midden, *O. occidentalis*, Bacterial diversity, *Bacillus*

INTRODUCTION

Earthworms are considered to be important ecosystem engineers in terrestrial ecosystem.[1] Earthworms are frequently associated with their ability to mix the soil, increase water infiltration rate, improve soil aeration, nutrient cycling, build soil structure and increase the biological activity of the soil.[2-8] The soil is used as a medium for plant growth whose productivity and stability greatly depends on the balance between living and non-living components. Crop plants stores the essential nutrients and energy from sun required for proper growth and are recycled in to the soil by the help of micro and macro-organisms through decomposition process.[9] Qualitative and quantitative microbial activities are the key factors for productivity and sustainability of soils health for maintenance of crop production.[10-12] Mechanical mixing of the mineral particles and organic matter in the digestive system of earthworms may result in increasing or decreasing the activity and number of beneficial or pathogenic microorganisms.[13]

Earthworm casts are hot spots of nutrient dynamism and microbial activity and may be used as model for studying the influence of earthworm soil microbial communities by making some changes in the patch structure of the microbial environment.[14] We studied the structure and activity of the bacterial communities in the soil and surrounding soil and measure implement in the agricultural field. The present study reports for the first time the microbial population from the midden of worm *Ocnerodrilus occidentalis.*

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MATERIAL & METHODS

Soil sample collection

Sampling was done to collect the earthworm *Ocnerodrilus occidentalis* Eisen from different agroecosystem sites in Ranchi, located between 21°58'N - 25°19, NL and 83°20'E- 88°4'EL and at a height of 629m above mean sea level (MSL) and study was carried out in

laboratory by culturing the earthworms in plastic container under oxygenated and moist condition. The middens were collected from the plastic container and used for microbial study.

Bacterial culture and isolation

The bacterial population in midden and soil was estimated by Dilution plate method.[15] The bacteria were isolated from soil samples by taking 1g of sample which was diluted with 9 ml of sterilized deionized water till 10^{-7} dilution. 1 mL inoculums of the primary suspension were taken for bacteria culture in a petriplate (diameter= 100mm) containing CzapekDox agar media and were inoculated at 37°C for 48h.[16] After that colony count were continued at every interval of 7 days till 42nd day.

Isolation of DNA and genomic analysis

Identification of the bacteria from midden and soil were done by isolation of DNA and genomic analysis. For the total DNA isolation from the pure culture, bacterial cells were washed with TESS buffer [10 mMtris/HCl, 1mM Na_2EDTA, 0.1 M NaCl and 0.1% Sarkosyl (N- lauroyl sarcosine)] and resuspended in TE buffer (10 mMTris/HCl, 1 mM Na_2EDTA). 50 mg lysozyme mL-1 and 0.1% SDS were used to lyse the cells. The subsequent phenol / chloroform extractions and ethanol precipitation were carried out as described by Sambrook *et al.* (1989)[17]. The quantity of the DNA was checked by running on 1.2% agarose gel. A single band of DNA with high molecular weight has been observed. The extracted genomic DNA of both isolates was used as template DNA for amplification of the 16S rDNA gene. PCR amplification of the Fragment of 16S rDNA gene was DONE. The purification of PCR amplicon was done to remove contaminants. Forward and reverse DNA sequencing was carried out by using BDT v 3.1 cycle sequencing kit on ABI 3730 X 1 genetic analyzer and consensus sequence was generated by Aligner software. The 16S rDNA gene sequence was used to carry out BLAST with nr database of NCBI gene bank database.[18,19]

RESULTS & DISCUSSION

The bacterial population in soil and in midden during the study period has been presented in Table 1. The, bacterial population in soil gradually decreased from $45.2\pm0.964X10^9$ to $4.8\pm0.665X10^9$ on 42 days of observation. On 7th day of observation bacterial population in midden was $47.9\pm0.568X10^9$ which gradually decreased to $11.5\pm0.568X10^9$. The bacterial population in midden was always higher than soil and is in conformity with the findings of Kumari *et al.*, (2010)[20]. The population showed

Table 1: Bacterial population in soil and midden

Days of observation	Bacterial population in soil (M±SD)	Bacterial population in midden (M±SD)	% Change
0	$45.2 \pm0.964\ X10^{9*}$	$57.2 \pm2.804X10^{9*}$	+26.54%
7	$35.1 \pm1.053X10^{9**}$ (-22.34)	$47.9 \pm0.568\ X10^{9**}$ (-16.25)	+36.46%
14	$19.1 \pm2.003\ X10^{9**}$ (-57.74)	$33.9 \pm1.738\ X10^{9**}$ (-40.73)	+77.48%
21	$14.6 \pm1.588X10^{9**}$ (-67.69)	$27.6 \pm0.450\ X10^{9**}$ (-51.74)	+89.04%
28	$12.3 \pm0.611\ X10^{9NS}$ (-72.78)	$25.2 \pm0.617X10^{9NS}$ (-55.94)	+104.87%
35	$6.7 \pm1.450\ X10^{9**}$ (-85.17)	$16.7 \pm0.929\ X10^{9**}$ (-70.80)	+149.25%
42	$4.8 \pm0.665\ X10^{9**}$ (-89.38)	$11.5 \pm0.568\ X10^{9**}$ (-79.89)	+139.58%

Values in parenthesis are percentage increase (+) or decrease (-) over initial value; Significant level *=pd"0.01; **=pd"0.001; NS=non-significant; n=3

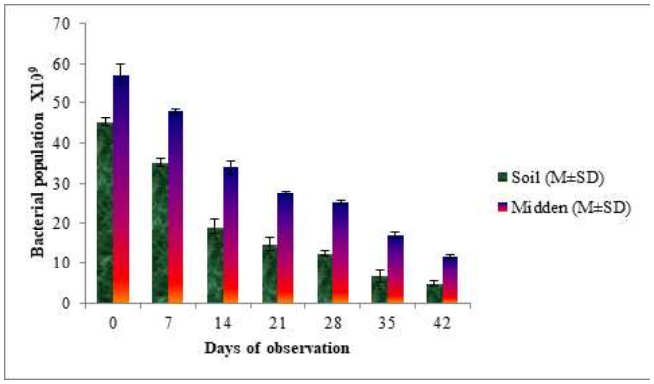


Figure 1- Bacterial population of soil and midden

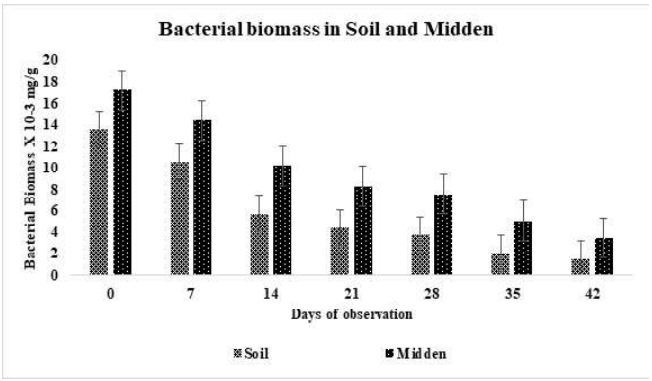


Figure 2- Bacterial Biomass of soil and midden.

significant increase from soil to midden ($p<0.001$) (Fig. 1). The maximum % change (149.25%) between bacterial population of soil and midden was observed in 35th day of observation. Initially percentage increase from bacterial population of soil to midden was found to be 26.54% which was gradually increased to 149.25%. The percentage decrease in bacterial population of soil over initial population was recorded as 22.34%, 57.74% and 67.69% on 7th, 14th and 21st day while more pronounced as 72.78%, 85.17% and 89.38% on 28th, 35th and 42nd day respectively (Table 1). The percentage decrease over initial bacterial population of midden was observed as 16.25% on 7th day which gradually increased due to aging of midden.

The wet weight (mg/g soil) of bacterial population in soil decreased by 22.34% (67.8±2.625 X 10^{-3} to 52.65±1.580 X 10^{-3}) on 7th day. On 14th and 21st day the decrease was in order of 57.74% and 67.69%. After 21st day, wet weight decreased was more pronounced 89.38% on 42nd day (Table 2). A similar trend of fall in biomass (mg/g soil) was observed. The initial biomass was observed as 13.56±0.525 X 10^{-3} which was gradually decreased to 1.44±0.199 X 10^{-3} on 42nd day (Fig 2). The initial wet weight (mg/g midden) of bacterial population in midden was 85.80±4.206 X 10^{-3} which decreased 16.25% on 7th day and 51.74% on 21st day (41.40±0.676 X 10^{-3}). A similar trend of fall in biomass was also observed in midden. The maximum biomass 17.16±0.841 X 10^{-3} was observed in first day and thereafter declined to 3.45±0.170 X 10^{-3} on last day of observation (Fig. 2).

Table 2: Wet weight of bacterial population of soil and midden

Days of observation	Soil (M±SD)	Midden (M±SD)	% Change
0	67.8±2.625 X10^{-3}	85.80±4.206 X10^{-3}	+26.54%
7	52.65±1.580X10^{-3} (-22.34)	71.85±0.852X10^{-3} (16.25)	+36.46%
14	28.65±3.004 X10^{-3} (-57.74)	50.85±2.608X10^{-3} (40.73)	+77.48%
21	21.90±2.382 X10^{-3} (-67.69)	41.40±0.676X10^{-3} (51.74)	+89.04%
28	18.45±0.916 X10^{-3} (-72.78)	37.8±0.916 X10^{-3} (-55.94)	+104.87%
35	10.05±2.175 X10^{-3} (-85.17)	25.05±1.393X10^{-3} (-70.80)	+149.25%
42	7.20±0.998 X10^{-3} (-89.38)	17.25±0.852X10^{-3} (-79.89)	+139.58%

Values in parenthesis are percentage increase (+) or decrease (-) over initial value

The literature review reveals some controversy on the role of earthworms on the size of the soil microbial biomass. Several studies have shown that earthworms reduce microbial biomass, mainly by consumption when the soil travels through the alimentary tract of earthworm.[21-28] In contrast, other studies have found earthworm induced increases in microbial biomass.[20,29-33] Devliegher and Verstraete (1995)[23] suggested that the net effect of earthworm on microbial biomass is a product of reductions in biomass during gut passage and stimulation due to mixing of organic matter into the soil profile. Brown *et al.*, (2000)[34] emphasized the importance of temporal and spatial scale when evaluating the effects of earthworms on the soil profile, suggesting the different behaviour of fresh earthworm casts than aged casts.

The main nutritional source of organic matter for earthworms is the plant detritus however a few microorganisms like have been sometimes found to be part of the diet of these worms.[35-37] This relationship between microorganisms and worms isn`t essentially restricted to a predatory process, as it has been reported that earthworms' digestive enzymes do not have significant impact on some microorganisms.[38]

On the basis of genomic analysis, identified 11 species of bacteria from the genus *Aeromonas, Bacillus, Sphingomonas, Acinetobacter, Methylobacterium, Brevundimonas, Pseudomonas* were found in the soil while one more *Kocuria* was found in the midden of *Ocnerodrilus occidentalis*. Seven species of bacteria of the genus *Bacillus* were identified (*B. insolitus, B. megaterium, B. brevis, B. pasteurii, B. sphaeriaus, B. thuringiensis* and *B. pabuli*) within the intestine of *Onychochaeta boincana.*[39]

An observation was made for the midden of the species *Ocnerodrilus occidentalis*, there was an increase in 572 colonies, further divided into 7 groups: *Aeromonas* 17%, *Bacillus* 53%, *Pseudomonas* 10%, *Sphingomonas* 4%, *Kocuria* 2%, *Acinetobacter* 11%, *Methylobacterium* 3% (Table 3). The genus *Bacillus* was dominant group found in the midden of *Ocnerodrilus occidentalis*. Hyun Jung *et al.* (2004)[37] reported that the genus *Bacillus* was

dominant group found in the intestine of the earthworm. Some species include *Bacillus cereus, B. subtilis, B. thuringiensis, Pseudomonas aerufasciens, P. putida*. In comparison to soil the genus *Bacillus, Aeromonas* were increased in the midden of *Ocnerodrilus occidentalis*.

Table 3: Percentage occurrence of bacteria isolated from soil and earthworm midden.

Identified species	Family	Soil	Midden
Aeromonas punctata strain JM10	Aeromonadaceae	16.5 %	17.4 %
Bacillus cereusprobio 32	Bacillaceae	26.7 %	31.4 %
Kocuria HO-9042	Micrococcaceae	-	1.7 %
Sphingomonas	Sphingomonadaceae	9.9 %	4.3 %
Pseudomonas	Pseudomonadaceae	9.5 %	9.9 %
Acinetobacter	Moraxellaceae	10.8 %	10.4 %
Brevundimonas	Caulobacteraceae	6.1 %	-
Methylobacterium	Methylobacter	-	2.6 %
Bacillus MBL13	Bacillaceae	7.9 %	7.8 %
Bacillus BFF-3	Bacillaceae	8.8 %	9.6 %
Bacillus HBUM 84231	Bacillaceae	2.8 %	4.3 %

ACKNOWLEDGEMENT

The first author (S. Kumari) thankfully acknowledges DST Govt. of India for financial support as INSPIRE fellowship and HOD Zoology for providing lab facilities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 71-76

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Costus speciosus: The "insulin plant" a boon for tribals in Jharkhand- A review article

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Received : 14th January, 2023 ; Revised : 26th February, 2023

Abstract- India, is going to be the diabetic capital of the world by 2035. About 42% of the total population of India is now suffering from hyperglycemia and type II diabetes. Jharkhand, the land of tribal having a population of about 4.06 crore among which about 14% of total population have been reported to suffer from diabetes. Tribal depends mostly upon the wild plants and vegetation for their ailment treatment. Herbal plants are now being extensively used as a therapeutic agent all over the world in both traditional and modern system of medicine. More than 2000 plants species are being used for drug formulation. *Costus speciosus*, an insulin plant is thus a wild variety plant of Jharkhand, belonging to the family Costaceae. It is believed that not only leaves but rhizomes and roots of this plant are used to control various ailments. Rhizome of this plant is a major source of diosgenin, a phytochemical used to lower blood glucose level. Thus, the present work is concerned with the review of the medicinal properties of insulin plant as a whole.

Key words: Tribal, ailment, *Costus speciosus*, phytochemicals, hyperglycemia

INTRODUCTION

The demand for herbal medicines is growing rapidly in both national as well as in international market. India is well known for its Ayurveda system of treatment. More than 2500 plant species are used to cure various ailments that include anti-carcinogenic activity, antifertility activity and spasmolytic activity.[1] Herbal product comes with various pharmacological activities such as anti-oxidant, anti-fungal, anti-microbial etc. Due to its numerous pharmacological properties natural products are more preferable among human beings as compared to the synthetic one. As synthetic drugs sometimes are also having limited access due to high cost of health case, socioeconomic condition or even regional availability, so *Costus* is considered as boon for tribal people. Apart from this, natural products are less toxic and shows no or very less side effects.[2-4]

According to WHO, about 80% of the total world population confide on Ayurvedic medicine system to cure diseases due to its easy availability.[5] Therefore, there is an urge to know about the potent and less toxic drugs that could be obtained from plants**.**

Costus speciosus (J.Koenig), one of the important herbal plant, now used to treat several diseases. *C. speciosus* also known as 'spiral ginger' or 'crepe ginger' belongs to family costaceae is now renamed as *Hellenia speciosa* by an Indian scientist named S.R Dutta.[6] It is native plant of South Africa, Indonesia, Australia, Asia and America.[7] More than 200 species of *Costus* were distributed throughout the world.

Costus speciosus (Koen.) is lately introduced in India and is now cultivated as an ornamental plant because of its beautiful flowers.[8]

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MORPHOLOGY

C. speciosus is a perennial rhizomatous erect herbal plant that grows upto 2.7m in height.[5,7,9] Leaves are mainly green with large and oblong shape and are spirally arranged over the stem.[10]

Costaceae family consists of 52 genera and each species of this family can be identified with their flower colour.[11]

It has large white colour flower emerges out from cone like red bracts which is ovate with pointed apex.

Costus speciosus is a rhizomatous plant that can be propagated through various vegetative methods or by seed dispersal methods.[12] Suitable environment for this plant cultivation is rainy season. Where the rhizome starts sprouting during the months of march to april. During the month of September, flower starts and lasts till October.[13]

Costus speciosus is commonly known as keukand (Hindi), kebuka (Sanskrit), keu and keumut (Bengali), tara (Assam), kashmeeramu in Telegu.[14]







Fig. 1- *Costus speciosus* (J.Koenig)

TAXONOMIC POSITION[15]

Kingdom	-	Plantae
Phyllum	-	Tracheophyta
Class	-	Liliopsida
Order	-	Zingiberales
Family	-	Costacaeae
Genus	-	*Costus*
Species	-	*speciosus*

MEDICINAL PROPERTIES

C. speciosus are widely used as a traditional system of medicine for treating various ailments. Not only leaves but entire parts of this plant are used in preparing medicines.

LEAF

It has been reported that leaves of *Costus* are known to possess anti-glycemic activity.[16,17] According to the studies of Agarwal *et al*., 2018, various pharmacological activities have been found in *C. speciosus* such as anti-microbial, anti-hyperglycemic, larvicidal, anti-inflammatory, analgesic, antigenotoxic, hypolipidemic, antipyretic.[18-21] Apart from this, leaves of *C. speciosus* are also being used to treat psychic disorders.[22] Maritim in one of his studies have reported that the leaf extract of *Costus speciosus* is used to treat mental disorder.

Instead of this, the diabetic patients who have consumed leaves of *Costus speciosus*, has also reported that the use of leaves may also lower blood glucose level. Therefore, it is believed that, consumption of one fresh *Costus* leaf daily may results to control hyperglycemia.[23]

It has been reported that aqueous leaf extracts of *Costus speciosus* is sprayed over mosquito larva to eradicate malaria.[24]

RHIZOME

Diseases like jaundice, pneumonia, bronchitis, constipation, skin diseases, dropsy and cough can also be cured using juice of *C. speciosus* rhizomes.[25-27] Sometimes paste of grounded rhizome is also applied over skin to reduce burning sensation.[28] It is reported that the juice of *Costus* rhizome is very effective in treating leprosy.[29-31] The rhizome of *C. speciosus* is bitter with aphrodiasiac, expectorant, astringent, anti-helminthic and hepatoprotective properties.[32-34]

Bhattacharya *et al.* (1973) has evaluated the anti-spasmodic and muscles relaxant property of ethanol extract of *Costus speciosus.*[35]

The rhizomes are a major reservoir of diosgenin, a steroid phytochemical known to possess anti-diabetic properties.[36,37] Therefore, in Indian traditional system, the leaves and rhizomes of *Costus speciosus* are now used as an alternative of synthetic drugs biguanides.

It has been observed that the rhizomes of this plant are also very effective and can be used to treat abdominal pain, headache, ear pain, liver problems and boils.[38,39] In East Asian countries like Japan, this plant is being used to control syphyllis.[40]

STEM

In Ayurveda, young stem paste is applied all over the blister.[41] Hafez *et al*. (2018) evaluated the anti-oxidant activity of methanolic extract of root and stem.[42]

Crude hexane extract of *C. speciosus* mitigate the liver damage and may also increase the total protein level in body.[43]

Stem decoction of *C. speciosus* plant is used to reduce high fever in patient.[44]

Phytochemical analysis of *C. speciosus* reveals the presence of phenols, flavonoids, glycosides, diosgenin, custonolide and tannins.[45-47]

CONCLUSION

Thus, from the above information it can be reviewed that the insulin plant, *Costus speciosus* is having a large number of beneficial constituents like flavones, resinoids, flavonoids, inulin, resins which may be responsible for the various medicinal properties and further exploration of medicinal properties and various phytochemical constituents responsible for the pharmacological actions are required to be done.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 77-80 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Histochemical detection of protein concentration in the gut lining of larval instars (Ist- IVth) of *Callosobruchus chinensis* Linn. propagating on moong.

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Received : 16th December, 2022 ; Revised : 16th January, 2023

Abstract- This study was carried out to assess the localization of protein in the gut of larval instars of *Callosobruchus chinensis* Linn. propagating in Moong. Histochemical analysis of the gut lining of the larval instars in *Callosobruchus chinensis* showed glycogen. However, the protein content was not the same in foregut, midgut and hindgut of larval instars.

Key words: Protein, Gut Lining, Larval instars, *C.chinensis.*

INTRODUCTION

The aim of the current study was to analyze histochemical characterization and the determination of protein in the gut lining of larval instars for better understanding for the presence of Protein. The gastrointestinal system is one of the most metabolically active system and prominent at all life stages due to energy utilization and perform digestion of the food, nutrient absorption and expelling waste products. Many compounds act mainly on the midgut of the insects interfering with larval development, even at no - lethal concentration. In this view, compounds can affect energy reserve (Sugar, Glycogen, Protein and Lipids) altering the development of the larvae and the adults.

In the larval phase, insect use carbohydrates and protein as the main source of energy for their development.[1] The gut sections of the larval instars subjected to HgBB histochemical staining technique have however displayed a concentric lining of protein deposition.

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MATERIALS & METHODS

I. Maintenance of laboratory culture of the pest:

Fresh adult pest assorted from the host specific sampling jars were released in pairs of both male and female in small size sterile multiplication jars of 200ml size containing 50gms of uninfected fresh seeds of Moong to facilitate their fresh reproduction and perpetuation. The multiplication jars were tag labeled as MPC (Moong Pest Culture) in order to get the accurate observation and results. Five replications of each culture were simultaneously maintained having the jar tags MPC I-IV respectively. In each unit jar of host specific culture of the pest beetle atleast 50gms of fresh uninfected and properly cleaned host grains was kept as the host food resources on which atleast 5 pairs of male and female adult beetles were release for propagation.

II. Collection of larval instars for histochemical studies:

As a matter of fact, the eggs laid on the outer seed coat of Moong were noticeable under hand lens or dissecting binocular microscope but the larval instars growing inside the cotyledons seed kernel of the grains were not easily observed for recording the emergence of

first larval instar and its successive growth through different instars totaling for a number in all the host varieties. The infected grains were soaked in lukewarm water and cut open with the help of scalpel. The larval instar growing inside the seed kernel making zigzag tunnel with the help of their strong mandibular teeth were carefully taken out from the tunnels with the help of sharp pointed needle and brush.

On the basis of the size and shape larval instars collected from the infected seed kernel were assorted into host wise plastic tube marked ML-I to ML-IV. These larvae procured from two different host culture jars, served as the essential raw material for the histochemical studies of the gut for localization of tissue proteins in relation to different host of leguminous grains.

Fixative Solution

1. Carnoy's Fixative

Rec. for carbohydrate and protein histochemistry; Fixation -3-6 hr.

Absolute alcohol ..... 60.00 ml
Chloroform ..... 30.00 ml
Glacial acetic acid ..... 10.00 ml
Wash 2-3 hrs. in absolute alcohol

Chapman's Mercury Bromophenol Blue Technique for Proteins.

Fixative - Carnoy's Fixative.

Procedure:

- The slides were deparaffinized in xylene for 10 min.
- The sections were hydrated by passing through absolute, 90%, 70%, 50%, 30% alcohol and distilled water: 5-10 min each or dips.
- The slides were stained in 1% $HgCl_2$ + 0.05% sodium bromophenol blue + 2% aqueous acetic acid for 10 - 15 min.
- The slides were transferred to 0.05% acetic acid for 2 changes of 5min.
- The slides were transferred to tertiary butyl alcohol for 2 changes of 30 min.
- The slides were transferred to 0.5ml n-butylamine in 100ml xylene until sections turn blue completely.
- The slides were transferred to xylene for 2 changes of 10 min.
- The slides were mounted in DPX.

OBSERVATION

HgBB histochemical staining for host specific detection of protein in the Foregut sections of larval instars (I^{st}-IVth) of *C.chinensis* L. feeding on Moong (Table 1).

The unstained section of the foregut of all the three larval instars developing on Moong host, when treated with the histochemical stains HgBB for locating the protein deposition have also given variegated degrees of colour reactions as follows. The second and third larval instars foregut wall display trace to moderate reaction in the form of light blue coloration in all the layers of the gut wall, more intense in the cuticular intima (Table-1).

On the contrary the IInd instar larval foregut derived from Moong host as given more prominent blue colour reaction which also shows (+++) higher amount of protein present in these sections, unfortunately the sections of the IVth instar could not give desired colour reaction in the larvae developing on Moong host except in the cuticular intima (+++) gives intense degree reaction due to absence of requisite protein (Table-1).

HgBB histochemical staining for host specific detection of protein in the Midgut sections of larval instars (I^{st}-IVth) of *C.chinensis* L. feeding on Moong (Table 2).

The status of staining reaction in the midgut section of the IInd larval instar feeding on Moong host has given greenish blue indicating the qualitative & quantitative differences of protein present in epithelial layers as well as layers of muscle fibres. Similar colour reaction has been also observed in the IIIrd larval instar midgut sections with prominent cryptic cells coloured blackish blue and light blue in the larval growing in Moong respectively(Table 2).

The IVth larval instar midgut section has almost given similar blue colour reaction without black or green tinse developing on Moong host (Table-2). It might be indicative of the presence of identical type of protein in these layers.

HgBB histochemical staining for host specific detection of protein in the Hindgut sections of larval instars (I^{st}-IVth) of *C.chinensis* L. feeding on Moong (Table 3).

The condition of histochemical reaction appears hindgut section of the larvae II, III & IV thriving on Moong host have provided more intense blue and green colour reaction in IInd and IVth instar respectively. The staining status of the IIIrd larval instar hindgut section on Moong host is disproportionate - some part of the epithelial follicle staining very light almost negative while other staining deep blue or highly positive (Table-3).

Table 1- HgBB histochemical staining reaction of the different layers of foregut of larval instars (Ist – IVth) of *C.chinensis* L. propagating on host Moong.

Host specific layer reaction.	Layers of Foregut of Larval Instar from Moong host					
Larval Instar	Peritonium	Muscles		Basement Membrane	Epithelium	Intima
		CM	LM			
Instar I	No result was achieved.					
Instar II	+	+++	+++	++	++	+++
Instar III	++	+	+	++	++	+++
Instar IV	++	+	+	++	++	+++

Legend: - = No reaction, + = Trace reaction, ++ = Moderate reaction, +++ = High reaction.

Table 2- HgBB Histochemical staining reaction of the different layers of Midgut of larval instars(Ist – IVth) of *C.chinensis* L. propagating on host Moong.

Host specific layer reaction.	Layers of Midgut of Larval Instar from Moong host					
Larval Instar	Peritonium	Muscles		Basement Membrane	Epithelium	Intima
		CM	LM			
Instar I	No result was achieved.					
Instar II	+	+	++	++	++	+++
Instar III	+	+++	+++	+++	++	+++
Instar IV	+	++	++	++	++	+++

Legend: - = No reaction, + = Trace reaction, ++ = Moderate reaction, +++ = High reaction.

Table 3- HgBB Histochemical staining reaction of the different layers of Hindgut of larval instars(Ist – IVth) of *C.chinensis* L. propagating on host Moong.

Host specific layer reaction.	Layers of Hindgut of Larval Instar from Moong host					
Larval Instar	Peritonium	Muscles		Basement Membrane	Epithelium	Intima
		CM	LM			
Instar I	No result was achieved.					
Instar II	+ +	+	++	-	+	+
Instar III	++	++	++	-	+	+
Instar IV	++	++	++	-	+	+

Legend: - = No reaction, + = Trace reaction, ++ = Moderate reaction, +++ = High reaction.

RESULT & DISCUSSION

The alimentary canal of insects is subdivided in Foregut, Midgut and Hindgut. The midgut is a region of absorption of nutrients and of synthesis and secretion of enzymes and hormones. This absorption is fundamental to the survival. The species, as any changes in this region may modify the growth and development of the insects.[2]

Histological sections were stained with Mercury Bromophenol blue for total protein, with the objective of targeting the differential deposition of protein in bound as well as free stage in the layers of the gut wall of larval instars II, III, & IV actively feeding on Moong, the histochemical staining technique Mercury bromophenol blue adopted in the present research work provided a very informative picture in respect to the host specific propagation of the pest. The presence polysaccharides by PAS indicates that the total protein (Bromophenol blue) of the mucus gland secretion in B., morio in glycoprotein.[3,4]

The Mercury Bromophenol blue for total protein showed that peritoneum layer of all the larval instar fore, mid- & hindgut region has given trace reaction (+) irrespective of the host varieties, nevertheless, the foregut & hindgut peritoneum of all the larval instar feeding on Moong host has given moderate reaction (++) Table 1,2,3. The intima layer in the foregut region of all the larval instar propagating on Moong host as well as the peritrophic membrane in the midgut region in place of the intima has given moderate to strong reaction indicative of high degree of protein deposition in these layers (Table 1,2)[5]. The high degree of protein deposition is observed in Moong host.

The histochemical findings showing better deposition of proteins in the foregut and midgut muscle layers in the larvae of *C.chinensis* propagating on Moong has given a clear indication of host specificity towards Moong seeds. Similar investigation in terms of characterization of midgut trypsin like protein, enzyme three trypsinogen factors from the lesser grain borer, has been found in the structure of *Rhizopertha dominica*.[6,7]

However, the histochemical result showed that the layers of hindgut in IInd, IIIrd and IVth instar feeding on Moong host has given moderate (++) reaction in peritoneum and muscular layer, almost negative (-) reaction in basement membrane except the epithelium and intima layer which has given trace (+) reaction in all the three instars. Microsection of the same gut area of the larvae derived from Moong has clearly given moderate to high reaction in the layers of peritoneal muscles and intima (Table 1,2,3). Lashman Sah & Srivastava have investigated the histochemical and histopathological changes in protein deposition in hindgut of *C.chinensis* treated with insecticide Gamma-BHC, fenitrothion and Endosulfan.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 81-84

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Sources of air pollutants in Ranchi, Jharkhand, India

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Received : 20th December, 2022 ; Revised : 20th January, 2023

Abstract- Air is one of the most important elements for the sustenance of life on this planet. But due to various anthropogenic activities air is being degraded drastically and very alarmingly. Air dust Particles size and their compositions are related to be significant factors in the assessments of the acute and chronic impacts of particulate matters on health. Studies examine the elements of particulate matters that exhibit serious health hazards on human health on long exposure and inhalation of these particulates. As a result, there is still a lot to learn about the sources of particulate matters, SO_2 & NO_2 that may regulate the concentration of these pollutants. Here, we assessed the primary pollutants releasing from the various sources in Ranchi and from the field data of four sampling stations, primary and secondary sources of oxidative potential and particulate matters in Ranchi. We found that the concentration of particulate matters, SO_2 & NO_2 were mainly governed by secondary aerosols, Auto-mobile emissions, building materials used in construction of roads, highways, and buildings. In contrast, most anthropogenic sources that contributed to the concentration of oxidative potential are fine-mode secondary organic aerosols, which were mostly produced by household biomass burning, and coarse-mode metals, and automotive non-exhaust emissions.

Key words: Anthropogenic, Exposure, Inhalation, Oxidative potential, Emissions.

INTRODUCTION

Despite the effectiveness of air quality regulations and advancements in pollution control techniques, the rise in urban migration has led to a growing population residing, working, and studying in close proximity to major sources of air pollution, including major roads, railyards, freight distribution canters, and industrial areas.

Characterizing air pollution sources in urban areas can be challenging, especially when it comes to transportation-related facilities. These facilities, such as rail yards, often consist of intricate emission sources within the site itself, while also supporting nearby freight movement activities and industries that contribute additional air pollution emissions.[1]

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Studies examining air quality measurements indicate that air pollution levels can be substantially higher in the vicinity of major transportation sources. Within the first 200-300 meters from the source, concentrations of various pollutants emitted by motor vehicles and other transportation sources are significantly elevated compared to background levels.[2-6]

The economic growth of a country is often attributed to urbanization and industrialization. However, the rapid urbanization has resulted in unregulated and disorderly industrial developments and increased vehicular emissions. These factors have emerged as significant environmental concerns globally, particularly in developing nations.[7]

Covid-19 Lockdown measures have had both positive and negative impacts on air pollution levels. In certain countries like China, India, and the USA, the concentrations

of particulate matter (PM), carbon monoxide (CO), nitrogen dioxide (NO_2), and sulfur dioxide (SO_2) have decreased as a result of reduced emissions from primary sources, leading to improved outdoor air quality.[8-10] However, other pollutants such as ozone (O_3) have increased in some European, Asian, and Latin American countries.[11-13] Moreover, lockdown measures have provided an opportunity to assess the short-term effects of economic activities on air pollution and identify associated impacts on human health.

Various anthropogenic activities, including industrial and automobile emissions, agricultural and biomass burning, construction activities, pyrotechnic displays, and related practices, constitute the primary sources of air pollutants.[14,15] Additionally, natural events such as dust and desert storms, forest fires, volcanic eruptions, sea spray aerosols, and pollen grains also have a significant impact on air quality.[16] These sources release a range of gaseous pollutants, including carbon monoxide (CO), sulphur dioxide (SO_2), oxides of nitrogen (NO, NO_2, NOx), ozone (O_3), particulate matter in varying sizes (PM10, PM2.5), and heavy metals.

STUDY AREA

Ranchi, located in the state of Jharkhand, India, is a bustling urban centre facing growing concerns about air pollution. As with many cities, Ranchi is confronted with a range of anthropogenic and natural activities that contribute to the emission of air pollutants. Industrial emissions, vehicle exhaust, agricultural and biomass burning, construction activities, as well as natural events like dust storms and forest fires, all play a role in the air pollution scenario in Ranchi. To ensure a healthier and cleaner environment, it is essential to address these diverse sources of air pollutants and implement effective measures to mitigate their impact on air quality in Ranchi. The latitude and longitude coordinates of Ranchi, Jharkhand is 23.3441°N latitude and 85.3096°E longitude. The total population of Ranchi city is currently more than 1.5 million.

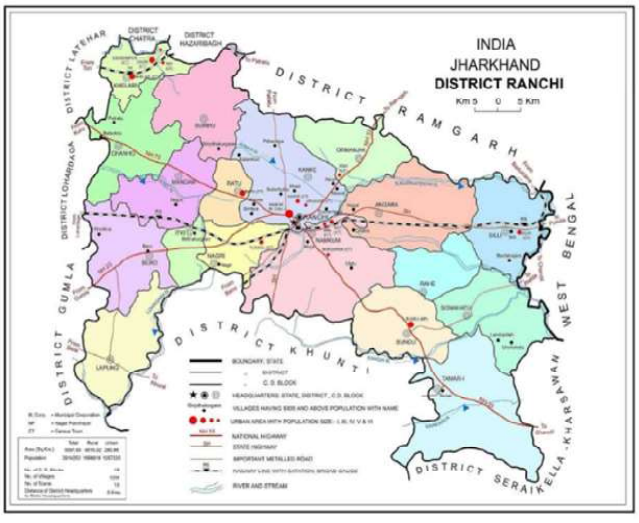


Figure 1- District Map of Ranchi.

RESEARCH & METHODOLOGY

AIR POLLUTANTS TRENDS IN RANCHI (2016-2020):

The analysis of annual pollutant concentrations (PM10, SO_2, and NO_2) from 2016 to 2020 was conducted to assess the trends and establish reduction targets for meeting national ambient air quality standards in Ranchi. The air quality data utilized in this study were obtained from the field sampling and by examining the trends of pollutants over the years, it becomes possible to advocate for a clean air action plan and establish specific targets. The long-term data collected from two manual monitoring stations in Ranchi for PM10, NO_2, and SO_2 are presented graphically in Figure 2, and the corresponding values are summarized in Table 2 and Table 3. It is evident that Particulate Matter (PM10) poses a significant challenge to Ranchi's air quality during the period of 2016-2021. The city consistently experiences elevated levels of PM10, although the concentrations of SO_2 and NO_2 remain within the prescribed standards. Table 1 indicating the various point and non- point sources of air pollutants emission form various industries, vehicle exhausts, emissions form power plants, constructions activities, agricultural waste burning, road dusts, residential emissions, forest fire, waste burning and so on. Table 2 is indicating the Air Quality Index (AQI) which was set by CPCB (Central Pollution Control Board) and their expected impact.

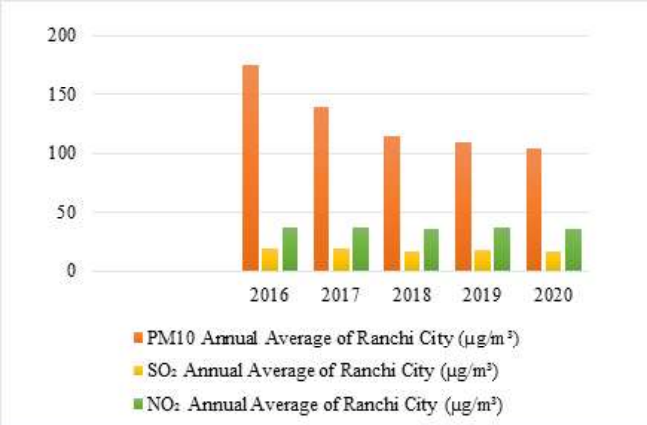


Figure 2- Average Air Pollution Trend in Ranchi City

Table 1- Point and non-point sources of air pollutants in Ranchi

Source	Type	Pollutants
Industrial emissions	Point source	Sulphur dioxide (SO_2), nitrogen oxides (NOx), particulate matter (PM), volatile organic compounds (VOCs), heavy metals
Vehicle exhaust	Point source	Carbon monoxide (CO), nitrogen oxides (NOx), volatile organic compounds (VOCs), particulate matter (PM)
Power plants	Point source	Sulphur dioxide (SO_2), nitrogen oxides (NOx), particulate matter (PM), mercury (Hg)
Construction activities	Point source	Particulate matter (PM), volatile organic compounds (VOCs), dust
Brick kilns	Point source	Particulate matter (PM), sulphur dioxide (SO_2), nitrogen oxides (NOx), carbon monoxide (CO)
Agricultural burning	Non-point source	Particulate matter (PM), carbon monoxide (CO), volatile organic compounds (VOCs)
Road dust	Non-point source	Particulate matter (PM), dust
Residential emissions	Non-point source	Particulate matter (PM), sulphur dioxide (SO_2), nitrogen oxides (NOx), volatile organic compounds (VOCs), carbon monoxide (CO)
Forest fires	Non-point source	Particulate matter (PM), carbon monoxide (CO), nitrogen oxides (NOx), volatile organic compounds (VOCs)
Waste burning	Non-point source	Particulate matter (PM), sulphur dioxide (SO_2), nitrogen oxides (NOx), carbon monoxide (CO), volatile organic compounds (VOCs)

Table 2- Categories of Air Quality Index along with the expected impact

Good (0-50)	Minimal Impact
Satisfactory (51-100)	Minor Breathing discomfort to sensitive people
Moderately Polluted (101-200)	Breathing discomfort to the people with lung, heart disease, children, and older adults
Poor (201-300)	Breathing discomfort to people on prolonged exposure
Very Poor (301-400)	Respiratory illness to the people on prolonged exposure
Severe (401-500)	Respiratory effects even on healthy people

Table 3- Average Air Pollution Trend in Ranchi City

Year	PM10 Annual Average of Ranchi City (µg/m³)	SO_2 Annual Average of Ranchi City (µg/m³)	NO_2 Annual Average of Ranchi City (µg/m³)
2016	173.98	17.91	35.58
2017	138.55	20.10	36.05
2018	114.54	18.29	36.60
2019	110.10	17.67	35.26
2020	108.48	16.61	34.80

CONCLUSION

The analysis of annual pollutant concentrations in Ranchi from 2016 to 2020 reveals a significant challenge posed by Particulate Matter (PM10) to the city's air quality. While the concentrations of SO_2 and NO_2 remain within prescribed standards, PM10 consistently reaches elevated levels. The data collected from monitoring stations and presented in Table 3 indicate a gradual decrease in PM10 levels over the years, suggesting some progress in addressing the issue. The identified sources of air pollutants, as presented in Table 1, include industrial emissions, vehicle exhaust, power plants, construction activities, agricultural burning, road dust, residential emissions, forest fires, and waste burning. These sources contribute to the overall air pollution in Ranchi, emphasizing the need for targeted interventions and reduction strategies. The Air Quality Index (AQI) categories described in Table 2 demonstrate the expected impact on human health based on different pollution levels. Ranchi has experienced air quality ranging from "Moderately Polluted" to "Very Poor" during the analysed period, potentially causing breathing discomfort, respiratory illness, and other adverse effects on vulnerable populations. It is worth noting that the COVID-19 pandemic had an unintended positive impact on air pollution levels, leading to a decrease in pollutant concentrations across the board. Figure 2 illustrates the average air pollution trend in Ranchi City, showcasing the decline in pollution levels during the pandemic. To ensure sustainable improvement in air quality, a comprehensive

clean air action plan is necessary. This plan should focus on reducing PM10 levels, addressing the identified sources of pollution, and promoting awareness and adoption of cleaner practices in industries, transportation, construction, and other relevant sectors. Additionally, continued monitoring and enforcement of air quality standards are crucial for tracking progress and maintaining accountability.

Efforts to mitigate air pollution in Ranchi should involve collaboration between government agencies, industries, communities, and other stakeholders. By implementing targeted reduction targets and adopting sustainable practices, Ranchi can strive towards achieving and maintaining national ambient air quality standards, thus safeguarding the health and well-being of its residents.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 85-92 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Report on the future of insects and ecosystems: Research and Management for sustainability

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Received : 22nd December, 2022 ; Revised : 23rd January, 2023

Abstract- The importance of insects in an ecosystem is well-established by different researchers. Due to the anthropogenic interventions, the relationship between insects and ecosystems mainly in agro-horticultural areas came under threat. Chemical fertilizers and insecticides have an impact on the life-table parameters of insects both pests and other non-pest insects. The objectives of this article are to focus on the core areas of research and management for the sustainability of integrated pest management. The causes of the diversity threat of insects were habitat loss, climate change, pollution, pesticides, invasive species, and overexploitation. The future of insect management depends on IPM strategies, biological control, genetic control, precision agriculture, sustainable farming practices, and citizen science. Fruitful utilization of recent technologies like RNA interference, autonomous robots, gene editing, nanotechnology, artificial intelligence (AI), and drones are game-changers in the management of insects and ecosystems.

Key words: Horticulture, Fertilizers, Pest insects, Integrated pest management, nanotechnology, AI

INTRODUCTION

A popular quote is - "*If all mankind were to disappear, the world would regenerate back to the rich state of equilibrium that existed ten thousand years ago. If insects were to vanish, the environment would collapse into chaos*."
(E. O. Wilson, 2007)

1. OVERVIEW OF THE IMPORTANCE OF INSECTS IN THE ECOSYSTEM

A class of animal kingdoms called Insecta carries most animal species on this earth.[1] It is cosmopolitan and is important because of its biodiversity, ecological practices, influence on pollination, and effects on human health and other natural resources. Their adaptation to the surrounding is commendable. The ecological factors like temperature, mean rainfall, humidity, food plants, and other minor factors.

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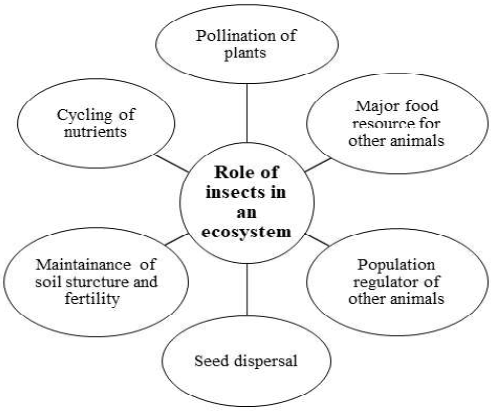


Fig. 1- Roles of insects in an ecosystem.

Moths, butterflies, bees and beetles are the most common pollinators. Insectivorous animals are belonging to different phyla of animal kingdom. Insects feed on both animals and plants. Insects participate in the dispersal of seeds of many plants.

This article covers analytical reports and reviews published recently on the subject concerned. It included **(a)** reviews of existing recent works of literature, **(b)** results from experimental studies, and **(c)** priorities for future research and design best practices for them.

1.1 What are the current issues with insect diversity?

The goal of this chapter is to document the harmonious interaction between insects and their ecosystem, the types of threats they face, and their impact on their diversity and ecosystem. What steps do researchers take to protect them for sustainable management practices?

1.2 Threats to insect diversity and ecosystem health

Insects are considered maintainers of the health of an ecosystem. About 58% of the earth's animals' biodiversity is insect diversity.[1] They serve as pollinators, food sources for many animals and some insectivorous plants, and decomposers of biological wastes. However, insects face different threats like biological, physical, chemical, environmental, natural, and fabricated threats for survival. It really affects their biodiversity and the functions of an ecosystem like energy flow, food pyramids, and ecological productions. Some of the major threats that the insect population faces nowadays are described under the following points:

(a) Habitat destruction and fragmentation: Several pieces of research clearly depicted the threat to the insect population by continuous habitat destruction and fragmentation. Manufactured activities like urbanization, deforestation, and agro-horticulture have led to the destruction and fragmentation of the natural habitat of insects. To meet the demand for the shelter of the increasing human population, forests, grasslands, and agricultural fields are fabricated and new infrastructural development may completely morph the area. Deforestation for agricultural, road infrastructure, and shelter purposes not only destructed the natural habitat of animals and plants but also the insects.

(b) Climate change: Climatic factors like temperature, rainfall, and humidity are important for the survival of insects. They vary from one place to another because of the different ecosystems and topography. Fabricated industries, burning of fossil fuels, and household items can alter the temperature of a place. It can lead to the loss of host plants, change of soil nature, and loss of animals in a food web. It may lead to the community-level cascade of decline and extinction of species.[2]

(c) Invasive Species: A non-native species of plants or animals introduced in an ecosystem for the purpose of economic growth purpose but become harmful to the environment, the economy, and human health is called invasive species. There are many examples reported by the scientists listed in Table 1.

Table 1- A list of invasive animals and plants and their effect on the insects and their ecosystem.

S.N.	Invasive animal/ plant	Effect
1.	Emerald Ash Borer	This beetle attacks and kills ash trees and also damage the local insects and the ecosystem.[3]
2.	Asian Longhorned Beetle	This beetle attacks hardwood trees and therefore, negative impact on both insects and the ecosystem. [3]
3.	Japanese Knotweed	A type of plant that outcompetes native plant, therefore, shows a negative impact on the insect population and local ecosystem. [4]
4.	Zebra Mussel	Shows a negative impact on both the aquatic ecosystems and insect population.

(d) Light pollution: Although the major factors, which affect insects in wider aspects, are habitat loss, use of insecticides, and climatic stress but artificial lighting at night (ALAN) has also a role. ALAN is created by different sources like buildings, streetlights that illuminate roads and highways, industrial sites, cultural programmes, markets malls, and other infrastructures. Still, spare research works have been done to correlate the effect of artificial lighting at night on the life-table parameters of insects and other nocturnal animals.[5] But the work done by many researchers have observed the effect of such illuminations on the nocturnally active insects like moths.[6,7] It affected the vital behaviours of insects that included feeding, migration, defence, and reproduction. Stewart (2021)[5] reported that 'white' LEDs with major short wavelength mimic daylight. It is well known that nowadays these LEDs are profusely used for ALAN. Owens reported that communication abilities like warming signals or courtship signals of bioluminescent taxa became obscured in ALAN. Whereas, the ultraviolet and blue radiations attract moths.[6]

(e) Overexploitation: The utilization of a natural resource in a manner where it loses sustainability is called the overexploitation of that resource. The resource can be a living or non-living material. In the case of insects, overexploitation can lead to significant damage to their population, diversity, and species richness of an ecosystem. There are many examples reported by scientists. A few of them are Wild Honey Bee, Lac Worm, Silk Worm, and Beetles. Due to commercialization, the Wild Honey Bee population has been depleted in recent years. It threw a negative impact on the pollination of plants. Beetle wings are collected for making decorative items. It created the overexploitation of the beetle population. Green Jewel Beetle (*Chrysophora chrysochlora*) population is declining therefore, the bird population feeding on them also declined.[7] Many colourful butterflies have been illegally exported to foreign countries for decoration like Southern Birdwing (*Triodes minos*). It declined the pollination of the plants.[8]

(f) Insecticides: Farmers control insects profusely and use various types of insecticides. Mostly synthetic types were used like organophosphates, Lindane organochlorines, carbamates, Cyclodiene, Pyrethroids, etc. They killed insects by damaging the nervous system, respiratory system, and digestive system. There are non-specific insecticides that also affected mammals, birds, microbes, and other beneficial insects. Insecticides are widely used to control pests and insects that harm crops, animals, and human health. However, the use of insecticides has raised concerns about their impact on human and ecosystem health. This report will provide an analysis of the impact of insecticides on human and ecosystem health. Impact on Human Health: Insecticides can have significant impacts on human health. Exposure to insecticides can cause acute or chronic poisoning, which can result in nausea, headaches, dizziness, seizures, and even death. Long-term exposure to insecticides has been linked to chronic health effects such as cancer, reproductive problems, neurological disorders, and developmental abnormalities. Pregnant women and children are particularly vulnerable to the harmful effects of insecticides. Exposure to insecticides during pregnancy can lead to birth defects, developmental disorders, and low birth weight. Children who are exposed to insecticides are at increased risk of developing behavioral problems, cognitive impairments, and respiratory illnesses. Impact on Ecosystem: Insecticides can also have significant impacts on ecosystem health. The widespread use of insecticides can disrupt ecosystems by killing off non-target organisms such as pollinators and natural enemies of pests. This disruption can lead to imbalances in the ecosystem, which can have cascading effects on other species and ecological processes. Insecticides can also contaminate soil, water, and air, leading to long-term environmental damage. Contaminated water and soil can affect plant growth and aquatic life, and contaminated air can affect the respiratory health of humans and animals. The use of insecticides can have significant impacts on human and ecosystem health. The harmful effects of insecticides on human health include acute and chronic poisoning, reproductive problems, neurological disorders, and developmental abnormalities. Insecticides can also disrupt ecosystems by killing off non-target organisms and contaminating soil, water, and air. To mitigate these impacts, it is important to minimize the use of insecticides and adopt alternative methods of pest control such as integrated pest management.

2. THE NEED FOR RESEARCH AND MANAGEMENT

About 80% of the animals found on earth are insects; their number and diversity reflect their intensity. Common people do not have much goodwill towards insects because many diseases are insect-borne. Almost all people have to deal with mosquitoes, flies, cockroaches, etc., for which many types of insecticides have to be used for prevention. Killing them generally gives happiness to people. In the same way, farmers also have negative feelings about insects because they harm their crops, as a result of which different types of chemical pesticides are available in the market and farmers buy them and use them in large quantities so that their crops can be saved. Stay and earn the benefit of a good yield. Lack of correct information, lack of training, and lack of effective pest management can be clearly felt. The reality is that about 35% of the world's food grains are produced only with the help of insects because it is the main polymer. Organic wastes are also eliminated by decomposing them. It is a source of food for many types of animals. For us, silk honey is also an important source of dyes, medicines, etc. It is necessary to understand the importance of insects and there should be a continuous effort to make them accessible to the common people. Globally, it is reported a decline in insect diversity and richness.

3. UNDERSTANDING INSECT DIVERSITY AND ECOSYSTEM FUNCTIONALITY

3.1 Methods for studying insect communities

Trapping and collection of insects are important for the proper study of their life cycle, host plants, host-pest interaction, the impact of climate change, and the effects of the insecticides applied on them. There are many methods used to collect them detailed as follows:

a. Sweep netting: this method is used to sweep flying or hopping insects by using the net in the field. Examples belonging to different orders are Coleoptera (beetles), Hymenoptera (bees and wasps), Diptera (flies), Lepidoptera (butterflies and moths), and Orthoptera (grasshoppers and crickets).

b. Pitfall trapping: Small holes are made and then cups or traps are placed there to catch insects that fall in. It is also used to collect samples of ground-dwelling insects. Insects of different orders are trapped by this method: Dilopoda (millipedes), Gryllidae (crickets), Araneae (spiders), Formidae (ants), and Carabidae (ground beetles).

c. Light trapping: It is used to trap nocturnal insects specially moths and coleoptiles. There are different types of this trap used by researchers.

d. Malaise trapping: It is made up by setting up a tent-like structure with a funnel-shaped trap. It directs flying insects into a container of alcohol or some other alternative preservatives. It is used to collect insects from different orders: Lepidoptera (butterflies and moths), Hemiptera (bugs), Hymenoptera (ants, bees, wasps), Coleoptera (beetles), and Diptera (flies).

e. Visual survey: researchers estimate the insect populations in a given area and time quickly use it.

f. Mark and recapture: Marking of insects with paint or tiny binds made by the researcher. Then they are released back into their environment. By counting marked and unmarked insects, the population size can be determined.

3.2 The Role of Insects in ecosystem services

Insect participates in different ecological services like pest control, pollination, wildlife nutrition, dung burial, and nitrogen recycling. Large animals excrete a great amount of dung may create a challenge the waste-management issues. Insects play a role in dung decomposition.[10] Dung beetles are popular decomposers that roll dung into balls. Flies and some species of moths lay their eggs in dung and their larvae feed on them. It recycles its nutrients. Some ant species collect and feed on dung. They are recycled dung.

3.3 The impact of climate change on insects and the ecosystem

The diversity of insects in an ecosystem is highly influenced by climatic factors like temperature, humidity, rainfall, light, food sources, etc.[11,12]

4. THE FUTURE OF INSECT MANAGEMENT

4.1 IPM strategies:

Integrated pest management (IPM) is considered an effective and sustainable approach to managing and regulating populations. It involves multiple tactics to control pests without making significant harm to the environment and non-target organisms. Major reaches are progressing towards the conclusion that IPM is a promising approach to managing insects and the environment. There are various alternative and traditional chemical-based pest control methods as explained below:

(a) Biological control: Natural predators, parasites and pathogens come under biological control to manage insect populations.[13] Parasitoid wasps like *Aphidius colemani* are widely used to control aphids like Green Peach Aphids in greenhouse crops. These wasps lay their eggs inside the aphid's body. Their larvae feed on the body of wasps and kill them. The use of predatory insects like Lady beetles and lacewings kills target insects in cropland. Other predatory bacteria and viruses also kill target insects.

(b) Genetic control: Genetic engineering is very helpful to control insect pests by introducing genetically modified insects.[14]

(c) Precision agriculture: Technological advancements like GPS, sensors and drones are used to gather information on cropland including crop health, pest populations and climatic conditions.

(d) Sustainable farming practices: Crop rotation, cover cropping and reduced tillage can help the farmer to get healthier soil and thus dependence on synthetic materials can be reduced.

(e) Citizen science: Common citizens are involved in scientific research with the help of digital devices like smartphones and other digital platforms. It is helping scientists to get valuable data for accurate analysis.[15]

4.2 Habitat restoration and conservation for insects:

Several reports say that due to unplanned developmental strategies in the last two hundred years damages have been made to a large extent. Different manuals of habitat restoration and conservation strategies. It includes the following steps:

(a) Restoration of a degraded landscape can help butterflies to regain status. A study in the Western Ghats region of India reported that the restored forest contained higher diversity of butterflies than the degraded forests. It was due to the increased host plants and microhabitats. Nature Conservation Foundation (NCF) Rajanikanth *et al.*, (2020)[16] also reported dragonfly conservation in the Western Ghats. This place is also known for the natural habitat for fireflies called bioluminescent insects.

(b) Restoration of grassland benefited the restoration of the population diversity of grasshoppers. A report published in 2019 said that grasslands in the Deccan Plateau region of India helped grasshopper populations to regain their richness.

(c) Agroforestry also emerged as a suitable place for different types of pollinators. In Tamil Nadu, it played a key role in increasing the diversity and abundance of pollinators. Das *et al.*, (2020)[17] reported the positive impact of the restoration of nectar plants on the population of butterflies.

(d) Restoration of wetlands in the Himalayan foothills of India is one of the best examples that depicted the benefit for the dragonflies' populations.

(e) For the diversity of beetles, the conservation of the Thar Desert in India was reported in 2016. Sengupta *et al.*, (2020)[18] reported the conservation of dung beetles in a tropical montane forest in the Southern Western Ghats.

(f) Conservation of tillage practices also found suitable for the increasing diversity of soil insects.

(g) Nilgiri Biosphere Reserve was announced World Heritage Site by UNESCO for butterfly conservation.[19]

(h) The Himalayan region is known as a natural habitat for many species of bees like the Himalayan honeybee and Giant honeybee. It was reported by *Apis* Himalayan Foundation (AHF) working for the conservation of bee habitats.

4.3 The use of technology for insect management:

(a) RNA interference: RNAi is a unique technology that is used to make a specific gene silent. In the pest control system, it can be used to silent target genes of pests and therefore, can control their population of them. One of the examples is controlling Diamondback moths. Researchers have developed an RNAi-based approach to control diamondback moths. They have identified a specific gene called V-ATPase that is essential for the moths' survival. By using RNAi to silence this gene, the researchers were able to significantly reduce the survival rate of the diamondback moths. The RNAi molecules were delivered to the moths by incorporating them into a transgenic broccoli plant, which the moths fed on.

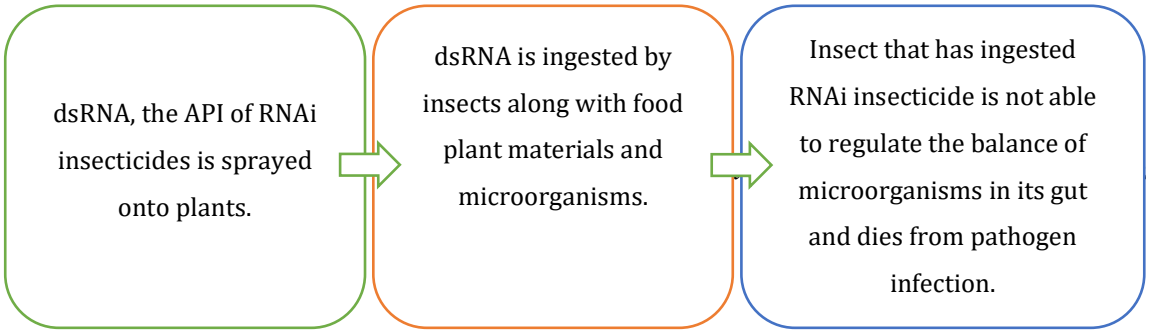


Figure 2: Scheme of the mechanism of RNAi action on controlling insect pests. The active pharmacological ingredient of RNAi insecticides is a double-stranded (ds) RNA. It is developed by the application of the RNAi mechanism. The sequence of the dsRNA is articulated specifically to make pair with a sequence within an essential gene or the survival of the pest. This assigned dsRNA is delivered to the insect midgut through diet. Once this dsRNA reached the cell, it starts the processing of its own RNAi pathway into a guide RNA (gRNA)- ssRNA. It disrupts the expression of the target gene and therefore insect dies.

(b) Autonomous robots: Robotics is a branch of engineering and technology that includes the conception, design, manufacture, and operation of specified robots. It works as an intelligent machine. It is very helpful in certain tasks of agriculture applications. A project H2020-funded Green Patrol worked on the detection and control of insect pests. It was very helpful to identify the insect pests in real-time. Chung *et al.*, (2014)[20] reported about an autonomous robot called Cabbot sampled adult whiteflies (*Bemisia tabaci*) on the host plant paprika plant (*Capsicum annuum*). It saved time in identifying and collecting insects on the host plants.

(c) Gene editing: Gene editing is one of the new biotechnological processes in which the specific nucleotide

sequences of an mRNA or DNA are edited purposely. One of them is known for its result and it is CRISPR-Cas9. It is used to develop a crop that is more resistant to pests. It is related to different categories of genes of insect pests.

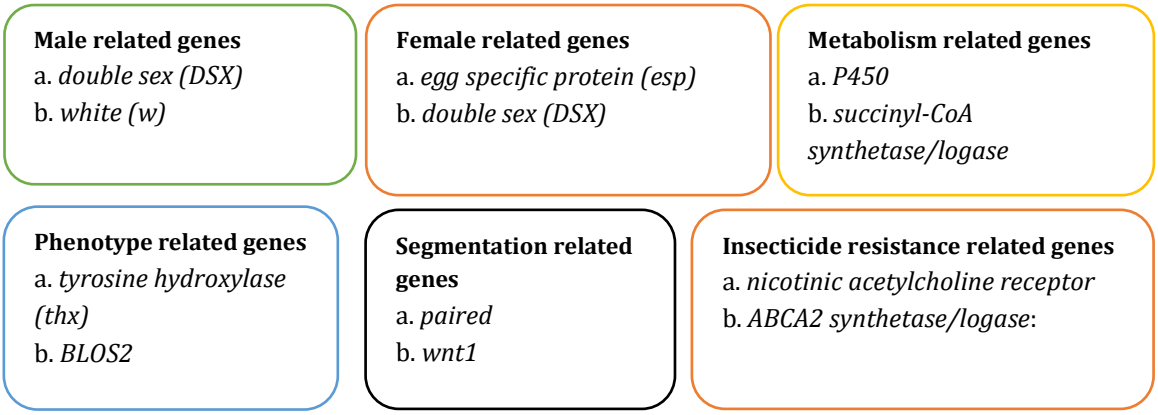


Figure 3: Various genes targeted by CASPER/Cos9 for insect pest control. It includes different categories of genes like male, and female, metabolism, phenotype, segmentation, and insecticide resistance-related genes.

(d) Nanotechnology: Nanotechnology has evolved as a potential tool to manage insect pests for their control more effectively and efficiently. Several researchers were helping the narrative to establish that nanoparticles can be an alternative method to control pests. Silver nanoparticles have been shown significant as insecticides that controlled mosquitoes, aphids, and termites. The research established the killing of bed bugs by the application of silver nanoparticles. Silver nanoparticles made from lemongrass oil were found effective to repel mosquitoes. Nano-sensors are also applied in cropland to detect the presence of pest larvae like the fall armyworm larvae were detected by the researchers in maize and other crops. Tomato plants were protected against whitefly infestation by the application of silicon dioxide nanoparticles. However, more research is needed to fully understand the effects of nanoparticles on pests, the environment, and non-targeted animals.

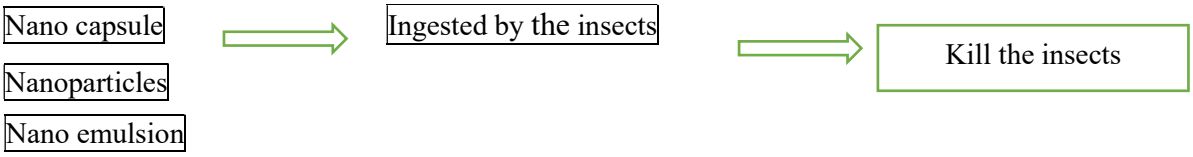


Fig. 4: The pest that leads to death can ingest different forms of insecticidal nanoparticles.

(e) Artificial intelligence (AI): AI is considered a technology of interest that can serve insect pest management more accurately without a great expense. Image AI is helping in the identification and control of insect pests. Singh *et al.*, (2019)[21] used AI to detect and classify tomato leaf diseases caused by insect pests effectively. It is also used in pest forecasting and early warning based on machine learning. It was reported by Lu *et al.*, (2019)[22] in China to control cotton bollworms.

(f) Drones: Drones are playing important role in insect pest management. They are used for surveying, monitoring, and spraying insecticides. They are well-equipped with cameras that can give accurate results in surveying the crop field. Drones can be equipped with sprayers or dispensers to deliver precise amounts of insecticides or other treatments directly to pest hotspots. This reduces the number of chemicals used and minimizes environmental impact. Drones can also be used to create detailed maps and models of crop fields, allowing farmers to identify areas of high pest activity and track changes over time. This information can be used to optimize pest control strategies and improve crop yields.

5. CHALLENGES AND OPPORTUNITIES FOR RESEARCH AND MANAGEMENT

5.1 Addressing knowledge gaps in Entomology

There are many fields in entomology specially related to insect pest management where extensive research work is required. Several scientists are working on different models of integrated pest management. Some of the knowledge gaps are belonging to Insect-pest interaction, Insect microbiomes, Insect conservation, Insecticide resistance, Insect social behaviour, Insect evolution, and

Insect genetics. Applications of Artificial intelligence and Robotics are a new field of research where machine learning and big data are helping researchers and farmers to get results that are more accurate.

5.2 Building Partnerships between Scientists, Policymakers, and Stakeholders

Making policies for effective and sustainable farming to fulfill the demand of the people as well as to maintain the ecosystem services and environmentally friendly practices are complicated. It needs active participation and strong scientific communication of stakeholders, scientists, and policymakers to build up effective science communication and evidence-based policies for the betterment of farmers and the environment. It can also point out the identification of shared goals, research questions, and objectives for the betterment of the community under a time frame. Sharing of data and research findings can also develop trust and accurate interpretation of a problem. These collaborative inputs may help in designing and implementing the research projects.

CONCLUSION

One of the major priority threats in the 21st century is to fulfill the need of hunger of more than eight billion human beings without altering the natural processes. The need for sustainable insect management practices to maintain a healthy ecosystem, enhance food security, and prevention of the spread of diseases. Various recent research outcomes have been discussed to address the challenges, the development of new technologies, the development of integrated pest management strategies, adoption of holistic approaches by taking care of environmental, economical, and social factors that influence the insect population. In this article, the author discussed the potential positive outcomes of ecosystem conservation and management in the field of providing quality food, clean water, and air, protecting against climate change, and mitigating climate change. Continuous research, engagement with social communities and stakeholders, and collaboration with different government and non-government institutions are required for effective integrated pest management.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 93-100 **Animal Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Impact of physico-chemical parameters on growth of IMCS (in both length and weight) in a mixed culture system

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Received : 23rd December, 2022 ; Revised : 24th January, 2023

Abstract- The current world population demands rich source of protein in their dietary management day by day. Fish is one of the major aquatic animals that is enriched with digestive protein and cost benefit to buy. The fish culture is a lucrative venture in the current era. As fish is aquatic vertebrate so impact of water quality in relation to physico-chemical parameters play a major role in life of fish. The major physico-chemical parameters of water quality management are like pH, Temperature, DO, Total Hardness, Salinity, Ammonia, Turbidity etc plays major role in fish production. The optimal range of these parameters leads to the maximum production of edible fishes like IMCs. The present work focuses on the optimal ranges of certain major water quality parameters for IMCS (catla, rohu, mrigal) culture in cement tanks (Artificial system) by intensive method to promote the fish farmers and culturists to do more and more fish culture which will be economically healthy for them by improving their financial status. Finding of this study shows that the average ranges of the above parameters are pH=7.35, temperature=31-33°C, DO=5.86 mg/l, Salinity=0.26 ppt, Ammonia=0.02 mg/l, Total Hardness=163.12 mg/l & Transparency=35.125 cm were the best suited for the growth of the IMCs. The SGR of Catla, Rohu & Mrigal was 13.08%, 13.28% & 12.75% respectively in 45 days culture period. The total survival rate in this study is 80%.

Key words: IMCS, Intensive mixed culture, Water quality parameters, Growth performance, Cement tank.

INTRODUCTION

Aquaculture is one of the most vital criteria in the progress of economy of a nation. The production of livestock in Aquaculture is comparatively higher than in Agriculture. Now, at present the "Blue Revolution" plays very important role in the Agro-based cultivation. The main advantage of Aquaculture is that it gives more production in lesser area than Agriculture. So, Aquaculture should be practiced more and more in the near future for healthy economy growth of a nation.

Water plays the most vital role in aquaculture as it carries out the fisheries/fish culture in it. Fish has a sufficient protein providing aquatic animal that strengthens the public health and economic condition of a country. Fish carry out all its physiological activities like swimming, reproduction, feeding, excretion etc. throughout water. So, the increase in fish culture is totally dependable on water and its quantities. Water-qualities refer to all the physical, chemical and biological parameters. Physico-Chemical parameters contain pH, temperature, salinity, turbidity, hardness, BOD, DO, COD, ammonia that affects the fish growth in water. The fish sectors contribute in the national GDP, agricultural GDP and foreign exchange by exporting both fish and its by-products. Fish culture also helps in the improvement of the employment of the nation.

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All living organisms grows best in tolerable limits of water quality parameters in which they show optimal performance.[1] So, any changes in these parameters can lead to the failure of desirable health of fish. Due to increasing population, the fish culture should be enhanced more and more to meet the requirement of food supply as well as to improve the economy of farmers and fish cultivators. Poor water quality can result in low product quality, low profit and potential human health risks.

In India, there are many edible species among which IMCs being a Cash Crop always take the leading position in fish culture. As the IMCs are natural feeder (Plankton Feeder), they require less amount of artificial feed that helps the farmer not spending so much money for feeding the fishes. The physico-chemical parameters play very crucial role in the growth of IMCs. For example, Indian Major Carps (IMCs) grow more quickly at the temperature range of 28-35°C and reach the sexual maturity at the age of 1.5-2 year.[2] The IMCs (Rohu, Catla & Mrigal) are the maximum edible fishes in Indian fish market and their cost ranges from Rs160-200/- per kg. So, IMCs' cultivation provides money both to its cultivators and sellers.

As the IMCs belong to different ecological niche, there is no competition for food among them. This helps the farmers to do their mixed cultivation in the same pond. As the population of India is increasing day by day, more cultivation of IMCs can meet the fish demand of India. IMCs can also be cultivated in smaller areas like tanks, as a result of which anyone can culture it. The proper growth of IMCs only depends on natural feed and Physico-Chemical parameters. But the Physico-Chemical parameters get changed several times because of the climatic changes in the aquatic environment. Any sudden/ drastic change in any parameter can lead to the death of fishes. So, it is very important for a IMCs cultivator to maintain the required amount of all the Physico-Chemical parameters in culture water.

Therefore, it is very important to study and maintain the effects of different types of Physico-Chemical parameters on the growth of IMCs in a mixed culture system for better commercial purpose.

MATERIALS & METHODS

1. STUDY AREA

Department of Fisheries Science, Centurion University of Technology and Management (CUTM) is situated in Paralakhemundi town in the district of Gajpati, Odisha having the co-ordinates 18.8069°N 84.1402°E on the north-east coast of India.

2. METHODOLOGY

At regular intervals of 3 days in 46 days (Dt.: 01/04/ 19-16/05/19) in the morning time, the water quality analysis was taken for the estimation of physico-chemical parameters of the water. Along with it, the measurement of both length & weight of IMCs (Catla, Rohu, Mrigal) were also taken at the same time.

The following water quality parameters were studied by the following methods:

1. Temperature= by Thermometer
2. pH=by Digital pH meter.
3. DO=using Standard Winkler Method by Titration.
4. Total Hardness= EDTA method
5. Transparency= Secchi Disk
6. Salinity=Normal Titration
7. Ammonia= By Ammonia Test Kit

3. EXPERIMENTAL BLUE PRINT

Cement tanks of 10×6×1.5 were selected for the intensive culture. A soil layer of 8 cm was given below the tank to avoid diseases and stress specially Argulosis. Ground water was used as the main water source for culture. There are 4 inlet plastic pipes and one large outlet source for water exchange.

4. STOCKING DENSITY

Stocking size= Fingerlings

Stocking rate= 3 sps. /m^2

So, total stocking= 3×60=180 sps.

Approximately 200 species were stocked.

Stocking rate= Catla: Rohu: Mrigal: 3:4:3

5. FEEDING

Total culture period= 50 days

Total sampling= 16 times

Sampling interval= 3 days

Table 1- Showing stocking density of IMCs

IMCS	STOCKING DENSITY
CATLA	60
ROHU	80
MRIGAL	60

The artificial feeds supplied to the Catla, Rohu and Mrigal were local made rice bran, mustard oil cakes and commercial floating feed (AVANTI COMPANY). The feeds were supplied in the required amount otherwise huge amount can also lead to the mortality of fingerlings. Feed waste and feed excreta were removed daily before feeding.[3]

Table 2- Showing feeding rate of IMCs

DAYS RANGE	PER BODY WEIGHT	AMOUNT OF FEED REQUIREMENT(g)/DAY
1-20	3%	650
21-40	2%	960
41-50	1%	600

6. FEED APPLICATION

Total amount of feed per day must be provided in morning and afternoon (or evening) time @ 50% total.

So, quantity of feed = 650/2= 325 gm (morning) & 325 gm (evening).

7. MEDICATION

i. After sampling, the species were treated under oxytetra cyclin solution as an antibiotic and stress-reliever.
ii. Disinfectant like $KMnO_4$ were used whenever required.
iii. Water exchange with fresh water supply=30%/day.
iv. Aeration was done regularly for sufficient oxygen supply.

8. GROWTH PARAMETERS

i. Mean Weight Gain (MWG) = W_f - W_i where W_i and W_f are initial and final body weights of fish (g) respectively.[4]
ii. Mean Length Gain (MLG) = L_2 - L_1 where L_2 and L_1 are final and initial body length of fish (cm) respectively.[5]
iii. Percentage Weight Gain (PWG) (%) = (Final Weight-Initial Weight) × 100/Initial Weight.[6]
iv. Percentage Length Gain (PLG) (%) = (Final Length-Initial Length) × 100/Initial Length.[6]
v. Specific Growth Rate (SGR) (%) = (lnw1-lnw0) x 100/Duration of experimental days, where w1 and w0 are final and initial body weights of fish (g) respectively and ln=natural logarithm.[7]
vi. Feed Conversion Ratio (FCR) = Feed intake by Fish (g)/Weight gain (g) where, weight gain=Final Weight-Initial weight.[4]
vii. Condition factor (K) = 100w/l3 where w and l are the observed total weight (g) and total length (cm) of a fish.[4]
viii. Survival rate (%) = no. of fish harvested/no. of fish stocked × 100.[5]
ix. Mortality (%) = no. of survival of fingerlings/no. of fingerlings stocked × 100

RESULTS

Economically fishes consider as a very important group of animals which provide a rich source of food, liver oil and a number of other by product like fish meal, fish manure, isin glass etc. The present work focuses on the optimal ranges of certain major water quality parameters for IMCS (catla, rohu, mrigal) culture in cement tanks (Artificial system) by intensive method to promote the fish farmers and culturists to do more and more fish culture which will be economically healthy for them by improving their financial status. Finding of this study shows that the average ranges of the above parameters are pH=7.35, temperature=31-33°C, DO=5.86 mg/l, Salinity=0.26 ppt, Ammonia=0.02 mg/l, Total Hardness=163.12 mg/l & Transparency=35.125 cm were the best suited for the growth of the IMCs. The SGR of Catla, Rohu & Mrigal was 13.08%, 13.28% & 12.75% respectively in 45 days culture period. The total survival rate in this study is 80%.

Table 3- Showing the range of different water quality parameters of culture water

Sl. No.	Sampling Date	Temperature °C	pH	DO mg/l	Salinity (ppt)	Ammonia mg/l	Total Hardness (mg/l.)	Transparency (cm.)
1.	1/4/19	29	6.9	6.7	0	0.0	110	42
2.	4/4/19	30	7.0	6.6	0	0.0	130	40
3.	7/4/19	30	7.0	6.5	0.1	0.0	150	40
4.	10/4/19	31	7.1	6.5	0.2	0.01	140	38
5.	13/4/19	32	7.2	6.4	0.1	0.01	140	35
6.	16/4/19	32	7.2	6.5	0.3	0.02	160	37
7.	19/4/19	30	7.0	6.4	0.5	0.02	180	35
8.	22/4/19	32	7.2	6.2	0.3	0.02	180	32
9.	25/4/19	33	7.4	6.0	0.4	0.03	160	30
10.	28/4/19	34	7.5	6.2	0.1	0.03	170	28
11.	1/5/19	33	7.4	5.9	0.2	0.04	180	32
12.	4/5/19	36	7.9	4.3	0.3	0.03	190	36
13.	7/5/19	35	7.7	5.8	0.5	0.03	190	34
14.	10/5/19	35	7.7	5.6	0.4	0.04	180	37
15.	13/5/19	35	7.7	5.5	0.4	0.04	170	33
16.	16/5/19	36	7.7	4.4	0.5	0.03	180	33
MEAN=		32.62°	7.35	5.86	0.26	0.02	163.125	35.125

Table 4- Showing growth (length & weight) of IMCS in mixed culture in cement tank

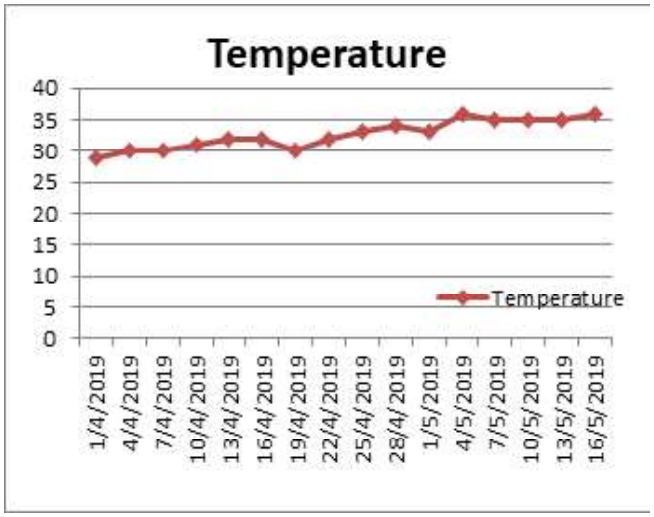
Sl. No.	Sampling Date	CATLA		ROHU		MRIGAL	
		Avg. Length (cm)	Avg. Weight (gm)	Avg. Length (cm)	Avg. Weight (gm)	Avg. Length (cm)	Avg. Weight (gm)
1.	1/4/19	10	75	8	80	6	60
2.	4/4/19	13	120	10	130	8.0	90
3.	7/4/19	17	160	12.5	170	9.5	110
4.	10/4/19	20	200	15.0	200	11.0	140
5.	13/4/19	22	230	17.0	227	13.0	160
6.	16/4/19	25	250	14.5	255	14.5	175
7.	19/4/19	27	270	22.0	280	16.0	185
8.	22/4/19	29.5	285	24.0	300	17.0	190
9.	25/4/19	31	300	25.5	330	18.0	195
10.	28/4/19	31.5	320	26.0	350	18.5	200
11.	1/5/19	32.0	335	26.5	365	19.0	210
12.	4/5/19	33	345	27.0	375	19.5	217
13.	7/5/19	33.5	350	27.25	383	20.0	225
14.	10/5/19	33.75	357	27.5	390	20.25	300
15.	13/5/19	34.0	362	28.0	395	20.5	305
16.	16/5/19	34.5	365	28.5	400	21.0	315
MEAN=		26.67	270.25	21.20	289.375	15.73	192.31

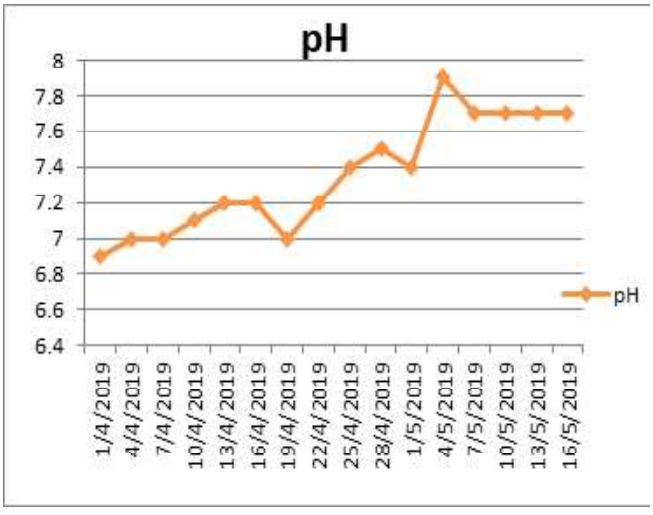
Table 5- Showing mean growth (length & weight) and percentage growth gain (length & weight)

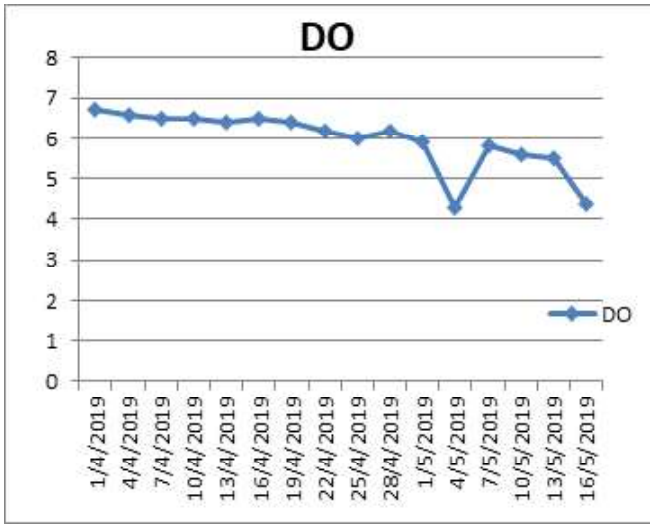
Sl. No.	Imcs	Avg.Initial Weight(G)	Avg.Final Weight(G)	Mean Weight Gain(G)	% Weight Gain (Pgw)	Avg. Initial Length (Cm)	Avg. Final Length (Cm)	Mean Length Gain(G)	% Length Gain (Plw)
1.	Catla	75	365	290	386.66	10	34.5	24.5	245.00
2.	Rohu	80	400	320	426.66	8	28.5	20.5	256.25
3.	Mrigal	60	315	255	340.00	6	21.0	15.0	250.00

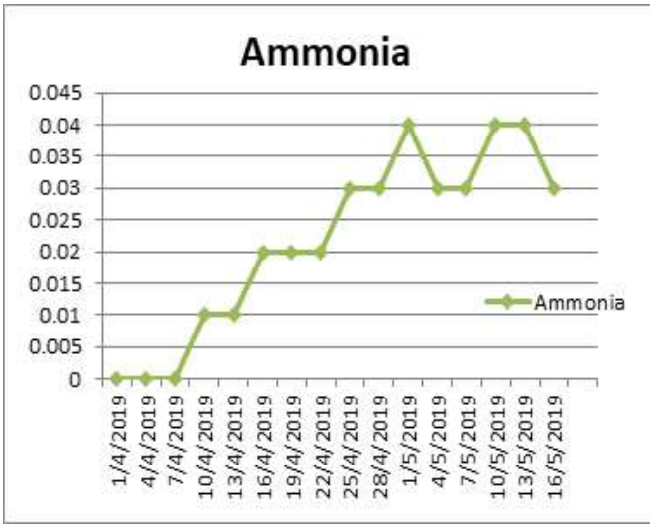
Table 6- Showing growth performance of IMCs in cement tank in mixed culture system

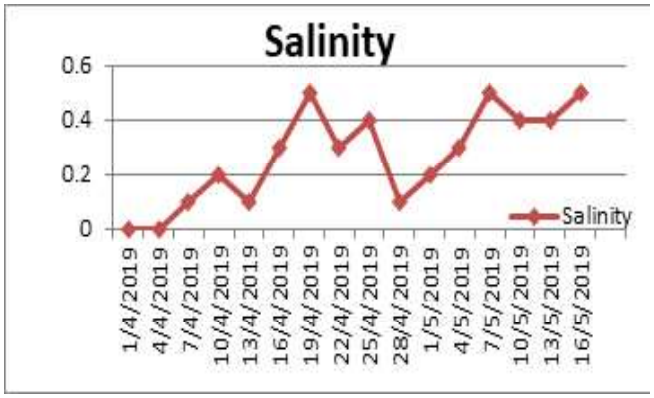
Sl. No.	Imcs	SGR (%)	Condition Factor (K)	Survival Rate (%)	No. of Survival	Mortality Rate (%)	No. of Mortality
1.	Catla	13.08	1.97	80	48	20	12
2.	Rohu	13.28	3.71		64		16
3.	Mrigala	12.75	7.55		48		12

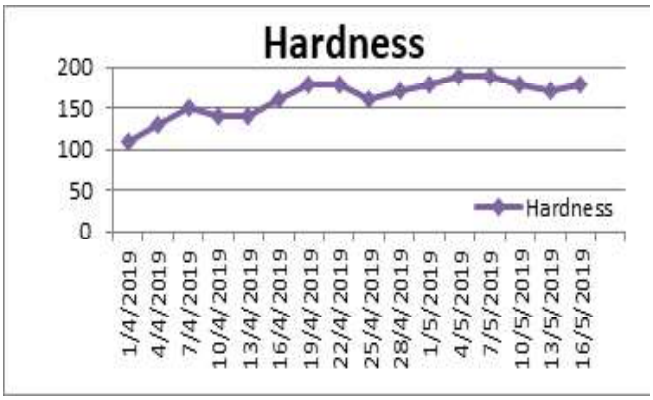


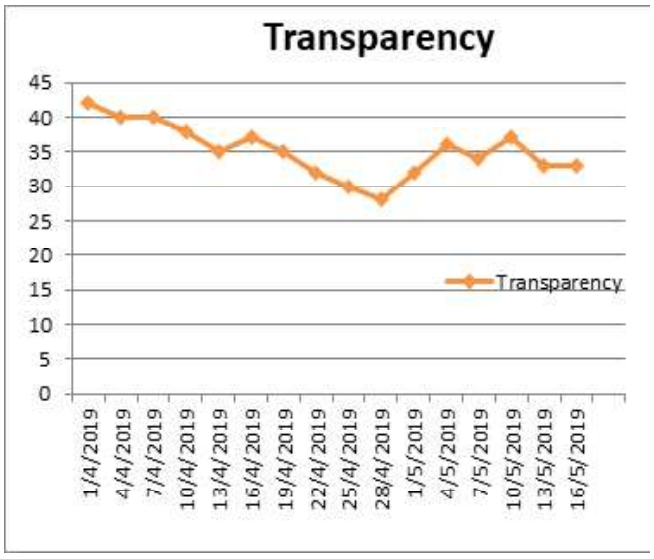


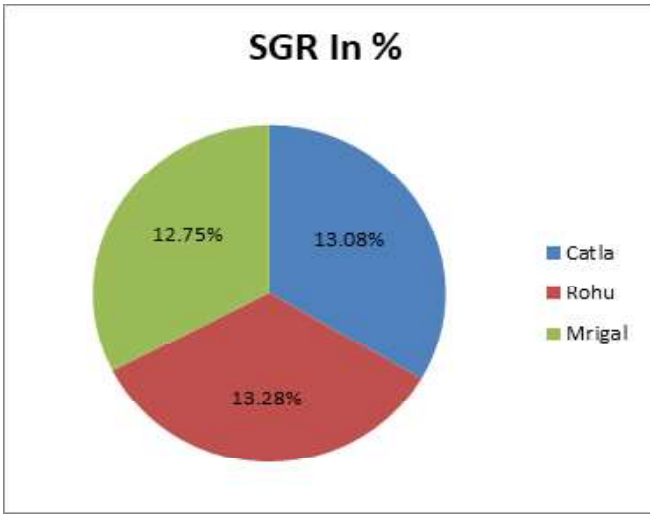












DISCUSSION

The initial weight of Catla, Rohu, Mrigal was 75gm, 80gm and 60gm, respectively and final weight was 365gm, 400gm and 315gm respectively.

Similarly, the initial length of Catla, Rohu & Mrigal was 10 cm, 8 cm, & 6 cm and the final length was also 34.5 cm, 28.5 cm & 21.0 cm respectively.

Verma and Mandal (2018)[5] reported a pH range in between 7.23-8.49. Rajkumar (2018)[8] stated that the pH ranged between 6.96-8.03 (average 7.47) was suitable for fish culture. According to Samad *et al.* (2017)[9], 7.07±0.22 is also suitable for fish pond. S. Samanta (CIFRI Manual) has noted that the productive water pH range is 6.5-9.0. The pH of the mixed culture tank was ranged from 6.9-7.9 which is totally in the favour of carp culture.

According to Premchand and Kiranmai (2017)[7], the water having DO in between 3.4-5.5 mg/L is moderately suitable for pisciculture. Bhoye *et al.* (2017)[3], stated that DO in between 4.24-4.43 (average) is less suitable for carp fingerlings. Verma and Mandal (2018)[5] showed that DO ranging between 5.0-6.2 was good for better productivity justifying our DO amount in tank water.

Supriya *et al.* (2016)[10], stated that IMCs can withstand salinity up to 10 ppt but for satisfactory growth salinity should not cross 5 ppt. Islam *et al.* (2014)[6] noted that the fish exhibited a normal response to treat between o-6% salinity level. At 8% salinity the fish showed moderate response. However, the good productivity of water should contain salinity<0.5 ppt. (CIFRI Manual). The range of salinity in our cultured tank was in between 0.0-0.5 ppt which is completely in favor of IMCs fingerlings growth.

Bhoye *et al.* (2017)[3] observed in experiment that the amount of Ammonia in water was in between 0.003-0.05 (average range). Kadhar *et al.* (2014)[1] showed that the Ammonia range can be 0.456-1.000 at 95% & 99% confident level respectively. In productive culture system, the nitrate content in Ammonia may go up to 2.5 ppm (CIFRI, Manual). The range of Ammonia in culture tank was ranging from 0.00-0.04 which was not toxic for IMCs culture.

According to Priyanka *et al.* (2018)[11], the IMCs perform maximum growth at 180 mg/L. Verma and Mandal (2018)[5], found ideal value of total hardness for IMCs fingerlings in mixed culture system in between 113.3-144.0 mg/L. S. Samant, (CIFRI Manual), stated that hardness is the measure of divalent basic cations(Ca & Mg mainly) in water. Ecosystem having moderately hard (62-120 mg/L) to hard (120-180 mg/L) water are productive. Yadav *et al.* (2017)[12] also showed the hardness can range from 170-198 mg/L. suitable for fish. The hardness level of water in mixed culture of IMCs was ranged from 110-190 mg/L in culture tank making the water completely suitable for fingerling survival and growth.

In the view of Verma and Mandal (2018)[5], the transparency of water in a culture system should be in between 16.0-72.3 cm. Samad *et al.* (2017)[9], in their experiment noted that the transparency of a good productive pond should be up to 40.83±1.34cm. In reservoir, the Secchi disk transparency is normally 100-150 cm while in the ponds it is 20-50 cm. (CIFRI Manual). So, the transparency of the mixed culture tank for IMCs fingerlings culture was ranged from 28-42 cm. This proves that the condition was completely suitable for the growth of fingerlings of IMCs in the mixed culture system in tank.

CONCLUSION

The present work shows that the physico-chemical parameters play very crucial role on IMCs growth (in both body length and weight). If we bring any sudden changes in the range of these parameters from their optimal value, it can lead to the mortality of cultured IMCs. This study also proves that farmers can also practice intensive culture of IMCs in their pond with high stocking density so that they will obtain maximum survival rate. So, they need to check all the water quality parameters mentioned above at regular interval for maximum yielding of IMCs.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 101-106 **Animal Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Studies on the impact of temperature changes on the life-table parameters of *Earias vittella* Fabricius (Lepidoptera: Noctuidae) in Ranchi, Jharkhand.

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Received : 26th December, 2022 ; Revised : 25th January, 2023

Abstract- The spotted bollworm, *Earias vitella* (Fabricius, 1794) is one of the major agricultural pests with a wide range of hosts like sorghum, sunflower, cotton, okra, etc. The current study has been done on the okra host to observe the effect of different constant temperatures (15, 18, 20, 28 ± 1°C with 70±5% RH) in laboratory conditions on the growth and reproduction of *Earias vitella* Fabricius using the age-stage, two-sex life table method. The results showed that *Earias vitella* could complete its entire life cycle on the okra hosts, in different constant temperatures with significantly different developmental and reproduction parameters. The result reveals that the durations of egg, larva, pupa, and adult development were significantly prolonged at lower temperatures whereas they became shorter at higher temperatures. The egg stage has the highest mortality in all temperatures. The incubation period at 15°C is longer (14.08±0.07), while the shortest incubation observed at 28°C (2.71± 0.06). The shortest larval period was calculated at 28°C with 12.54± 0.4 whereas it becomes longest at 15°C with 38.39± 0.41. The oviposition recorded more at 28°C (310.21± 1.23) and lowest fecundity was observed at 20°C (97.66± 0.75). No fecundity was observed at 15 and 18°C. The intrinsic rate of increase and finite rate of increase were significantly different at both temperatures (20 and 28°C). The Net reproductive rate and Gross reproductive rate are also significantly different. The mean generation time decreases as the temperature increases. The results indicate that *Earias vitella* can grow and reproduce normally at higher temperatures, but no reproduction is observed at lower temperatures. (15 and 18°C). Nonetheless, the data show that this pest is harmful when the population density is high.

Key words: *Earias vitella*, Life table, Temperature, Okra, Spotted bollworm, Lepidoptera,

INTRODUCTION

Climate change will be more evident in tropical regions of developing countries like India which are already facing staid challenges for food security and economic development.[1] Future changes in temperature are expected to vary spatially and temporally within the geographical region. This variation in temperature will affect many natural systems that are sensitive to temperature including insect establishment, abundance, and development.[2-6] The insect population growth can be disrupted by the low temperature because a certain threshold temperature is required in each life stage of any insect for its complete growth and development. The life table is a powerful and necessary tool for analyzing the effects of environmental factors on insect survival, growth, development, and reproductive capacity. Insects are able to function faster and more efficiently at higher temperatures and they can feed, develop, reproduce, and disperse more rapidly when the climate is warm.[7] It is well known that temperature directly influences the rate of biochemical reactions and

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strongly influences the growth rate and development of insects.[8] As the insects are poikilothermic in nature, many difficulties can occur when collecting data for the life table of any insect population under multiple conditions.

Moths and butterflies belong to the order Lepidoptera. Among them, moths are common pests on plants.[9] The adults of *Earias vitella* are noctuid moths, and the larvae are caterpillars responsible for the damage (borers). It was reported that *Earias* spp. bore in the plant tissue and are difficult to control once they enter the fruits.[10] *Earias vitella* is reported to cause the most serious losses in okra compared to other pests.[11] A life table concisely summarizes certain vital population statistics.[12] Life tables are useful in the study of insect population dynamics, in that they provide a convenient method for recording and accounting for population changes in the life cycle of an insect. The thermal optimum is the temperature at which a species develops, reproduces and survives optimally.[13] Temperatures lower or higher than the optimum temperature lead to a decrease in the development rate. Temperature in?uences the duration of each instar and the number of instars those larvae go through before reaching the adult stage.[14] During the present investigation, okra and artificial diet suggested by Tamhankar *et al.* (1992)[15] and Gupta *et al.* (2005)[16] were screened for life-table and fecundity of *Earias vitella* when larvae are reared under laboratory conditions. The objective of the present study was to collect appropriate information for constructing life tables for spotted bollworms at different temperatures.

MATERIALS & METHODS

Insect culture

Continuous laboratory cultures of *Earias vitella* were established from the larvae of different larval stages collected from the okra field of ICAR Plandu (Namkum, Ranchi), BAU, Ranchi, and different local markets of Ranchi, Jharkhand. The culture was reared in environmentally controlled conditions under four different temperatures 15, 18, 20, and 28 ± 1°C with 70±5% RH. The insects were supplied with okra (*Abelmoschus esculentus*). And the food was changed on a daily basis to develop into pupa and further into adult ones. Larvae pupated in the white papers kept in the jar, were transferred into another glass beaker. When adults emerged, the moth was reared in pair in a large glass beaker and provided will a cotton ball soaked with 5% honey water solution.

Experimental design

The Insect culture was established at the Laboratory of the University Department of Zoology, Ranchi University, and ICAR Research Centre of Eastern Region, Plandu Ranchi (Jharkhand). The laboratory culture of *Earias vitella* Fabricius was maintained at 15, 18, 20, and 28±1°C with (70±5%) relative humidity (RH).

In order to evaluate the development and reproduction of *Earias vitella* at different constant temperatures, five replicates of 30 eggs (150 total) were placed in 500 mL glass jar and petri dishes with an artificial diet suggested by Gupta *et al.* (2005)[16]. At each temperature treatment, the hatching and survival rates were recorded. Hatched larvae were reared in groups of 20 per 500 mL glass jars on an artificial diet until pupation. Excreta was removed and food was replaced on a daily basis. The duration of larval developmental time from hatching to pupation was recorded and the number of surviving larvae was noted daily. Pupal survival was calculated based on the number of emerging adults. Male and female adults that emerged on the same day were paired and placed in the large glass beaker. The fresh healthy twig of okra was also kept in a conical flask containing water that was placed into the glass beaker. Glass beakers were also provided with white paper for oviposition and 5% honey solution for adult feeding, both were replaced daily. The number of eggs laid per female and adult mortality were recorded daily until all adults died.

Life table

The age-stage, two-sex life table approach was used to analyze the raw life-history data for *Earias vitella*.[17,18] The age-stage-specific survival rate (s_{xj}), Female age-stage-specific fecundity (f_{xj}, daily number of eggs produced by a female at age x and stage j), age-specific fecundity of the total population (m_x, daily number of eggs produced by all females at age x), and age-specific survival rate (lx, probability that the newly oviposited egg will survive to age x) were calculated for each cohort from the daily record of the survival and fecundity of all individuals in the cohort.

Data analysis

According to the age-stage two-sex life table theory,[18] the developmental period survival rate, longevity, and female daily oviposition were analyzed by using the computer program, TWOSEX-MS Chart.[19] (Chi 2016b)7. The parameters of s_{xj}, l_x, f_{xj}, m_x, r, λ, T and R0 were calculated using the bootstrap method included in the

computer programme TWOSEX MS Chart.[20] This program enormously reduces the otherwise complications and simplifies the time-consuming processes involved in the calculation of many parameters.

RESULT

Development time

Earias vitella successfully developed to adulthood in all different constant temperatures from 15°C to 28°C on their natural diet as well as on an artificial diet, however, no egg laying and hatching were observed in 15 and 28°C, indicating that 15°C exceeded the lower threshold. The lowest temperature 15°C caused a significant decrease in the egg hatchability when compared with the 28°C. The incubation period of the egg is longer observed at 15°C, 14.08±0.07, whereas at 28°C it was 2.71 ± 0.036. The larval developmental time varied from 38.39± 0.41 at 15°C to 12.54 ± 0.4 at 28°C. According to the results, larval development of *Earias vitella* was completed in five instars at 15, 18, 20, and 28°C. The Pupal period varied from 50.00± 0.14 at 18°C to 11.05 ± 0.09 at 28°C. Mean total time of development significantly decreased with increasing temperature.

Table 1. Mean developmental time (Days SE) of immature stages of *Earias vitella* reared under different constant temperature

Temperature (ºC)	Egg	L1	L2	L3	L4	L5	Pupa
15	14.08±0.07	10.11±0.09	4.89± 0.08	8.03± 0.08	8.36± 0.07	7.00± 0.09	50.00± 0.14
18	7.59±0.04	8.5 ±0.07	4.41±0.07	6.22±0.09	6.24±0.08	4.69±0.09	48.04±0.09
20	4.26± 0.04	3.51± 0.05	4.1± 0.07	5.03± 0.09	4.66± 0.07	3.93± 0.10	42.06± 0.08
28	2.71± 0.06	2.22± 0.06	2.25± 0.07	2.85± 0.08	2.3± 0.08	2.92± 0.11	11.05± 0.09

The length of the Adult pre-oviposition period (APOP), Total pre-oviposition period (TPOP), and fecundity of *Earias vitella* Fabricius under three constant temperatures were presented in Table 2. The length of APOP was longest at 20°C while the shortest was observed at 28°C, with 20°C > 28°C in turn. In addition, the TPOP was longest at 20°C, with 20°C > 28°C in turn, whereas the shortest length of TPOP was recorded at 28°C. Highest fecundity was recorded at 28°C while the lowest fecundity occurred at 20°C. Many researchers reported different findings about the numbers of an egg laid.

Table 3 shows the life table analysis of various parameters for *Earias vitella* Fabricius at different constant temperatures. 'r' is the useful parameter for describing population dynamics, which includes survival, development, and reproduction.

The 'r' and 'λ' increased significantly with increase in temperature from 20°C to 28°C. R0 of *Earias vitella* was higher at 28°C by recording 57.93 ± 9.88 and lowest at 20°C (22.77 ± 3.37). The reduction of 'T' was recorded at 20°C (30.62 ± 0.31) as compared with 28°C (76.86± 0.25). The higher value of GRR was recorded for 28°C (202.03 ± 23.05) while lower at 20°C (58.28 ± 6.07).

The life span of the male moth was recorded as 6, 13, 19, and 22 days while for the female moth, it was 8, 16, 23, and 25 days respectively on 28, 20, 18, and 15 ± 1°C. The life span of adult females of *Earias vitella* was a little bit longer than that of a male adult. The duration of the life cycle is directly correlated with temperature. The minimum duration of the life cycle was 36-37 days at 28°C and the maximum time was observed 128-129 days at 15°C.

Table 2. APOP, TPOP and Fecundity (Days) of *Earias vitella* under different constant temperature

Temperature (ºC)	APOP	TPOP	Fecundity
20	6.17± 0.13	74.23± 0.24	97.66± 0.75
28	2.07± 0.04	28.32±0.00	310.21± 1.23

(APOP-Adult pre-oviposition period, TPOP- Total pre-oviposition period)

Table 3. Life table parameters of *Earias vitella* at a different constant temperature

Temperature (ºC)	r	λ	R0	T	GRR
20	0.04 ± 0.00	1.04 ± 0.00	22.77 ± 3.37	76.86 ± 0.25	58.28 ± 6.07
28	0.13 ± 0.00	1.14 ± 0.00	57.93 ± 9.88	30.62 ± 0.31	202.03 ± 23.05

(r = intrinsic rate of increase, ë= finite rate of increase, R0 = net reproductive rate, T = mean generation time, GRR = Gross Reproductive Rate)

Table 4. Adult lifespan and complete lifecycle (Days) of *Earias vitella* Fabricius.

Temperatures (°C)	Adult life span (Days)		Complete life cycle (Days)	
	Male	Female	Male	Female
15	22	25	128	129
18	19	23	107	111
20	13	16	83	86
28	6	8	36	37

DISCUSSION

The construction of life tables is an important tool for understanding the population dynamics of an insect. It serves as a framework for organizing dates on mortality and natality. It brings about simple summary statistics such as life expectancy and reproduction rate. From the angle of pest management, it is very useful to know when (and why) a pest population suffers high mortality. This is usually the time when it is the most vulnerable. By knowing such vulnerable stages from the life table, we can make time-based applications of insecticides for insect pest management.

The population stage structures of insect pests are susceptible to various environmental factors like temperature, humidity, rainfall, pesticides, food sources, natural enemies, etc. The adult male and females, and larvae are the most common developmental stages that are widely studied to correlate such effectiveness.[21] The author used an array of environmentally controlled conditions of four different rearing temperatures 15, 18, 20, and 28 ± 1°C with 70±5% RH to examine the effects on the life table parameters of *E. vittella.* Results clearly depicted that the temperature significantly influence the entire lifespan of insects. The entire lifespan of males, females, and unknown sex in the experimental cohort gradually decreased as temperatures increased from 15 to 28°C. Various reports depicted that as temperature increases the increase in female fecundity occur in various other insects.[22,23] At 15 and 18°C the fecundity of *E. vitella* was not observed but as temperature increases by 20 and 28°C the fecundity increases (Table 2). Huang and Chi (2012)[24] said that the *r value* was introduced as a useful concept for studying insect populations. According to the life table theory, a population was increasing only when $Ro > 1$ and $r > 0$.[25] In the present study, $Ro > 1$ and $r > 0$ only from 20°C to 28°C, indicating that the *E. vitella* population increased in this temperature range; the *E. vitella* population could not increase, however, when $r < 0$ at 15 and 18°C (Table 3). The bootstrap technique was used to estimate the means and variance of the population parameters. The highest *Ro* value for *E. vitella* (57.93) was found in the 28°C experimental group. Among the other population parameters, the intrinsic rate of increase (r) is a critical demographic element for determining levels of environmental resistance to insects.[26] Contrasting the *Ro* and *r values* invariably yields considerable insight beyond that obtainable from independent analysis of individual life-history parameters.[27] Conversely, the life table values found in the 15°C, and 18°C temperature regimes revealed that these conditions were unfavorable to the pest because their life cycle was completed in about 129 and 111 days, respectively in females, while in males it was 128 and 107 days, respectively. Therefore, the temperature proved to be a vital parameter for the fecundity of *E. vitella* as per the data in the above tables of this article.

CONCLUSION

The effect of changing the temperature of the insect affects the life cycle of the insect. Based on the result of this research, it is concluded that the female fecundity of *E. vitella* is reflected in the special range of temperature.

ACKNOWLEDGMENT

The author acknowledged the support and all possible help of the Head, University Department of Zoology, Ranchi University, and the Director, ICAR, Plandu, Ranchi.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 107-110

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Study on the zooplankton diversity of Hasanpur Barahi, Swamp, Madhepura

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Received : 28th December, 2022 ; Revised : 29th January, 2023

Abstract- The Swamp of Hasanpur Barahi, Madhepura district in the state of Bihar is where the current work was done in 2022. It is known that planktonic organisms can interact with a variety of water contaminants. According to data analysis, the months of October 2022 during the observation period had the highest monthly abundance of zooplankton. Copepods were the most prevalent group of zooplankton. When rotifers populations are high in a given month, it means that untreated residential sewage from the catchment region has entered directly and contaminated the organic matter. Since May 2022 was when the greatest and minimum diversity (D) was discovered, it is clear that the community in this area is generally stable. According to the current study, the winter population maxima of copepods can be related to favourable temperature and the abundance of food, including bacteria, nano-plankton, and suspended detritus.

Key words: Hasanpur Barahi Swamp, zooplankton, population maxima

INTRODUCTION

A wide group of microscopic organisms known as plankton drift in aquatic settings, primarily in lakes and oceans. These species are essential to aquatic environments and are at the base of the marine food chain. Phytoplankton and zooplankton are the two main kinds of plankton that can be roughly categorised.

Phytoplanktons are microscopic, photosynthetic creatures that are primarily made up of different kinds of bacteria and algae. Photosynthesis, which results in the production of oxygen and organic matter, is carried out by phytoplankton using sunlight and nutrients like nitrogen and phosphorus. They act as the main food sources for many marine creatures and are the principal producers in aquatic habitats. Zooplankton is a term used to describe small, drifting organisms such microscopic crustaceans, fish larvae, and other tiny invertebrates. They consume phytoplankton or other zooplankton and are either herbivores or carnivores. Zooplankton are a crucial link in the food chain that transfers energy from primary producers (phytoplankton) to larger marine creatures and fish at higher trophic levels.

Ocean currents and water currents, which can carry planktonic creatures' great distances, are in charge of them. Seasonal variations in their number and make-up, as well as changes in the availability of nutrients and temperature, can affect both. Monitoring plankton populations is essential for comprehending the dynamics and health of aquatic ecosystems. Plankton populations can also act as markers of climatic and environmental change.

The responses of various plankton species to seasonal variations in the physical and chemical properties of water, as well as their frequency and timing of occurrence, vary. Every organism in a water body, whether it be a plant or an

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animal, tiny or large, plays a crucial part in the flow of energy in the system, and as a result, the current study will be lacking without having a thorough understanding of primary producers. Consequently, an effort was conducted to assess the plankton of Hasanpur Barahi Swamp.

MATERIALS & METHODS

Study Area- The Hasanpur Barahi Swamp being looked into is an elderly Swamp. The Swamp receives water during floods and is closed off after the wet season, providing flood control that can be used for home and agricultural needs. Fish culture is currently not being fully utilised in Swamp. However, it has the potential to generate revenue and employment for the region through fish farming.(Fig 1)



Fig. 1. Hasanpur Barahi Swamp, Madhepura

During the year 2022, samples of plankton were gathered from two stations in the Hasanpur Barahi Swamp at monthly intervals using a plankton net made of ordinary bolting silk cloth. Plankton analysis, both quantitative and qualitative, was performed on the material. Then, quantitative estimates were made. Samples were collected from 0.5 m depth by 1l plastic bottle. Samples were fixed using lugo's solution.

Qualitative analysis was done by Needham and Needham (1963)[1] & APHA (2005)[2]. Quantitative analysis was done by counting in Sedwick-Rafter (S-R) counting chamber as described in APHA (2005)[2]. Expressed by zooplankton unit per litter.

RESULT & DISCUSSION

Bihar's freshwater ecology is home to an abundance of plankton. The quantity and quality of other higher organisms that belong to various trophic levels of the water body's food chain are determined by the quality and quantity of planktons. As a result, plankton is crucial to the systems' energy flow. An essential part of the ecosystem, phytoplankton adapts to changes in the environment quickly. This is because the plankton is crucial to the movement of organic materials and energy throughout the ecosystem.[3]

Zooplankton

There were Seven different zooplankton species identified, and three of those species *Cyclops* sp., *Mesocyclops* sp., and *Diaptomes* sp., were copepods. Cladocera included the genera *Daphnia* and *Ceriodaphnia* sp.. *Keratella* sp. and *Bmchiones* sp. belonged to the Monogononta.

During the time of observations, among the Monogononta zooplankton, *Keratella* sp. and *Bmchiones* sp. consistently showed up in all months. Among the Copepod, *Cyclops* sp. and *Mesocyclops* sp. consistently

Table1. Puopulation of zooplankton (unit per litre) during the study year 2022

Genus/ Months	Copepods			Cladocera		Monogononta	
	Cyclops	*Mesocyclops*	*Diaptomes*	*Daphnia*	*Ceriodaphnia*	*Keratella*	*Bmchiones*
January	96	30	200	99	76	51	48
February	125	58	126	85	8	89	58
March	89	69	156	56	79	50	122
April	56	45	89	46	100	52	45
May	199	82	49	89	126	175	199
June	134	65	85	0	89	188	58
July	89	56	0	49	15	40	69
August	56	122	59	76	102	45	84
September	48	28	87	89	59	85	38
October	111	10	122	99	76	46	46
November	89	868	36	88	89	151	89
December	87	94	81	58	83	45	57

appeared throughout the entire observation period in all the months. Except for July 2022, when it was absent, *Diaptomes* sp. consistently appeared in all other months. *Daphnia* sp. was consistently present throughout the whole observation period, with the exception of June 2022, when it was discovered to be less so. February 2022 and July 2022 both saw low numbers of *Ceriodaphnia* sp. (Table 1)

Pollutants from the discharge of sewage and industrial water with a range of characteristics load are received by the dam. Singh *et al.* (2012)[4] suggests that there are noticeable seasonal fluctuations in the appearance and abundance of different plankton. Seasonal variation in the zooplanktons in temperature as well as tropical climates are a frequent element of any aquatic ecosystem. It appears that two relatively high concentrations of nutrients, such as phosphate and nitrates, are directly associated to the higher phytoplankton population during monsoon months. Workers have emphasised the use of diatoms as markers of contamination.

In the current study, it was discovered that various phytoplankton grow at various temperatures. Despite their immense relevance, zooplankton play a significant role in fish production by providing food for fish in freshwater ponds and lakes. Two taxa of Rotifers, three taxa of Copepods, and two taxa of cladocerans have been identified in the current study. This is more in line with the hypothesis put forth by Pennok (1957)[5], who claimed that the plankton in open water rarely contains more than 1 to 3 taxa of copepods, 2 to 4 species of cladoceram, and 3 to 7 taxa of rotifers.

According to observations, the density and diversity of rotifers are significantly influenced by parameters such as water temperature, turbidity, transparency, and others throughout the summer and monsoon seasons. Similar to the findings of Shalini *et al.* (2017)[6], the largest number of *Keratella* sp. in the summer and early rainy season shows eupophic condition of Konar Dam. The current analysis supports Ugale *et al.* (2005)[7] conclusion that the greatest number of species were present during the winter season as opposed to the summer and monsoon season.

One explanation for the fewer species is that there are fewer nutrients in the dam, which leads to lower productivity. Another explanation is that vital components like pH and dissolved oxygen are being depleted. The majority of cladoceran species are primary consumers that ingest tiny algae and four types of particulate matter found in detritus. As a result, they have an impact on the benthic food chain of a lake ecosystem's cycling of matter and energy. The present study's cladoceran population maxima throughout the winter can be linked to the favourable temperature and amount of food, including bacteria, nanoplankton, and suspended detritus.

Rotifers react to environmental changes far more swiftly than other planktonic species. In municipal and industrial discharges, rotifers predominated while copepods and cladocerans were less common. Since the water temperature and the availability of food to organisms actually affected the copepod population during the summer, Choubey (1997)[8] reported a high density of copepods during October. According to Pullie and Khan (2003)[9], the presence of cladocerans throughout the winter months may be caused by suitable weather conditions, the availability of food in the form of nanoplankton, and suspended debris. The diversity and density of cladocerans are also significantly influenced by physicochemical parameters as DO, water temperature, and turbidity.

According to Jain *et al.* (1997)[10], temperature affects the physical, chemical, and biological characteristics of the ponds, while pH measures the water's acid-base balance. An important part of the biology of cultured organisms is determined by the amount of gaseous oxygen dissolved in an aqueous solution, or DO. According to McLeay *et al.* (2001)[11], a high alkalinity value denotes the existence of both weak and strong bases like carbonates, bicarbonates, and hydroxides. Water's overall hardness is influenced by pH and dissolved solids. Groundwater and surface water nitrate concentrations can be linked to biochemical and anthropogenic sources such fertilisers used in agriculture. The dominance of copepods and rotifers is influenced by the availability of nutrients (Kumar *et al.*, 2006)[12]. The summer months had the highest concentration of zooplankton, and the monsoon season saw the lowest concentration. The present data show a fast decrease in the zooplankton population during the rainy season, which may be related to a sudden drop in temperature and a dilution in the concentration of minerals and salts in marsh water. Cladocera's dominance of the zooplankton peak was discovered to be caused by ideal temperature and nutritional conditions and a decreased oxygen content during summer nights.

Cladocerans, however, were prevalent from March to June and either absent or present in very small quantities

during the remaining months, according to Bohra and Kumar (2004)[13]. According to Winkler (2002)[14], an increase in atmospheric temperature increased the rate of evaporation, and the copepods' favourable relationship with temperature suggested that warm seasons after winter were better for copepod development. According to Pandey *et al.* (2009)[15], the race between cladocerans and other zooplankton groups may be to blame for the drop in cladoceran numbers during rainy months. However, cladoceran richness was also reported to be highest in summer and lowest in winter.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 111-114

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Effects of atrazine exposure on some biochemical parameters in mice: A comprehensive study

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Received : 30th December, 2022 ; Revised : 30th January, 2023

Abstract- This study intends to look into how atrazine exposure could affect different metabolic markers in mice. Concerns concerning the herbicide atrazine's possible negative effects on non-target organisms, such as animals, have been raised. In this study, we performed a thorough investigation of biochemical markers in mice exposed to various atrazine concentrations for a predetermined amount of time. The findings give information on the potential physiological effects of atrazine exposure by highlighting changes in important biochemical indicators. Atrazine was administered to adult male mice for ten days. The animals were slaughtered by cervical dislocation at the conclusion of the tenth day; the liver and kidney. For histological analysis and biochemical evaluations, blood glucose was taken. Measurements were made of the body's initial weight, final weight, liver weight, kidney weight, and blood glucose.

Key words: Atrazine, biochemical markers, Anthron method, Ellman's method, Ohkawa method

INTRODUCTION

Atrazine, (2-chloro-4-ethylamino-6-isoprop-ylamino-s-triazine) herbicide, is widely used across the world to control broadleaf and grassy weeds in maize, sorghum, sugarcane, cotton, pineapple crops and landscape vegetation, so that atrazine and its metabolites are widely persistent in water and are mostly found in soil especially in farming seasons.[1] Atrazine is still used in significant amounts across the globe today, despite being outlawed in the European Union and restricted in other nations. It is one of the agricultural pesticides that are most commonly used in the United States, and its use in Asian countries has been expanding.[2] As a result, atrazine exposure poses a risk to both people and wildlife. Clarifying the toxicological mechanisms by which environmental toxins affect the reproductive system of humans and other mammals has become increasingly important in recent decades. According to a number of reports, atrazine might negatively impact reproductive health.[3] Atrazine can pollute the environment's water supplies in a number of ways, including runoff and drainage from treated regions, precipitation, and pesticide water spills. The toxicological response of numerous different animals exposed to atrazine has been studied. The knowledge of how atrazine might affect environmental creatures has undergone numerous advancements and alterations. The potential danger of atrazine to vertebrates and invertebrates that might be harmed by an unintentional exposure to the chemical has been assessed.

Atrazine's chemical characteristics make it prone to runoff from agricultural and urban applications as a result of rainfall and inadequate irrigation management.[4] One of

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the chemicals that is frequently used to control weeds in maize or other crops, such as green vegetables, is atrazine. Due of its well-defined moment within soil, it has the potential to mix with water. According to reports, atrazine extends to water as a result of the immediate agronomic needs of aquatic areas or occasionally as a result of improper management in particular contexts. When they enter that environment, bacteria cannot damage them. Atrazine has been a persistent water contaminant for years due to the extensive usage of its metabolites, deethylatrazine, and other derivatives.[5] The half-life of atrazine is between 95 and 350 days, and it is resistant to degradation.

Exposure to atrazine has been linked to serious health issues include cancer, neurological disorders, dermatological conditions, and respiratory problems. Therefore, atrazine's harmful effects are of particular concern to and interest to researchers all over the world. Due to its widespread use, widespread contamination of ground and surface waters, usage pattern, high persistence, and probable biological influence on the environment, atrazine has drawn a lot of attention.[6]

Additionally, after oral exposure to atrazine, male rats showed negative effects on the quantity of sperm in the testicles and epididymis, motility, viability, morphology, and daily sperm result.[7] Atrazine exposure in rats was reported to have caused liver damage.[8]

The endocrine (hormonal) system is atrazine's main target in both people and animals. Adults (humans and experimental animals) have experienced effects such as ovarian histopathology (changes in ovarian tissue), shortening of the estrous cycle, attenuation of the LH (luteinizing hormone) surge, decreases in pituitary hormone levels, and liver effects such as elevated serum lipids, liver enzymes, and liver histopathology. The endocrine system may be affected by the behaviour of atrazine as an endocrine disruptor.[9]

Adults has also experienced effects on their immune systems, cardiovascular systems, and central nervous systems. Adult individuals who have been exposed to atrazine may develop certain non-Hodgkin's lymphomas. Atrazine concentrations in drinking water were associated with a significantly increased risk of preterm delivery, intrauterine growth retardation, and lower birth weight. Therefore current study intends to look into how atrazine exposure could affect different metabolic markers in mice.

MATERIALS & METHODS

Three groups (G1,G2,G3) of three adult male mice were taken each weighed 156.20 ± 6.19 gm on an average (table 1). Nine adult male mice were obtained and housed. The animals were given water and food, and were kept in a 10:10 cycle of light and darkness. Prior to receiving doses of the test drugs, the mice were housed for a minimum of 1 week to acclimatise. The study's experimental plans and techniques were compliant with the industry-recognized manual for the handling and use of laboratory animals.

The mice that were kept overnight were put to death by cervical dislocation at the conclusion of the experiment. Blood was drawn, allowed to clot, and then the serum was extracted by centrifuging the blood samples at 1500 rpm for 15 minutes. Each mouse's heart, liver, brain, and kidney were rapidly removed from each batch of slain mice, freed of any remaining fat, rubbed in a cool solution of 1.15% potassium chloride, and weighed. Prior to analysis, the separated serum and tissues were kept at 80°C. Each mice had a preserved kidney, piece of the liver, brain, and cerebellum in neutral buffered formalin. Hematoxylin and eosin staining of tissue sections was a standard procedure for microscopy.

The levels of lipid peroxidation and reduced glutathione were measured using the Ellman's method and the Ohkawa method, respectively. Blood glucose will be estimated by Anthron method. Using commercially accessible assays, serum parameters such as amino transferase, alkaline phosphatase, and urea were determined. The kit approach was used to determine blood glucose levels. The values were given as mg/dL.

RESULTS

Animals exposed to pesticides typically lose weight overall and in their organs. Atrazine administration in the current investigation led to a rise in body weight in all experimental mice over the course of the trial. When compared to untreated control mice, the weight of the liver and kidneys increased in atrazine-treated animals. There have also been reports of body weights remaining the same or dropping after atrazine administration. Reduced meal intake or necrotic alterations in various body tissues may be to blame for the decreased body weight and organ weight (liver and kidney) after atrazine medication. In contrast, an increase in body weight was seen in our study when atrazine was administered. Our study's body weight gain,

especially in atrazine-treated mice, Our findings concur with those of a prior study by Gojmerac *et al.,* (1995)[10] who noted increased body weights in rats following chronic atrazine administration at low doses. In general, atrazine is not thought to be particularly harmful.[11] According to Trentacoste *et al.* (2001)[12], rats treated with atrazine at a dose of 100 mg/kg per day experienced an average body weight drop of roughly 9%. This demonstrated that the observed abnormalities in the male reproductive tract were caused by the treated rats' reduced food intake rather than by the direct effects of atrazine on the reproductive system. The immune system acts as a tightly knit system and offers significant defences against the effects of chemical exposure, according to research on the long-term immunotoxicity of atrazine.[13]

Table 1- showing effect of atrazine on body weight, liver, kidney and blood glucose

Group of Mice		Initial body weight	Final Body weight	Weight of Liver	Weight of Kidney	Blood glucose
G1	A1	156.20 ± 1.25	226.30 ± 1.23	7.46 ± 0.35	1.53 ± 0.79	85.81 ± 1.03
	A2	150.30 ± 1.65	230.55 ± 1.62	7.18 ± 0.36	1.48 ± 0.89	82.57 ± 1.45
	A3	160.35 ± 2.01	250.26 ± 2.36	7.66 ± 0.46	1.57 ± 0.51	88.09 ± 1.11
G2	B1	150.35 ± 1.36	231.02 ± 1.80	7.18 ± 0.89	1.48 ± 0.76	82.59 ± 0.89
	B2	161.25 ± 0.85	248.25 ± 1.12	7.70 ± 0.78	1.58 ± 0.46	88.58 ± 1.45
	B3	156.36 ± 1.23	227.01 ± 2.03	7.47 ± 0.58	1.54 ± 0.58	85.90 ± 1.82
G3	C1	158.68 ± 2.03	241.96 ± 1.01	7.58 ± 0.64	1.56 ± 0.09	87.17 ± 1.23
	C2	150.01 ± 2.11	231.46 ± 0.31	7.17 ± 0.21	1.47 ± 0.43	82.41 ± 1.46
	C3	162.39 ± 0.26	255.32 ± 1.28	7.76 ± 0.84	1.59 ± 0.13	89.21 ± 1.70

Blood glucose level values are in mg/dl, Weights are mentioned in grams

CONCLUSION

The kidney and liver saw the majority of the histological alterations found in this investigation, respectively. It's also feasible that the atrazine-treated mice's minor hepatocyte degeneration and/or mild periportal cellular infiltration by mononuclear cells were enough to trigger the release of the serum indicators from the liver into the bloodstream. The anatomical and physiologic variations of this organ can be partly responsible for its vulnerability to the harmful effects of noxious substances. Therefore, to cause significant histopathological alterations in the brain, different experimental paradigms may be needed, such as a longer exposure duration or greater atrazine dose. The symptoms displayed by these experimental animals, such as lethargy, a loss of body weight, and ultimately death, lead us to believe that their health circumstances are abnormal and suggest the presence of systemic toxicity. Therefore, under the same experimental conditions, a bioactive substance like quercetin that protects a target tissue from chemically induced oxidative damage may actually predispose or not protect another tissue from damage caused by the toxic chemical.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 115-118 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

An extensive investigation on the current state of freshwater cage aquaculture in India

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Received : 04th January, 2023 ; Revised : 05th February, 2023

Abstract- About 13% of the world's population gets their animal protein cheaply from aquaculture, which also employs an estimated 24 million people. By 2024, the sector is expected to have filled the worldwide fish protein gap as a result of static marine fish catches. Cage culture has been identified as one of the most effective methods to utilise underutilised water resources, notwithstanding the sector's diversity in terms of the species cultivated and farming techniques. The earliest attempts were made in India utilising Indian big carps in the Ganga and Yamuna rivers and air-breathing catfish in marshes. Later, the Powai, Govindsagar, Halali, Tandula, and Dimbe reservoirs were used by the Central Institute of Fisheries Education to conduct cage culture for the development of fingerlings and table-sized fish. The last ten years have seen a considerable success in cage culture after numerous attempts with average success. The National Mission for Protein Supplements and other state government programmes have supported this method, which has become well-established in Tamil Nadu and Jharkhand. A decision was made to expand fish production through cage culture.

Key words: Cage Culture, Fish Production, underutilised water resources,

INTRODUCTION

A brief description of cage culture

Due to the presence of over 2.25 million hectares of ponds and tanks, 3.0 million hectares of reservoirs, and 0.78 million hectares of beels, swamps, and other water bodies, India has a large potential market for cage culture, particularly in freshwaters. Cyprinids, perches, snakeheads, and catfish have all been successfully bred in cages thus far. In some aquatic bodies, cage culture can be carried out in extended, semi-intensive, and intensive ways. As a thorough method, cage culture uses phytoplankton and zooplankton from the water bodies rather than supplemental feed.[1] As a semi-intensive practice, cage culture calls for a low protein supplementary food (30%), and if the feed's protein concentration rises above that threshold, the approach can be viewed as being intensive. Farmers need to be trained in cage culture because it is a relatively new aquaculture method in India. Unplanned growth in cage culture can have negative environmental effects, thus it's important to establish appropriate rules to ensure this industry grows sustainably without having any negative effects on the environment.

Choosing a location for cages

One of the most crucial aspects of cage aquaculture practices is site selection, which takes into account parameters like water quality, water depth, and currents needed for the cage culture units. Important water quality parameters like dissolved oxygen, temperature, pH, and ammonia should be periodically checked when cultivating.

Since photosynthesis and wind action are the main sources of dissolved oxygen, a location with better wind and wave action will ensure that proper oxygen levels are maintained for the cultured animals. Dissolved oxygen

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should not go below 5 mg/l. Less than 4 mg/l of dissolved oxygen may result in low survival rates, poor growth, and fatality. A key physical factor governing molecular dynamics and biological processes is temperature. Unfavourable temperatures have an impact on the health, feeding habits, and rate of reproduction of fish. The ideal temperature range for the majority of warm water species is between 24 and 30°C. Another crucial factor is pH, which can change as a result of CO_2 absorption and release during photosynthesis, respiration, and decomposition. An ideal pH range of 6 to 9 is required for improved fish growth, while high pH can cause water bodies' ammonia levels to rise.[2] Ionised and unionised ammonia are the two types of ammonia. Ammonia with ions is more harmful to fish. Ammonia concentrations above their optimal (maximum) range of 0.02 to 0.05 mg/l can be influenced by temperature and pH.

About 6 metres of water is the ideal depth for successful cage aquaculture. The bottom of a cage must be at least 1 m away from the water's surface in order to prevent negative environmental effects, according to earlier cage culture experiences. To maintain appropriate water quality conditions inside the cages, the chosen area should have water current speeds between 20 and 30 cm/second. When the flow rate is more than 40 cm per second, feed may leak from the cages into the ecosystem, and the shape of the cage may alter, potentially stressing the fish and lowering yield.[3] As a result, careful consideration of the water current's speed and pattern is also crucial while placing cages. The location should be free from strong winds and waves because these factors might make it difficult to maintain the cages and, in the case of severe weather, even cause the cage system to collapse. Avoid areas that are prone to flooding, fish breeding grounds, local fishing areas, areas that receive industrial effluent, and areas where there are large concentrations of fish.

Benefits and Drawbacks

Compared to traditional pond culture, cage aquaculture techniques are simple to master and take up a lot less room. Cage culture enables the effective use of large quantities of water, such as lakes and reservoirs. Fish health monitoring, disease detection, and health management are made simple in cages. Individual cages also offer the potential for an efficient and thorough harvest, making it more economically viable than other extensive fish growing techniques.[4] There are certain restrictions, though. Natural disasters might bring the culture system to a halt, and the high density of the fish could cause disease to spread more quickly than in more broad culture systems. Problems with cages can arise with other stakeholder groups, such fishermen.

Construction of Cages

It is possible to practice with four different types of cages: surface floating, fixed, submersible, and submerged cages. The floating cage is the most popular, practical, and environment-friendly of these, especially in India. The location of the cage in the water can be simply changed depending on the water level. Aluminium, galvanised iron, stiff PVC, and HDPP pipes are examples of building materials.[5] The cages are floated using empty barrels, HDPP jerry cans, airtight PVC pipes, and fibreglass. Materials for netting and cages should be sturdy, resilient, lightweight, weather- and corrosion-resistant, fouling-resistant, easily repairable, affordable, and widely accessible.



Cage grouping and mesh size

Similar to the spacing between the cages, cage grouping is another crucial element. Perhaps 20 to 24 units, grouped in two rows of 10 to 12 cages each, can make up a cage battery. To enable water exchange and sustain dissolved oxygen levels, a certain distance must be kept between the cages. The cage culture units must be at least 20 to 25 metres apart.



The mesh size can be decided taking into account size of ûsh stocked. Generally maintaining larger mesh is better for ûsh stock as it allows for improved water

exchange and dissolved oxygen levels and less fouling. Knotless mesh is recommended, as it allows better water exchange than normal knotted mesh. A mesh size of 4 – 6 mm^2 for fry rearing and 16 – 20 mm^2 for grow out are recommended.[6]

A net should be placed over cages to stop fish from jumping out or being eaten by birds. Due to consistent water quality that is less easily impacted by fish waste, larger water bodies are better suited for cage culture than smaller ones. Cages can also be installed in ponds that are between 0.5 and 2.5 hectares in size.

Species selection

The species chosen for cage culture depends on consumer demand, high market value, and seed availability on the local and regional markets. As a result, species with a higher market value can bring in more money. Poor species selection by the fish producer might result in poor marketability and loss because the most popular species in one market may be the least popular in another market.[7] Another crucial factor is seed availability; if there are no hatcheries for a species in the area, it is best to avoid using that species because it can force you to rely solely on wild gathering or suffer high shipping costs if you buy it from elsewhere.

The chosen species should be able to tolerate overcrowding, have good flesh, exhibit rapid growth, higher survival rates, lower food conversion ratios, ready acceptance of feed, rapid adoption of artificial diets, and exhibit higher disease resistance. The species that have been determined to be the best candidates for cage culture are the air-breathing fish, Indian major carps, *Tilapia*, *Pangasius*, etc.

***Pangasius*:-** The ideal *Pangasius* stocking density for fry (20 mm) and fingerlings (50 mm) is 500 to 600 individuals/m^3 and 80 to 100 individuals/m^3, respectively. In order to evaluate the fish's growth rate and overall health, periodic sampling is necessary. The average length of cage cultivation is six to eight months.

Indian major Carps:- With a minimum suggested stocking size of 15g, the ideal stocking density for Indian major carps is approximately 80 fish/m^3. Indian major carp fingerlings and advanced fry up to 25mm in size are suitable for harvesting. Minor carps (*Labeo catla*, *Labeo rohita*, *Labeo calbasu*, *Labeo gonius*, *Labeo bata*, and *Cirrhinus mrigala*) have a stocking size of roughly 1.3 to 2.6 g (40 - 65 mm) and can be harvested for about 15 to 36 g with a survival rate of about 70%.

Transportation and seed supply

To achieve excellent survival and development rates, seed should be purchased from a reputable fish hatchery after being quality-checked. Farmers should therefore observe aggressive swimming and quick acceptance of natural or artificial feed when buying seeds to determine the quality of the seed. The seed should be conveyed using the shortest route possible and in the morning or late afternoon. The seed should be placed in plastic bags with 1/3 water and 2/3 oxygen. When you get to the cage location, float the plastic bags in a water-filled container there for 15 to 20 minutes. To equalize the water temperature in the bag and the container, open the plastic bags, slowly splash small amounts of water into each, and then let the fish swim out on their own. Then, before releasing the seed into the cages, give it a dip treatment with salt (5%) or potassium permanganate (3-5%) to get rid of ectoparasites.

Stocking Density

The quantity of fish stocked per cubic metre in cages is referred to as stocking density, which affects fish development and survival as well as the water's quality and overall health. The carrying capacity, fish feeding habits, and stocking size all affect stocking density. Additionally, it is advised to stock animals of an even size to facilitate handling, prevent predation and cannibalism, and ease harvesting challenges. In order to achieve good development and survival, an ideal stocking density is necessary. Fish size and stocking density are inversely correlated. Higher stocking densities make it difficult for feed to be accessed; however this problem can be solved by evenly distributing feed around the cage. When managed according to optimum management practices, 15g fish with an ideal stocking density of 80 fish/m^3 can guarantee a survival rate of 98 to 100%.

Feed and feeding

In the shape of granules or pellets, feed supplies the fish with the balanced nutrients they require. For cage culture, floating pellets are preferable to sinking pellets. Farmers can also examine how fish respond to feeding and their general health thanks to floating pellets. When using sinking feed, extra care must be taken to prevent waste and maximise use. To prevent feed waste while using automatic feeders, floating feed is advised. The right nutritional elements must be included in fish feed to meet the needs of the fish. In semi-intensive cage culture, larger fish require just 20–30% of the total protein, though this

varies depending on the species. However, for little tilapia and catfish (1 to 25g size), 32–36% of protein is required. Additionally, the feeding must last for at least 6 to 8 hours, and two feedings a day are sufficient.

To reduce feed wastes, it is necessary to maintain good feeding practices. Feeding should not be more than 3 to 5% of body weight. Through sampling, the feeding ration must be changed a few times every month. While the amount of the meal grew with the number of fish culture days, the percentage of feeding could be decreased. Mash-type feed should not be used because it can be dropped out of the cage, go uneaten, and affect the water's quality.

Management practices

Water samples from inside the cages should be taken routinely to assess the water quality parameters such dissolved oxygen, ammonia, and phosphate, and periodic records should be kept to track changes and alter management as necessary. In order to prevent excessive algal growth, which could restrict water exchange, cages should be cleaned as needed. Fish that have been stocked should have their health checked on a regular basis; if an illness is found, it must be quarantined and treated in accordance with the symptoms.

Harvest and marketing

Feeding should be discontinued one or two days prior to harvest. Harvesting should be done in stages, with larger fish being sold first and then smaller fish. This guarantees the crop's highest possible returns. The ability to harvest the appropriate quantity with minimal labour and keep the remaining animals alive at a minimal cost is cage culture's advantage.

In contrast to the current production of 16.24 Million Tonnes, India's expected demand for fish in 2022 is 19 million tonnes. The National Fisheries Development Board has implemented several programmes, including the fish brood bank, aquaculture intensification in ponds and tanks, reservoir fisheries development, cage and pen culture, infrastructure for post-harvest, modern fish markets, and a disease surveillance programme, with the goal of tripling the country's current fish production. The most effective method to fulfill the rising need for fish production, though, has been found to be cage culture.

CONCLUSION

The use of water resources is essential for cage cultivation. About 43,000 ponds and tanks of various sizes, covering a total of about 65,000 acres, are where the majority of Bihar's culture fisheries resources can be found. If cage culture is implemented in around 70% of the current reservoirs, tanks, and ponds, fish production is projected to increase by over 94,000 tonnes. This demonstrates unequivocally that cage culture is a successful method for raising fish productivity and that it also provides rural areas with economic opportunities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 119-122 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Study on the impact of deltamethrin on a few hematopoietic variables in freshwater fish *Heteropneustes fossilis* (Bloch)

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Received : 05th January, 2023 ; Revised : 05th February, 2023

Abstract- In the agricultural area (Paddy), it has been discovered that many pests have developed resistance to the usual pesticides, such as deltamethrin (a pyrethroid), which are typically used to treat the crops to make them insect-free. However, the pond also serves as an excellent refuge for fish that live at the bottom of the water column, Heteropneustes fossilis (Bloch), which are frequently subjected to chemicals used to eradicate insects. After 30 and 40 days of exposure, there was a dramatic peak in opercular beat as well as a gradual increasing trend of 20 to 30%, all of which contributed to a significant mortality rate. Using accepted techniques, the current experiment was conducted to determine how deltamethrin affected the hematopoietic variables in the Asian stinging catfish *Heteropneustes fossilis*. Fish were treated for 96 hours to a sub-lethal dose of deltamethrin. The blood was drawn from both control and experimental fish, and the haematological parameters total leucocyte count, total erythrocyte count and haemoglobin percentage was measured. The findings showed that while total leucocyte count somewhat increased in experimental fish, total erythrocyte count, haemoglobin percentage, dramatically dropped. As a result of exposure to deltamethrin, the study found that Asian stinging catfish *Heteropneustes fossilis* blood underwent noticeable modifications.

Key words: Deltamethrin, *Heteropneustes fossilis*, hematopoietic variables, chemicals

INTRODUCTION

Heteropneustes fossilis, also known as the Asian stinging catfish or fossil cat, is a type of airsac catfish that can be found in India, Pakistan, Bangladesh, Bhutan, Nepal, Sri Lanka, Myanmar and Thailand. Additionally, it has been introduced to Iran's Tigris River Basin.[1] Air breathing fictitiously, Adults are typically found in ponds, ditches, swamps, and marshes, though they can occasionally be seen in murky rivers. Known to have been or be grown in rice fields. Capable of enduring slightly brackish water, it is omnivorous. When enough rainwater collects, they can spawn in ponds, abandoned ponds, and ditches but only during the monsoon season. Like other members of the same family, oviparous, unique pairing is possible. Due to its therapeutic properties, it is in high demand. *H. fossilis* is primarily found in ponds, ditches, swamps, and marshes, while it can also be found in murky rivers. It can stand water that is a little bit brackish. It eats everything. This species can spawn in ponds, abandoned ponds, and ditches when enough rainwater collects, but it only does so during the monsoon season. It is highly sought after because of its purported medical benefits.[2]

A typical synthetic pyrethroid pesticide used in agricultural and pest management is deltamethrin. Aquatic habitats, especially freshwater species like *Heteropneustes fossilis* (sometimes called the stinging catfish or sting fish), may suffer as a result of its use.[3] Because of the chemical's extreme toxicity to aquatic life, deltamethrin exposure can

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have a variety of physiological and biochemical effects, including changes to haematological markers. The current study discusses a few hematopoietic variables that deltamethrin may affect in *Heteropneustes fossilis*.

MATERIALS & METHODS

Without regard to gender, 30 adults live *Heteropneustes fossilis* fish weighing 150–220 g were procured from the neighbourhood market of Madhepura and placed in the control tank in the lab, given a thorough wash using tap water, and then allowed to acclimatise for 20-40 days. They were given a typical fish diet. 25 fish were distributed in the group of 5, which were then moved to six separate tanks having 20 litre capacities for the experimental treatment and exposed to 1/4th sub lethal concentration of deltamethrin, with 5 fish remaining in the reserve tank as the control. A drug called ethylenediamine tetraacetic acid (EDTA) is used to manage and treat heavy metal poisoning. It belongs to the chelating drug subclass. The benefits of using EDTA as an effective treatment for lead toxicity are discussed in this activity, along with its side effects and indications. By severing the caudal peduncle and using EDTA as an anticoagulant, blood was collected in a glass tube at the conclusion of the exposure period. Only 2 ml blood samples for hematopoietic variables were taken. Total leucocyte count, total erythrocyte count and haemoglobin percentage were calculated as hematopoietic variables.

Table 1- Effect of deltamethrin on total leucocytes count (× 10^3/mm^3) in *Heteropneustes fossilis*

Group	Body weight (gm)	20 days	30 days	40 days
Control	200 ± 1.23	4.16 ± 0.25	4.16 ± 0.25	4.16 ± 0.25
Group 1	187 ± 0.18	5.01 ± 0.46	4.87 ± 0.21	5.11 ± 0.10
Group 2	199 ± 0.11	4.99 ± 0.52	4.82 ± 0.12	4.81 ± 0.08
Group 3	202 ± 0.16	5.81 ± 0.12	4.96 ± 0.34	5.26 ± 0.09
Group 4	218 ± 0.08	5.42 ± 0.11	4.67 ± 0.45	4.87 ± 0.08
Group 5	150 ± 0.09	4.45 ± 0.20	4.50 ± 0.11	5.69 ± 0.07

Table 2- Effect of deltamethrin on total erythrocytes count (× 10^6/mm^3) in *Heteropneustes fossilis*

Group	Body weight (gm)	20 days	30 days	40 days
Control	200 ± 1.23	2.61 ± 0.4	2.61 ± 0.4	2.61 ± 0.4
Group 1	167 ± 0.16	0.98 ± 0.52	1.69 ± 0.45	1.11 ± 0.09
Group 2	193 ± 0.17	1.12 ± 0.49	1.58 ± 0.15	0.99 ± 0.05
Group 3	203 ± 0.09	1.67 ± 0.84	1.23 ± 0.63	1.87 ± 0.09
Group 4	198 ± 0.14	1.36 ± 0.36	1.78 ± 0.17	1.78 ± 0.07
Group 5	213 ± 0.09	1.17 ± 0.45	1.16 ± 0.12	1.58 ± 0.06

Table 3- Effect of deltamethrin on haemoglobin percentage (g%) in *Heteropneustes fossilis*

Group	Body weight (gm)	20 days	30 days	40 days
Control	200 ± 1.23	10.56 ± 0.97	10.56 ± 0.97	10.56 ± 0.97
Group 1	205 ± 0.11	9.1 ± 1.7	8.54 ± 0.67	6.42 ± 0.42
Group 2	197 ± 0.13	8.9 ± 1.2	7.42 ± 0.59	5.23 ± 0.63
Group 3	152 ± 0.18	6.2 ± 1.8	5.31 ± 0.74	3.62 ± 0.48
Group 4	159 ± 0.19	9.9 ± 1.4	8.33 ± 0.82	7.99 ± 0.85
Group 5	220 ± 0.02	8.8 ± 1.1	7.41 ± 0.69	6.09 ± 0.28

RESULTS & DISCUSION

Effect of deltamethrin on total leucocytes count :-

Fish subjected to sub-lethal concentrations of 1/4th of deltamethrin for 20 days, 30 days and 40 days in group 1 were found to have total leucocytes count of 5.01 ± 0.46, 4.87 ± 0.21 and 5.11 ± 0.10, respectively, in group 2 were found to have total leucocytes count of 4.99 ± 0.52, 4.87 ± 0.21 and 4.81 ± 0.08, respectively, in group 3 were found to have total leucocytes count of 5.81 ± 0.12, 4.96 ± 0.34 and 5.26 ± 0.09, respectively, in group 4 were found to have total leucocytes count of 5.42 ± 0.11, 4.67 ± 0.45 and 4.87 ± 0.08, respectively, in group 5 were found to have total leucocytes count of 4.45 ± 0.20, 4.50

± 0.11 and 5.69 ± 0.07, respectively, while the mean control was 4.16 ± 0.25g/dl. Over the control, the haemoglobin level was considerably higher for all exposure times.

Effect of deltamethrin on total erythrocytes count :-

Fish subjected to sub-lethal concentrations of 1/4th of deltamethrin for 20 days, 30 days and 40 days in group 1 were found to have total erythrocytes count of 0.98 ± 0.52, 1.69 ± 0.45 and 1.11 ± 0.09, respectively, in group 2 were found to have total erythrocytes count of 1.12 ± 0.49, 1.58 ± 0.15 and 0.99 ± 0.05, respectively, in group 3 were found to have total erythrocytes count of 1.67 ± 0.84, 1.23 ± 0.63 and 1.87 ± 0.09, respectively, in group 4 were found to have total erythrocytes count of 1.36 ± 0.36, 1.78 ± 0.17 and 1.78 ± 0.07, respectively, in group 5 were found to have total erythrocytes count of 1.17 ± 0.45, 1.16 ± 0.12 and 1.58 ± 0.06, respectively, while the mean control was 2.61 ± 0.4 g/dl. Over the control, the haemoglobin level was drastically lower for all exposure times.

Effect of deltamethrin on haemoglobin percentage :-

Fish subjected to sub-lethal concentrations of 1/4th of deltamethrin for 20 days, 30 days and 40 days in group 1 were found to have haemoglobin contents of 9.1 ± 1.7, 8.54 ± 0.67 and 6.42 ± 0.42, respectively, in group 2 were found to have haemoglobin contents of 8.9 ± 1.2, 7.42 ± 0.59 and 5.23 ± 0.63, respectively, in group 3 were found to have haemoglobin contents of 6.2 ± 1.8, 7.42 ± 0.59 and 3.62 ± 0.48, respectively, in group 4 were found to have haemoglobin contents of 9.9 ± 1.4, 8.33 ± 0.82 and 7.99 ± 0.85, respectively, in group 5 were found to have haemoglobin contents of 8.8 ± 1.1, 7.41 ± 0.69 and 6.09 ± 0.28, respectively, while the mean control was 10.56 ± 0.97g/dl. Over the control, the haemoglobin level was considerably lower for all exposure times.

Anaemia: Fish exposed to deltamethrin may become anaemic. A reduction in the quantity of red blood cells (erythrocytes) or a drop in haemoglobin concentration, are two symptoms of anaemia. The blood's ability to carry oxygen may be diminished as a result, depriving tissues of oxygen.[4]

Hematocrit is the proportion of red blood cells to total blood volume. Hematocrit levels can change after exposure to deltamethrin. A higher hematocrit may be a sign of stress or dehydration, whereas a lower value may be indicative of anaemia. Leukocyte count in fish can change as a result of exposure to delamethrin. It may cause white blood cell counts to either rise (leukocytosis) or fall (leukopenia). These variations in leukocyte count may be a sign of inflammatory or stress-related immune system reactions. Hemolysis, or the rupture or destruction of red blood cells, is a potential side effect of deltamethrin. Serum haemoglobin levels may rise as a result of hemolysis, which releases haemoglobin into the blood.[5]

Thrombocytopenia: Thrombocytopenia is a condition in which there is a reduction in the number of platelets (thrombocytes) in the blood as a result of exposure to deltamethrin. A decrease in platelets can make it more difficult for the fish to control bleeding because platelets are essential for blood clotting. Changes in Differential Blood Cell Counts: Exposure to deltamethrin may also alter the proportions of certain white blood cells, such as neutrophils, lymphocytes, and monocytes. Changes in these differential counts can reveal information about the immune system and general health of the fish.[6,7]

The specific effects of deltamethrin on haematological markers in *Heteropneustes fossilis* can differ based on a number of variables, including the concentration and length of exposure, the age and health of the fish, and environmental circumstances.[8,9] Studies on the effects of deltamethrin on this species may also offer more thorough and data particular to this species. In conclusion, exposure to deltamethrin can significantly harm freshwater fish like *Heteropneustes fossilis'* haematological characteristics. In regions where deltamethrin is used or where pesticide runoff is a possibility, monitoring these metrics is crucial for determining the health and wellbeing of fish populations. The main hematopoietic reactions of rainbow trout in the study were somewhat different from the results of some other authors who examined the reactions of rainbow trout and common carp to exposure to synthetic pyrethroid.[10] After acute exposure to deltamethrin, carp showed significantly decreased RBC, Hb, and PVC values (P 0.01) due to probable haematopoiesis disruption, although there were no changes in the white blood cell profiles.[11] PVC, Hb, LEU, and RBC levels were found to be lower in carp after cypermethrin poisoning,[10] and total leucocyte count and neutrophil granulocyte count were shown to be lower in carp after acute permethrin poisoning.[12]

CONCLUSION

The results of the current study indicate that deltamethrin decreased the haematological parameters in

Heteropneustes fossilis, which suggests that the synthetic deltamethrin (pyrethroids) may impair immunological function and cause serious physiological issues that finally lead to fish mortality. Farmers who come into direct touch with deltamethrin, a synthetic pyrethroid used to protect many fruits, vegetables, and field crops from disease, could suffer health effects.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 123-126 **Animal Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Study on the impact of *Annona squamosa* on the biological control of *Spilosoma obliqua* on the jute plant

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Received : 06th January, 2023 ; Revised : 05th February, 2023

Abstract- Agriculture, which is the science of cultivating the soil, growing crops, and raising livestock, is the key to development in the rise of sedentary human civilization. A significant problem now is to expand food production without harming the environment. Synthetic pesticides that are sold commercially can harm both human civilization and the environment as a whole. Pesticides are administered, and more than 85% of them end up in the environment, posing serious health risks. Commercially available synthetic insecticides harm both human civilization and the entire environment. Synthetic pesticides were used, which resulted in bioaccumulation in agricultural fields, which raised resistance and decreased soil biodiversity. Therefore, it is important to promote natural pesticides and insecticides that are safe for the environment. *Spilosoma obliqua* is a polyphagous pest that causes significant damage to plants. In this study *Annona squamosa* active ingredients are employed as biopesticides to reduce *Spilosoma obliqua*. When compared to Control, the larval population of *Spilosoma obliqua* in *A. squamosa* dropped by a maximum significant amount of 7.09 and 11.02%, respectively, at 1 and 2 weeks, respectively. After three and four weeks, the findings showed that *A. squamosa* had the highest increase in the reduction of the larval population (22.26 and 47.23%, respectively) when compared to Control (0.00%). The greatest increase in *A. squamosa* larval population after 5 weeks was 82.52%, which was declared non-significant when compared to control's minimum larval population decline of 0.00%.

Key words: ***Annona squamosa, Spilosoma obliqua,*** **bioaccumulation, Synthetic pesticides**

INTRODUCTION

Jute production is one of India's oldest and most well-known industries. However, due to the greater number of mills, West Bengal is where the Indian jute sector is most dependent. About 3.5 lakh people are employed in the sector, including those who depend on the jute industry's forward and backward linkages. 800,000 hectares of land are typically used for the cultivation of raw jute and mesta. The jute industry plays a significant role in India's national economy. The golden fibre, jute, is natural, renewable, biodegradable, and satisfies all requirements for "safe" packaging. With an estimated 1.968 million tonnes of annual production, India is the world's largest producer of jute. Improvements in crop cultivation techniques and the use of technology to the jute farming industry are credited with India's prominence in the world's jute production. Jute is produced in Bihar, which is India's second-largest producer. West Bengal is at the top. Due to its hardiness and ability to withstand moisture stress, it is grown in a

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variety of agro-climatic conditions as opposed to jute, which is only grown in eastern and north eastern States. The soil is acidic in nature across a large portion of the Jute region, especially in North Bengal, Bihar, and North-Eastern States.

Spilosoma obliqua (Walker), the Bihar hairy caterpillar, is a polyphagous pest that belongs to the lepidoptera order of class of insect with a chewing type of mouthpart. It is only found in the oriental region and is known to severely harm a number of crops of agricultural and horticultural value. *Spilosoma obliqua*, a jute hairy caterpillar, is the most prevalent and harmful pest. The Bihar hairy caterpillar, *Spilosoma obliqua* seriously harms a number of crops. Their attack significantly lowers the yield and lowers the fibre quality. Jute hairy caterpillar larvae devour the leaves voraciously, skeletonizing them, which reduces photosynthesis, growth, and fibre production. One of the most dangerous polyphagous and widely dispersed insect pests, *Spilosoma obliqua*, often known as the jute hairy caterpillar or the Bihar hairy caterpillar, damages a variety of cultivated and non-cultivated plant species. In a crowded environment, the adult female cluster-lays eggs on the underside of the epidermis of the leaves. While later stages of the larvae consume the leaves from the margin, giving the plant's leaves the appearance of a net or web, the young caterpillars are gregarious and they feed on green, soft tissues while leaving behind the veins, skeletonizing the leaves. Larvae of *S. obliqua* go through six instars. Early instars of *S. obliqua* have a gregarious feeding behaviour, and later instars are voracious feeders, which significantly harm the developing plant. It first infests jute in the month of April, when the plants are approximately two to three feet tall, but the heavy infestation happens in the months of June and July. After the eggs hatch, the young caterpillars start feeding in clusters on the lower epidermis of the leaves. The adult female deposits her eggs in clusters on the lower surface of the leaves. Early leaf deterioration takes on a strange membranous aspect. If not quickly handled after emerging, the jute hairy caterpillar can be quite harmful. The final element of an integrated pest management system is chemical control. Chemical control is typically an efficient way of control, particularly in times of urgency when the pest population surpasses the economic threshold. The polyphagous behaviour of S. *obliqua* larvae has, however, received little research in Bihar. Knowing which host plants, the pest prefers and at when stages of the life cycle they are vulnerable may assist reduce the insect's population and lessen the harm it causes.

MATERIALS & METHODS

Areas from Saharsa's rural regions were worked out for this investigation. Jute fields were mostly explored during the investigation. The fields that were chosen were from various parts of Bihar's Saharsa district. Samples were collected by touring different jute fields. The hairy caterpillar was collected in its mature and immature phases in the field and treated with biopesticides. In cage containing jute plants, several male and female adults were gathered.

The fresh leaves of *A. squamosa* were crushed with a small amount of fresh water, and then the resulting paste was sieved through a piece of cloth. Water was then added to the resulting sieved solution. As stock solutions, these solutions were employed in the treatment processes. 20 ml of leaf extracts of *A. squamosa* /80 ml of water were combined to create the needed spray at a 20 percent concentration, and it was then used. A 2-3 week old fungal culture was employed for a variety of tests. By scraping the culture's surface with a sterile loop and distilled water, conidia containing mycelia were collected, and the concentration was set at 20%.

After 1, 2, 3, 4, and 5 week of *A. squamosa* solution treatment on jute plants in plastic jars, the Abbott's formula was used to compute the % mortality of larvae and pupae.

$$\text{Corrected \%} = 1 - \frac{\text{n in T after treatment}}{\text{n in Co after treatment}} x\ 100$$

Where: n = Insect population,
T = treated,
Co = control

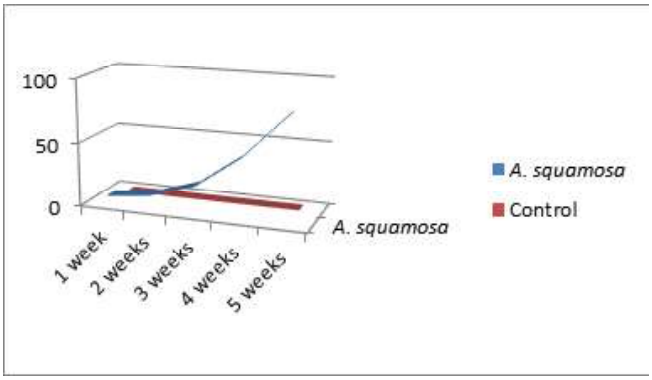
RESULTS & DISCUSSION

After 1, 2, 3, 4, and 5 week, the impact of treatments with *A. squamosa* on the population of *Spilosoma obliqua* larvae following findings were observed. The results after 1 and 2 weeks showed that the larval population of *Spilosoma obliqua* in *A. squamosa* decreased by a maximum meaningful amount of 7.03, 11.02% respectively as compared to Control. The data showed a highest rise in the reduction of the larval population in *A. squamosa* 22.26, 47.23% respectively as compared to Control, 0.00%, after 3 and 4 weeks. After 5 weeks, the results indicated a

maximum rise in *A. squamosa* decrease of the larval population of 82.52%, which was deemed non-significant when compared to Control's, minimum reduction of the larval population of 0.00%. (Table 1 & Graph 1)

Table 1- Effect of *A. squamosa* on larval population of *S.obliqua*

Duration	*A. squamosa*	Control
1 week	7.03	0
2 weeks	11.02	0
3 weeks	22.26	0
4 weeks	47.23	0
5 weeks	82.52	0



Graph 1- Effect of *A. squamosa* on larval population of *S.obliqua*



Fig. 1- Egg of *Spilosoma obliqua*



Fig. 2- Larval (1st instar) population of *Spilosoma obliqua*



Fig. 3- Larval (3rd instar) population of *Spilosoma obliqua*



Fig. 4- Jute field infected by *Spilosoma obliqua*

According to the results of the current study, *S. obliqua* larvae infected plants become dormant and stop feeding when treated with leaf extracts of *A. squamosa*. The larvae eventually die, generally within days, after becoming pale yellow and black, becoming flaccid. As the body parts decay, they shift from brown to black. This inhibits development, transformation, and procreation. In a holistic pest management strategy, leaf extracts of *A. squamosa* can be extremely effective against the *Spilosoma obliqua* Bihar hairy caterpillar. For organic gardeners, caterpillars can provide a significant challenge. Depending on the species, caterpillars have the power to wreak havoc on ornamental plants, crops, shrubs, and even trees. Pesticides can eliminate caterpillars, but the drawback is that they also eliminate helpful insects.

The current study found that treatment with leaf extracts of *A. squamosa* causes *S. obliqua* larvae-infected plants to go dormant and stop feeding. Eventually, the larvae die, usually within a few days after becoming pale yellow and black and becoming flaccid. The body components change colour from brown to black as they deteriorate. This prevents growth, transformation, and

reproduction. *A. squamosa* leaf extracts can be very efficient against the *Spilosoma obliqua* Bihar hairy caterpillar when used as part of a comprehensive pest management plan. Caterpillars can present a big barrier for organic gardeners. Caterpillars can destroy attractive plants, crops, shrubs, and even trees, depending on the species. Although pesticides can kill caterpillars, they also kill beneficial insects, which is a negative.

The effectiveness of *Annona squamosa* against the Bihar hairy caterpillar (*Spilosoma obliqua*) was also examined by Srimannarayana *et al.* (2000)[7]. The best antifeedants were discovered to be Annona oil-based formulations, followed by *Annona* seed extracts and leaf-based formulations. The antifeedant *Annona* products were more effective against silkworm than against Bihar hairy caterpillar at the same concentration when compared to a neem-based pesticide. Similar studies were also done on cabbage leaves in plastic jars, the effectiveness of botanical extracts against *Spilosoma obliqua* caterpillars was investigated. These extracts included Neem, Sarifa, bioagent *Beauveria bassiana*, and insecticide (Cypermethrin 0.07%).[8]

CONCLUSION

Most developing nations today rely heavily on the use of imported insecticides. The output of the agricultural sector is declining as a result of this dependence. The growth is good at the moment when we use such synthetic pesticides, but the fertility of the land declines. Additionally, it might cause bugs to become more resilient. Although the body can easily metabolise and remove the majority of highly hazardous pesticides, acute short-term exposure might result in their buildup. Pesticides' active components, transporters, solvents, and emulsifiers can have harmful side effects. Biopesticides can be a more effective alternative to synthetic pesticides in terms of regulated release of the active components, stability, and sustainability.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 127-130 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Studies on heterogeneity of hydrographic macro invertebrate fauna of Koshi River

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Received : 15th January, 2023 ; Revised : 18th February, 2023

Abstract- The aquatic ecosystem's overall biodiversity can be measured using the hydrographic macroinvertebrates. Numerous researchers have emphasised the significance of bottom fauna as a connection in the energy flow from primary productivity to fish output in the lotic ecosystem's productivity. The results of a thorough investigation of the variety of macro- hydrographic invertebrate fauna in the Koshi River near Saharsa, Bihar have been done in the present study. The current study was worked on for a full year, from November 2021 to October 2022. Twenty different species of macroinvertebrate were identified over the course of the study. This research paper presents a comprehensive study on the hydrographic macro invertebrate diversity and distribution patterns in the Koshi River basin. The Koshi River, a major river system in the region, supports a diverse range of fish species that play important ecological and economic roles. This study aims to document the heterogeneity of hydrographic macro invertebrate fauna, analyze their distribution patterns, and discuss the potential factors influencing fish diversity in the Koshi River.

Key words:mperature, Okra, Spotted bollworm, Lepidoptera,

INTRODUCTION

Indian, Chinese and Nepalese territory are all crossed by the transboundary Kosi or Koshi river. In Tibet and Nepal, it drains the southern and northern Himalayan slopes, respectively. Averaging 74,500 km^2 (28,800 sq mi), the Kosi River drains land in Tibet, Nepal, and Bihar. It is 720 km (450 mi) long. Previous studies suggested that the river's path had changed over the previous 200 years, moving more than 133 km (83 mi) from east to west. The moving was random and fluctuating in character, however, as shown by an analysis of 28 historical maps from the period of 1760 to 1960, which showed a minor eastward trend for a long time.

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The basin has a significant potential for the production of hydropower and is crucial for the irrigation of areas downriver. But the basin is very vulnerable to erosion, sedimentation, and other natural disasters, which could get worse with changes in the climate and other factors. The constant supply of water from the Koshi River and its tributaries is necessary for agriculture, hydropower production, and other important activities. In addition, the freshwater ecosystems in the basin are used for irrigation, drinking water treatment, and fisheries.

Aquatic creatures which literally means "bottom-dwelling", without backbones that are large enough to be seen without a microscope are known as hydrographic macroinvertebrates. Snails, worms, crustaceans, larval stages of water insects like nymphs of the stonefly and

mayfly are among them. Hydrographic macroinvertebrates spend a portion of their life in or near bodies of water. Hydrographic macroinvertebrates are typically found buried in the sand and silt at the bottom or stuck to rocks, vegetation, logs, and sticks. Hydrographic macroinvertebrates are potent water quality predictors. These species are less able to escape the consequences of pollution and sedimentation since they are less mobile than fish. In order to survive, many species of mayfly nymphs, caddisfly larvae, and stonefly larvae need rapid, chilly water that is also well-oxygenated. In general, their presence indicates high-quality water. Hellgrammites, freshwater clams, and water pennies are some more species that are not tolerant to pollution.[1, 2]

They are trustworthy indicators since they live their entire or a large portion of their lives in water, are simple to gather, and have varying levels of pollution tolerance. Macroinvertebrates, unlike fish, have limited mobility, often survive for more than a year, are reasonably simple to identify in the lab, and respond to human disturbance in fairly predictable ways.[3] Macroinvertebrates can actually internalise the impacts of the stressors they are exposed to over time and in combination since they are unable to flee pollution. For many years, biologists have been examining the condition and make-up of hydrographic macroinvertebrate communities.

We can determine the biological health of a waterbody by analysing the variety and number of hydrographic macroinvertebrates there. Generally speaking, macro invertebrate species of a wide diversity and great number, including those that are intolerant of pollution, are supported by waterbodies in robust biological state. Only species that can tolerate pollution, or samples with very limited diversity or abundance, may point to a less healthy waterbody. The most complete indicator of the health of a waterbody is its biological status. The chemical and physical elements of a waterbody are normally in good shape when the biology of the waterbody is. In addition to hydrographic macroinvertebrates, fish and algae populations are examined by scientists to produce reliable estimations of biological state.

MATERIALS & METHODS:

Study Area: In the eastern region of the Indian state of Bihar, in the Saharsa District, is Saharsa, a city and municipal corporation. It is located close to the Kosi River's eastern banks. It serves as the divisional headquarters of the Kosi Division and the administrative centre for the Saharsa District.

Collection of samples: From the Koshi River, hydrographic macroinvertebrates were gathered. Bring the bottom sediment you've gathered as soon as you can to the lab. The residue was then placed in an enamel tray with a little amount of sugar solution (approximately 250 ml and 10 gms of sugar). The benthic species were picked up using a dropper and forceps, preserved in 70% alcohol, and identified using standard keys as a result of an increase in water density.

Table 1- Sampled species from Koshi River near Saharsa, Bihar

Sl. No.	ANNELIDA
1.	*Pristine aquiseta,*
2.	*Tubifex tubifex,*
3.	*Limnodrillus sp.*
4.	*Dero sp.*
	INSECTS
1.	*Tabanus larvae,*
2.	*Odonata nymphs,*
3.	*Chironomous tentum,*
4.	*Chironomous tendipetiformes,*
5.	*Limnophora larvae,*
6.	*Berosus larvae,*
7.	*Hydrocanthus iricolor,*
8.	*Pelocoris femoratus*
9.	*Notonecta sp*
10.	*Culex larva*
	GASTROPODA
1.	*Vivipara bengalensis,*
2.	*Faunus ater,*
3.	*Bellamya sp*
4.	*Pila globosa*

RESULTS & DISCUSSION

Numerous aquatic invertebrates are constrained to habitats with tight habitat requirements and little seasonal variation. They are essential for secondary production and the dynamics of energy flow in aquatic ecosystem maintenance. The majority of aquatic ecosystems are home to an abundance and diversity of invertebrates, which are also relatively simple to collect and study. Benthic macroinvertebrates have a significant impact on the nutrient cycle, primary productivity, material degradation, and movement. Annelids, insects, and gastropods were among

the benthic macroinvertebrates that were gathered for the current piece of research. There were 20 species total, divided into four main groups. There are 6 species of Annelida, 10 species of insects, and 4 species of gastropods. The Koshi River included more *Limnodrillus* and *Dero* species than any other Annelid species. Insects such as *Chironomous* larvae, an indicator species that reveals how dirty the ecosystem is, were abundant in Koshi River. *Odonata, Notonecta, Gerris* spp. and *Chironomid* larvae were also dominant.

Life and water are inextricably linked in a certain way. It is essential for the existence of all living things because it is the source of all life. Rivers are important freshwater systems with varying physicochemical compositions that are essential for the survival of aquatic life. A severe threat to both civilization and the survival of flora and animals has been posed by the alteration of the quality and appearance of natural water resources. Additionally, *Melania* spp. from Santhal Pargana, Bihar, was described by Arvind and Chandan (1999)[4] as a useful bioindicator of sewage-borne severe pollution and hypereutrophication. Numerous aquatic invertebrates are constrained to habitats with tight habitat requirements and little seasonal variation. Others are more adaptable and can endure in a variety of environment types.[5] The bottom fauna of some contaminated rivers were researched by Purdy (1926)[6], who came to the conclusion that *Limnodrillus* favours heavily polluted environments. Additional research has revealed a connection between elevated Oligochaet and Chironomid population densities and organically polluted lotic habitats.[7]

CONCLUSION

According to the study, the number of indicator species including *Chironomous* larvae and *Limnodrillus*, in the Koshi River suggests that the ecosystem is contaminated. Aquatic ecosystems that support life are threatened by anthropogenic use of freshwater for a variety of purposes, making planned development the only option accessible to emerging nations to prevent environmental deterioration or resource depletion

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 131-138 **Interdisciplinary Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Analysis of changes in land use land cover and seasonal land surface temperature using remote sensing data and GIS in PMC Area, Patna, Bihar

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Received : 13th December, 2022 ; Revised : 14th January, 2023

Abstract- Land use land cover (LULC) has significantly affected land surface temperature (LST) due to rapid regional and global urbanisation. Due to the urban heat effect, the dynamic changes in land surface temperature. This research examines the analysis of modification in land use land cover and LST conditions in the PMC, Patna, Bihar. We have employed the Remote Sensing (RS) approach and GIS methods on satellite-driven data of the study area. Further ground truthing has been done to authenticate the changes concerning LST and LULC from 2007-2017. It examines its association with changes in LULC. Land use land cover map of PMC, Patna was developed by using Indian Remote Sensing (IRS) LISS IV Satellite images of the year 2007-2018 with 5.8-meter spatial resolution. For LULC, the maximum likelihood equation and the supervised classification techniques were implemented to classified two multi-date or multi-temporal satellite images. Then temporal changes were identified by comparing two LULC maps formed independently between 2007 and 2017. The seasonal variations of land surface temperature (LST) maps were developed applying Moderate Resolution Spectroradiometer (MODIS) LST data of 2007 and 2017 acquired from NASA LPDAC to address the variability of land surface temperature. In the recent past, due to imprudent land management, rapid population expansion and their activities related to urban/ economic development caused significant natural landscape changes. Thus, PMC Patna has become environmentally sensitive to ecological and environmental deterioration risks. This study aims to analyse changes in LULC and LST in Patna Municipal Corporation, Patna. It is decisive for updating significant modification of LULC and variation in LST over time 2007-2017 in the study area. The research analysis reveals the substantial changes in land use, land cover, and LST developed in the PMC area, Patna. The result shows significant changes designed in the built-up area by increasing (+) 21.86 %, and cropland and vegetation area declined by (-) 8.95 % and (-) 5.8 %, respectively, between 2007-2017. The LST results indicated land surface temperatures varied from 1.02°C to 3.08°C from 2007 to 2017. These results acknowledged that the changes in LULC have significantly influenced the LST in the PMC area, Patna. It could result from cropland and vegetation cover changes between 2007 and 2017, as acknowledged in the LULC change. The enlargement or extension of built-up area and decrease in the vegetal landscape is human-induced, resulting in temperature changes since 2007. The study shows that changes in different types of LULC due to an increase in anthropogenic activities with the expansion of urban landscape, along with changes in cropping patterns, can degrade the spatial pattern of urban land surface temperature (LST). This study will benefit future land use and urban development action plans. The study will provide helpful guidelines for environmental engineers and planners to consider expanding urban built-up area enlargement and spatial configurations without proper LULC planning and conserve natural ecological sustainability.

Key words: False Color Composite, Geographical Information System, Image processing, Land use Land cover, Land Surface Temperature, Remote sensing, Resourcesat, Satellite Imagery.

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INTRODUCTION

LULC and LST change analysis are essential to land development and management activities. LULC and LST will provide helpful guidelines for environmental engineers and planners to consider expanding urban built-up area enlargement and spatial configurations without proper LULC planning and conserve natural ecological sustainability. The importance of PMC, Patna, the capital of Bihar, is placed on the southern bank of the river Ganges. The PMC, Patna, is a fast-developing city and an imminent place for trade and business in the recent past. The movement of people from different locations of Bihar to search for jobs, education facilities and work caused the rapid population inflation of PMC, Patna.

The hypothesis of applying RS digital data in change analysis gives the result of the differences in the landscape due to modification in the digital or pixel values and the LST variation/distributions in a particular area.[1] The RS and GIS techniques contribute to distinguishing the outcome of changes in LULC and LST after analysing satellite images.[1] The analysis of LULC and LST mapping using RS and GIS approach due to its favour of time-consuming and cost benefits rather than the traditional field methods. Ground information can be captured instantly, directly, and repetitively, and image analysis is done more quickly and cost-effectively than traditional field methods.[2]

The rapid modification of LULC caused a significant impact on the local environment to the regional and global climate due to human intervention in the terrestrial environment[3]. The annual temperatures have continuously increased worldwide for the last 150 years.[3] Some study reveals that alteration of land and changes in temperature caused a negative impact on climatic condition on a global scale.[4] The environmental deterioration is due to rapid urbanisation and city expansion, which depletes the vegetation, agricultural land, and water bodies[3]. Recent studies show climate change is happening because of modifications/changes in LULC.[3] Anthropogenic activities significantly contribute to urban expansion, adversely affecting the natural environment.[3] The analysis of this research has notable changes in LULC and LST that developed in the study area between 2007-2017, associated with the enlargement of built-up land and economic growth.[5] These modifications rate will advance in the forthcoming as expected.[1] These rapid modifications in the landscape will influence nature and human sustainability.[1]

The specific and comprehensive data on LULC and LST is essential for many developing activities.[6] This study aims to determine the changes in LULC and LST between 2007 to 2017. We applied RS and GIS methods to correlate two disparate years of satellite imagery of the same sensor.[7] LISS IV satellite imagery of 2007 and 2018 with 5.8 m spatial resolution were used to examine the temporal changes in LULC.[8] For analysis of changes in Land Surface Temperature (LST/E) of the study area, Moderate Resolution Spectroradiometer (MODIS) LST satellite data for 2007 to 2017 was procured from the NASA LPDAC online to addressed the variability of land surface temperature.[9,10] The supervised maximum classification method was adopted to analyze two multi-date satellite imageries and develop LULC maps. Then temporal changes were identified by contrasting two LULC maps of 2007 and 2017. The LULC maps were composed separately.[11] MODIS LST satellite data were applied to determine Land Surface Temperature (LST) changes. To generate LST variations of the study area during the study period, bands 31 and 32 were for daytime observation of Land Surface Temperature. The main aimed to analyse the LST variations to show the relationship between the changes in LULC and LST. The outcome of the analysis indicated that the changes in LULC features are influencing the LST over time in Patna Municipal Corporation (PMC), Patna, between 2007-2017/18.[10,11] This study will benefit future land use and urban development action plans. The study will provide helpful guidelines for environmental engineers and planners to consider expanding urban built-up area enlargement and spatial configurations without proper LULC planning and conserve natural ecological sustainability.

METHODOLOGY

Location of the Study Area

PMC, Patna is the most populous growing city in India, and it is the headquarters of Patna district and the capital of Bihar state. The area is located on the southern bank of the river Ganga at a Latitude between 25° 33’22’’N to 25°39’20’’N and a Longitude between 85°04’50’’E to 85°16’03.55’’E. The study area exists in SOI toposheet nos. 72G/2 & 72G/6. Physiographically the topography of the study area is flat, and a part of the alluvial floodplain

deposits part of the Ganga basin at the height of 53 m/ 173 feet above MSL: Figure 1 and Figure 2 show the location of the study area. The total area of PMC, Patna, is 109.218 sq. km.[12] The total population of PMC, Patna is 16, 87,828 as per the 2011 census with its seventy-two wards, as per 2011 census.[12,13]

Satellite data obtained from NRSC/NDC data centre/ User Order Processing System [UOPS]. IRS-P6 LISS-IV data of 2007 and IRS R2A LISS-IV data of 2018 (Figure.1 and Figure.2) with spatial resolution 5.8 m was used to examine the temporal changes in LULC types. The images were geo-rectified WGS 84 Datum and UTM Projection.

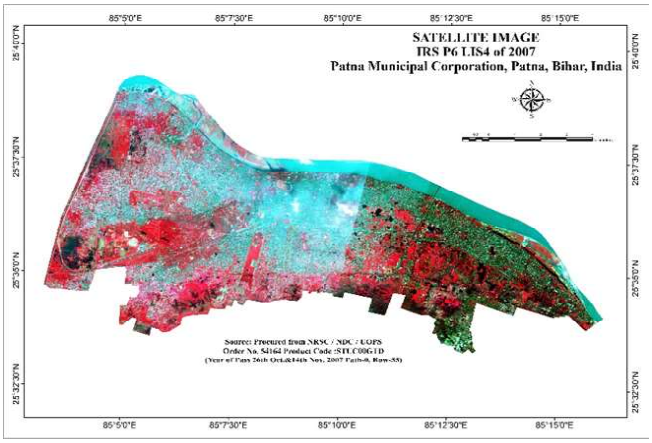


Fig.1- Study Area Map (LISS-1V Image/2007)

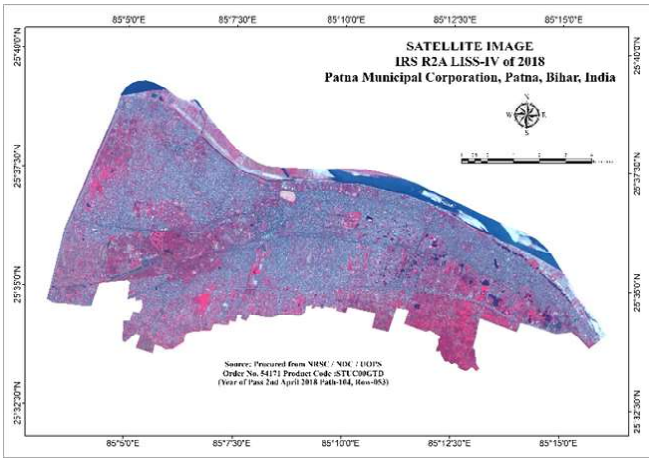


Fig.2- Study Area Map (LISS-1V Image/2018)
Source: Procured from NRSC/NDC/UOPS

IMAGE PRE-PROCESSING

Image Classification

The Image Classification method was used to analyse multi-pixel satellite images. The classification procedure can arrange or categorise the DN values in an image into separate classes.[11,12]

Supervised Classification

We applied a supervised classification method for ground inspection at particular locations in the study area. In the supervised process, first established the training set and derived a pixel value of the spectral image. The classified pixels developed into different LULC types.[12]

Maximum Likelihood Classifier

The supervised maximum classification procedure was used to generate five different LULC maps. This Likelihood Classifier method is worked on a probability function. This activity determines the variance and covariance of each class.[11,12] The maximum classifier is described as follows.[12,14]

$$L_k(X) = \frac{1}{(2\pi)^{\frac{n}{2}}|\Sigma_k|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(X-\mu_k)\Sigma_k^{-1}(X-\mu_k)^t\right]$$

Where n: represents the number of bands
X: n bands of image data
Lk(X): class k likelihood of X
μk: class k mean vector
Σk: matrix of class k variance-covariance
|Σk| : determinant of Σk

LST Method

We applied GIS Spatial Analyst mathematical tool and calculated the surface emissivity for each image. MODIS LST satellite data (bands 31 and 32) were used for daytime observation to address the variability of land surface temperature.[9,10] This study uses the 250m resolution datasets to estimate Land Surface Temperature (LST).[10] The temporal distributions of LST were evaluated at the season and month scales to understand the better relationship in the study area. The procured MODIS data were reprojected UTM projection and datum WGS 84. The coordinates system of the satellite data was converted

Table 1- Technical Specification of Satellite Data

Satellite	Sensor	Acquisition Date	Spectral Band	Resolution (m)	Path/Row
(IRS P6)	LISS IV	26.10.2007 14.11.2007	B2- 0.52-0.59 B3- 0.62-0.68 B4- 0.77-0.86	5.8	102/56 102/55
(IRS R2A)	LISS IV	02.04.2018	B2- 0.52-0.59 B3- 0.62-0.68 B4- 0.77-0.86	5.8	104/53

into degrees Celsius (°C). We extracted pixels' surface emissivity from satellite-derived surface emissivity. Then the measured temperature is transformed into a standardised temperature by dividing them with satellite-derived surface emissivity at each pixel. The standardised temperatures were used to interpolate LST using the Kriging method. Finally, we converted interpolated temperature map to a surface temperature map by multiplying it with the satellite-derived surface emissivity.

CHANGE ANALYSIS OF LULC

Classification/Post-Classification Change Analysis

The pre and post-classified LULC images were used to examine the changes and the most crucial technique of getting the changes of LULC. The significance of this technique which reduces the impacts of LULC between multi-temporal data and develops a better matrix of land use land cover changes.

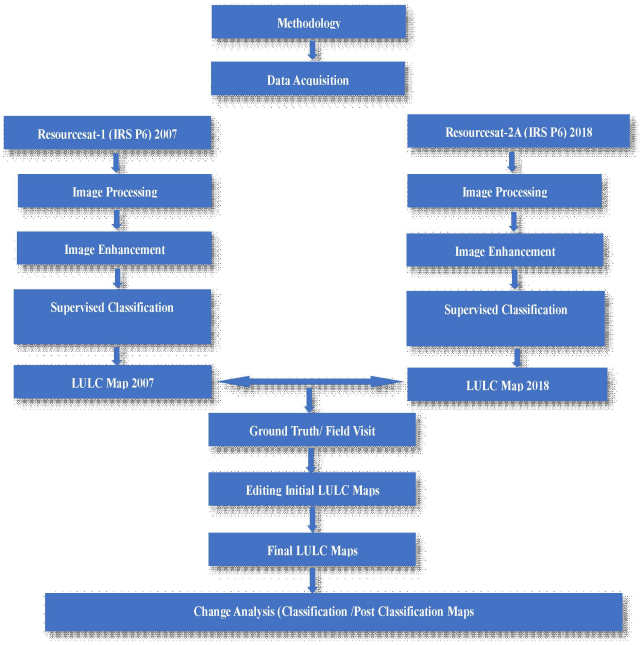


Fig.3- Flow Chart of LULC Method

LST Estimation and Change Analysis

The seasonal variations of land surface temperature (LST) were developed using Moderate Resolution Spectroradiometer (MODIS) LST satellite data from 2007 and 2017. To address the variability of the land surface temperature detected by comparing two LST-derived images, which were produced independently. Validation was done through punctual and spatially distributed for accuracy assessment procedures and through the interpolation Inverse Distance Weighting (IDW) method.

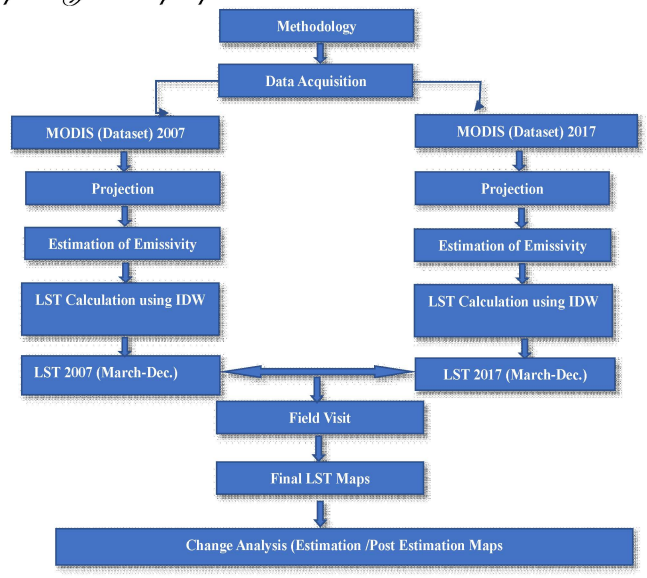


Fig.4- Flow Chart (Methodology)

RESULTS & DISCUSSIONS

Land use Land Cover (LULC)

The outcome of the analysis indicates that the LULC features of the study area have been modified between 2007-2017. The Built-up land has increased by 21.86 %, whereas decreased Agriculture included Fallow Land by 8.95 % and vegetation by 5.8 %. Water bodies and wasteland is also dropping between 2007-2017. These changes are significantly contributed by human concern such as social and economic activities due to improper LULC management. The relative changes are shown below in (Table-2).

The supervised maximum classification techniques were applied to classified two multi-temporal imageries. We categorised different LULC classes such as built-up area waterbodies, agriculture included fallow land, wasteland, and vegetation. Then temporal changes were identified by the contrast between two LULC maps of 2007 (Figure.5) and 2018 (Figure.6). The LULC maps were prepared independently. The accuracy of the LULC map (Pre classified/post-classified) was done through field verification and satellite Image (Google).

Land Surface Temperature (LST)

In this study area, the seasonal (March & December) LST estimation reflected a significant change from 2007 to 2017. The maximum LST was 24.07 °C, whereas the minimum LST was 23.09 °C in March 2007. In March 2017, the maximum LST showed 27.15 °C, whereas the minimum LST was observed at 25.92 °C. The results

Table 2- The comparative study of LULC between 2007-2018

Class Name	LULC Area (2007) (Sq.Km.)	% of LULC Area (2007)	LULC Area (2018) (Sq.Km.)	% of LULC Area (2018)	% Change Between 2007 & 2018
Water Bodies	8.9	7.43	5.18	4.33	(-) 3.1
Agriculture/Fallow Land	31.92	26.66	21.18	17.71	(-) 8.95
Waste Land	16.56	13.83	11.86	9.92	(-) 3.91
Built-up Land	44.67	37.36	70.80	59.22	(+) 21.86
Vegetations	17.49	14.60	10.52	8.80	(-) 5.8
Total area	**119.54**	**100**	**119.54**	**100**	

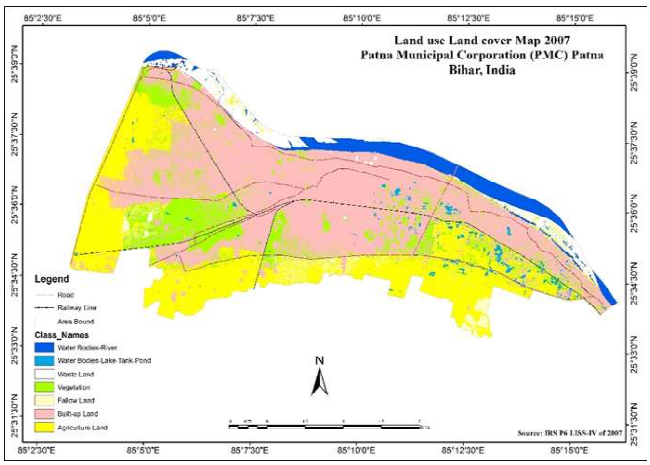


Fig.5- LULC Map (2007) of the Study Area
Source: Resourcesat-1 LISS IV of 2007

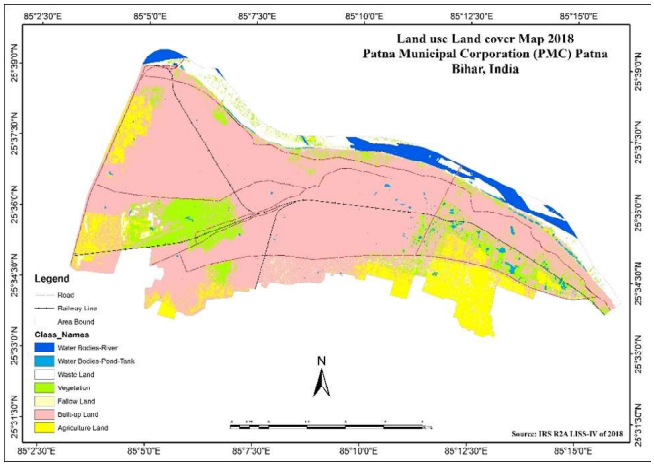


Fig.6- LULC Map-2018 of the Study Area
Source: Resourcesat-2A LISS IV of 2018

Table 3- The comparative seasonal changes of LST between 2007-2017

Land Surface Temp ^{o}C	March 2007 (LST ^{o}C)	March 2017 (LST ^{o}C)	LST Change 2007 to 2017 in March (LST ^{o}C)	December (2007) (LST ^{o}C)	December (2017) (LST ^{o}C)	LST Change 2007 to 2017 in Dec (LST ^{o}C)
Maximum Temp ^{o}C	24.07	27.15	+3.08	17.71	18.73	+1.02
Minimum Temp ^{o}C	23.09	25.92	+2.83	17.17	17.97	+0.8

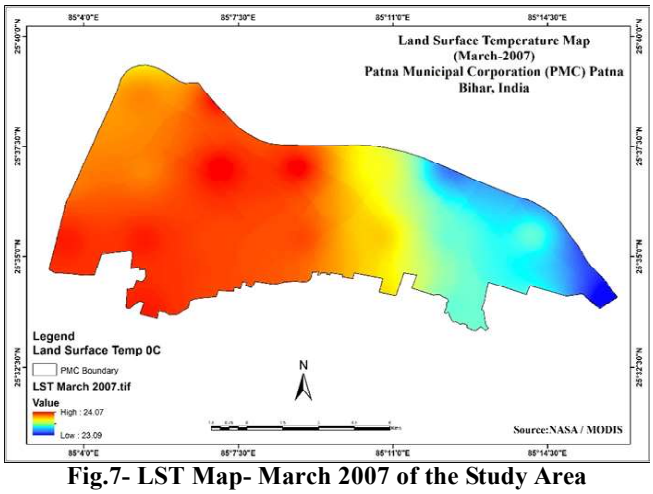


Fig.7- LST Map- March 2007 of the Study Area
Source: Moderate Resolution Spectroradiometer (MODIS) LST satellite data of 2007

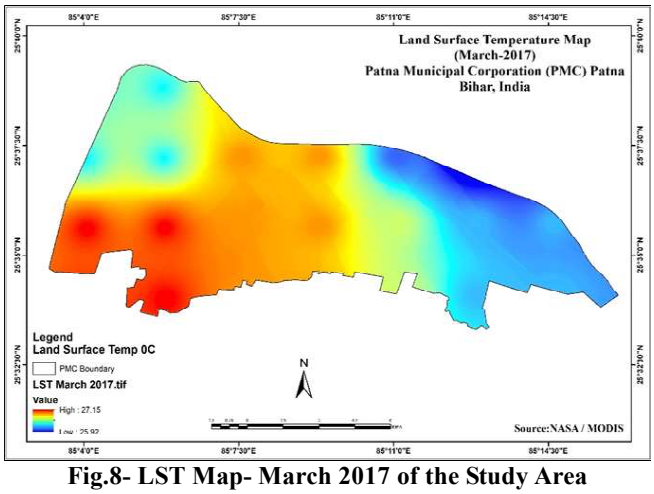


Fig.8- LST Map- March 2017 of the Study Area
Source: Moderate Resolution Spectroradiometer (MODIS) LST satellite data of 2017

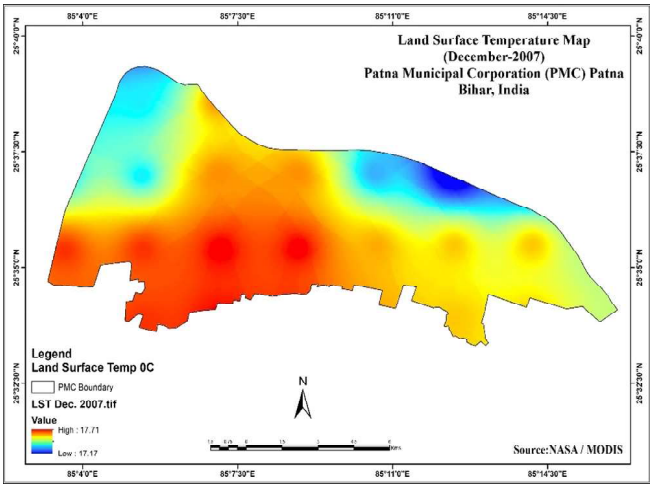


Fig.9- LST Map- December 2007 of the Study Area Source: Moderate Resolution Spectroradiometer (MODIS) LST satellite data of 2007

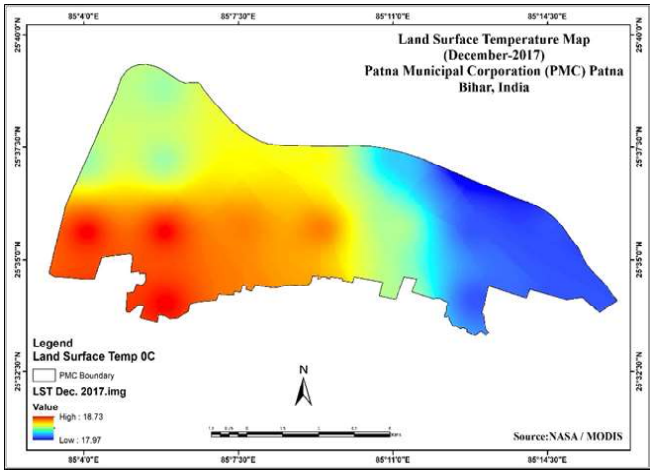


Fig.10- LST Map- December 2017 of the Study Area Source: Moderate Resolution Spectroradiometer (MODIS) LST satellite data of 2017

indicated that LST estimation significantly changed by +3.08°C from March 2007 to 2017. In December 2007, the LST estimation revealed that the maximum and minimum values were 17.71°C and 17.17°C, respectively; in 2017, the maximum and minimum LST was 18.73°C and 17.97°C, respectively. The results estimated that LST estimation significantly changed by +1.02°C in December from 2007 to 2017. The study observed that changes in LULC or landforms, primarily expanding built-up areas and reducing vegetation/ agriculture land related to social and economic concerns, can significantly change LST. The Built-up land has increased by 21.86 %, whereas decreased Agriculture included Fallow Land by 8.95 % and vegetation by 5.8 %. Water bodies and wasteland is also dropping between 2007-2017. It is estimated that significant changes in these classes of LULC are contributing to the rising LST of the study area. Most importantly concluded that replacing natural vegetation and agricultural land with non-evaporating, non-transpiring surfaces like built-up material such as concrete, metal, stone and urban construction can increase LST or surface radiant temperature.[15]

CONCLUSION

PMC, Patna, one of the most vulnerable areas due to rapid changes in LULC due to urban and economic development, will generate or affect Land Surface Temperature. LST increases when the natural LULC is modified or mismanaged without proper planning because LULC is directly proportional to LST. In this study, multi-temporal LISS IV data for LULC classification and MODIS LST/E datasets were used for LST estimation and then detected the changes between 2007-2017, which may be correlated with the findings of the results to understand better LULC influences LST. Change analysis was done by comparing pre-classified/post-classified maps and ground verification. The result showed the main negative changes of LULC that have occurred in the agriculture/ fallow class, vegetation, water bodies and wasteland due to the urbanisation or urban expansion and ultimately influencing the LST. Based on the findings, these negative changes mainly observed rapid enlargement of the built-up area without proper planning and economic concerns. Avoiding LULC changes without appropriate planning, conserving natural ecological sustainability and growing more vegetation would help reduce LST in PMC, Patna.

ACKNOWLEDGEMENT

The author would like to thank Magadh University for granting the PhD research work. The Department of Environmental Science, A.N. College, Patna of the Magadh University, is highly appreciated for allowing the GIS laboratory work. The author is also profoundly grateful to the National Remote Sensing Center (NRSC), Indian Space Research Organisation (ISRO), and Govt. of India for their guidance during the Satellite data procurement.

FUNDING

There is no funding or financial support for this research work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 139-144 **Interdisciplinary Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Implications for prevention and management of lead poisoning based on lead protoporphyrin risk assessment

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Received : 04th January, 2023 ; Revised : 04th February, 2023

Abstract- Developmental delays, neurological impairment, and cardiovascular disease are just a few of the negative health consequences of lead poisoning, a serious public health concern. Porphyrin synthesis, namely delta-aminolevulinic acid (ALA) and porphobilinogen, is frequently linked to lead toxicity (PBG). This review includes a collection of case studies that explore the connection between lead poisoning and lead porphyrin synthesis. The case studies offer a thorough review of the clinical traits, laboratory results, and therapeutic approaches for lead poisoning brought on by lead porphyrin synthesis. The review also examines the limitation and difficulties of employing lead porphyrin synthesis as a biomarker for lead poisoning, including the possibility of false positives and the requirement for additional confirmatory tests. The evaluation finishes by emphasizing the necessity for more investigation to better comprehend the connection between lead toxicity and lead porphyrin synthesis as well as to create fresh lead toxicity detection and therapy methods.

Key words: ALA, Prophyrin, case studies, health

INTRODUCTION

Lead is a chemical element with the symbol Pb (from the Latin plumbum) and has the atomic number 82 in the periodic table. One of the seven metals of antiquity was lead. It was discovered at least 3500 B.C. Despite being common in nature, lead is not known to have any physiological role in the body. There are various places where people might be exposed to lead, including paint and battery recycling.[1] Lead is a persistent metal that may be found in dust, soil, and water. It is retained in both soft tissues and bones after being absorbed through the respiratory system or gastrointestinal tract (GIT). Lead tends to damage the brain system and kidneys, two organ systems in the body that lead tend to damage the most. Inhalations of lead-contaminated air, ingestion of lead-tainted food or water, or direct contact with lead-polluted soil are the most prevalent routes of exposure. The precise processes by which lead exposure affects the development and function of the central nervous system are yet to be determined. Lead neurotoxicity has been linked to oxidative stress, calcium signalling deregulation, and aberrant neuronal transmission/gene expression pathways.[2,3] However, Porphyria is a Greek word that signifies purple. When there is enough porphyrin in the urine, it takes on a reddish-purple, reddish-brown, or "port wine" colour. Porphyrin excess in the urine has been seen in patients of lead poisoning since 1895. Several research on the effects of lead has shown that it inhibits heat production at multiple locations. This route is complicated and multi-step, including eight separate protein enzymes (proteins that mediate and speed up biological events by binding chemicals together in the appropriate orientation to react, and so on).

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HISTORY

The history of lead poisoning is extremely lengthy. By the first century A.D. Lead poisoning, sometimes known as saturnism, and the prevalence of "palsies" were noted during the Middle Ages. In 1839, a detailed clinical description of lead poisoning was provided, Chronic lead consumption was linked to stomach colic, skin pallor, and swelling of the body. In 1839 the first comprehensive clinical description of lead poisoning was published, and in 1840, red cell counts were used to establish the anaemia caused by lead poisoning for the first time. Red blood cell punctate basophilia was first seen in 1899. The discovery of "hematoporphyrin" in the urine and feces of a patient with acute lead poisoning is believed to have been made for the first time by Stokvis (1889). As early as 1895, instances of lead poisoning were shown to have an excess of porphyrins in the urine. Inhibition at multiple locations of heam biosynthesis has been found in investigations on the impacts of lead. The most frequent ways to become exposed to lead are by inhaling lead-tainted air, consuming lead-tainted food or drink, or coming into direct touch with lead-contaminated soil.[4,5]

HEMATOLOGICAL EFFECT

Heme is a protein's important prosthetic group that serves as a subcellular compartment for many biological functions such as haemoglobin and myoglobin. Heme is synthesised in the cytosol and mitochondria, and it is acquired by intestinal absorption and intercellular transfer. Porphyria can be caused by a variety of 'genetic' factors, including the most prevalent type of Acute Hepatic Porphyria (AHP), Acute Intermittent Porphyria (AIP), which is caused by mutations in the pathway's third enzyme, porphobilinogen deaminase (PBGD) (aka hydroxymethylbilane synthase). One of the organs that lead poisoning targets is the haematological system. Succinyl coenzyme (CoA) and glycine start the chain of events that results in heme synthesis, which is completed by the insertion of a Fe^{2+} ion into a proto-porphyrin molecule.[4] The enzyme aminolevulinic acid (ALA) synthase catalyses the creation of ALA from glycine and succinyl CoA with in the mitochondrial matrix as the first stage of heme production. The impact of lead on the enzymes responsible in the heme biosynthesis pathway is one of the most significant mechanisms of lead poisoning. ALAD and ferrochelatase, both of which are inhibited by lead, are the enzymes in the biosynthesis route of heme in which the effects of lead are of the most clinical importance. Heme biosynthesis is disrupted as a result of lead poisoning.

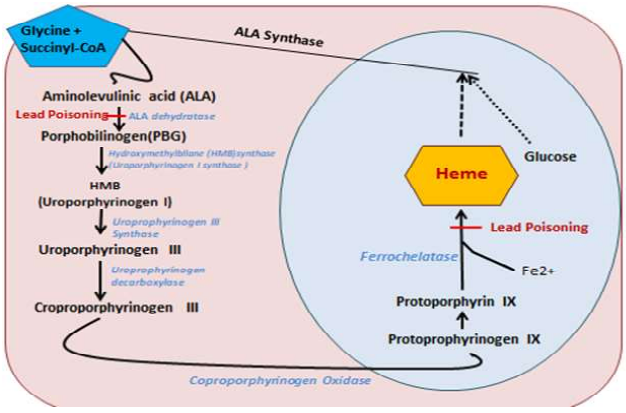


Fig. 1- A Pathway for lead porphyria formation

LEAD PROPHYRIN

A category of genetically based metabolic illnesses are collectively known as porphyrias. The heme biosynthesis pathway's unique enzymes' poor activity results in the buildup of process intermediate in each individual form of porphyria.[6]

The table provides a concise summary of lead porphyrin levels in different populations and study designs. The studies included in the table highlight the variability in lead porphyrin levels across different populations, as well as the range of sample sizes and study designs used to assess lead porphyrin levels.

HUMAN HEALTH

The category of compounds known as lead prophyrins has the potential to have a significant negative impact on human health. Numerous anthropogenic and natural sources can expose people to lead prophylins, which is harmful to their health. Among other things, lead-based paints, petrol, and industrial activities are frequently cited as anthropogenic sources of lead prophyrins. The consumption of tainted food, water, and soil are considered natural sources of lead prophyrins. Mining, smelting, battery manufacturing, and recycling are some of the industries that have been noted as important sources of lead prophyrin exposure. It is crucial to provide a variety of guidelines to control lead prophyrin exposure for human health because of the variety of sources. For example, companies that generate large amounts of lead must adhere to strict rules.

The table gives an overview summary of the lead porphyrin concentrations in various demographics and research types. The research included in the table demonstrate the wide diversity of sample sizes and study designs used to measure lead porphyrin levels, as well as the variation in lead porphyrin levels among various populations.

Table 1: Key findings on Lead Protoporphyrin (PbPP) Levels in Humans.

Sl. No.	Study Tittle	Location	Sample size	Finding	Reference
1.	Alterations in urinary metabolomics profiles due to lead exposure from a lead–acid battery recycling site.	Vietnam	164	Disruption of heme biosynthetic pathway by which Pb related of Health effects.	7
2.	Oxidation DNA damage and oxidative stress in Lead exposed Worker	Poland	100	Lead exposure may raise the risk of cancer development in workers due to lead-induced DNA damage.	8
3.	Aminolevulinic Acid Dehydratase Genotype Mediates Plasma Levels of the Neurotoxin, 5-Aminolevulinic Acid, in Lead-Exposed Workers	South korea	307	Lead-associated subclinical renal damage has been observed by employees who had modest levels of lead exposure.	9

CASE STUDIES

A 45-year-old Korean male who had smoked for 25 years, was ok until late 1991, when he acquired a chronic cough, sputum production, and dyspnea. Acute intermittent porphyria was diagnosed during a 3-week hospitalization based on symptoms and a high level of urinary α aminolevulinicacid (378 µmol/L [4.95 mg/dL]). The proper diagnosis was made after the finding of a high blood lead level (3.7 µmol/L [76 µg/dL]). 4 weeks before becoming unwell, the patient reported drinking tea made from lead-adulterated Chinese herbal medicine. The presence of zinc protoporphyrin in circulating red blood cells was discovered in 70% of the patient's cells, which was compatible with the period of lead exposure and the influence of lead on heme production.[10]

37-year-old man, had a history of stomach pain, anorexia, and malaise. He had as of late returned from a trip to India where he had been taking two distinctive home grown tonics. He have no history of smoking or drinking and had no prior stomach pain, surgery, hepatitis, blood transfusion, or high-risk sexual behaviour. His entire family was in superb health. Laboratory investigations revealed that the levels of plasma urea, electrolytes, amylase, and calcium were normal. Liver function tests revealed a predominant hepatic picture, with alkaline phosphatase 150 µ/L (normal range, 30-120 µ/L), and alanine aminotransferase 270 µ/L alanine aminotransferase level is equal to 40 u/L, a 200 µ/L aspartate aminotransferase level is equal to 30 u/L, and a 45 µmol/L bilirubin level is equal to 20 µmol/L. Haemoglobin was 11.6, white blood cells were 8.6, and platelets were 250 x 10'/L on the blood film; the reticulocyte count was 4%.The blood lead concentration was high(lead excretion was 2.3 µmol/24 h (normal<0.37 µmol/24 h), and analysis of the herbal tablets revealed a very high lead content. Lead poisoning was diagnosed in this case based on the anemia with accompanying basophilic stippling. Whereas the abnormal liver function resulted in a mild hepatitis condition, which was initially assumed to be caused by infection, which was the focus of the initial laboratory tests, however, the abnormal liver enzyme pattern could also be related to acute lead poisoning-associated toxic hepatitis.[11]

An elderly male patient with abdominal pain and constipation was admitted to the hospital. X-rays of the abdomen revealed a radiopaque mass in the caecum and progressive anemia. After discharge, the patient was diagnosed with bowel obstruction. Due to the symptoms persisting, he was hospitalized again; a radiographic examination revealed radiopaque particles in the colon and a haemoglobin level of 8.7 g/dl. Lead and zinc protoporphryin levels in the blood were 106.7 microg/dl and 6.6 microg/gHb, respectively. In this case, the route

and vehicle of ingestion of the toxic were not clear. Acute lead poisoning caused by food ingestion is extremely uncommon in adults and is often related to mental illnesses. Probably due to the ingesting of a single bolus of lead dust, the lead body burden was shown as a caecal mass. This mass was estimated to weigh several grams, based on the density and size of the mass.[12]

A 42-year-old woman was admitted to the hospital with severe stomach pain. Pancreatitis was suspected, but test results were inconclusive. A normochromic anemia was discovered, with a hemoglobin level of 7.4 to 9.4 g/dl and 3.31 106 erythrocytes/B. The percentage of reticulocytes in the blood was increased to 5.6 percent. There were no signs of hemolysis. The leucocyte count was increased to 12800. Hematologic findings (Hb 15.3 g/100 ml, MCH 31.2 pg, red blood cell counts 4.7 mill., hematocrit 45.8%, leucocytes 7 200) and general clinical chemical data as well blood lead content and lead excretion.[13]

Autistic A 13-year-old kid was taken to the emergency room of Kasturba Medical College Hospital, Attavar, Mangalore, India with a high grade fever, vomiting, intense colicky stomach discomfort, and constipation that had lasted three days. All test was conduct and the total leucocyte count was 3500 cu mm, the hemoglobin concentration was 7%, and the platelet count was 1,33,000 cu mm. Microcytic hypochromic anemia was seen on the peripheral smear. Both the CSF tests and the brain CT scan were normal. Electrolytes in the serum were normal. Using the ESA model 3010B lead analyser (ESA, Inc., Chelmsford, MA) and the differential pulse anodic stripping voltammetry approach, the blood lead (PbB) level was calculated to be 59.5 g/dl (DPASV). Patient received symptomatic treatment. To treat anemia, two units of packed cells were administered. The patient did not experience any other seizure episodes and did not need anticonvulsant therapy. The clinician's decision to ask the parents for a full history in response to the kid's increased lead levels led to the discovery that the youngster may not have been exposed to other potential sources of lead, such as toys, paint, or dirt. However, they acknowledged that over the previous two protect the public from the harms of lead prophyrin, local medicine had been given as syrup. Parents were instructed to quit the traditional medication once the patient was discharged from the hospital. If a child exhibits acute porphyria-like symptoms and tests positive for porphyrin in the urine, heavy metal poisoning should be considered.[5]

DISCUSSION

The results of lead protoporphyrin (PbPP) tests depend on the amount of PbPP present in a person's blood. Higher levels of PbPP often suggest increased lead exposure and the risk for lead poisoning.

If a person's PbPP levels are elevated, further testing to look for signs of lead poisoning may be advised. Tests for blood lead levels or other indicators of lead exposure and toxicity may be necessary.

Especially in children and pregnant women, lead poisoning can have major negative health repercussions. It can have an impact on a foetus' development in pregnant women and cause behavioural issues, learning impairments, and developmental delays in children. The kidneys, neurological system, and reproductive system can all be harmed by prolonged lead exposure.

The best method for preventing lead poisoning is to avoid exposure to lead. To do this, one can avoid lead-containing items, check the water and soil for lead, and use safe working procedures in jobs that expose one to lead. When lead poisoning is thought to be present, it's critical to get medical help very once and eliminate the source of exposure.

CONCLUSION

Lead prophyrin and its health implications is of utmost significance due to the serious health effects it can cause in humans. Exposure to lead prophyrin can lead to a range of health issues including neurological damage, developmental delays, cognitive impairment, anemia, and even death. It is therefore critical to take preventative measures and limit exposure to lead prophyrin as much as possible. Based on the available research, it is apparent that the negative health effects of lead prophyrin must be taken seriously and addressed in order to minimize the negative impact on human health. Further research is necessary to fully understand the mechanisms through which lead prophyrin leads to health problems, and to develop effective strategies for prevention and treatment. It is crucial that policymakers and public health officials prioritize efforts to reduce lead exposure and protect the public from it.

ACKNOWLEDGEMENT

I thank the staff of the Center of occupational and environment health, Maulana Azad Medical College, and the faculty of Assam downtown, University for their help and support. This article is part of the Ph.D. work of the first author.

CONFLICT OF INTEREST

This review paper has no conflict of interest and is not funded through any sources.

FINANCIAL DISCLOSURE

No funding was received for this paper

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 145-148 **Interdisciplinary Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Microwave assisted synthesis and study of some chromium (III) complexes of p-cresol with TBC

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Received : 05th January, 2023 ; Revised : 05th February, 2023

Abstract- p-cresol dissolved in tetrahydrofuran and I,4 dioxane, when oxidized with TBC in varying molar ratios 1:1, 1:2, 3:1, gave six Cr (III) complexes which were analyzed for elementary composition and characterized by FTIR spectroscopy. The Process of oxidation was carried out in micro-oven for different time interval and at different wattage.

Key words: p-cresol, TBC, FTIR spectroscopy,

INTRODUCTION

p-Cresol is a phenolic chemical compound having chemical formula C_7H_8O with molar mass of 108.14g/mole which is used widely in industries like pharmaceuticals, leather and cosmetics industries.[1,2] p-Cresol has been used for over past 120 years in the production of resins, plastic, medicines, dyes and other materials.[3-6] TBC is a versatile oxidant.[7,8] It is prepared by dissolving CrO_3 in tertiary butyl alcohol.[9,10] The compound was prepared for the first time by Wienhaus but R. V. Oppenauer and H. Oberrauch introduced it as a new oxidizing agent.

MATERIALS & METHODS

CHEMICALS USED

1,4 Dioxane, tertiary butyl alcohol, chromium trioxide, acetone.

EXPERIMENTAL

(a) Oxidant : Substrate (1:1) –PC101

1 gm p-cresol was added to oxidant, TBC prepared by dissolving 1 gm of CrO_3 in 10 mL of tertiary butyl alcohol. The mixture was stirred vigourously and heated in microwave oven at 16°C for 90 seconds. The brown-coloured product was washed with acetone and collected as sample R1.

(b) Oxidant: substrate (1:2) –PC112

2.2 gm p-cresol was added to oxidant, TBC prepared by dissolving 1 gm of CrO_3 in 10 mL tertiary butyl alcohol. The mixture was stirred vigourously and heated in microwave oven at 16°C for 200 seconds. The brown coloured product was washed with acetone and collected as sample R2.

(c) Oxidant: Substrate (1:3) –PC113

3.3 gm p-cresol was added to oxidant, TBC prepared by dissolving 1 gm of CrO_3 in 10 mL tertiary butyl alcohol. The mixture was stirred vigourously and heated in microwave oven at 16°C for 100 seconds. The brown coloured product was washed with acetone and collected as sample R3.

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Table 1- FTIR Results (sample no.01)

PEAK	NATURE OF PEAK	GROUP ASSIGN
335.94	weak	O-H Stretching
1600.92	sharp	C=C stretching
1369.46	medium	O-C
1307.74	weak	O-H
1242.16	medium	C-O stretching
1172.72	weak	C-O stretching
1153.43	weak	C-O stretching
968.27	medium	C-O stretching
844.82	weak	O-H
771.53	weak	C-H
570.93	weak	M-O
536.21	weak	M-O stretchings

Table 2- FTIR Results (sample no.02)

PEAK	NATURE OF PEAK	GROUP ASSIGN
3294.42	Weak	O-H
2893.22	Broad	O-H
1604.77	Sharp	C=O
1489.05	Sharp	C-O-O
1369.46	Medium	O-C
1300.2	Medium	O-C
1230.58	Weak	C-O
1149.57	Weak	C-O
1072.42	Weak	C-O
964.41	Medium	C-O
844.82	Weak	O-H
771.53	Weak	O-H
680.66	Medium	M-O
532.35	Weak	M-O
459	Weak	M-O

Table 3- FTIR Results (sample no.03)

PEAK	NATURE OF PEAK	GROUP ASSIGN
3298.28	Weak	O-H
2889.37	Broad	O-H
1604.77	Sharp	C=O
1489.05	Sharp	C-O-H
1369.46	Medium	O-C
1300.02	Medium	O-C
1222.87	Weak	C-O
1149.57	Weak	C-O
1072.42	Weak	C-O
964.41	Medium	C-O
844.82	Weak	C-O
771.53	Weak	O-H
686.66	Medium	M-O
540.07	Weak	M-O

Table 4- Thermogravimetric results (sample no.01)

Temperature	Formulation sequence showing the change	Percentage Theoretical	Loss percentage
31.4 -150^{0}C	$Cr_2(CH_3CO_2)_4\ (H_2O)_2$ |-2H_2O	18	18.32
150 -460^{0}C	$Cr_2(CH_3CO_2)_4$ |-2$(CH_3CO_2)_4$	47.55	47.73
460 -600^{0}C	Mixed oxides of Cr		

Table 5- Thermogravimetric results (sample no.02)

Temperature	Formulation sequence showing the change	Percentage Theoretical	Loss percentage
31.98 -140^{0}C	2CrO.2H_2O.$CH_2COOH.CH_2(COOH)_2$ |CH_2COOH	17.85	17.50
140 -360^{0}C	2CrO.2H_2O.$CH_2(COOH)_2$ |2H_2O	10.55	12.73
360 -700^{0}C	2CrO.$CH_2(COOH)_2$ |$CH_2(COOH)_2$ 2CrO and other oxides	30.87	29.34

RESULT

The FTIR curves and DTA TGA analysis support the following formulation for the sample PC1, PC2 and PC3.

PC1—$Cr_2(CH_3CO_2)_4\ (H_2O)_2$

PC2—2CrO.2H_2O.$CH_2COOH.CH_2(COOH)_2$

PC3—$Cr_2O.CH_3COOH.CH_3COCOOH.2H_2O$

The FTIR Curves (table-1) of sample R1 contain almost all the peaks which are expected for the formulation.

Similarly, the proposed formulation of sample R2, on the basis of empirical formulation, is strongly supported by the FTIR curves as well as TGA-DTA analysis $Cr_2O_3.2(COOH)_2.4H_2O$.

The expected broad peak for O-H stretching (H-bonded) at 3402.3 cm, bidendate carboxylic acid functioning as ligand at 1562.3 cm, C-O stretching at 1280.0 cm, C-C stretching at 950.7 cm^{-1} and M-O stretching at 622.0 cm^{-1} are present in the curve. Similarly, the loss pattern is just what we expected for the formulation as shown. We observed that the FTIR curves (table -2) of sample no.02 contain almost all the peaks which are expected for its formulation as shown. We observe that the FTIR curves contains almost all the peaks which are expected for its formulation 2CrO.2H_2O.$CH_3COOH.CH_2$ $(COOH)_2$. It is supported by the peaks at 33766.8cm (O-

H stretching), 2370.2cm (C-H stretching) 1565.4 cm (bidentate carbxylate ligand), 1434 cm (-COOH co-ordinated) 809.8 cm (C-C stretching) etc. Also the loss pattern in TGA-DTA curve supports the proposed formulation.

ACKNOWLEDGEMENT

The authors are thankful to the ISM Dhanbad, Department of Chemistry for FTIR spectroscopic measurements and BIT Mesra for TGA/DTA Analysis.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 149-154 **Interdisciplinary Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Studies on health status of fluorosis suffering inhabitants of Rajauli, Bihar

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Received : 05th January, 2023 ; Revised : 05th February, 2023

Abstract- The intake of fluoride rich water above the maximum permissible limit of 1.5 mg/L leads to a disease called fluorosis. Rajauli block of Nawada district, Bihar is one of the worst fluorosis endemic areas. Inhabitants suffering from dental, skeletal and non-skeletal fluorosis have been found in the villages of Rajauli block. Out of the 44 samples analysed from four villages of Rajauli, 41 water samples were contaminated with fluoride. To identify the fluorosis affected people, health status survey was done. The height and weight of the villagers (who were above 6 years of age) were measured and the BMI (Body Mass Index) values were calculated. During the health survey many people of the village reported dental problems, muscular pains, joint pains, bone deformities and many more ailments related to fluorosis. Some cases of knock knee and bow legs were also observed. Many of the inhabitants had their BMI level less than 18.5.

Key words: fluorosis, endemic, contaminated, deformities, ailments

INTRODUCTION

Water is one of the most important resources needed by all living forms to sustain life on the earth. Out of the available global water only 0.5% is available for domestic purposes.[1] Water from wells, tube wells, bore wells, lakes, rivers etc. which are sources of fresh water are used directly or indirectly for domestic purposes. However there are many water sources that have got contaminated due to water pollution.[2] Not only the surface water but also the groundwater is facing problem due to contamination by many chemicals, compounds, elements, ions, etc.[3] One such contamination in groundwater is caused due to fluoride (F-) ions. The maximum permissible limit of fluoride in drinking water as per the WHO guidelines is 1.5 mg/L.[4] Intake of fluoride rich water above the permissible limit of 1.5 mg/L leads to a disease called fluorosis- a severe bone disease.[5] The district of Nawada is one among the 11 districts of Bihar where groundwater is contaminated by fluoride ions. The Rajauli block of Nawada district is one of the worst fluorosis endemic areas.[6]

Many villages in Rajauli including Hardia, Hanuman Nagar, Singar Khas, Bhaunr and Kachhariadih have visible cases of fluorosis.[7,8] The health status of the fluorosis affected inhabitants was not good. People suffering from dental fluorosis (moderate and severe), skeletal fluorosis (knock-knee and bow legs) and non-skeletal fluorosis were found in the villages. The BMI values of many inhabitants was found lower than 18.5. The intake of nutritious food by the villagers was very poor.

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MATERIALS & METHODS

Collection Of Water Samples

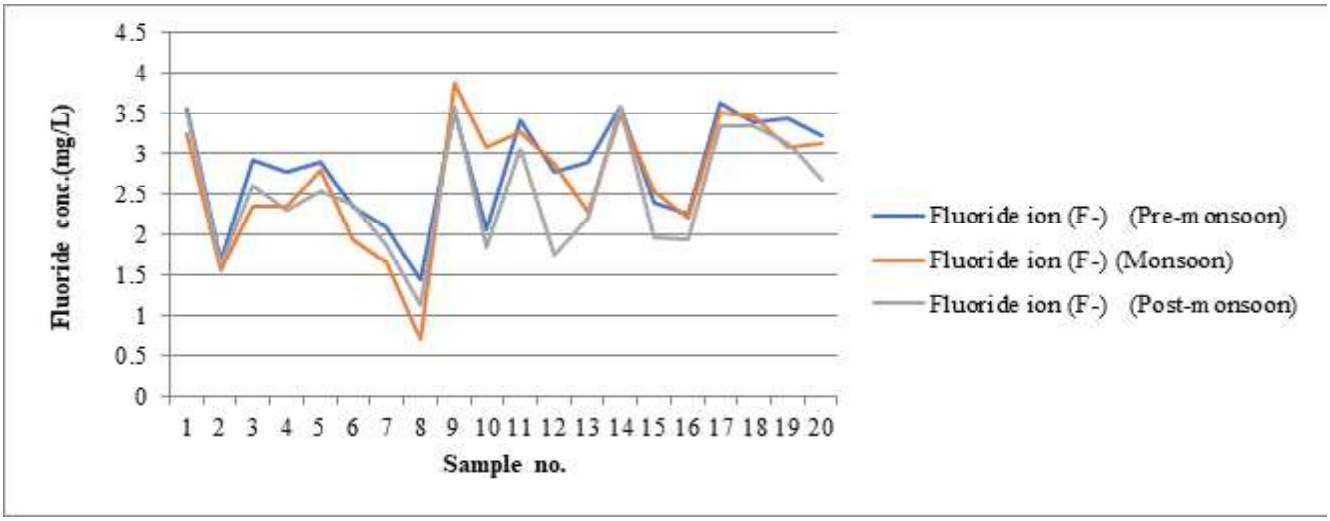
The different villages of Rajauli where the groundwater is contaminated by fluoride was identified by sampling and testing the groundwater sources by the 'Aquasol Kit'. The samples were collected from Jazpur (Hardia Sector D), Hanuman Nagar , Singer Khas , Bhaunr villages of Rajauli block of Nawada district. Water from different groundwater sources were collected during 2018 and 2019. Water was collected from hand pumps and bore wells -the depth of which ranged from 80 ft to 130 ft.

Testing of the samples for fluoride contamination

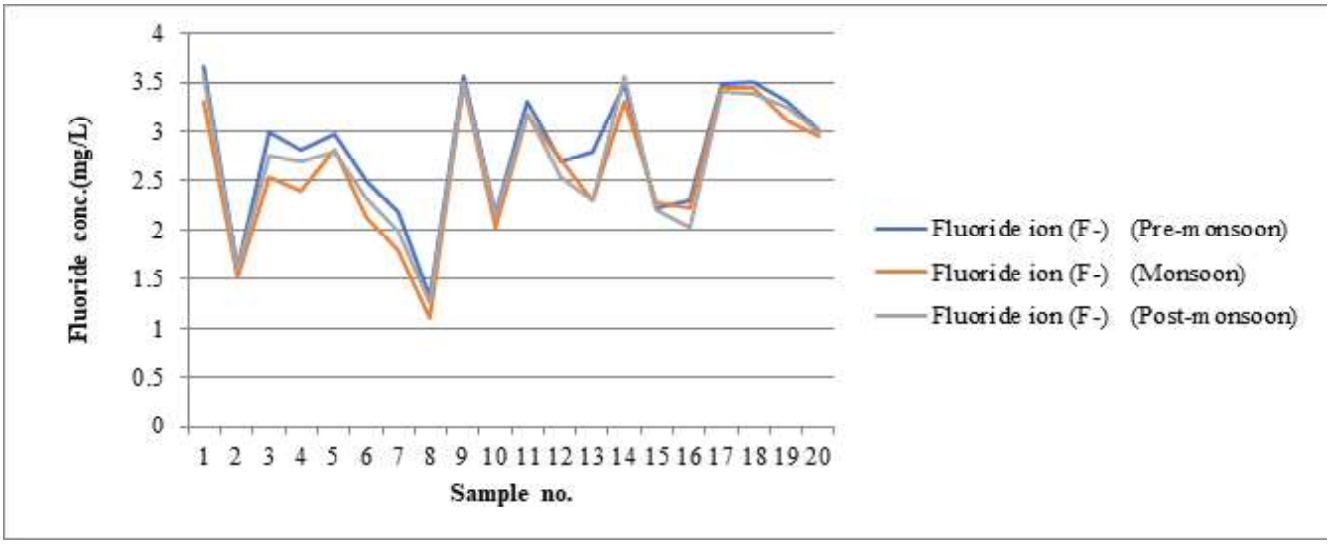
Water samples were collected in 1 L size plastic bottles from the villages. The water samples were collected during pre-monsoon, monsoon and post-monsoon periods of 2018 and 2019. The samples were then analysed by the Ion Selective Electrode Method in the Centre for Fluorosis Research, Department of Chemistry, A.N. College, Patna.[9,10] Other physico-chemical parameters like pH, Electrical Conductivity, TDS, Total Hardness, Calcium Hardness, Magnesium Hardness, Alkalinity, Turbidity, Chloride, Sodium ion, Potassium ion, Sulphate and Iron were also analysed by the standard methods described by APHA and NEERI-2012.

Study on health status of the people

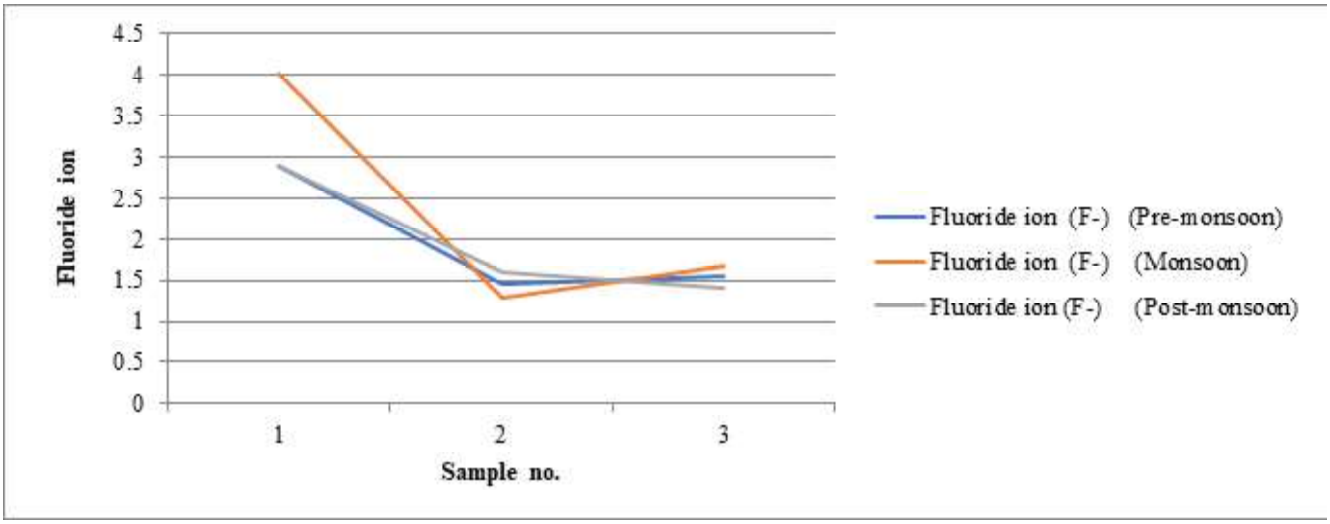
Health status survey was done in the affected villages. The health status survey included the study of their daily food intake and their Body Mass Index (BMI) values.[11] Most of the people had dental problems, some had muscular and joint pains. Very severe cases of bone deformity were found in the villages. The BMI was also found low in many of the villagers.



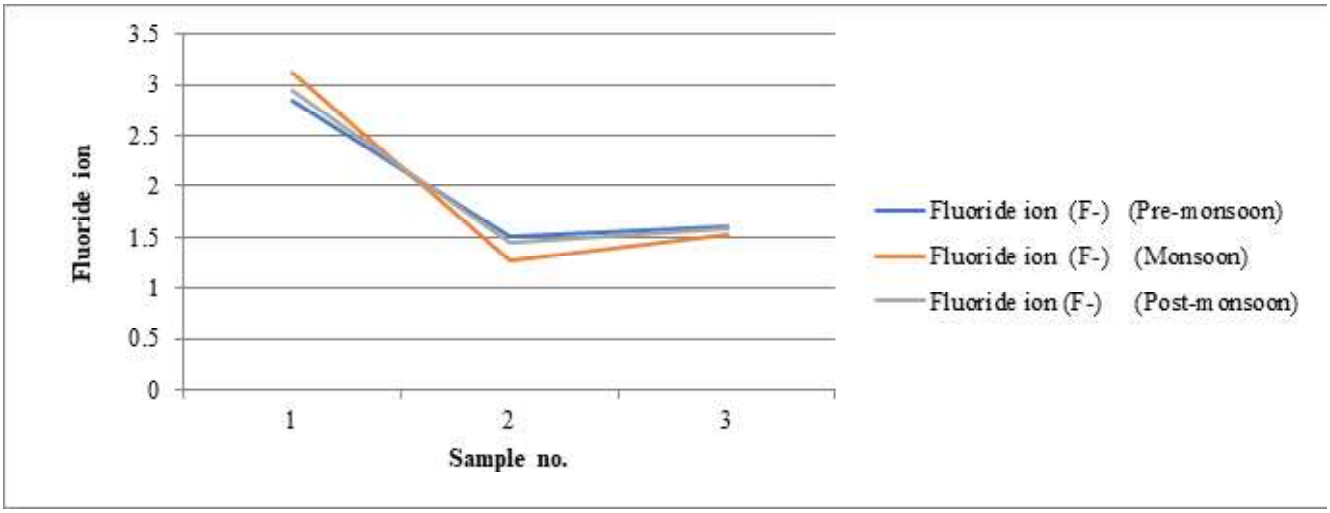
Graph 1- Fluoride conc. in water samples of Hardia Sector D (2018)



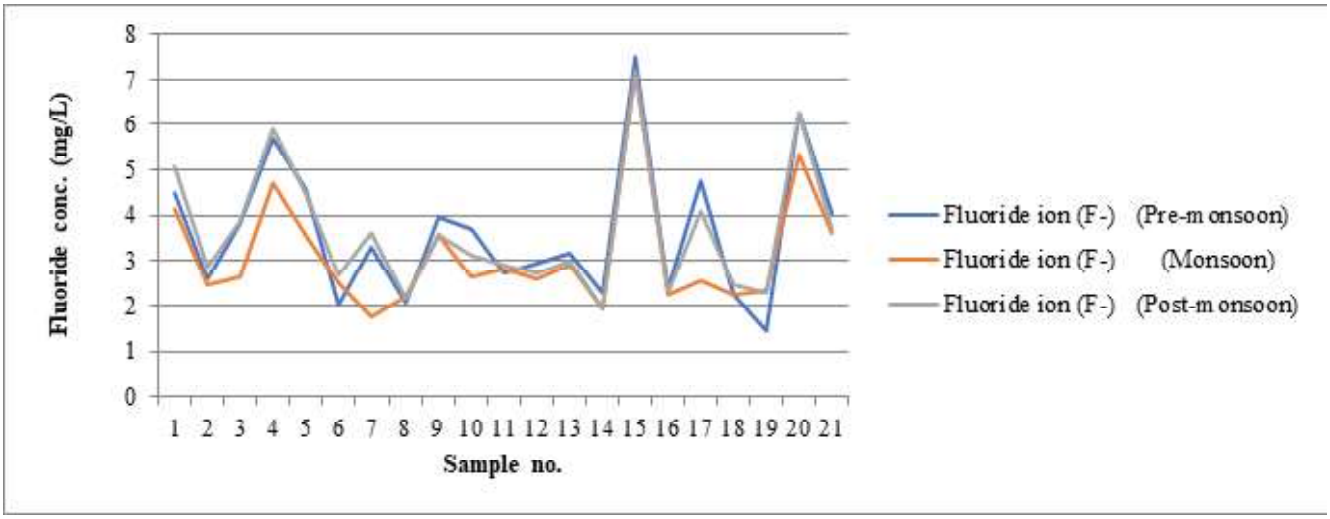
Graph 2- Fluoride conc. in water samples of Hardia Sector D (2019)



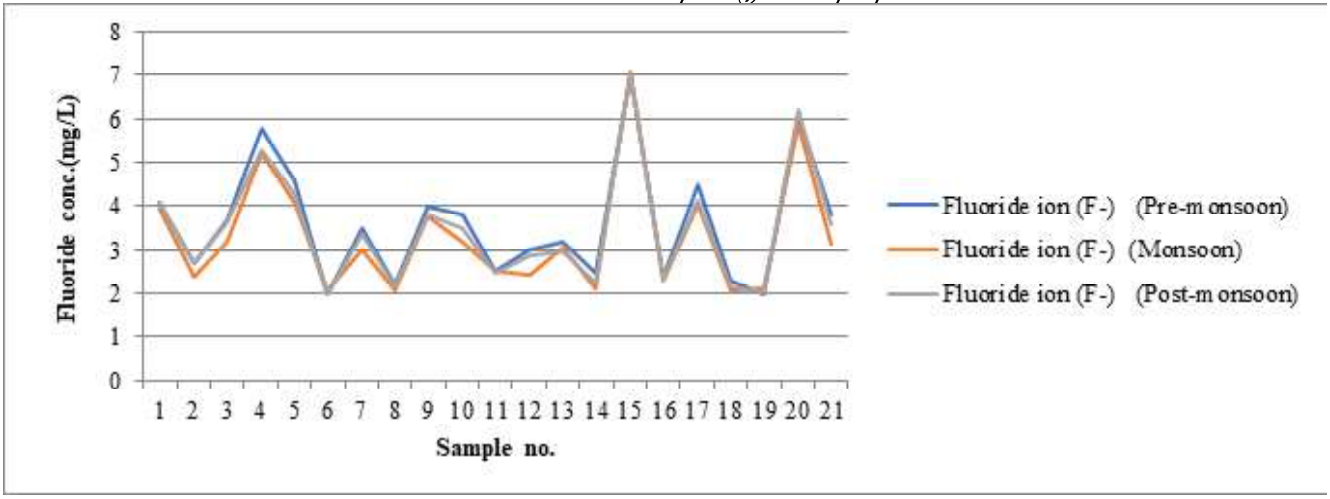
Graph 3- Fluoride ion conc. in water samples of Singar Khas and Hanuman Nagar (2018)



Graph 4- Fluoride ion conc. in water samples of Singar Khas and Hanuman Nagar (2019)



Graph 5- Fluoride conc. in water samples of Bhaunr (2018)



Graph 6- Fluoride conc. of water samples of Bhaunr (2019)

RESULTS & DISCUSSION

On testing the samples for fluoride it was found that maximum groundwater sources of the villages were contaminated with fluoride. Most of the groundwater sources had the fluoride level more than the permissible limit of 1.5 mg/L.

Jajpur (Hardia Sector D)

The groundwater sources in the village were contaminated with fluoride.

The peak value found was 3.65 mg/L which was during the pre- monsoon season of 2019. The inhabitants of that area reported the problems of joint pains, muscular pains and dental problems. All the three types of fluorosis are prevalent in that village. Few cases of knock-knee and few cases of bow legs were observed in the village. Inhabitants were also found to be suffering from non-skeletal fluorosis. Few people of the age group 15-40 years had abdominal or joint pains.

The BMI values of many inhabitants was found lower than 18. The intake of nutritious food by the villagers was very poor.





Fig 1. Skeletal Fluorosis cases in Hardia Sector D

Hanuman Nagar and Singar Khas

The groundwater sources in these villages were contaminated with fluoride.

The peak value found was 4.02 mg/L which was during the monsoon period of 2018. More than half of the inhabitants surveyed reported the problems of dental fluorosis. A few severe cases of dental fluorosis were also seen. Severe cases were also seen in children. Very few cases of skeletal fluorosis affected people were found in the village. There were no cases of knock-knee found. However few cases of people with bow legs were found in the villages. Maximum villagers above 40 years of age had the problem of joint and abdominal pains.

Bhaunr

Most of the groundwater sources in Bhaunr village were highly contaminated with fluoride.

Out of the 21 sources tested almost all the drinking water sources showed the presence of fluoride above the permissible limit. In one of the groundwater sources the concentration of fluoride was as high as 7.51 mg/L which

is much higher than the maximum permissible limit of 1.5 mg/L.

The fluoride concentration in most sources showed the peak value during the pre-monsoon period and minimum during the monsoon period. The inhabitants of that area reported the problems of joint pains, muscular pains and dental problems. Maximum people were normal and had no bone deformities. Except a few bow leg cases, no other inhabitants had skeletal problems. Since most of the people get sufficient nutrition, so the severity of the disease is less in this village as compared with the neighbouring villages where the severity of fluorosis is high due to lack of nutrition.



Fig 2. A case of Dental Fluorosis in Hanuman Nagar



Fig 3. A case of Dental Fluorosis in Bhaunr



Fig 4. A case of Skeletal Fluorosis in Bhaunr

CONCLUSION

The present work confirms that the high concentration of fluoride in drinking water is responsible for the disease fluorosis in the four villages- Hardia Sector D, Singar Khas, Hanuman Nagar and Bhaunr. High level of fluoride is responsible for physical deformities among the inhabitants of the affected villages. The severity of fluorosis was higher where there was lack of sufficient nutrition.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 155-164

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Quantitative ethnobotanical assessment of Barda Sanctuary, Gujarat, India.

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Received : 3rd January, 2023 ; Revised : 2nd February, 2023

Abstract- For a long time, ethnomedicinal plants have been crucial for curing many ailments in the world. Maldharies of Barda Sanctuary depending on for the purpose of medicine on plants and most of them having traditional knowledge on how to use different medicinal plants as remedy to treat like cough, fever, asthma, headache and other diseases. This study was started for the identification of medicine men who have been using plants as medicines in Barda Sanctuary, Gujarat, India. And also, to document their traditional knowledge quantitatively on how they utilize ethnomedicinal plants. Field study was performed over the duration of 3 years in Barda Sanctuary. Total information related to the utilization of medicinal plants was gathered with the help of questionnaires and structured interviews with the traditional healers. After collecting the data, it was analysed with the help of use value (UV), informant consensus factor (Fic) and fidelity level (FL). Finally, 51 species of plants having medicinal properties distributing in 47 genus which belonged to 31 plant families were commonly used by Maldhari medicine men of Barda Sanctuary. These plants cured 39 ailments. Total 12 categories of ailment were identified. Most used plant part was leaf. Fic values of this study showed that healers had agreed in using the medicinal plants in curing hair care, liver problems and Genito-urinary problems. Dermatological infections had highest use-reports and 32 plant species had fidelity level of 100% being the highest. From the result the plants, *Maytenus emarginata, Mitragyna parvifolia, Alstonia scholaris, Phyllanthus emblica, Eucalyptus camaldulensis, Pterocarpus marsupium* and *Mimusops elengi* (having high UV), *Zizyphus nummularia, Acacia chundra, Carissa congesta, Syzygium cumini* and *Tamarindus indica* (having high FL) are recommended for future studies in ethnopharmacology field for discovering medicines.

Key words: Data analysis, Ethnomedicine, Maldharies, Barda sanctuary

INTRODUCTION

According to WHO "Roughly 65-80% of the population of the world in developing countries depends mainly on plants for meeting up their primary health care requirements which is because of the case of poverty and not having the access of modern medicine".[1] Recently, information of ethnobotany in the research of medicinal plant research has increased in getting a lot of attention in scientific community.[2] People have started to gain interest in medicinal plants because prescribed drugs are very costly for maintaining own and well-being and also to bio prospect new drugs derived from plants.[3]

Looking into History, it is found that most of the medicines were made from plant, it may have been plant parts or much more complex forms like extracts or mixtures. The most important benefits of using medicines which are made directly from plant are that they are comparatively safer than synthetically made drugs. They offer maximum benefits in therapy and they provide affordable treatment.[4] For nearly 200 years, pharmacopoeia having been

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dominated by medicines made from herbs and rough estimate of worldwide prescribed drugs that is 25% drugs actually came by the use of plants. As WHO said "Out of 252 drugs, which were considered basic and essential, 11% are of exclusive plant origin".[5,6]

Last few decades showed that in India the study of traditionally used ethnomedicinal plants has increased a lot. Also, recent study shows increasing use reports of plants which are being utilised by traditional healers of the nomadic tribes.[7-11] Ethnomedicinal uses of plants in Barda Sanctuary were available in scarce amount and all of the studies were having deficiency in analysing the data by not approaching in the quantitative way.[12,13] This particular research has been done by keeping aim to recognise medicine men among the Maldharies of Barda Sanctuary in Gujarat, India and to quantitatively analyse their traditional knowledge of ethnomedicine with the help of different ethnobotanical tools on how they utilize medicinal plants which are commonly used.

MATERIALS & METHODS

Study area and people

Barda sanctuary present in two districts of Gujarat, namely Porbandar and Jamnagar. This sanctuary cover about 192 sq. km and located between 21°40'-21°55' N latitude and 69°40'-69°50' E longitude (Fig. 1). The vegetation is dry and no surplus water is available throughout the year. The study was conducted in ten ness of Barda sanctuary (Adityana, Ajmapat, Dhedhia, Ghumli, Godhana, Kileshwar, Mota jungle, Naliadhar, Sat serda and Sat virda) which are inhibited by Maldharies.

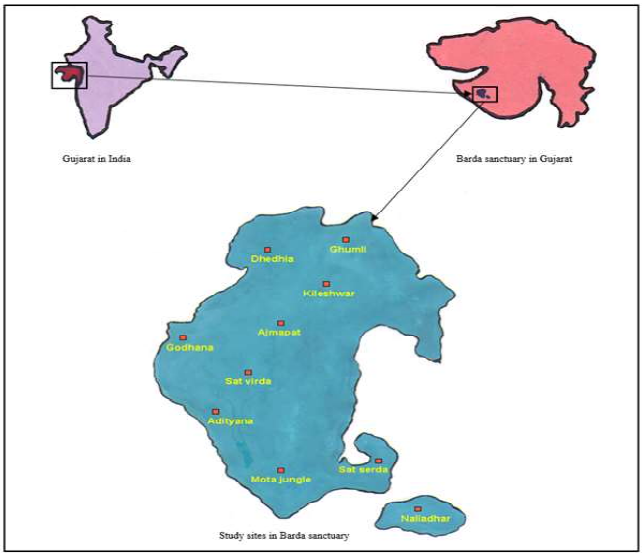


Figure 1. Barda Sanctuary

Maldharies are the indigenous people of the study area, nomads of Gujarat, living predominantly in Barda sanctuary. Maldharies have general knowledge of ethnomedicinal plants. They also know how to make medicines from plants for curing ailments like cough, head ache, fever, etc.

Data collection

The study area investigation was performed for acquiring information from medicine men related to the use of plants. Minimum 7 days were spent with the Maldharies of the Barda Sanctuary during the field visits. It was required to document their traditional knowledge of ethnomedicinal plants, for that a total of 10 field surveys were done during 3 years in Barda sanctuary. 9 traditional healers were interviewed with the help of questionnaires and oral conversations. They have adequate knowledge of ethnomedicinal plants found which they use. Information was recorded about the plants like their local name and which part is utilised to cure disease. Herbarium specimens were collected for the deposition at Department of Life Science of Bhavnagar University, Gujarat, India.

Ailment categories

Total 12 ailment categories were identified by gathering the information from the medicine men. (Table 1).

Table 1. Ailment categories

Ailment categories	Biomedical terms	Gujarati terms
Circulatory system/cardiovascular diseases (CSCD)	Heart pain	Hraday no dukhavo
	Heart strength	Hraday ni shakti
	Blood purification	Lohi ni shudhee
	Memory power	Yad shakti
Dental care (DC)	Mouth heat	Mo ni garmee
	Mouth ulcers	Mo na chanda
	Tooth ache	Dant no dukhavo
	Foul odour	Mo ni durgandh
	Teeth strength	Dant ni shaktee
	Worms in gums & teeth	Mo ma jivat

Dermatological infections/diseases (DID)	Wounds	Gha
	Burns	Dajvu
	Itching	Khanjval
	Skin ulcers	Chamdi ma chanda
	Cuts	Vadhiya
	Pimples	Kheel
	Boils	Gumda
Ear, nose, throat problems (ENT)	Nasal infection	Nak ma chep
	Ear ache	Kan no dukhavo
	Cataracts	Motiyo
	Ophthalmia	Aankh no sojo
Endocrinal disorders (ED)	Diabetes	Madhuprameh
Gastro-intestinal ailments (GIA)	Stomach pain	Pet no dukhavo
	Constipation	Kabajiyat
	Gastric complaints	Vat ni takleef
	Dysentery	Mardo
	Indigestion	Apacho
	Intestinal ulcer	Aantarda ma chanda
Fever (Fvr)	Fever	Tav
Genito-urinary ailments (GUA)	Burning during urination	Peshab ma baltra
	Delivery pain	Prasav ni peeda
Hair care (HC)	Dandruff	Khod
	Hair loss	Val kharva
Liver problems (LP)	Jaundice	Kamlo
Respiratory system diseases (RSD)	Cough	Kaf
	Asthma	Dam
Skeleto-muscular system disorders (SMSD)	Head ache	Matha no dukhavo
	Rheumatism	Sandhiva
	Swelling	Soja

Data analysis

Informant consensus factor (F_{ic})

This index was used to determine the agreement in how all the medicine men used the plants to cure ailments. F_{ic} calculation has been performed by formula:[14]

"$F_{ic} = N_{ur} – N_t / N_{ur} – 1$"[14]

Where, "N_{ur} = number of use-reports for an ailment category, N_t = number of species used for an ailment category by all informants".[14] Value of Fic can be obtained between 0 to 1. High value (close to 1.0) means informants used less species for curing a particular ailment. Low value means informants do not agree about the species that can be used for treating a particular illness of the ailment category.

Use Value (UV)

Plant species which is locally used as medicine and its importance can be calculated by use value (UV). UV calculation has been performed by formula:[15]

"$UV = \Sigma U / n$"[15]

Where, "UV = use value of a species, ΣU = summation of number of use-reports given by informants for a given plant species and n = total number of informants interviewed for a given plant".[15] Index UV is for determining the highest use of a particular plant which indicates most frequently used plants. High UV means there are maximum use- reports for plant, low UV means there are minimum use-reports for plant.

Fidelity level (FL)

It is used to identify the plants which were used most frequently by the medicine men for treating an ailment. FL calculation has been performed by formula:[16]

"$FL (\%) = N_p / N \times 100$"[16]

Where, "N_p = number of use-reports given for a species for a particular ailment category and N = the number of use-reports given for any species".[16] High FL means that all the use-reports showed same way of the way of utilization of plants. Low FL means plant species used for various purposes.[17]

RESULTS & DISCUSSION

Documentation of indigenous ethnomedicinal knowledge

This study showed that medicine men of Barda Sanctuary used 51 plant species, distributed in 47 genus and belonging to 31 plant families. These plant species used in maximum amount by Maldharies traditional healers for treating ailments of 39 types. Plant family Fabaceae was found to be most important which had six plant species. We have given botanical name of plants, local (Gujarati) name of the plants, use value, life form, parts of plant used and ailment treated (Table 2).

Life forms and parts used

Trees were found to be the primary source from which medicine men of Barda Sanctuary made medicines (51%) which was followed by herbs (25%) and then shrubs (24%) (Fig. 2). The leaves (28%) among all the other plant parts used, were used most frequently for preparing the

medicines. It was then followed by fruit (25 %), stem bark (20 %), root (16 %), flower (6 %), seed (4 %) and latex (1%). (Fig. 3). Other tribal communities in other parts of the world used leaves for preparing medicines from plants as well.[17-25] It can be attributed to a fact that it is quite easy to collect leaves. This is the reason why leaves are used most frequently in making medicines by tribal people.[26] Scientific perspective provides information on leaves participating in the process of photosynthesis and metabolites production.[27]

Table 2. List of commonly used medicinal plants by Maldharies in Barda Sanctuary

Botanical name (Family)	Local name	Life form	Use value	Parts used	Ailment category: no. of use-repots (ailments treated)
Abrus precatorius L. (Fabaceae)	Chanothi	He	0.89	Lf	ENT: 5 (Cataracts)
Abrus precatorius L. (Fabaceae)	Chanothi	He	0.89	Lf Lf	ENT: 5 (Cataracts) GU: 3 (Stomach pain)
Abutilon indicum (L.) Sweet, Hort. (Malvaceae)	Khapat	Sh	0.78	Sb	DC: 7 (Mouth heat)
Acacia chundra (Roxb. ex Rottl.) Willd. (Fabaceae)	Kher	Tr	0.89	Sb	DC: 8 (Mouth ulcers)
Acacia concinna Sensu auct., non (Willd.) DC. (Fabaceae)	Shikakai	Sh	0.67	Fr	LP: 6 (Jaundice)
Acacia leucophloea (Fabaceae)	Harmo baval	Tr	0.78	Sb	RSD: 7 (Cough)
Acacia sengel (L.) Willd. (Fabaceae)	Gorad	Tr	1.67	Rt	ED: 2 (Diabetes)
				Rt	GUA: 5 (Burning during urination)
				Lf	DID: 3 (Boils)
				Lf	SMSD: 2 (Swelling)
				Lf	LP: 1 (Jaundice)
				Lx	RSD: 2 (Cough)
Achyranthes aspera L. var. *aspera* (Amaranthaceae)	Anghedi	He	1.89	Sb	DC: 2 (Teeth strength)
				Lf	GUA: 9 (Burning during urination)
				Lf	RSD: 6 (Asthma)
Adina cordifolia (Roxb.) Bth. & Hk. f. ex Brandis (Rubiaceae)	Haldarvo	Tr	1.67	Fl	SMSD: 4 (Head ache)
				Sb	DID: 5 (Boils)
				Fl	RSD: 6 (Cough)
Aegle marmelos (L.) Corr. (Rutaceae)	Bili	Tr	3.00	Fr	ENT: 5 (Ear ache)
				Fr	GIA: 9 (Dysentery)
				Lf	SMSD: 5 (Rheumatism)
				Lf	DID: 6 (Burns)
				Rt	CSCD: 2 (Heart strength)
Alianthus excelsa Roxb. (Simaroubaceae)	Arduso	Tr	1.56	Sb	GIA: 3 (Stomach pain)
				Sb	Fvr: 2 (Fever)
				Sb	RSD: 3 (Cough)
				Sb	DID: 2 (Boils)
				Sb	SMSD: 4 (Rheumatism)
Allium sativum L. (Liliaceae)	Lasan	He	0.89	Lf	Fvr: 8 (Fever)
Aloe barbadensis Mill. (Liliaceae)	Kuvarpathu	Sh	1.00	Lf	DID: 9 (Burns)
Alstonia scholaris (L.) R. Br. (Apocynaceae)	Saptparni	Tr	2.78	Lf	DID: 2 (Pimples)
				Sb	GIA: 5 (Dysentery)
				Sb	Fvr: 3 (Fever)
				Fl	RSD: 6 (Asthma)
				Fr	ED: 4 (Diabetes)
				Fr	CSCD: 5 (Blood purification)
Andrographis paniculata (Burm. F.) Wall. ex Ness (Acanthaceae)	Lilu kariyatu	He	0.67	Lf	ED: 6 (Diabetes)
Argemone mexicana L. (Papaveraceae)	Darudi	He	0.56	Sd	DID: 5 (Skin ulcers)
Buchnania lanzan Spreng. (Anacardiaceae)	Charodi	Tr	0.44	Sd	CSCD: 4 (Memory power)
Butea monosperma (Lam.) Taub. (Fabaceae)	Khakhro	Tr	1.11	Sb	DC: 2 (Tooth ache)
				Rt	RSD: 3 (Cough)
				Rt	DID: 5 (Boils)

Carissa congesta Wt. (Apocynaceae)	Karamda	Sh	1.00	Fr	DID: 9 (Itching)
Clerodendrum phlomoides Hort. Ital. ex DC. (Lamiaceae)	Arni	Sh	1.00	Lf	GUA: 9 (Delivery pain)
Coccinia grandis (L.) Voigt. (Cucurbitaceae)	Tindori	He	0.67	Fr	ED: 6 (Diabetes)
Commiphora wightii Arn. Bhandari (Burseraceae)	Gugal	Tr	0.44	Sb	SMSD: 4 (Rheumatism)
Datura metel L. (Solanaceae)	Kalo dhaturo	He	0.44	Sd	SMSD: 4 (Swelling)
Enicostema hyssopifolium (Willd.) Verdoon (Gentiaceae)	Mamejvo	He	1.00	Lf	Fvr: 9 (Fever)
Eucalyptus camaldulensis Dehnh. (Myrtaceae)	Nilgiri	Tr	2.44	Lf	ENT: 5 (Nasal infection)
				Sb	RSD: 9 (Cough)
				Sb	SMSD: 8 (Rheumatism)
Garcinia indica (Thouras) Chiosy. (Clusiaceae)	Kokam	Tr	0.78	Fr	DID: 7 (Cuts)
Helicteres isora L. (Sterculiaceae)	Marda shing	Tr	1.67	Fr	GIA: 3 (Dysentery)
				Rt	ED: 5 (Diabetes)
				Rt	Fvr: 7 (Fever)
Jatropha curcas L. (Euphorbiaceae)	Ratanjyot	Sh	0.67	Lf	DC: 6 (Foul odour)
Justicia adhatoda L. (Acanthaceae)	Ardusi	Sh	0.56	Lf	ENT: 5 (Ophthalmia)
Maytenus emarginata (Willd.) Ding. Hou. (Celastraceae)	Vikdo	Sh	3.11	Lf	LP: 7 (Jaundice)
				Sb	DID: 6 (Boils)
				Rt	SMSD: 4 (Head ache)
				Lf	DC: 3 (Tooth ache)
				Lf	Fvr: 6 (Fever)
				Fr	CSCD: 2 (Blood purification)
Mimusops elengi L. (Sapotaceae)	Borsali	Tr	2.22	Sd	GIA: 3 (Intestinal ulcer)
				Rt	SMSD: 7 (Swelling)
				Fl	RSD: 8 (Cough)
				Fl	CSCD: 2 (Heart strength)
Mitragyna parvifolia Roxb. Koth. (Rubiaceae)	Kalam	Tr	2.89	Sb	ED: 2 (Diabetes)
				Rt	SMSD: 7 (Rheumatism)
				Rt	Fvr: 1 (Fever)
				Rt	GIA: 6 (Gastric complaints)
				Fr	DID: 4 (Skin ulcers)
				Fr	CSCD: 1 (Blood purification)
				Fr	RSD: 5 (Cough)
Momordica dioica Roxb. ex Willd. (Cucurbitaceae)	Kantola	He	0.56	Fr	DC: 5 (Mouth heat)
Murraya paniculata (L.) Jack. (Rutaceae)	Kamini	Sh	0.78	Lf	Fvr: 7 (Fever)
Nyctanthes arbortristis L. (Oleaceae)	Parijatak	Sh	0.44	Lf	HC: 4 (Dandruff)
Ocimum basilicum L. (Lamiaceae)	Damro	He	0.89	Lf	ENT: 4 (Ear ache)
				Fl	SMSD: 4 (Head ache)
Oroxylum indicum L. Vent. (Bignoniaceae)	Tetu	Tr	0.89	Fr	CSCD: 8 (Heart strength)
Pergularia daemia (Frosk.) Chiov (Asclepiadaceae)	Chamar dudhli	He	0.78	Lf	DID: 7 (Boils)
Phyllanthus emblica L. (Phyllanthaceae)	Aamla	Tr	2.56	Fr	HC: 9 (Hair loss)
				Rt	SMSD: 2 (Swelling)
				Rt	DID: 6 (Burns)
				Fr	CSCD: 1 (Blood purification)
				Fr	GIA: 5 (Constipation)
Pongamia pinnata (L.) Pierre. (Fabaceae)	Karanj	Tr	1.44	Sb	DC: 2 (Tooth ache)
				Rt	DID: 6 (Swelling)
				Sd	SMSD: 5 (Rheumatism)
Pterocarpus marsupium Roxb. var. *acuminatus* Prain (Fabaceae)	Biyo	Tr	2.33	Sb	DC: 1 (Worms in gum & teeth)
				Sb	ED: 5 (Diabetes)
				Lf	DID: 7 (Cuts)
				Lf	RSD: 6 (Cough)
				Lf	CSCD: 2 (Blood purification)

Sapindus laurifolius Vahl (Sapindaceae)	Aritha	Tr	1.56	Fr	HC: 9 (Dandruff)
				Lf	DID: 1 (Boils)
				Fr	SMSD: 3 (Head ache)
				Fr	GUA: 1 (Delivery pain)
Senna italica Mill. (Fabaceae)	Mindhi aaval	Tr	0.67	Fr	GIA: 6 (Gastric complaints)
Syzygium cumini (L.) Skeels (Myrtaceae)	Jambudo	Tr	0.56	Fr	ED: 5 (Diabetes)
Tamarindus indica L. (Fabaceae)	Aamli	Tr	1.00	Fr	GIA: 9 (Constipation)
Terminalia arjuna (Roxb.) W. & A. (Combretaceae)	Arjun Sadad	Tr	0.89	Sb	CSCD: 8(Heart pain)
Terminalia chebula Retz. (Combretaceae)	Harde	Tr	1.00	Fr	GIA: 9 (Indigestion)
Trachyspermum ammi (L.) Sprague (Apiaceae)	Ajmo	He	0.89	Fr	Fvr: 8 (Fever)
Typha angustata Bory & Chaub. (Typhaceae)	Ghabajariyu	He	1.00	Lf	DID: 9 (Wounds)
Withania somnifera (L.) Dunal (Solanaceae)	Ashwgandha	Sh	0.89	Rt	CSCD: 8 (Blood purification)
Wrightia tinctoria R. Br. (Apocynaceae)	Dudhlo	Tr	1.11	Lf	DID: 4 (Boils)
				Rt	SMSD: 6 (Rheumatism)
Zizyphus nummularia (Burm. F.) Wight. & Arn. (Rhamnaceae)	Chani bor	Sh	0.67	Fr	CSCD: 6 (Heart pain)

Parts used: Lf – Leaf, Sb – Stem bark, Fr – Fruit, Sd – Seed, Lx – Latex, Fl – Flower, Rt – Root
Life forms: He – Herb, Sh – Shrub, Tr – Tree

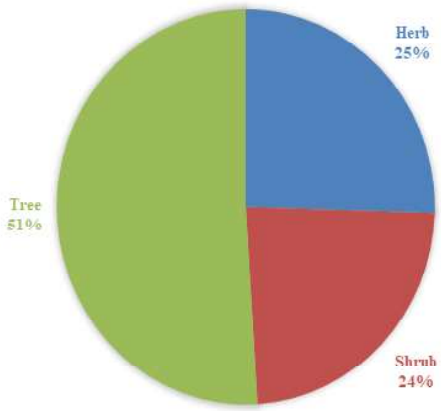


Figure 2. Life forms percentage used by Maldhari medicine men

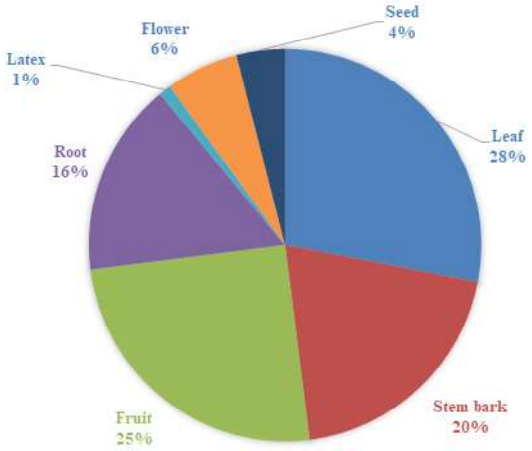


Figure 3. Plant parts percentage used by Maldhari medicine men

Plant use values

Plant species used most was *Maytenus emarginata* having 28 use-reports given by 9 informants, which gave the highest use value of 3.11. *Maytenus emarginata* was used in treating many ailments and it was recognised by every traditional healer as a plant species for treatment of jaundice.

Other most important species of plants having high use values were *Aegle marmelos* (27 use-reports, 9 informants, UV 3.00), *Mitragyna parvifolia* (26 use-reports, 9 informants, UV = 2.89), *Alstonia scholaris* (25 use-reports, 9 informants, UV = 2.78), *Phyllanthus emblica* (23 use-reports, 9 informants, UV = 2.56), *Eucalyptus camaldulensis* (22 use-reports, 9 informants, UV = 2.44), *Pterocarpus marsupium* (21 use-reports, 9 informants, UV = 2.33) and *Mimusops elengi* (20 use-reports, 9 informants, UV = 2.22).

The plant with very low use values was *Nyctanthes arbortristis* with a UV of 0.44. If the plant species is scarcely available in any particular area, it can then lead to low UV.[28] In this study, plants reported with a low use value (below 1.00) were *Abrus precatorius*, *Acacia chundra*, *Allium sativum*, *Ocimum basilicum*, *Oroxylum indicum*, *Terminalia arjuna*, *Trachyspermum ammi*, *Withania somnifera* (8 use-reports by 9 informants with a UV of 0.89), *Abutilon indicum*, *Acacia leucophloea*, *Garcinia indica*, *Murraya paniculata*, *Pergularia daemia* (7 use-reports by 9 informants with a UV of 0.78), *Acacia concinna*, *Andrographis paniculata*, *Coccinia grandis*, *Jatropha curcas*, *Senna italica*, *Zizyphus nummularia* (6 use-reports by 9 informants with a UV of 0.67), *Argemone mexicana*, *Justicia adhatoda*, *Momordica dioica*, *Syzygium cumini* (5 use-reports by 9 informants with a UV of 0.56).

Informant consensus factor

Informant consensus factor (F_{ic}) results depends on how the medicine men of the study are treat a particular ailment and also whether particular plant is available in the study area or not.[29] We classified illness into ailment categories for using index informant consensus factor (F_{ic}). In our study area the (F_{ic}) values ranged from 0.77 to 0.90. 103 use-reports in dermatological infections/diseases (cured by 19 plant species). 69 use-reports in skeleto-muscular system disorders (cured by 15 plant species), 61 use-reports in respiratory system diseases (cured by 11 species), 58 use-reports in gastro-intestinal ailments (cured by 10 plant species), 51 use-reports in fever (cured by 9 species) (Fig. 4). F_{ic} value of 0.90 which is the highest value, was obtained in the treatment of hair care. F_{ic} value 0.77 which is the lowest value, was obtained in circulatory system/cardiovascular system disorders with the lowest F_{ic} of 0.77. For treating ailments of infections/diseases 19 plant species were used (Table 3).

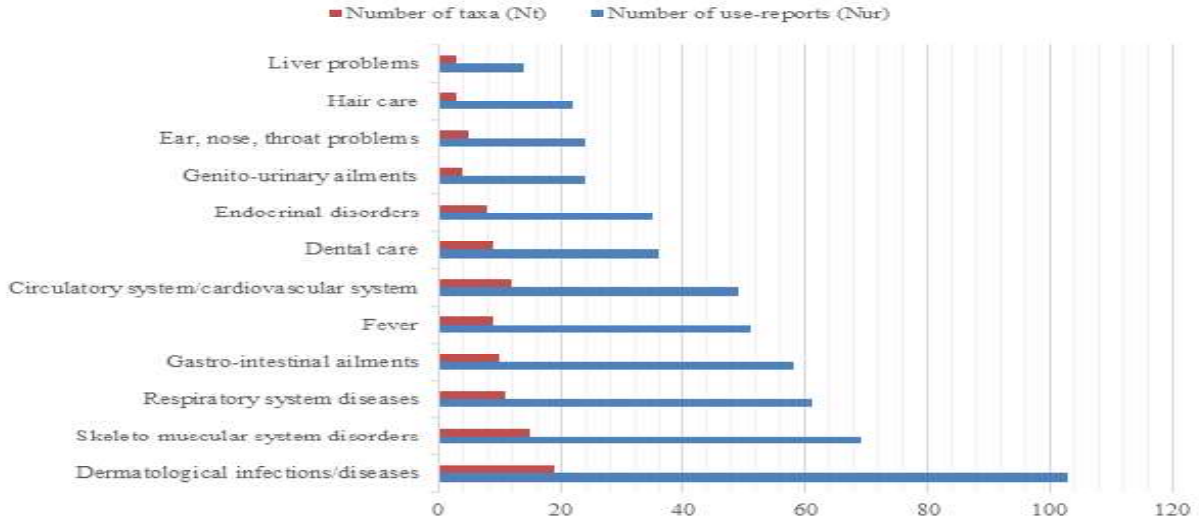


Figure 4. Ailment categories arranged by number of use-reports

Table 3. Informant consensus factor for commonly used medicinal plants

Ailment category	Number of use-reports (N_{ur})	Number of species (N_t)	Informant Consensus Factor (F_{ic})
Circulatory system/cardiovascular system disorders	49	12	0.77
Dental care	36	9	0.77
Dermatological infections/diseases	103	19	0.82
Ear, nose, throat problems	24	5	0.82
Endocrinal disorders	35	8	0.79
Fever	51	9	0.84
Gastro-intestinal ailments	58	10	0.84
Genito-urinary ailments	24	4	0.86
Hair care	22	3	0.90
Liver problems	14	3	0.84
Respiratory system diseases	61	11	0.83
Skeleto-muscular system disorders	69	15	0.79
Total	546	108*	

* a species may be reported in more than one ailment category

Fidelity level

This index identifies most important plants in each ailment category (Table 4). Many informants used 32 species in the treatment of single ailment category and these plant species had 100% (highest) fidelity level. These plant species are *Buchnania lanzan*, *Oroxylum indicum*, *Terminalia arjuna*, *Zizyphus nummularia*, *Withania somnifera* (CSCD), *Abutilon indicum*, *Momordica dioica*, Acacia *chundra*, *Jatropha curcas* (DC), *Aloe barbadensis*, *Argemone mexicana*, *Carissa congesta*, *Garcinia indica*, *Pergularia daemia*, *Typha angustata* (DID), *Justicia adhatoda* (ENT), *Andrographis paniculata*, *Coccinia grandis*, *Syzygium cumini* (ED), *Allium sativum*, *Enicostema hyssopifolium*, *Murraya paniculata*, *Trachyspermum ammi* (Fvr), *Senna italica*, *Tamarindus indica*, *Terminalia chebula* (GIA), *Clerodendrum phlomoides* (GUA), *Nyctanthes arbortristis* (HC), *Acacia concinna* (LP), *Acacia leucophloea* (RSD), *Commiphora wightii*, *Datura metel* (SMSD).

The maximum FL of 100% of these plants showed choices of the interviewed traditional healers who made medicines from plants for the treatment specific ailments. This may have been an indication that they have huge healing potential.[18]

Table 4. Fidelity level (FL) values for common medicinal plants used by Maldhari medicine men by ailment category

Ailment categories	Most preferred species with specific ailments	FL (%)
Circulatory system/cardiovascular system diseases	*Buchnania lanzan* (Memory power)	100.00
	Oroxylum indicum (Heart strength)	100.00
	Terminalia arjuna, *Zizyphus nummularia* (Heart pain)	100.00
	Withania somnifera (Blood purification)	100.00
Dental care	*Abutilon indicum*, *Momordica dioica* (Mouth heat)	100.00
	Acacia chundra (Mouth ulcers)	100.00
	Jatropha curcas (Foul odour)	100.00
Dermatological infections/diseases	*Aloe barbadensis* (Burns)	100.00
	Argemone mexicana (Skin ulcers)	100.00
	Carissa congesta (Itching)	100.00
	Garcinia indica (Cuts)	100.00
	Pergularia daemia (Boils)	100.00
	Typha angustata (Wounds)	100.00
Ear, nose, throat problems	*Justicia adhatoda* (Ophthalmia)	100.00
Endocrinal disorders	*Andrographis paniculata*, *Coccinia grandis*, *Syzygium cumini* (Diabetes)	100.00
Fever	*Allium sativum*, *Enicostema hyssopifolium*, *Murraya paniculata*, *Trachyspermum ammi* (Fever)	100.00
Gastro-intestinal ailments	*Senna italica* (Gastric complaints)	100.00
	Tamarindus indica (Constipation)	100.00
	Terminalia chebula (Indigestion)	100.00
Genito-urinary ailments	*Clerodendrum phlomoides* (Delivery pain)	100.00
Hair care	*Nyctanthes arbortristis* (Dandruff)	100.00
Liver problems	*Acacia concinna* (Jaundice)	100.00
Respiratory system diseases	*Acacia leucophloea* (Cough)	100.00
Skeleto-muscular system disorders	*Commiphora wightii* (Rheumatism)	100.00
	Datura metel (Swelling)	100.00

CONCLUSION

This study showed that Maldhaires were still using ethnomedicines. Only a few people of the study area had ethnomedicinal knowledge. Therefore, it is required to collect this knowledge possessed by Maldhareis. By doing so, our investigation will help to prevent this knowledge from getting glost of the Maldhari nomads. Ethnomedicinal plants which showed highest fidelity level and highest use values in investigation, might possibly be indicating important phytochemicals, therefore these plant species require a thorough research for having the potential of new drugs for treating of many ailments.

These ethnomedicinal plants which showed high values of indices should be evaluated for further phytochemical and pharmacological studies for checking their efficacy. As a result of this study, we suggest these plants *Maytenus emarginata*, *Aegle marmelos*, *Mitragyna parvifolia*, *Alstonia scholaris*, *Phyllanthus emblica*, *Eucalyptus camaldulensis*, *Pterocarpus marsupium* and

Mimusops elengi for further studying the ethno pharmacological properties, because these plant species had the high use values. Plants *Zizyphus nummularia*, *Acacia chundra*, *Carissa congesta*, *Syzygium cumini* and *Tamarindus indica* with highest fidelity level of 100% for particular ailment category could be analysed further in the studies of pharmacoloy.

ACKNOWLEDGEMENT

We are very grateful to the entire staff of Department of Life Sciences of Bhavnagar University, Bhavnagar, Gujarat, India and the entire staff of Department of Life Sciences of Bhakta Kavi Narsinh Mehta University, Junagadh, Gujarat, India.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 165-170

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

A comprehensive review on various uses of *Lantana camara* L.

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Received : 23rd November, 2022 ; Revised : 24th December, 2022

Abstract- *Lantana camara* is more famous as poisonous weed instead of medicinal plant in the majority of the nations. The camara plant can propagate even in the outrageous extreme climate of tropical as well as sub-tropical areas and thus get naturalized overall as a decorative plant counting India. Steroid, flavonoids, saponins, alkaloids, triterpenoids and tannins are some phytochemicals that are identified from the *L. camara* L. Besides, this plant is also its contribution in the form of oil- generating plants, and the accessible fundamental oil available in the market is known by the name of *Lantana* oil. However it also shows its properties in treating variety of diseases viz. bronchitis, stomach pain, gastrointestinal diseases, etc. The natural resources are misused and have drawn nearer to the edges of earth's capability. Mill made paper's life cycle really damages the atmosphere from beginning and lasts for the end. Thus this study highlights that how an alternative eco-safe material and method can help in the production of handmade paper. The key purpose is to emphasize the potential of invasive plants and weeds those makes the negative effect on the other side by crops. Thus, these weeds are used as non-wood raw material for paper production.

Key words: Medicinal, Poisonous, Ornamental, Disease, Invasive plant, Natural Resources, Handmade paper.

INTRODUCTION

Northern European (Sweden, Finland, and North-West Russia), North American (Canada and United States), and the East Asian nations (such as China, South Korea, East Siberian Russia and Japan) has the major dominance on the pulp and paper production. China is the world's leading producer of paper followed by the United States.[1] In India, the paper industry is a combination of small, medium and large mills. Paper industry in India is the largest industry and stood on twentieth position among the world's paper producing industry.[2] According to the Indian Paper Manufacturers Association (IPMA) in 2009. From the total world's production of paper and paper board, the Indian paper industry represents around 1.6%.

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A paper is shaped when wet cellulose filaments got from different sources are squeezed and pressed together and dried into thin, adaptable sheets. This versatile material has various purposes such as for recording as a hard copy, cleaning, bundling, and printing and also in various modern industrial and development processes. As demonstrated by the report on Ecology Global Network, the use of paper has become over 400% in the beyond 40 years worldwide. As of now, in each continent, the pulp and paper industries utilize just about 4 billion trees or 35% of the total trees cut all through the world. Thus, to reduce or to prevent the rate of deforestation there is a strong need of non-woody raw materials. Also, in the nations with lacking forest resources; the need to find non-wood plant fibres, seems appropriate for papermaking and is realized more strongly. Likewise, global warming and limitations on carbon

dioxide discharges increases gradually due to wood cutting and this creates the troublesome to the environment.[3] Generally, paper can be described as the extraction of cellulose from the plant materials whose quality normally depends upon fineness and brilliance of the fibers. A variety of synthetic fibers, such as Polyethylene and Polypropylene can be incorporated into the paper to find the desirable physical properties as the fibers are commonly sourced naturally by origin. For paper making the most commonly and recognized source of the natural fiber is considered as wood pulp from pulpwood trees.[4-6]

According to various studies – an invasive alien species is those which spreads beyond its zone of distribution.[7-9] Invasive species are considered as either native or is non-native species that can intensely colonize a specific living habitat.[10] They are likewise generally dispersed in a wide range of biological systems all through the globe. Invasive alien species have the capability to make themselves, invade, adapt and compete natives and also make dominance at the new environment in few weeks only.[11] Invasive weeds are presently influencing each and every ecosystem types in the world and they are the second most noteworthy global threat to the biodiversity after habitat destruction.[9,12] IAS competes the native species for resources such as - breeding sites and food indirectly by changing habitat and adjusts nutrient cycling, hydrology and other ecosystem processes. Their effects are on local biodiversity in ecosystem like farming and range lands, national parks, streams, lakes, waterways, power dams, street sides and metropolitan green spaces with immense economical as well as social consequences. This drastically change the ecosystem in both positively and negatively ways.[13-15]

This plant species belongs to the family Verbenaceae and order Lamiales as shown in figure 1. *Lantana* is a genus of about 150 species of perennial flowering plants.[16] *L. camara* species was first represented and acknowledged its binomial name by Linnaeus in 1753.[17] *L. camara* grows in tropical, subtropical, and mild areas at the height of up to 2000 m.[18] *Lantana camara* is sometimes also known as "Red (Yellow, Wild) Sage", despite of its classification that belongs to family sage (Lamiaceae), and order sagebrush (Asterales).[19]

Lantana camara grows in diverse habitats and on the various types of soils. *Lantana* mostly grows supreme in open, un-shaded conditions such as wildernesses, edges of the rain forests, beachfronts, scrub/shrub lands, urban areas, agricultural areas, grasslands, riparian zones and forests that are at the age of recovery from logging or fire.[19] It grows up to 3 m of height. In Himachal Pradesh, Uttar Pradesh, Uttarakhand, Maharashtra, Madhya Pradesh, and north-eastern States of India, this weed species has its wide dominance.[20] The life cycle of *L.camara* is shown in Figure. 2.



Fig.1. *Lantana camara.*

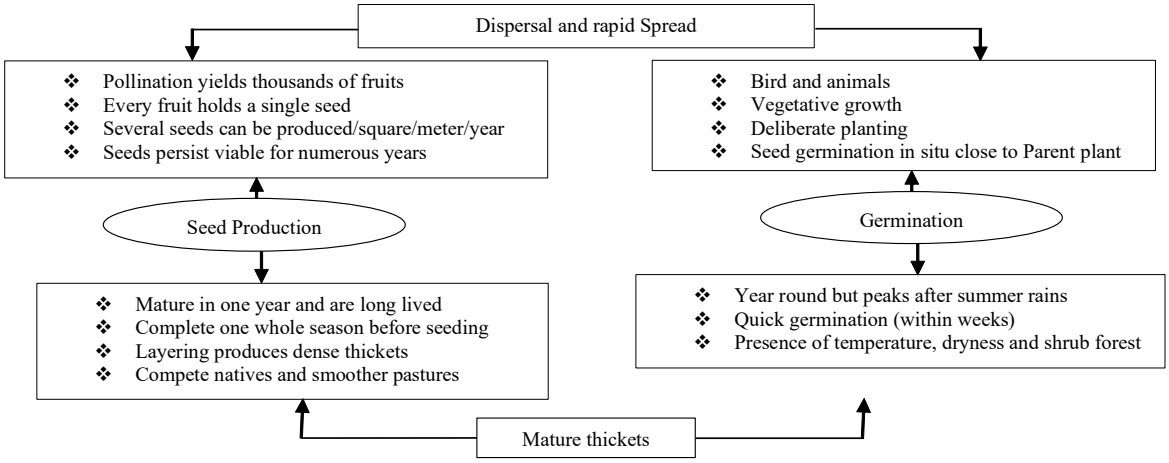


Fig. 2- Life cycle of *Lantana camara*

Pharmacology of *Lantana camara*

An amazing number of modern drugs have been isolated from natural sources.[19] The leaves' infusion of *Lantana camara* are utilized to treat gastrointestinal sicknesses, as emmenagogue, diuretic, expectorant and antirheumatic[21] to treat wounds[22] and is used as a tonic for stomach torments as it can be used as an insecticide.[23] *Lantana camara* plant has been accounted for to have various pharmacological properties as like antipyretic, antithrombin, hostile to inflammatory, antimicrobial, antimutagenic, antitumor, inhibitors of the enzymes acetylcholinesterase furthermore, antinoceptive.[21,24-27] *L.camara* has helpful potential due to different bioactive components, including steroids – lancamarone, triterpenoids, oligosaccharides, iridoid glycosides, naphthoquinones, and phenylpropanoid glycosides.[28]

The leaf extricates were found to have most of compound constituents including Triterpenoids, alkaloids, flavonoids, tannins, saponins, glycosides.[28-30] In stem and fruit- tannins, saponins, flavonoids, and terpenoids are reported.[30] Root of the plant has significant bioactive compound i.e., 'Oleanolic acid'.[31] From different parts of the *Lantana camara v*arious compounds are obtained which are presented in Table 1.[32]

Table 1- Parts of *Lantana camara* with their useful compounds

Sl. No.	Part	Compounds	Action
1.	Leaves and stem	Oleanonic acid	Anti-inflammatory
2.	Leaves	Lactones comprising euphanes	Anti-thrombin
3.	Aerial parts	Lantanoside and Camarinic acid	Nematicidal
4.	Roots, stem and leaves	Oleanonic acid	Antitumor, Antimicrobial, anti-inflammatory
5.	Leaves	Apigenin	Anti-proliferative
6.	Branches and leaves	Martynoside	Cardioactive
7.	Leaves	Camaraside	Anti-tumour

Toxicity

Lantana camara is among the top ten invasive weeds and noxious plant present on the earth, thus, this plant is considered as a poisonous plant.[19] *Lantana camara* L., is generally the weed of Central and South American origin and has a gigantic effect to the native composition of the terrestrial ecosystem.[20,33] In India some regions including pasture fallow land, farm, and forest elements, etc., has its unpredictable very random spread and existence.[34-36]

Its attack is involved in broad loss of local species variety by means of limitation, competition, recruitment, and alteration to the structure and function of ecosystem.[33,37,38] According to the World Conservation Union, after the habitat destruction invasive species are typically considered as the second utmost prominent threat to the biodiversity.[39,40] In their new ecosystems, these become predators, parasites, competitors, hybridizers, and causes disease to the native and domesticated plants and animals.[33,41] In its foliage Lantadenes are present which has harmful and toxic pentacyclic triterpenoids. Hepatotoxicity and photosensitivity are caused in some grazing animals such as goat, sheep, horses and bovines by Lantadene A and B.[19,42] If cattle feed these plant parts, then it can cause pink muzzle, nose disease and jaundice.[43]

***Lantana camara* as a papermaking material:**

Handmade paper making is the art of tradition that has been experienced from generations together by a specific class of society. The knowledge of this art transfers from generations to generations. These craft persons are generally known by the name of "Kagzi's".[44] The traditional forest based raw materials viz. wood and bamboo can be replaced by potential non wood sustainable raw material. A very few reports available in literature who have illustrated that *Lantana camara* has the potential to be used as a raw material in for making paper. Hence, for the sustainable management of *Lantana camara* it is essential to flourish a management framework keeping in purview advantages as well as limitations of various control strategies.[45]

From the studied sample, it was seen that the combination of *Lantana camara* and rice straw have yielded the pulp and paper of superior or greater quality.

Fibre characteristics of *Lantana camara* was done for analysis the pulp and paper making. The following derived values of fibre: coefficient flexibility, Runkel ratio and fineness ratio of the *Lantana* species was considered as very significant source for making paper, thus it can be used as a potential substitute in the papermaking industries especially for *Gmelina arborea* or other species.[4]

The results of proximate chemical analysis reflected the suitability of *L. camara* for paper making. Plant material of *L. camara* contains 66.06% holocellulose and 26.93% lignin. The pentosan was found to be 13.69% and solubility in hot water recorded as 7.25% and alcohol benzene ratio was 5.10%. For the assessment of soundness of wood for decay 1% NaOH solubility test is performed. In future for further researches the value that ranges in between 10-30% is usually considered as suitable. 1% NaOH solubility test recorded in this study was 18.75% and the ash content was 2.30%**.** It shows its suitability in the form of raw material for making the pulp and paper. The values of anatomical studies of *L. camara* observed 684 minimum, 1134.20 maximum, 912 as average. Fibre diameter contains 22, lumen diameter is 10. Wall thickness is 3.06, runkel ration has 0.43, and length/width has 0.68 and Shape factor is 0.36. This record shows that lantana has shorter fibre length. The smaller the fibres, the uniform and more perfect will be the paper or sheet formation.

Soda, soda sulphite, and Kraft pulping had been reported to be performed in stainless steel digester in an electrically heated along with regulator- controlled system. With the help of C-Ep-H-H sequence unbleached pulps are made bleached**.** Bleached pulp yield was recorded maximum (45.6%) at 12% chlorine demand with 81.6% brightness. In case of soda – sulphite and Kraft process bleached pulp yield was 44.5 and 46.2% with brightness 82.5% and 73.5%. The physical strength properties of bleached pulps express that initial freeness was maximum (26 OSR) in case of pulp recorded by the sulphite process while minimum (22 OSR) in case of pulp recorded with 18% NaOH. As compared to the soda and sulphite bleached pulp, bleached kraft pulp possess higher strength properties. Each pulp possessed sufficient strength properties and thus can be utilized for the production of varieties of paper and products. According to a report it has been observed that *Lantana camara* possess hollo-cellulose - 75.03%, Alcohol benzene Extractive- 8.461%, lignin- 18.21% and silica- 2.31% which shows the satisfactory potential as a raw material for making paper. However, there is a justified need to conduct a detailed study from the specific point of view in utilization of specific processing techniques in the field of handmade papermaking.

Through consideration of the above mentioned, the ongoing review is being proposed to evaluate the complete potential of the plant for making handmade paper sheets in ecofriendly manner so as develop a simple and cost-effective method of the sustainable management of the common weed plant and invasive plant species of *Lantana camara.*

CONCLUSION

The business of paper completely depends on the forest. Mill paper industry uses the forest- based woody raw material due to which not only the deforestation but also the pollution problem increases. Natural resources start to diminish gradually and its capacity comes closer to the limits of earth. The complete life cycle of mill made paper actually harms the environment continuously from beginning till end. It just starts from the cutting of tree and ends with the emission of toxic chemicals as like carbon dioxide, sulphur compounds and other harmful chemicals in the environment. Thus, mill sector makes the adverse effect on environment. For the production of pulp and paper it is required to discover the alternate technique and that can be the handmade paper.

Thus, this *Lantana camara* can be an option for the production of paper which will provide better opportunities for the management of weeds and by using these sources significant products can also be produced. One of the main advantages is that it is 100 % recycled and wood free. It is of potential economic value as it contains appreciable number of cellulosic fibres, have incredible strength properties. If such weeds cum agricultural wastes are managed in the proper manner, then it can benefit both farmers as well as the environment. Thus, only the utilization of these plant species can be the powerful strategy for dealing such weeds. Thus, environment can get rid of the adverse consequences of *Lantana camara* through this proper utilization approach and can also help in the nation's economic upliftment. Such weeds provide the opportunities to generate the employment along with the management of weed waste at the community level.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 171-176 **Plant Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Centella asiatica and *Oldenlandia corymbosa* as a rich source of bioactive compounds

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Received : 23rd December, 2022 ; Revised : 24th January, 2023

Abstract- *Centella asiatica* and *Oldenlandia corymbosa* are in traditional medicinal use for thousands of years. *Centella asiatica* is mainly used for wound healing and treating different skin condition whereas *Oldenlandia corymbosa* is used for the removing toxins as well as activating blood circulation etc. The present study was designed to identify different active compound present in both the plants and to look for their biological activity. GC-MS analysis was performed to identify different active compounds and their activity was identified using the previous literature.

Key words: ***Centella asiatica, Oldenlandia corymbosa,*** **GC-MS, bioactive compound, traditional medicine**

INTRODUCTION

North Eastern part of India is a biodiversity hotspot. The rich natural resources of this region have been used as indigenous traditional knowledge in the form of herbal medicines since ages.[1,2] The medicinal properties of *Centella asiatica* include its role as immune system enhancers, antioxidant molecule, antimicrobial agent[3], skincare formulations[4], an oral rinse[5], a lower limb ulcer treatment[6], a memory enhancer in Alzheimer's disease[7], and a topical hair composition[8] are well documented. A study from Malaysia has also demonstrated that the method of extraction of bioactive compounds plays an important role in detection and antimicrobial affect of the plant extract.[9] Two such plants studied in current research were *Centella asiatica* and *Oldenlandia corymbosa* whose medicinal properties have been studied. There are very few studies on documentation of bioactive compounds for these two studied plant from this part of the country. However, Different studies published worldwide have reported that as many as seventy compounds extracted from *C. asiatica*. Some of the most abundant bioactive compounds reported in *C. asiatica* were nmely asiatic, asiaticoside, madecassoside and madecassic acid from the triterpene class.[10-13] Compounds from the saponins group that could be found in a smaller quantity were brahmoside, centelloside, glycosides and alkaloids.[10,14]

The whole plant of *Oldenlandia corymbosa* is associated with compounds such as geniposide, 6 alpha-hydroxygeniposide, scandoside methyl ester (6 beta-hydroxygeniposide), asperulosidic acid, deacetylasperul oside, asperuloside, 10-O-benzoylscandoside methyl ester, 10-O-p-hydroxybenzoylscandoside methylester, (+)-lyoniresinol-3 alpha-O-beta-glucopyranoside, and rutin a study carried out from Thailand.[15] Another study from China has elucidated the antiproliferative properties of *Oldenlandia corymbosa* based on animal cell line study.[16]

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There are very scanty of reports on this particular plant species. However, other species of *Oldenlandia* has been studied for chemical composition[17] and presence of important peptides[18]. A detailed study on *O. corymbosa* pertaining to its antioxidant properties, presence of phytochemicals and bioactive compounds was carried out by Datta *et al.*, (2019)[19].

However, the key factor of the research work was to analyses the active compounds which known biological role from these plants.

METHODS

The dry samples of the studied plant extracted were sent for GC-MS analysis and the compounds present in the extracts were identified based on library research. Their 3D structures were extracted from Pub chem compound database.

RESULT

The results are tabulated below (table 1) reflecting the compounds chemical formulae, molecular weight and their beneficial role. In the study, Neophytadiene was detected in *Centella asiatica* which is known to have anti-inflammatory, antioxidant and cardio protective properties as described previously by an extensive study carried out by Bhardwaj *et al.*, (2020)[20] based on *in vivo* and coupled with *in vitro* study. The antimicrobial activity of the compound has been established by another study carried out by a Turkish study where a methanolic extract at a plant extract containing 70% of Neophytadien content (based on AC/MS data) was found to show effectiveness against certain studied microorganisms.[21] Further antioxidant and anti cancerous properties at the same molecule (isolated from *Menthe pulegium* L.) has been well documented to

Table 1- Showing the compounds chemical formulae, molecular weight and their beneficial role.

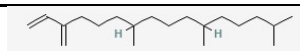
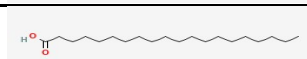
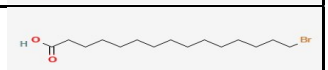
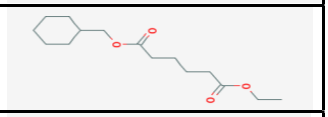
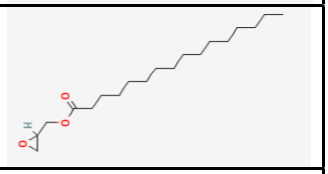
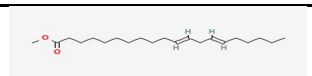
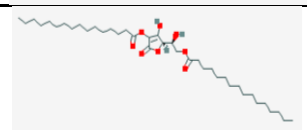
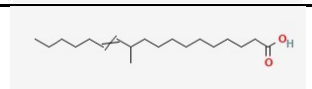
Sl. No.	Name of the compound	Molecular formula	Molecular weight (g/mol)	Peak area (%)	Structure
Plant name: Centella asiatica					
1.	NEOPHYTADIENE	$C_{20}H_{38}$	278	3.794	
2.	EICOSANOIC ACID	$C_{20}H_{40}O_2$	312.5	17.888	
3.	PENTADECANOIC ACID, 15-BROMO-	$C_{15}H_{29}BrO_2$	321.29	0.332	
4.	ADIPIC ACID, CYCLOHEXYLMETHYL ETHYL ESTER	$C_{15}H_{26O4}$	270.36	15.37	
5.	GLYCIDYL PALMITATE	$C_{19}H_{36}O_3$	312.5	2.762	
Plant name: Oldenlandia corymbosa					
1	11,14-EICOSADIENOIC ACID, METHYL ESTER	$C_{21}H_{38}O_2$	322.5	1.998	
2	L-(+)-ASCORBIC ACID 2,6-DIHEXADECANOATE	$C_{38}H_{68}O_8$	652.9	0.781	
3	GLYCIDYL OLEATE	$C_{21}H_{38}O_3$	338.5	1.914	
4	11-METHYLOCTADEC-12-ENOIC ACID	$C_{19}H_{36}O_2$	296.5	1.265	

exhibit anti-oxidant protection and free radical scavenging activity, antimicrobial activity, anti coagulating potential as well as anti cancerous affects.[22]

The next key compound found in the *Centella asiatica* extract was Eicosanoic acid and its anti inflammatory and anti therogenic properties have been represented by a previous study conduct in 2015.[23] The same compound has been also found to have surfactants properties as recorded by another study carried out in India.[24]

Glycidyl Palmitate (GP) in the current plant extract was found to be associated with anti-cancerous activity as demonstrated by a study carried out earlier.[25] However, a contrarian study was also carried out to reserve at using activated bleaching earth during the manufacturing of edible oils.[26]

Apoptosis inhibitory role of GP has been represented by few studies carried out earlier on indigenous medicinal plants.[27-29] Again, larvicidal, nematicidal and pesticidal properties at the same compound have been reported by a previous study conducted on Sharma *et al.* (2021)[29], which was an extracted AC-MS analysis of *Azadiracta indica* leaf extract. Compounds such as *Pentadecanoi caced*, 15- Is Bromo, adipic acid and cyclohexylme thyl ethyle ester were also observed in the studied extract which are not directly found to have biological activity, but might have synergistic effect along with other biological active compounds. The record plant extract evaluated was 11,14- Eicosadienoic acid, methyl-ester which has reported antimicrobial activity.[30]

The study by Suresh *et al.* (2014)[30], could see antimicrobial activity of fractions of fatty acid methyl ester against a range of clinically pathogenic bacteria's as well as few fungal species.

The second compound detected in the same sample was ascorbic acid 2,6 dihex- decanoate and is found to be associated with antioxidant, cardio protective properties as well as cancer preventive nature.[31-33] The same study also reported anti- infertility characters of the studied compound. The flavor and the presence of bioactive compounds are commercialized as food or herbal medicine looking into its capacity to fight against pathological conditions.[34] Transporters like properties along with anti-inflamatory, anti- scorbutic and wound healing properties of the compound has also been previously reported.[35] A range of health benefits ranging from anti-allergic, anti-anxiety, anti-cancers to neuro transmitter and neuroprotective properties are associated with the studied compound.[36-37]

Glycidyl Oleate, another compound isolated from *Oledenlandia corymbosa*, extract has been reported to have insectifugal properties.[38] The same compound is also recorded to have anti-arthritic as well as anti- inflammatory properties as studied earlier.[39]

Compounds like methyloctade C-12 enoic acid reported in the current study do not have any reported activity just like many other compounds (not shown in table) documented in the current study. The biological role of such compounds alone or their synergistic affect may elucidate further knowledge to their ever evolving field of traditional knowledge.

CONCLUSION

To conclude the current research findings, it will be at utmost importance to document the bioactive compounds of the traditional plants of Indian sub- continent as a whole and NE-India in particular. The study findings may be of interests for pharmaceutical companies as well as companies dealing with herbal medicine ecosystem.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 177-180

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Land use changes and their implications in Gir Sanctuary, Gujarat, India.

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Received : 7th January, 2023 ; Revised : 6th February, 2023

Abstract- The understanding of land use pattern is important for any area whether it be a protected area, urban area or even a small village. This study was initiated with the aim of studying the changes in the land use pattern of Gir Sanctuary of Gujarat from the year 2018 to 2021. The data analysis was performed with the help of QGIS software and the necessary data was downloaded with the help of semi-automatic classification plugin. This study revealed that out of the major six categories, two of them have shown noticeable changes. Over the course of four years, trees have increased from 3.1015% to 10.5578% and scrublands have decreased from 92.1266% to 84.8109%. This study concludes that as a protected area of Gujarat state, the increase in the tree diversity is a very good indicator for the conservation aspect of the flora of Gir Sanctuary.

Key words: Gir Sanctuary, Land Use Pattern, Semi-automatic classification, QGIS

INTRODUCTION

In the world, most of the parts have seen the changes in the land use.[1] Most of the time changes in the land use is associated with the people of a particular area.[1] Study of changes in land use is required because it provides a view on how resources are being used.[2] Land use changes help to understand how a particular ecosystem service is functioning.[3] One of the reason to study changes in the land use is because it provides an understanding that how people have changed the way they have been using the land over the years.[3] By studying the land use changes, it can help to prepare better strategies of sustainable development.[1] Land use changes also helps to predict the changes in a particular ecosystem.[3] Some of the negative changes related to land use for example: run-off, over-grazing, degradation of water quality can also be understood by studying the changes in the land use.[1]

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In India, Gir Sanctuary is known as "The Home of Asiatic Lion". It is distributed in 1412 km^2 of Gujarat state, covering three districts: Junagadh, Gi-Somnath and Amreli. It is also the home of nomadic tribes known as "Maldharies", as well as many floras and fauna. Most of the Gir Sanctuary Forest is dry deciduous. Because of its versatility in being distributed in three districts, different climatic conditions and weather makes it a rich region for studying the land use pattern.

MATERIALS & METHODS

This study was undertaken with the aim to use remote sensing with the help of computer software QGIS and also by the means of personal observation to classify the land use pattern of Gir Sanctuary from the year 2018 to 2021. Methods by which land use changes can be studied consists of remote sensing.[2] The necessary data was obtained by the use of QGIS software. All the processing related to studying the changes in land use was performed with the help of Semi-automatic classification plugin.[4] A total of

four maps consisting of land use changes pattern of Gir sanctuary in the years: 2018, 2019, 2020 and 2021 were created with the help of QGIS software.

RESULTS

This study reveals that the land use pattern of Gir Sanctuary can be categorized into six major categories: Water bodies, Trees, Grasses, cultivated land, Built up, Scrubland. In this study, it has been observed that in the course of four years: from 2018 to 2021, water bodies have increased from 0.4649% to 0.6917%, trees have increased from 3.1015% to 10.5578%, grasses have increased from 0.0002% to 0.0205%, built up has increased from 0.1599% to 0.1848%, cultivated land has decreased from 3.8531% to 3.7341%, and scrubland has decreased from 92.4149% to 84.8109%.

DISCUSSION

Some of the research has been done to study the changes in land use changes in the different parts of the world which includes Golpayegan city, Iran.[10] Mid hills, Himachal Pradesh, India.[5] Cameron Highlands, Malysia.[6] Kajiado Country, Kenya.[5] Gangetic Plains, Uttar Pradesh, India.8 Nainital district, India.[9]

Similarities have been observed with previous findings with respect to increase of grasses and built up.[5,7,9,10] Contradictions have been observed with the previous findings with respect to decrease in cultivated land.[5,10] Also changes from shrubland to trees to has been observed with previous research.[11]

CONCLUSION

Gir Sanctuary has a very diverse climate and because of this it harbors some of the very important plant and animal species of India. Because of the findings in our study, with respect to increase in water bodies, grasses and tress of the Gir Sanctuary, it shows a positive impact on the ecology of the area. It shows that the ecosystem is progressing in the right direction and this leads to sustainable ecosystem services. It is also a good indicator related to the conservation aspect of this protected area. The results which have been provided here shows the importance of studying the changes in land use pattern and also benefits of using remote sensing. Changes from trees to shrubland indicates formation of new patch of the woody species and it can lead to positive changes in new ecosystem properties and soil quality.

Table 1. Changes in the land use pattern of Gir Sanctuary: 2018 to 2021

Land use category	2018		2019		2020		2021		Changes from 2018 to 2021	
	km²	%	km²	%	km²	%	km²	%	km²	%
Water bodies	6.5643	0.4649	7.3621	0.5214	9.8444	0.6972	9.7668	0.6917	+3.2025	+0.2268
Trees	43.7931	3.1015	61.0167	4.3213	127.6038	9.0371	149.0761	10.5578	+105.2830	+7.4563
Grasses	0.0028	0.0002	0.0593	0.0042	0.3177	0.0225	0.2894	0.0205	+0.2866	+0.0203
Cultivated land	54.4057	3.8531	54.4961	3.8595	51.3600	3.6374	52.7254	3.7341	-1.6803	-0.1190
Built-up	2.2577	0.1599	2.3298	0.1650	2.5345	0.1795	2.6093	0.1848	+0.3516	+0.0249
Scrubland	1304.8983	92.4149	1286.7075	91.1266	1220.0597	86.4065	1197.5299	84.8109	-107.3684	-7.6040

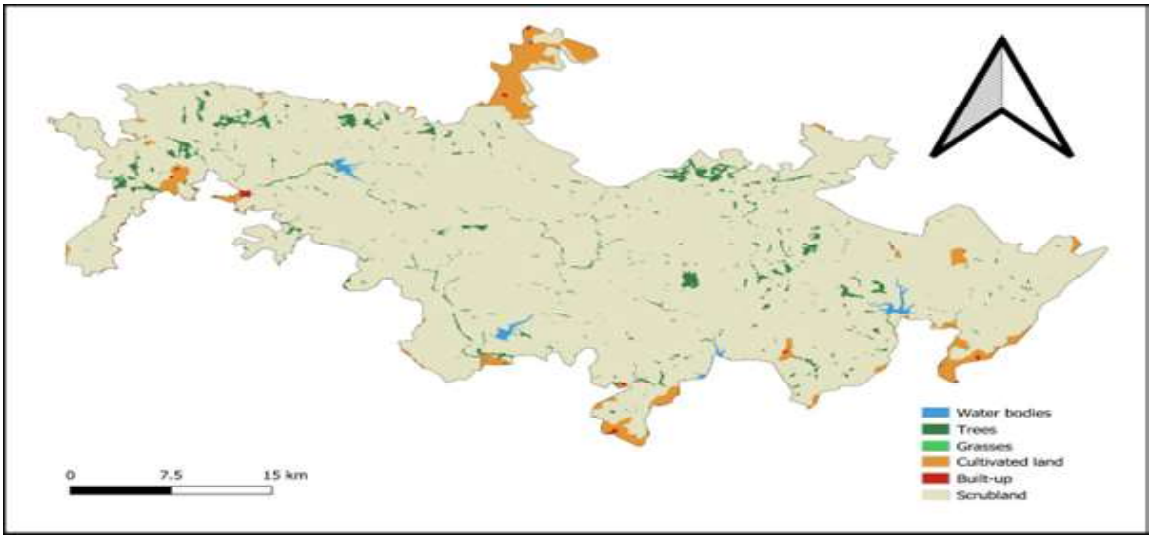


Figure 1. Land use pattern of Gir Sanctuary, 2018

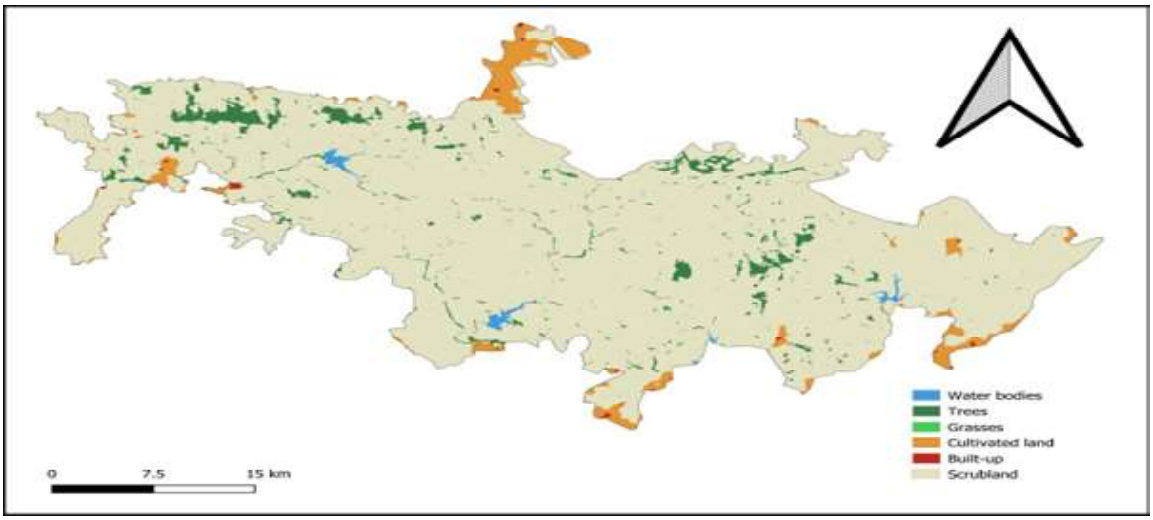


Figure 2. Land use pattern of Gir Sanctuary, 2019

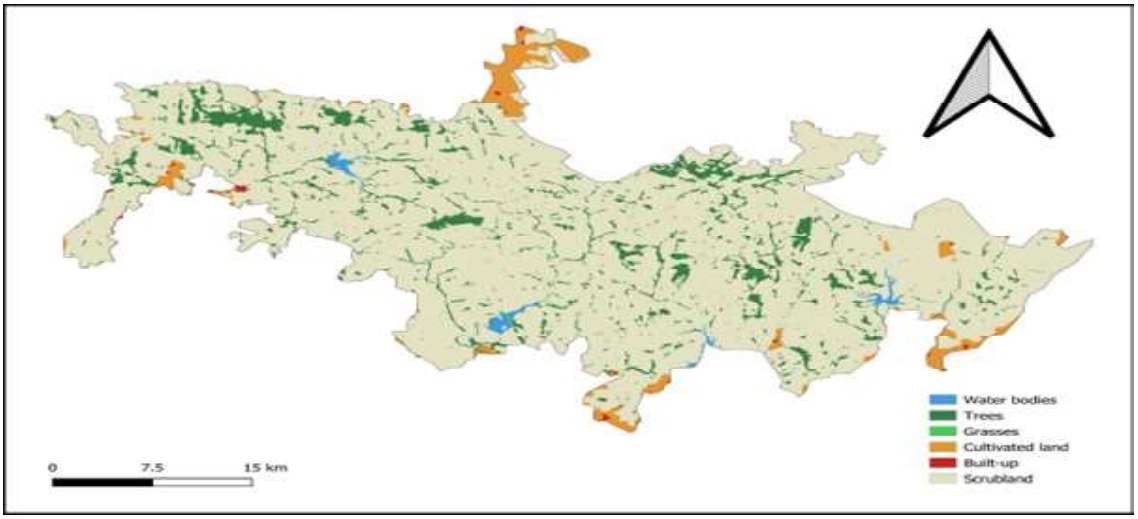


Figure 3. Land use pattern of Gir Sanctuary, 2020

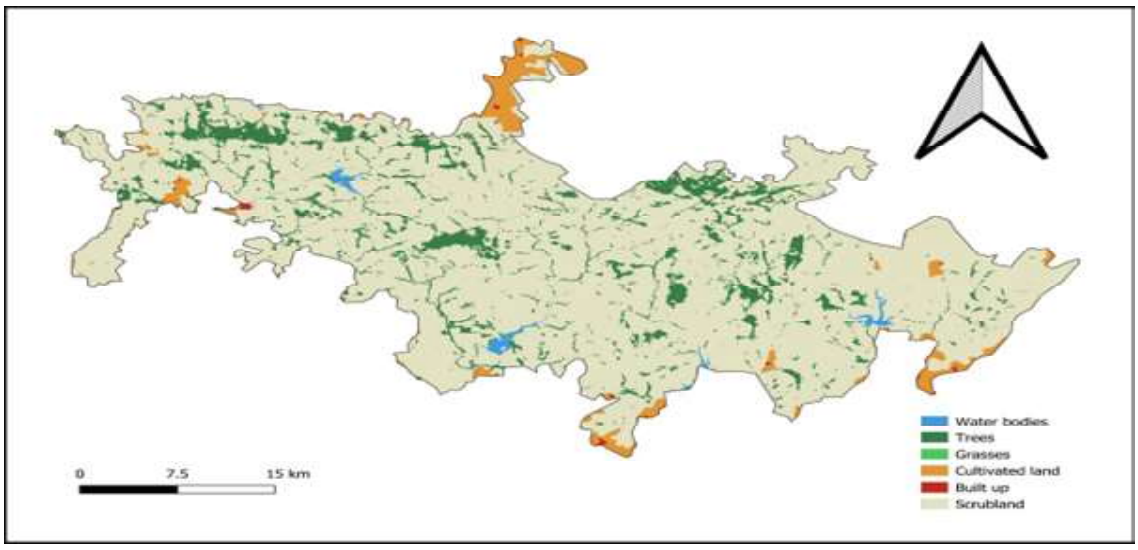


Figure 4. Land use pattern of Gir Sanctuary, 2021

ACKNOWLEDGEMENT

The authors are grateful to the entire staff of Department of Life Sciences, Bhakta Kavi Narsinh Mehta University and SHODH scholarship provided to the Ph.D. scholars by Government of Gujarat.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 181-184

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Physico-chemical studies of aquatic fern *Marsilea quadrifolia* in Ranchi District, Jharkhand

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Received : 27th December, 2022 ; Revised : 27th January, 2023

Abstract- The present investigation was carried out to study the physico-chemical parameters of the medicinally important aquatic fern *Marsilea quadrifolia*. It is known by Sushnisaag among the tribal of Jharkhand, West Bengal and Bangladesh. Tribal and local people use it as a food for more than 3000 years. Many research studies confirmed that this fern has anti-inflammatory, diuretic, febrifuge, depurative and refrigerant properties.

Key words: *Marsilea quadrifolia,* Fern, Extractive value, Solvents, Moisture Content.

INTRODUCTION

It is an aquatic fern belonging to the family Marsileaceae. It is also known by several names in several other states like Aaraikeerai in Tamil, Cauptiya in Hindi, Sunsuniya in Jharkhand and West Bengal, Neeraral in Malayalam, European Water clover in English. It is found throughout in India in marshy places and in rivers, canals and ponds[1] and it is widely distributed in temperate and tropical regions of the world. In submerged water it is rooted in the bottom of clayey soil.[2] In Red Data Book of the International union for conservation of Nature (IUCN)[3] *Marsilea quadrifolia* is enlisted. It bears four parted leaf resembling to four leaf clover so it is called as water clover. It has a long-stalked petiole with four clovers like lobes which are either held above the water or are submerged.

MATERIAL & METHODS

Marsilea quadrifolia was collected from the paddy fields of Piska Mode, Ranchi. The identification of this plant was done with the help of flora book "The Botany of Bihar and Orissa" Vol VI.[4] The whole plant was washed under running tap water to remove all the soil particles, dust from it, 3- times, then washed it with distilled water also. The plant was shade dried for 21 -24 days or dried it in an oven at 55°C-60°C, until it gets completely dried. Made a coarse powder by grinding it in mortar and pestle. Different solvents were used to made a plant extract.



Fig. 1. *Marsilea quadrifolia*

Physico-chemical parameters:-

a) Extractive value
b) Moisture content
c) Ash value
d) Foaming index

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a) Extractive value:- Powdered plant parts were soaked in five different solvents that is in benzene, chloroform, methanol, ethanol, distilled water. Covered all the beakers with Aluminum foil and left it for 24-48 hours. In five different weighed beakers the extract were filtered by using Whatman filter paper. The liquid extract was left to evaporate and after evaporation measured the weight of beakers.

$$\text{Extractive value} = \frac{\text{Extractive weight in solvent}}{\text{weight of drug material} \times 100}$$

b) Moisture content: - Heating the plant powder at 120°C in a hot air oven. The difference between the initial weight and final weight after drying is the moisture content.

Moisture content = Initial weight of plant powder − Final weight of plant the powder after drying.

c) Ash value: - In an earthen pot the plant powder was heated continuously for an hour on heater.

Ash value = Initial weight of the plant powder- Final weight.

Ash value (%) = weight of Ash/weight of leaf×100.

d) Foaming index: - In a ratio the decoction of plant material and the water was put in ten test tubes, now shake the test tubes. Measured with scale when the foam appeared.

RESULT & DISCUSSION

a. Extractive value

	Methanol	Ethanol	Chloroform	Benzene	Distilled water
Initial weight	47.458	50.787	49.130	53.487	42.466
Final weight	47.491	50.839	49.141	53.498	42.525
Difference	0.033	0.052	0.011	0.011	0.059
Percentage	3.3	5.2	1.1	1.1	5.9

b. Moisture content

Moisture % = 9.6%

c. Ash value

Ash value of 100gm

7.4%

d. Foaming index

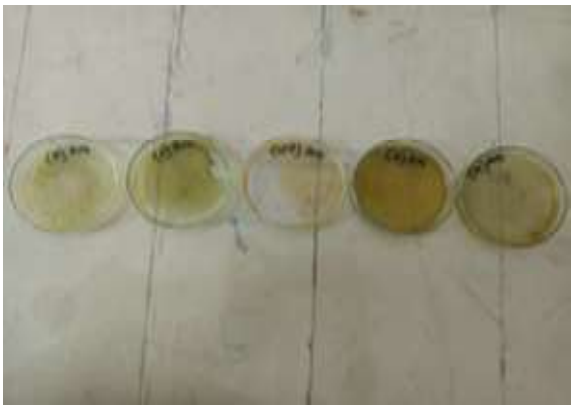
Height of the foam was measured less than 1cm in each test tube. So the foaming index was found to be less than 100.

After performing physico-chemical parameters it was found that distilled water soluble extractive value was found to be 5.9% which signifies that large amount of constituents was soluble in distilled water and less in benzene and chloroform that was 1.1% each. The moisture content was found to be 9.6%. The ash value was found to be 7.4%, a large value of ash indicates the presence of more inorganic matter. The foaming index was found to be less than 100.

In the present investigation it was found that the moisture content was found to be 9.6% which was greater than the results of Mousumi Das (2021) having moisture content 9.2% and Joy *et al.* (2022)[5] having moisture content 6.82%.

The total ash content obtained in this present investigation was found to be 7.4% which was smaller than the findings of Mousumi Das (2021) having total ash content 11.53% and Joy *et al.* (2022)[5] having total ash content 8%.

Parameters	Results
1) Extractive Values	
a. Methanol Soluble Extractive	3.3%
b. Ethanol Soluble Extractive	5.2%
c. Chloroform Soluble Extractive	1.1%
d. Benzene Soluble Extractive	1.1%
e. Distilled Water-Soluble Extractive	5.9%
2) Moisture Content	9.6%
3) Ash Value	7.4%
4) Foaming Index	Less than 100



Extractive value



Moisture content



Total ash content



Foaming index

CONCLUSION

The investigated plant is having many medicinal importance. Regular consumption of this saag relieves in sleep disorders, hypertension and headache. Local people and tribal eat this saag by cooking its petiole and leaves in oil with some salts and masalas. Juices made from leaves is used as a diuretic and febrifuge. It is used to treat snake bite and is applied on abscesses. The plant is diuretic, depurative anti-inflammatory, refrigerant. The plant is also used in eye diseases, cough, bronchitis, diabetes, diarrhoea and skin disease.

ACKNOWLEDGEMENT

I would like to express my gratitude to Head, University Department of Botany, Ranchi University, Ranchi, for the suggestions, constant encouragement and providing necessary facilities for completion of my work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 185-186 **Plant Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Standardization of quality control parameters of some ethnomedicinal plants

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Abstract- Medicinal plants have been used as traditional treatment for numerous human diseases since thousands of years. Medicinal properties of plants are due to the active chemical constituents present in different parts of plants. The present investigation was carried out to study one of the physicochemical parameters of some ethnomedicinal plants. Standardization of the plants were done with the help of extractive values of leaves. Water-soluble extractive value is higher in *Adhatoda vasica* and *Vitex negundo* which is 23.18 ± 1.83, 8.54 ± 2.0 respectively. In *Centella asiatica* methanol-soluble extractive value 17.70 ± 0.70 is higher. The result of the present study may be used for the identification and standardization of the plant material and phytochemical screening for determination of active constituents.

Key words: Ethnomedicinal, physicochemical, extractive values.

INTRODUCTION

Mankind has been using plants as therapeutic agents for thousands of years and continue to rely on them or health care needs. According to a WHO estimate, around 80% of the world's inhabitants depend on traditional medicine or their primary health care, majority of whom use plants or their active principles. Plants used in traditional medicine contain a wide range of ingredients that can be used to treat chronic as well as infectious diseases.[1]

Tribal's of Jharkhand are also utilizing a large number of plant species as herbal remedies in various disease and ailments. The present study has brought light on some interesting plants having the potential of ability to control ailments.

MATERIALS & METHODS

Plants were collected from local areas. The collected specimens were botanically identified with the help of flora Botany of Bihar and Orissa Vol.1, Vol. 2, Vol.3.

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The collected plant was washed with tap water. The plant was air dried thoroughly under shade at room temperature for 2 weeks to avoid direct loss of phytoconstituents from sunlight. The shade dried material was powdered and was stored in the air tight container and a portion of it was used for further analysis.

For the determination of extractive value accurately weighed 5.0 g of coarsely powdered air-dried material was placed in a glass-beaker and macerated with 50 ml of the solvent for 6 h, shaking frequently, and then allowed to stand for 24 to 96 h. the mixture was filtered rapidly in five different weight beakers, taking care not to lose any solvent. The liquid extracts were then left to evaporate. After evaporation the final weight of the beakers were taken. The difference in weight was calculated to determine the respective value.

$$\text{Extractive value \%} = \frac{\text{Extractive weight in solvent}}{\text{Weight of drug material}} x100$$

RESULTS & DISCUSSION

The study of extractive values can serve as a valuable source of information and provide suitable standards to

determine the quality of plant material in future investigations or application.

Extractive values of *Adhatoda vasica, Vitex negundo* and *Centella asiatica* leaves are depicted in Table 1, 2 and 3 respectively. Which shows highest 23.18 ± 1.83 water-soluble extractive value and lowest 3.64 ± 0.89 benzene-soluble extractive value of plant *Adhatoda vasica*, highest 8.54 ± 2.0 water-soluble extractive value and lowest 3.47 ± 1.71 benzene-soluble extractive value of plant Vitex negundo, for the plant *Centella asiatica* 17.70 ± 0.70 methanol-soluble extractive value is highest and benzene-soluble extractive value of 2.56 ± 0.98 is lowest. Higher water-soluble extractive value of *Adhatoda vasica* and *Vitex negundo* leaves implies that water is a better solvent of extraction, but in case of plant *Centella asiatica* methanol-soluble extractive value is higher than the water-soluble extractive indicating presence of more methanol soluble components in the drug.

In the present study, the criteria to choose suitable solvent depend upon the extractive potential of a solvent.[2] Estimation of extractive values indicates the amount of phytochemicals present when extracted with a particular solvent. The compositions of phytoconstituents vary with the type of solvent used.[3]

Table 1: Extractive values of *Adhatoda vasica* leaves

S.N.	Nature of Solvent	Colour	Extractive value (%)
1.	Ethanol	Green	11.74 ± 1.84
2.	Methanol	Brownish green	12.58 ± 1.24
3.	Acetone	Green	4.96 ± 0.28
4.	Benzene	Green	3.64 ± 0.89
5.	Aqueous	Brown	23.18 ± 1.83

Values are presented in average of three replicate ± standard deviation

Table 2: Extractive values of *Vitex negundo* leaves

S.N.	Nature of Solvent	Colour	Extractive value (%)
1.	Ethanol	Green	7.38 ± 0.98
2.	Methanol	Green	7.63 ± 2.02
3.	Acetone	Green	3.92 ± 1.61
4.	Benzene	Brown	3.47 ± 1.71
5.	Aqueous	Brownish yellow	8.54 ± 2.0

Values are presented in average of three replicate ± standard deviation

Table 3: Extractive values of *Centella asiatica* leaves

S.N.	Nature of Solvent	Colour	Extractive value (%)
1.	Ethanol	Green	12.23 ± 0.95
2.	Methanol	Green	17.70 ± 0.70
3.	Acetone	Green	3.4 ± 1.20
4.	Benzene	Brownish green	2.56 ± 0.98
5.	Aqueous	Yellowish	17.23 ± 2.09

Values are presented in average of three replicate ± standard deviation

CONCLUSION

In the present work, a medicinally useful plant in the Indian system of medicine were selected. Extractive values can be used as reliable aid for detecting adulteration. These simple but reliable standards will be useful to a person in using the drug as a home remedy. Also the manufacturers can utilize them for identification and selection of the raw material for drug production.

Crude drugs contain a number of constituents and these have a selective solubility in different solvents. Water, alcohol, alcohol/water mixtures, generally 45%, 60%, 90% ethanol, ether are used as solvents. Extractive values are primarily useful for the determination of exhausted or adulterated drugs. The extractive value of the crude drug determines the quality as well as purity of the drug. Water-soluble extractive value plays an important role in evaluation of crude drugs. Less extractive value indicates addition of exhausted material, adulteration or incorrect processing during drying or storage or formulating.

ACKNOWLEDGEMENT

The authors are thankful to the authorities of Department of Botany, St. Xavier's College Ranchi, for providing all the facilities and support.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 187-190

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Ecological study of sacred grove (Jaher) of Golpur village of Dumka district and its ethnobotanical significance

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Received : 29th December, 2022 ; Revised : 29th January, 2023

Abstract- The present study was conducted among Santals, a tribe mostly centered in Santal Pargana, one of the five administrative division of Jharkhand to document their traditional knowledge of various plants and their association with Sacred groves which are ecological conservation reserves. Sacred groves popularly known as Jaher or Jaherthan by Santals are the traditional communal sites of *in-situ* conservation. A significant change has been noticed regarding its conservation in recent years. Field observation, secondary data and several interviews have been conducted with the prominent village people like Pradhan, the village head and Naeke, the village priest who carries out all religious rituals, and other elderly knowledgeable tribals to gather information. The socio-cultural activities associated with the Sacred groves and various ethnobotanical uses of important plants used by Santal tribe have been discussed in this paper.

Key words: Sacred groves, Dumka, Santals, Santal Pargana

INTRODUCTION

Sacred groves are group of trees considered sacred for a particular community. They are the forest patches communally protected by local people due to their religious association with them. Such grove must have at least five Sal (*Shorea robusta*) trees since Santals consider them as their deities namely Sin' bonga, Maran Buru, Jaher Era (Lady of the grove), Moreko - Turuiko and others. Sin' bonga is the supreme deity. A significant change has been noticed regarding its conservation in recent years. Sacred groves are the natural gene pool preserver and example of habitat preservation through community participation.[1] The Santals have a culture of their own which they have preserved unchanged from time immemorial.[2] Worship of sacred groves is the traditional practice of various group of the society. It is mandatory to compile the ethnobotanical presently existing among the diverse community before its values are completely vanished.[3] There are many studies entitled to further quantify these ethics which leads to biodiversity conservation and sustainable ecosystem.[4] Sacred grove is rich heritage among the tribal communities which played significant role in religious and socio- cultural life among the local tribal people.[5] In recent years traditional ethnobotanical studies have received much attention due to their wide local acceptability and clues for new or less known medicinal plants.[6] Thus, there is now urgency for ethnobotanical research amongst aboriginal people.[7]

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MATERIALS & METHODS

The study area, Dumka is located in the center of Santal Pargana in the eastern part of Jharkhand. It lies between 24° 30' 00" N latitude and 87° 30' 00" E longitude with an average elevation of 137 meters (449 feet). It covers a total area of 12,601 km2.

The present study considers the ecological survey of vegetation inside the sacred grove of Golpur village of Dumka and the inter-relationship of indigenous people (Santals) with Sacred grove and its floral structure. Regular visits and surveys were made to collect plant specimens both in flowering and fruiting stages. Field observation on habit, habitat, medicinal and socio-cultural uses have been recorded in the field at the time of collection.

RESULTS & DISCUSSION

Topography:Rocky elevated land Date- 11.02.2021
Boundary: Present Human interference: Partial

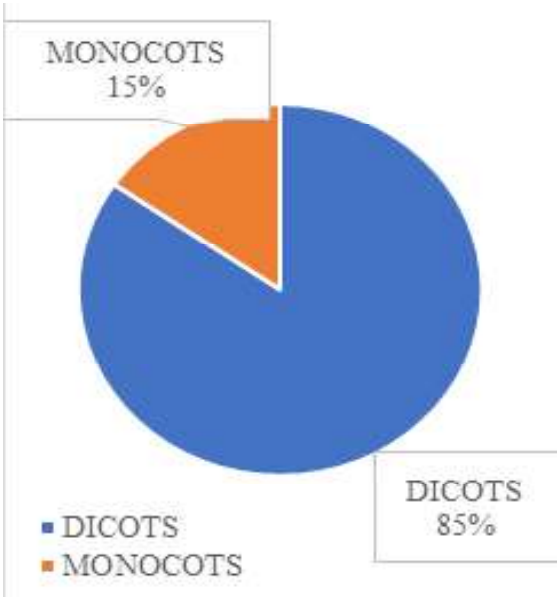
Table 1. Vegetation cover of Golpur SG

S.N.	Species name	Family	Local name	Hindi name	English name
1	*Shorea robusta*	Dipterocarpaceae	Sarjom	Sakhu, Sakhua	Sal
2	*Alangium salviifolium*	Cornaceae	Dhela	Dhera, Akol	Sage leaved alangium
3	*Phoenix dactylifera*	Arecaceae	Khijur	Khajur	Date palm
4	*Azadirachta indica*	Meliaceae	Neem	Neem	Margosa
5	*Ziziphus mauritiana*	Rhamnaceae	Janum	Ber	Indian jujube
6	*Annona squamosa*	Annonaceae	Mandargom	Sharifa	Custard apple
7	*Clerodendrum infortunatum*	Lamiaceae	Bharni	Bhant	Hill glory bower
8	*Lantana camara*	Verbenaceae	Putus	Raimuni	Sage
9	*Cynodon dactylon*	Poaceae	Dhubighaas	Dhoob	Bermuda grass
10	*Solanum virginianum*	Solanaceae	Ragaini	Bhatkataiya	Thorny nightshade
11	*Kyllinga bulbosa*	Cyperaceae	Mutha ghaas	Anuang, Nirbishi	White water sedge
12	*Dentella repens*	Rubiaceae	Kantha arak'	Cherumaneli	Creeping lickstoop
13	*Desmodium triflorum*	Fabaceae	Sukrichakap'	Kudaliya	Creeping Tick Trefoil
14	*Lathyrus hirsutus*	Fabaceae	Gaighura		Hairy vetchling

Table 2. Vegetation cover of Golpur SG

S. N.	Name of Species	No. of individuals in each quad.										Total no. of individuals of each Sp.(X)	Total no. of quad. of occurrences (Y)	Total no. of quad. studied(Z)	F%= (Y/Z * 100)	frequency class	Density(X/Z)	Abundance (X/Y)
		1	2	3	4	5	6	7	8	9	10							
	UPPER STOREY																	
1	***Shorea robusta***	7	6	3	7	6	7	6		8		50	8	10	80	C	5	6.3
	MIDDLE STOREY																	
2	***Azadirachta indica***				1			1				2	2	10	20	A	0.2	1.0
3	***Phoenix dactylifera***				1							1	1	10	10	A	0.1	1.0
4	***Ziziphus mauritiana***			2		3		3		2		10	4	10	40	B	1	2.5
5	***Alangium salviifolium***				1		2		1			4	3	10	30	B	0.4	1.3
6	***Annona squamosa***						2					2	1	10	10	A	0.2	2.0
	SHRUB LAYER																	
7	***Lantana camara***				1		2		1			4	3	10	30	B	0.4	1.3
8	***Clerodendrum infortunatum***		1	3		1		2		3		10	5	10	50	C	1	2.0
	GROUND FLORA																	
9	***Lathyrus hirsutus***			2	1	2	3	4				12	5	10	50	C	1.2	2.4
10	***Cynodon dactylon***					3	3		2		2	10	4	10	40	B	1	2.5
11	***Solanum virginianum***		1		1		4		2	2	5	15	6	10	60	A	1.5	2.5
12	***Kyllinga bulbosa***			1	2		1					4	3	10	30	B	0.4	1.3
13	***Dentella repens***				2		3		4	3	1	13	5	10	50	A	1.3	2.6
14	***Desmodium triflorum***			2		3		3		2		10	4	10	40	B	1	2.5

Fig 1. Representation of Dicot and Monocot families Golpur SG



Since the total number of families are 14, out of which 11 belong to dicot families and 3 belong to monocot families (Arecaceae, Cyperaceae, Poaceae), we get Fig.18.2.

Significance of Sacred groves: The concept of Sacred groves (Jaher) has been a part of rich traditions and diverse culture for many generations. Sometimes, they are also known as natural museums of giant trees, treasure houses of threatened species, dispensaries of medicinal plant, regulators of water sheds, recreation centers for urban life, veritable gardens for botanists, gene banks of economic species and recreation and spiritual retreat.

The ethnobotanical plants recorded in the Sacred groves of Santals primarily consisted of religious plants like *Shorea robusta* Gaertn. Tribals traditional belief system prohibits them to cut or pluck any plant parts before



Fig 1. Sacred grove of village Golpur



Fig 2. Sacred grove of Santals (Jaher)

time, thus promoting successful flowering process. These giant trees harbour great genetic diversity like the ectomycorrhizal fungi species. Medicinal and edible plant species are food not only for the indigenous people residing here but also for other organisms, thus sustaining a large number of species survival and helps in maintaining the food chain and ecological balance.

CONCLUSION

These Sacred groves (Jaher) are declining day by day. Some places where once these Sacred groves existed are now extinct. However, during this modernization and globalization, the recent past years has transformed and weakened religious, cultural and biological integrity. Changes in social belief, modernization and erosion of cultural practices are some of the major factors contributing towards degradation of the ancient heritage which need to be looked into for keeping up the ecological system and biodiversity conservation.

ACKNOWLEDGEMENT

We express our sincere gratitude towards our prominent tribals (Manjhi, Naeke) and native villagers who supported us in gathering ethnobotanical and socio-cultural information.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 191-192

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Persicaria limbata (Meisn.) H. Hara (Polygonaceae): An addition to flora of Allahabad District, Uttar Pradesh

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Received : 29th December, 2022 ; Revised : 28th January, 2023

Abstract- *Persicaria limbata* (Meisn.) H. Hara of Family Polygonaceae is a new addition to the flora of Allahabad, Prayagraj district of Uttar Pradesh. The present paper deals with its detailed description, updated citation, phenology, habit, distribution and photographs.

Key words: *Persicaria limbata* (Meisn.) H. Hara, Polygonaceae, Flora, Allahabad district, Uttar Pradesh.

INTRODUCTION

Family polygonaceae have been explored extensively in Uttar Pradesh. This family commonly known as knotweed family or smartweed-buckwheat family. The family comprises about 1200 species of 56 genera distributed globally. Flora of British India[1] published with 96 species of 7 genera. The flora of upper gangetic plain includes 19 species of 2 genera[2] and supplement to Duthie's flora of upper gangatic plane[3] has been published 9 species of single genera. In Family Polygonaceae in India[4] have revealed the occurrence of 121 species and 29 varieties belonging to 12 genera. Flora of Uttar Pradesh[5] is represented by 25 species of 6 genera and flora of Allahabad[6] by 6 species of 3 genera. The specimen has been collected from Iradatganj Ghurpur Prayagraj (U.P.). Prayagraj district is located between 24º472 & 25º472 N latitude and 81º09 & 82º212 E longitude. It is identified as *Persicaria limbata* (Meisn) H. Hara and found to be new record of Flora of Allahabad (Prayagraj) district U.P.

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MATERIALS & METHODS

Several field trips were organised for the study of family Polygonaceae in district Prayagraj of Uttar Pradesh. The specimens were identified with the help of various Flora[7] and the e-Flora of India. Localities and other necessary field information were recorded in field diary. Specimens are recorded with their name, place, date and made herbarium by standard methodology.[8] These Herbarium specimens were deposited at Duthie Herbarium, Department of Botany, University of Allahabad, Prayagraj.

Persicaria limbata (Meisn.) H. Hara, Fl. E. Himalaya: 71. 1966. *Polygonum limbatum* Meisn. in DC., Prodr. 14(1): 123. 1856; Hook. f., Fl. Brit. India 5:30.1886; Duthie, Fl. Upper Gangetic plain 3(1): 34. 1915.

An erect or procumbent annual herb, stems reddish-brown and green in young stage, up to 2 m or more long, sometimes rooting at the lower nodes, longitudinally striate, hollow in the lower part. Leaves 8–16 × 0.2–1.7 (3.5) cm, linear-lanceolate, acute and apiculate at the apex, narrowed to the base. Ochrea brownish, 1–1.5 cm long, glabrous, membranous, with a spreading or recurved, herbaceous, 1–7 mm long limb, ciliate at the apex and hairy outside. Flowers pedicellate, in fascicles of 2–3. Peduncles up to

8-12 cm long, hairy; bracts hairy or glabrous, with or without a ciliate margin. Perianth white-pink, pink or white, 4-5 mm long, 5-lobed, lobes 2.5-3 mm long. Stamens 7; filaments 2–2.5 mm long, filiform; anthers pink, 0.4 × 0.2 mm. Ovary 1.5 × 1.5 mm, compressed; styles 2, 1–2 mm long, united for half their length. Stigma bifid. Nut dark red-brown to black, shiny, smooth, 2.5–3 × 2.5–3 mm, convex, the persistent style base forming a beak.

Ecology: Growing in moist places, often in pond.

Flowering and fruiting: May to November Ghurpur, Prayagraj, Deepak Khare: 33204 (DUTHIE)

Type: Syntype from Senegal and Nile Delta

Distribution: Tropical Africa, South Africa, Egypt and India.

In India: Bihar, Karnataka, Uttarakhand, Uttar Pradesh, West Bengal, Madhya Pradesh.

Specimen examined: Jaunpur, N.P.Singh & C.L. Malhotra: 19504 (BSD); Kannauj, B.K. Sinha: 52145 (BSA); Lucknow, J.G. Srivastava: 46981(LWG); Saharanpur, Som Dev: 3048(BSD)

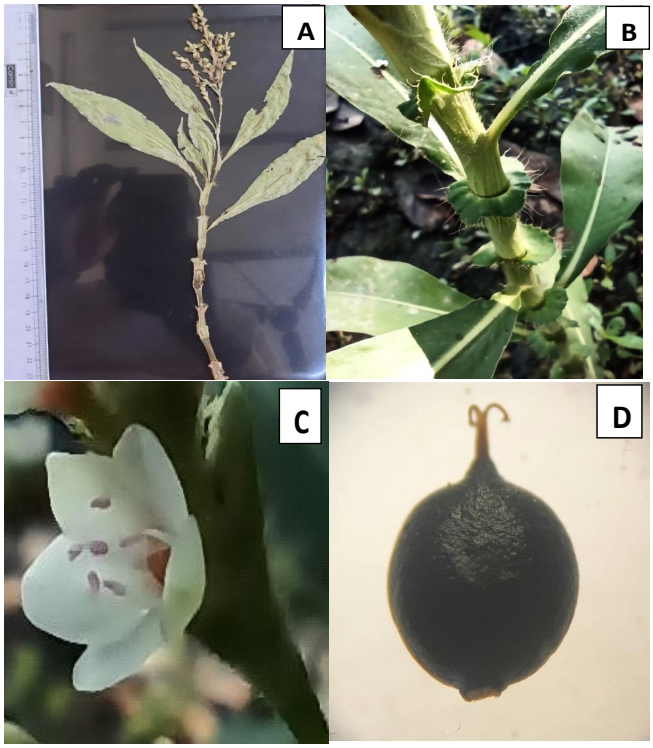


Plate: *Persicaria limbata* (Meisn.) H. Hara: A-Whole plant, B- Ochrea, C- Stamens, D-Nut with style and stigma

RESULT & DISCUSSION

During identification process recured scabrid ciliated limb are found which is characteristic feature of P*ersicaria limbata*. Biconvex nuts are found. This character is mentioned in Flora of Uttar Pradesh and Flora of Madjya Pradesh.

ACKNOWLEDGEMENT

The authors are thankful to the Head, Department of Botany, University of Allahabad, for facilities and University Grant Commission for financial assistance. The authors are also thankful to the Dr. Arti garg, Scientist-E, Botanical Survey of India, Central circle, Prayagraj (Allahabad) for providing access to library and herbarium facilities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 193-198

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Economic importance of Hydrophytes: A Review

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Received : 30th December, 2022 ; Revised : 30th January, 2023

Abstract- The present paper deals with role of water loving plants, i.e., Hydrophytes in different fields. As we know, most of the aquatic plants are considered as weeds, being troublesome both for aquatic habitat as well as mankind because the people are unaware about the great potential and economic importance of these plants. During the recent past, a number of instances and experiments have proved that these highly productive plants are generally more productive than conventional terrestrial crops. These plants can be used for making bio-gas, bio- fertilizers, fodder, as fish feed, and several plant-based industries. These plants are the source of a number of chemicals, which has immense medicinal value. Many of these plants are sold in the urban vegetable market for its important food values, as vegetable or as medicine. The present work is a holistic approach of aquatic plants with reference to their economic importance. The study was done in Pakur district of Jharkhand state covering some important water bodies and the plants were identified and collected as per BSI guidelines.

Key words: Hydrophytes, Aquatic, economic importance, Pakur.

INTRODUCTION

Hydrophytes refer to the aquatic plant bodies that live freely, entirely or partially submerged in different aqueous environments. They are the plant bodies that adapt themselves to survive in an aquatic ecosystem by modifying their leaves, stems and roots due to the modifications in their cellular components. The hydrophyte plants can live in diverse habitats like marine, lentic and lotic water sources. Hydrophyte is a term that has been originated from the Greek word, Hydro means water and phyton means plants.

There were many scientists whose contribution made it possible to study the nature, existence, adaptations and role of hydrophytic plants. Firstly, the hydrophyte plants were regarded as aquatic plants, growing in the water. The term hydrophyte was first given by the two scientists named Warming and Raunkiaer. Warming organized the plant species by the degree of soil wetness. Then, Weaver and Cements were the two scientists who classified hydrophyte plants into three major groups; floating, submerged and amphibious types.

Hydrophytes are those plants which live in water and adjust with their surroundings. They either remain fully submerged in the water like *Hydrilla, Vallisneria*, etc. or most of their body parts remain under the water like Trapa, lotus, etc. water lilies, sedges, crow foots are other important water plants. Many hydrophytes absorb water and gases over the whole surface and have no stomata like the spiked water milfoil (*Myriophyllum spicata*), which is completely submerged in water.

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Aquatic and marshy amphibious plants are still regarded by general mass of people as 'nuisance' because they are not yet aware of the great potential and economic value of these profusely growing uncontrollable plants. These are highly productive as compared to most terrestrial crops.

Hydrophytes play a significant role in the stabilization of shoreline, in the removal of nitrogen, phosphorus and other toxic water pollutants, weed control etc. They also function as the producers for aquatic organisms like fish, tadpole, insect larvae etc. It also traps sediments and limits erosion. As a matter of fact, these plants have been proved to be good source of livestock feed, human food, fish feed, biofertiliser, energy, fiber and paper. Furthermore, they have the capacity to purify the wastewater through the uptake of dissolved nutrients, including trace metals. Macrophytes play an important role in the ecological functioning of lentic ecosystems. They also play a key role in determining the structure and function of ecosystems by making available energy and nutrients for higher trophic levels. Aquatic macrophytes have played very important role in restoration of shallow eutrophic lakes. Therefore, the evaluation of the life cycle of macrophytes is a key issue before using them in restoration of waterbodies. Some sporadic mentions on the flora of the lentic waterbodies were made by earlier workers.[1-7]

MATERIALS & METHODS

The present work is based on collection of samples from various water bodies of Pakur district in the state of Jharkhand. Pakur district is situated in the north east corner of Jharkhand State. It is located at 23°40' to 25°18' latitude and 86°25' to 87°57' E. longitude, being part of Chotanagpur plateau. It is situated on the north-eastern corner of Jharkhand state, being dominated by tribal communities and aquatic habitats with fabulous treasure of aquatic and semi aquatic vegetation. The present work includes regular visit of the sampling sites for collection and identification of plants. Some of the local people and ethnic communities were also contacted to know how common mass of people use these plants in their daily life to arrange their bread and butter.

RESULT & DISCUSSION

Most of the aquatic plants are considered as weeds; the reason behind this is that the people are unaware about the great potential and economic importance of these plants. During the recent past, a number of instances and experiments have proved that these highly productive plants are generally more productive than conventional terrestrial crops. These plants can be used for making bio-gas, bio-fertilizers, fodder, as fish feed, and several plant-based industries. These plants are the source of a number of chemicals, which has immense medicinal value. A number of aquatic plants are used as food. During scarcity this group of plants is consumed by the poor people as staple food. Many of these plants are sold in the urban vegetable market for its important food values, as vegetable or as medicine.

Further research on economic importance of Aquatic plants will open new vistas to mankind for their foods and also for several articles of commerce. Economic importance of certain hydrophytes has been discussed as below.

FOOD VALUE

Some of the shoreline plants are good source of cattle feeds like Typha, some wetland grasses, even *Eichhornia crassipes* is consumed by the cattle, as these are the good source of protein supplement. Most of the water birds feed on these aquatic plants.

Aquatic plants act as the life line for most of the fish reared in the water bodies, especially the carps like silver carp, Milan carp, Chinese carp etc. in one way the aquatic plants serve as food for the fish and in other way, fish are good source of controlling weeds biologically. *Potamogeton, ceretophyllum, Utricularia, Eichhornia, Najas, Hydrilla and* other duck weed sps are good source of fish food.

A number of common aquatic plants are used as human food. *Trapa bispinosa,* Rhizomes of *Nelumbo nucifera* and *Nymphea* species called as Kamal Kakdi is a good source of carbohydrate used by tribals and rural people as main food. Rhizomes of *Scirpus grossus* is also used as food. Beside these, some of the aquatic plants like, *Alternanthera philoxeroides* (Mart.) Griseb., *Alternanthera sessilis*, *Bacopa monniere* (L) Wttst., *Centella asciatica* L., *Chenopodium album* L., *Enhydra fluctuans* L., *Ipomoea aquatica* Forsk., *Ludwigia perennis* Linn., *Hygrophila auriculata* (K. Schum) Heine., *Sagittaria sagittifolia* Linn. etc. are used as pot herbs and are very common in Indian kitchens.

SOURCE OF COMPOST AND FIBER

Luxuriant growth of plants under nutrient rich water bodies act as a good source of compost, for e.g., *Eichhornia, Pistia, Spirodella, Lemna,* and *Wolfia* are good source of compost and bio-fertilizer when composted in proper way.

In recent times *Eichhornia* fiber is being used to make folders, baskets, mats and other useful goods. Similar possibilities are there to use the fibers of *Nelumbo* and *Nymphaea.* There is need of further research in this aspect to explore more and more fiber yielding aquatic plants.

Medicinal importance of Aquatic Plants

The rural and tribal people still have faith on their traditional system of treatment. During the course of studies following plants have been reported to be used against different diseases. The brief ethnomedicinal descriptions of such plants have been mentioned below.

1. *Acorus calamus* Linn.

Fam:- ARACEAE

Local Name:-Ghorbaz

Used against cough cold & fever. Rhizomes after heating along with mustered oil used for body massage. Rhizome is used as stringent by the local athletes.

2. *Aeschynomene indica* Linn.

Fam - FABACEAE

Local Name – Phulan

The extract of the plant is used as spermicidal agents. Decoction of the leaves along with leaves of *Andrographis paniculata*, root of *Thespesia populnea* and stem bark of *Strichnos nuxvomica* are taken internally for 40 days to cure snakebite.

3. *Alternanthera sessilis* (L.) R.Br.

Fam:- AMARANTACEAE

Whole plant leaves and shoots are accredited with galactogogue; a good fodder increases the flow of milk in the cattle. Young shoots are nutritious, contain protein and iron.

4. *Ammania baccifera* Linn.

Fam- LYTHRACEAE

Local Name – Dad Mari

Bitter, appetizer, laxative, stomachic, aphrodisiac, removes blood troubles, strangury, Leaves are acrid, used for ring worm and other skin diseases. Herb is reported to posses' anti- tubercular properties.

5. *Anagallis arvensis* Linn.

Fam:- PRIMULACEAE

Local Name – Krishna neel

Whole plant is used in leprosy, dropsy, epilepsy, hydrophobia, fish poisoning and snake bite. It is diuretic, diaphoretic, expectorant and useful in rheumatism, cerebral, hepatic and renal complaints. The extract of plant is hypertensive, antiviral and diuretic inhibits herpes and poliomyelitis virus.

6. *Bacopa monniere* (L) Wttst.

Fam:- SCROPHULARIACEAE

Local Name: - Brahmi

Against nervous disorder, constipation, bronchitis in children. It is used against mental ailment. It is also used as brain Tonic.

7. *Blumea lacera* (Burm.F.) DC.

Fam:- Asteraceae

Local Name – Jangali muli

It is hot, pungent, bitter, anti- pyretic; cures bronchitis, blood diseases. Juices of leaves are used as an anthelmintic, febrifuge, astringent, and diuretic; it is given in bleeding piles. Root with black pepper is given in cholera. The root kept in the mouth is said to cure diseases of the month. Essential oil from Blumea has been shown analgesic, hypothermic and tranquilizing activities.

8.*Caesulia axaillaris* Roxb

Fam:- Asteraceae

Local Name – Gurguza

Plant is used in baldness. Bark of fresh herb is crushed and applied on wounds. The essential oil from plant is antimicrobial on human pathogenic bacteria and fungi.

9. *Centella asiatica* Linn

Fam:- APIACEAE

Local Name –Thankuni

Plants is acid, bitter; digestible, tonic, cooling, Laxative, alterative, alexiteric, antipyretic; improves appetite, voice, memory; cures Leucoderma, anemia, Urinary discharges, blood dieses, bronchitis' inflammations' Fever, Spleen enlargement, thirst, asthma., small pox, As a remedy for leprosy it has a considerable repute. Powder of dried leaves with milk is an alternative tonic and said to improve memory. The root contains the major portions of the active principle "vellarin".

10. *Chenopodium album* Linn

Fam- CHENOPODIACEAE

Local Name – Bhatua Sag

The Plant is considered as laxative and anthelminitic; Leaf powder is used externally as an antiseptic around genitalia of children. The infusion used for curing intestinal ulcer, and considered useful in piles, throat and eye troubles. Decoction is tonic, diuretic and aphrodisiac, given in biliousness, hepatic disorders and spleen enlargement. Flowers and buds are used in stomach troubles, Weakness in children. Seeds are used for abortion.

11. *Commelina benghalensis* Linn

Fam- COMMELINACEAE

Local Name – Kanteri

The drug is bitter, emollient, demulcent, refrigerant, and laxative, beneficial in leprosy. The plant is used in sores and snakebite.

12. *Commelina diffusa* Burm.F.

Fam- COMMELINACEAE

Local Name – Kansura

Whole plant crushed and applied on burns, itches and boils. Leaves are used for poulticing sores.

13. *Cyperus iria* Linn.

FAM- CYPERACEAE

Local Name – Galmotha

The Plant is used as astringent, stimulant, tonic and stomachic.

14. *Cyperus rotundus Linn*

FAM- CYPERACEAE

Local Name – Motha

The root is pungent, acrid, cooling, astringent, appetizer, stomachic, diuretic, emmenagogue, vulnerary; Useful in Leprosy, Thirst, Fever, dysentery, prurits, Pain, Vomiting, epilepsy, Opthalmia, erysipelas, dyspepsia and urinary contractions. The Whole plant along with young leaves of *Azadirachta indica* and black pepper with leaves of *Ocimum sanctum* is made into decoction and the vapour is used to cure malarial fever. The decoction is given three times a day for the same purpose. Tubers diuretic, Root paste is applied for healing wound and sores and also used in stomach complaints.

15. *Eclipta alba* (L.) Hassk

Fam: -ASTERACEAE

Local Name – Bhengaria

The juice is taken to cure diarrhoea, throat pain and to reduce fever. Root is tonic and alterative, emetic and purgative and also applied to the wounds of cattle. Juice of leaf is hepatic tonic. Source of a black stain, enters into preparations for darkening hair. Fresh plant is applied with sesame oil in elephantiasis, and juice in affections of liver and dropsy. Root is used as tonic in hepatic and spleen enlargements and in various chronic skin troubles; it is given internally in scalding of urine. It relieves headache when applied with oil. Farmers use the herbal juice for sickle wounds. The root is used as tonic for darkening the hair. The leaf juice is used for external application on cattle wounds to prevent pus formation and bacterial infection. Root is used in snakebite and scorpion string. Leaf paste is applied to cure white spot formed due to burning. The oil prepared from the herb is a great repute as hair dye and has cooling affect on the brain.

16. *Enhydra fluctuans* Lour

Fam:- ASTERACEAE

Local Name:-Muchri Sag

Consumed in the form of potherbs (Sag) as blood purifier and the leaf along with honey is consumed against anaemia.

17. *Heliotropium Indicum* Linn.

Fam- BORAGINACEAE

Local Name – Hanthi Sunrh

The Juice of the leaves is used as an application to wounds and roses, to boils and to the bites of scorpions and stings of insects. Decoction of leaves is used in urticaria and fever, and that of roots in charge and fever. Flower is considered as emmenogogue in small doses. Seeds masticated as stomachic.

18. *Hygrophila auriculata* (K.Schum) Heine

Fam:- ACANTHACEAE

Local Name: -Ramdana.

Plant is used against body swelling, itching and waist pain. Root used in hydrocele. The decoction of the plant is used against jaundice.

19. *Ipomea auqatica* Forsk.

Fam *:-* CONVOLVULACEAE

Local Name *:-* Karmi Sag.

It is given to nursing mother to promote lactation. It promotes sperm formation too. It is also used against cough, anemia and diarrhoea.

20. *Ludwigia perennis* Linn

Fam :- ONAGRACEAE

Local Name :- Prasuti ghas

Decoction of whole plant is given in Prasuti (Post natal) and in fever.

21. *Monochoria vaginalis* (Burms.F.) Pres

Fam- PONTEDERIACEAE

Local Name – Monochoria

Leaf juice is used in cough; root of the plant is useful in stomach and liver complaints. Root is also used in asthma and tooth ache.

22.***Nelumbo nucifera*** Gaertn,

Fam :- NYMPHACEAE

Local Name :- Kamal

Rhizomes in powdered form are used against. Menorrohoea & piles. Flowers are used against fever and liver diseases. Seeds are used against Skin diseases.

23. *Nymphaea nouchalli* Burm. f.

Fam*:* - NYMPHACEAE

Local Name*:* - Kumudni

It is used against cough, vomiting & leucorrhoea; It is also given to person suffering from premature ejaculation.

24. *Nymphoides indicum* (L.) O. Kuntze,

Fam : MENYANTHACEAE

It is given against fever and jaundice.

25. *Oldenlandia corymbosa* Linn.

Fam- RUBIACEAE

It is considered as stomachic and refrigerant; decoration is prescribed in remittent fever with gastric irritability and also used in jaundice and other liver trouble. Juice is given internally with little milk and sugar in the burning at the stomach pit and to cure heat eruptions.

26. *Oxalis corniculata* Linn.

Fam- OXALIDACEAE

Local Name – Tin Patia

Leaves have been used in fever, dysentery and scurvy.

27. *Polygonum glabrum* Willd.

Fam *:* POLYGONACEAE

Local Name *:* Dardmari

Leaves are used against Colic pain and Whole plant is used against itching.

28. *Portulaca oleracea* Linn.

Fam- PORTULACEAE

Local Name – Golgola Sag

Herb is refringent, vulnerary, antiscorbutic, aperient and diuretic; used in scurvy and dieses of liver, spleen, Kidney, and bladder; also employed in cardio- vascular disease, dysuria, haematuria, in dysentery, sore nipples and ulceration of herb is employed to stimulate gastic secretion.

29. *Rungia repens* Linn. Nees

Fam- ACANTHACEAE

Herb is given in cough, fever; as vermifuge and diuretic. The plant is also given in skin infections. Fresh leaves are bruised and mixed with castor oil and applied to scale in cases of *Tinea capitis*, a scaly fungal infection.

30. *Sphaerantus indicas* Linn.

Local Name – Gorakh Mundi

Fam- Asterceae

Hot laxative, digestible, tonic, Fattening, alternative, alexipharmic; used in insanity, tuberculous glands, indigestion, bronchitis, spleen disease, elephantiasis, anemia, pains in uterus and vagina, piles, strangury, epileptic convulsion, asthma, leucoderma, dysentery, vomiting, urinary discharges, rectal pain, Looseness of breasts, hemi crania.

31. *Spilanthes acmella* Murr.

Fam- ASTERACEAE

Local Name – Akakara

Leaves and flower heads are chewed to relieve toothache and infection of throat and gums. Leaves are rubbed on the skin to smothe the itching.

32. *Vallisnaria spiralis* Linn

Fam *:* HYDROCHARITACEAE

Local Name*:* Shenwar

It is used against diarrhoea, leucorrhoea and syphilis.

Other Uses

Treatment of waste water and Source of Energy

Some fast-growing aquatic plants absorbs heavy metal and purify water, e.g., *Eichhornia, Typha, Phragmites*, *Spirodella, Lemna,* and *wolfia* are successfully used to purify water. These plants are also the good source for Bio-Gas

CONCLUSION

On the basis of above discussion, it may be concluded that hydrophytes are so potent herbs having immense nutritional and medicinal values and useful in so many ways. There is need to cultivate these plants and proper marketing management, so that common mass of people will get employment and entire human population will be benefitted in addition to biodiversity conservation.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 199-202

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Taxonomy and economic importance of *Clerodendrum phlomidis* L.f. in Rajasthan

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Received : 29th December, 2022 ; Revised : 23rd January, 2023

Abstract- *Clerodendrum phlomidis* L.f. (Verbenaceae) is important desert plant. There are three species in the genus *Clerodendrum* namely *C. aculeatum* (L.) Griseb, *C. indicum* (L.) Kuntze and *C. phlomidis* L.f. *C. phlomidis* L.f. plays a beneficial role in the treatment of many ailments and in ecological as well as medicinal area. The plant is employed in ethnomedicine, veterinary care, fodder for pet animals, ecological significance as a soil binder, local fuel and the preservation of grains against worms and fungal diseases.

Key words: *Clerodendrum phlomidis*, Ethnomedicine, Rajasthan, Soil binder

INTRODUCTION

Taxonomy deals with nomenclature, classification and identification of plant. *Clerodendrum* belongs to family Verbenaceae under the Lamiales order. Verbenaceae is identified as an aromatic family with serrate and opposite leaves.

Clerodendrum phlomidis L.f. commonly known by Arna, Arni or Kakad. It is important shrub of desertic areas in Rajasthan. Species is commonly occurred in Sikar, Jhunjhunu, Churu, Barmer and Jodhpur. It is mainly used as fodder for camel, goat etc. and local fuel. Besides this *C. phlomidis* have medicinal properties like analgesic, anti-microbial etc. as well as some ecological significance like soil binder, local fuel and walls of "Jhopdas" (Huts) for aeration or retain moisture.

Rajasthan is located in the Northwest of India with a geographical area of 3,42,239 sq. km. Between 23°3' and 30°12' North latitude and 69°30' to 78°17' East longitude, the state expands. The Aravalli ranges stretch from North-East to South-West divides the state approximately into two major regions; one is the North-Western arid plain and the second is the Eastern semi-arid plain and the South-Eastern plateau regions, which are radically different in terms of geologic structure. Physiographically, the state is separated into 4 plains. The western desert: with barren hills, rocky plains and sandy plains, the Aravalli hills: stretch South-west to North-east starting from Gujarat and ending in Delhi, the eastern plains: with rich alluvial soils and the South-eastern plateau (Fig-1).

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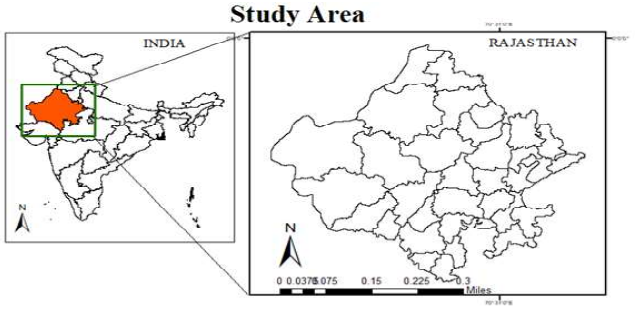


Fig. 1- Study Area

MATERIALS & METHODS

Extensive field surveys of the study area were conducted between 2019– 2022. Field work was carried out throughout the year. Prepare a list of plant species with date wise, occurring in different seasons especially in rainy and winter. Focus has been made on wild and cultivated area of Sikar, Churu and Jhunjhunu district. All the collected specimens were identified with the help of Flora of Rajasthan[1] and Flora of the Indian Desert[2]. Field notes eg. Date, Locality, Habit and Habitat of the plant have been recorded along with ecological as well Ethnoveterinary and Ethnomedicinal significance with the help of local people. The Specimens examined by the herbarium, Department of Botany, University of Rajasthan, Jaipur (India).

OBSERVATIONS

In Rajasthan, there are 3 species of *Clerodendrum* L., reported as a wild, namely *C. aculeatum* (L.) Griseb, *C. indicum* (L.) Kuntze and *C. phlomidis* L.f**.** Species are identified mainly by vegetative characters of stem and leaf. *C. phlomidis* is unique feature i.e. opposite leaf with corolla-tube (more than 2 cm long).

1’ Plants armed……*C. aculeatum*

1’ Plants unarmed:

2 Stem fistular. Leaves verticillate, sessile. Corolla-tube upto 1 cm long……..*C. indicum*

2’ Stem solid. Leaves opposite, petiolate. Corolla-tube more than 2 cm long…….*C. phlomidis*

Nomenclatural Status

Clerodendrum phlomidis L.f. Suppl. 292. 1781; Wight, Ic. 4(3): t. 1473. 1849; Clarke in Hook f. Fl. Brit. India 4: 590. 1885; Duthie, Fl. Gangetic Plain 2: 225. 1911. *Volkameria multiflora* Burm**.**f. Fl. Indica 137 1768. *Clerodendrum phlomidis* var. *rubrum* Roxb. ex Voigt Hort. Suburb. Calcutt. 465. 1845. *Clerodendrum multiflorum* (Burm.f.) Kuntze Revis. Gen. Pl. 2: 526. 1891**.** *Clerodendrum phlomidis* f. *rubrum* (Roxb. ex. Voigt) Moldenke Phytologia 22: 6. 1971.

Botanical Description

Plant erects with subscandent branches, 1-3 m tall. Stems and branches whitish pubescent. Leaves petiolate, 1-8 × 1-6 cm, rhomboid-ovate, acute apex, base truncate to acute, margin coarsely serrate or subentireor undulate, glabrescent. Flowers white or creamish-white or pinkish, dichasial cyme inflorescence.



Fig. 2- a. Plant Habit and **b.** Flowers

Calyx-lobes 1-2 cm long, acute, with few, sessile glands outside. Corolla-tube upto 3.5 cm long, much exserted; lobes 5, sub-equal, obtuse or acute. Drupes 0.5-1.25 cm long, obovoid, 4-lobed, glabrous, blackish-brown, with persistent calyx. Seeds 5-7 mm long, oblong dirty white (Fig- 2).

Flowering and fruiting period is August-March.

Distribution

C. phlomidis is distributed throughout the desertic habitat. In Sikar district the plant observed from agricultural field boundaries and sand dunes.

Economic importance

Fodder and local fuel

Plant leaves are used as a source of fodder for grazing pet animals. Fodder is especially used for goats and camels. People also use the plant roots and stem for local fuel.

Ecological uses

The plant important role plays in sand dune stabilizer. Soil erosion prevents by spreading roots and densely branching pattern used as a barrier for wind flow. Commonly found well grown in hedge along the agricultural field boundaries which protects farm crops from outer side grazing animals. It is used as a wall material in “Jhopdas” (Hut) for aeration.

Ethnoveterinary

The plant leaves used to control worms and digestive issues in cattle Leaves juice also used to kill lice on the body of domestic animals.

Ethnomedicinal uses

Plants roots used in curing various human diseases and health problems such as digestive disorders, piles, constipation, swelling of the body.

DISCUSSION

The genus *Clerodendrum* identified by Carl Linnaeus in the book Species Plantarum (1753). *Clerodendrum phlomidis* was first named by Carl Linnaeus in 1781. Wight (1849), Clarke (1885) and Duthie (1911) revealed similar opinion. Previously same plant described by Burman, Nicolaas Laurens under the name of *Volkameria multiflora* in 1768. *Clerodendrum phlomidis* f. *rubrum* name given by Moldenke in 1971.

C. phlomidis is observed in sandy to rocky wasteland habitats. It is found well grown in hedge along the boundaries of cultivated fields.[1] Verbenaceae is identified by- Plants aromatic, leaves opposite, serrate, stem often angular, non-glandular hairs if present unicellular, flowers zygomorphic, in racemes, spikes or heads, pollen exine thickened near apertures, style simple with bilobed stigma, stigmatic area conspicuously swollen and glandular, ovary with four ovules, ovules attached to the margin of false septa.[3] *Clerodendrum* possess simple leaf, cymose inflorescence, stamens didynamous, exserted and inserted with corolla-tube.[1]

Plant is distributed across India's arid regions.[4] The plant distribution in Rajasthan has already been documented in the following districts: Jaipur, Ajmer, Barmer, Banswara, Jhalawar, Jodhpur and Pali. According to our survey observation, the plant is present in some districts i.e. Sikar, Jhunjhunu, Churu, Barmer and Jodhpur.

C. phlomidis has achieved wide acceptance for its significance against many disorders. *C. phlomiodis* finds a lot of applications in Indian veterinary practices. The plant is used similarly by local people and traditional communities in other parts of the country. The traditional knowledge of the above plant species to cure a particular ailment of livestock is confined to primitive people and passes from one generation to another. The plant has great economic importance for various purposes including ethnomedicinal to treat various human diseases, ethnoveterinary use to treat animal diseases and pesticidal uses like fungicidal and nematicidal. The plant is used for cattle for diarrhoea and worms or when the stomach swells.[5] Leaves are used to kill lice on the body of domestic animals, Foot and mouth diseases and Skin problems.[6] Plants are used in curing many diseases and health problems such as digestive disorders, acidity, gas, diarrhoea, laxative, liver tonic and general health tonic. Roots are used for inflammation, swelling, jaundice, piles, constipation, slowness of digestion, swelling of the body, painful discharge of urine and urinary disease.[7] Leaves used for stomach pain, dyspepsia, digestive disorders, lung diseases, rheumatism, inflammatory diseases, swellings, locally tied for the treatment of guinea worms[8,9] and anti-obesity[10].

C. phlomidis is mentioned in the application of various tree disorders. The whole plant used as a pesticide. Leaf extracts are used for preserving grains[11], protecting from fungal infections[12] and controlling green worms[13]. Additionally, the plant is also used for local fuel needs and fodder primarily for camel and goats. It is best impacts on soil binder, huts wall formation for aeration. In present conditions the depletion of the plant population by the removal of farm boundaries and overgrazing by animals.

CONCLUSION

Family Verbenaceae is occupied by 10 genera in Rajasthan. The *Clerodendrum* L. has observed 3 species including *C. aculeatum* (L.) Griseb*, C. indicum* (L.) Kuntze and *C. phlomidis* L.f. *C. phlomidis* is desertic wild shrub. It is located in sand dunes and agricultural field boundaries. Plant has useful for ecological importance as well as medicinal properties.

ACKNOWLEDGEMENTS

We are thankful to UGC-CSIR, New Delhi for providing fellowship for research and Head, Department of Botany, University of Rajasthan, Jaipur for providing necessary facilities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 203-206

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Allelopathic effect of weed (*Phalaris minor* Retz.) extract on seed germination, growth & vigour of wheat (*Triticum aestivum* L.) var. "vidisha

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Received : 31st December, 2022 ; Revised :31st January, 2023

Abstract- Weeds are plants which grow elsewhere and are generally unwanted in the crop fields. In the wheat fields of Kosi region in Bihar, weeds are found growing luxuriantly and affect wheat crop in many ways. The present paper is based on experiment conducted to examine the Allelopathic effect of weed (*Phalaris minor* Retz.) extract on wheat (*Triticum aestivum* L.) var. "Vidisha". Various concentrations of weed (*Phalaris minor* Retz.) extracts i.e., 5%, 10%, 15%, 20% were prepared in distilled water and used for the treatment of seeds of wheat (*Triticum aestivum* L.) var. "Vidisha" in triplicate for seven days. The said concentrations were applied to examine per cent seed germination, growth and vigour of the seedlings. It was found that weed extracts reduced the percent seed germination over control (0%), while growth of the seedlings showed increased root and shoot lengths.

Key words: Weeds, Allelopathy, Allelochemicals, Seed Germination

INTRODUCTION

Weeds are plants which grow elsewhere and are unwanted in the crop fields. They interfere with seed germination, growth and yield of crop plants. Allelopathic effects of weeds may be inhibitory or stimulatory. Saraswat (1987)[1] reported beneficial effect on wheat growth. Earlier, it was reported that root and shoot length increased by the aquous extract of weed (*Phalaris minor* Retz.) extract. Bhatia *et al.* (1982)[2] also found benificial effect on the growth and vigour of wheat by the allelopathic effect of extract of *Phalaris minor* Retz.

Weeds are an important factor in the management of lands and water resources but their effect is greatest on agriculture.[3,4] In North Bihar, particularly in Kosi region wheat fields are infested with varieties of weed species which affect the productivity adversely. As a result, every year wheat yield decreases considerably. The loss in wheat productivity due to presence of weed species in the wheat field had estimated up to 25%.[5] Weeds not only compete with crop plants for water, sunlight and nutrients but also produce some chemicals called "Allelochemicals" and their effect is called as Allelopathy.[6] *Phalaris minor* Retz. is an annual grass weed which is most troublesome in the wheat fields in Kosi region of North Bihar.

Keeping in view of the above mention facts the present paper is based on the examination of allelopathic effect of various concentrations of weed (*Phalaris minor* Retz.) extracts on the seed germination, growth and vigour of the wheat (*Triticum aestivum* L.) var. "Vidisha"

MATERIAL & METHODS

Wheat seed var. "Vidisha" was procured from local seed seller and weed species (*Phalaris minor* Retz.) was

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collected from wheat field lying around B.N.M.U. North Campus, Madhepura.

Stem and leaves of weed (*Phalaris minor* Retz.) were dried in oven at 75°C for 24 hours. Stem and leaves of weed plants were separated and chopped into small pieces for the preparation of extracts. 10 g of these plant organs were crushed in grinder adding 100 ml distilled water. The mixture was kept for 72 hours at room temperature. Aqueous extracts thus obtained were filtered with Whatman filter paper no 1 and the filtrate was made up to 100 ml with addition of distilled water. In this way pure extract so obtained was stored in refrigerator. Different concentrations i.e., 5%, 10%, 15%, 20% were prepared from the pure extract by diluting it with distilled water as required. Before subjecting wheat seeds to the treatment with various concentrations of the extract solution, seeds were treated with 0.01% Hg_2C_{12} for few minutes. Seeds were then taken for each treatment in Petri dishes (10 cm diam.) with filter paper soaked with 10 ml of extract solution of different concentrations. For the control pure distilled water was used. Each treatment was carried out in triplicate. Seeds were allowed to germinate under room temperature (26°C±1°C) for seven days.

For the observation of per cent seed germination daily records of seed germination were maintained. For seedling establishment, 5 seedlings of seven days old were taken out from the Petri dishes from each concentration and planted in earthen pots containing soil and manure in the ratio 4:1. Pots were irrigated with water. Establishment of seedlings were recorded only after a fortnight. Growth and vigour of the seedlings were studied by measuring the linear length of the root and shoot after seven days of seed germination and seedling emergence. The data of various observations obtained were analyzed statistically according to Snedecor & Cochran (1967)[7].

OBSERVATION

Table 1- Effect of aqueous extracts of weed (*Phalaris minor* Retz.) on the seed germination, growth and vigour (In terms of length of root and shoot) of wheat (*Triticum aestivum* L.) var. "Vidisha"

Plant species	Concentration (%)	Germination (%)	Linear growth (cm)	
			Root	Shoot
Wheat (*Triticum aestivumL.*) Var.Vidisha	Control-0% (pure distilled water)	80±0.48	5.00±1.50	2.50±1.20
	5%	32.00±0.48	6.00±0.90	6.50±0.75
	10%	38.5±0.50	7.25±1.25	6.40±0.56
	15%	10.00±0.30	7.00±1.20	7.50±0.15
	20%	38.50±0.50	5.60±0.80	6.30±0.45

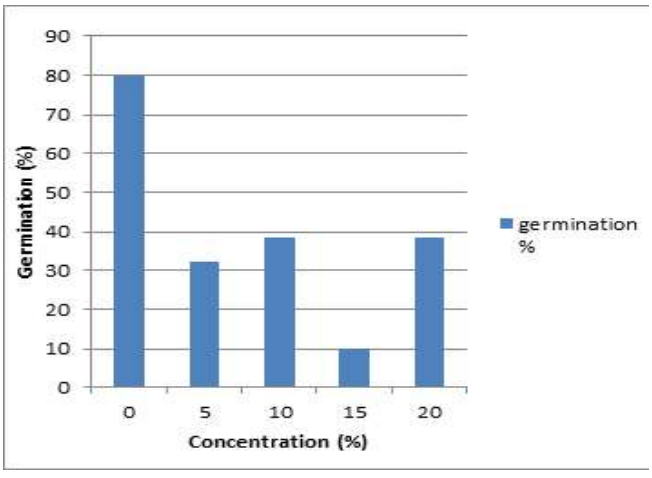


Fig. 1- Graphical representation of per cent seed germination of wheat (*T.aestivum* L.) in various concentrations of weed (*P.minor* Retz.) extracts.

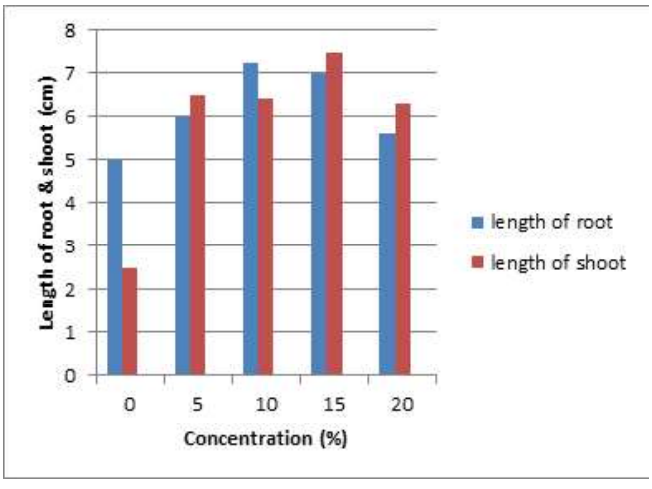


Fig. 2- Graphical representation of root and shoot length of wheat (*T.aestivum* L.) seedling in various concentrations of weed (*P.minor* Retz.)

RESULTS & DISCUSSION

The aqueous extracts of weed (*Phalaris minor* Retz.) showed inhibitory effect on wheat seed germination while, the effect on root and shoot length were found to be stimulatory.

The weed extracts inhibit per cent seed germination of wheat (*Triticum aestivum* L.) var. "Vidisha". The results obtained shown that maximum per cent of seed germination i.e., 80% was found in control (pure distilled water-0%) and the minimum per cent of seed germination was found in 15% aq. extract concentration corresponding to 10% germination. While the value of other treatments- 5%, 10%, 20% corresponded to 32%, 38.5%, 38.5% of seed germination respectively.

Regarding the growth and vigour of the seedlings, the linear lengths of root and shoot were found to be increased showing stimulatory effects in different concentrations of weed extract.

The linear length of root and shoot were maximum under 10% and 15% concentrations of weed extract corresponding to 7.25 and 7.50 cm respectively. The minimum length of root and shoot were recorded as 5.00 and 2.50 cm respectively in 0% concentration of control (pure water). The result obtained clearly indicated inhibitory effects on seed germination and stimulatory effect on growths and vigour of the seedling.

Wheat seed germination in various concentrations was inhibited. In the weed extract concentration of 15%, it was inhibited up to 90% germination, while in 5%, 10% and 20% seed germination inhibited upto 68%, 61.5%, 61.5% respectively.

These results are supported by previous works of Le taurneau *et al.* (1956)[8]. Gupta and Mittal (2012)[9] also found reduction in the per cent seed germination of wheat after treatment with leaf extract of *Phalaris minor* Retz.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 207-212

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Study of medicinal trees in Pandwa block of Palamu District

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Received : 2nd January, 2023 ; Revised : 2nd February, 2023

Abstract- Comprehensive study of medicinal trees has been taken from Pandwa block of district Palamu from where about 45 medicinal trees species which belonged to 24 families are recorded by help of vaidh, native peoples, knowledgeable persons and tribal men that are being used trees parts such as bark, leaves, fruits, stem etc, for the treatment of several diseases viz. malaria, dysentery, anemia, diabetes etc. Koraiya (*Wrightia tinctoria* Roxb.) species is found as a rare species belonged to Apocyanaceae family as well as Palas (*Butea monosperma* L.) is the dominating species that belonged to family Fabaceae and widely used in the treatment of various ailments. Almost 32 tribal communities are belonged to Palamu district as far as people of Pandwa block of the district Palamu are very conscious about treating properties of medicinal trees. Pandwa is prevalent as a major production of minerals and using of great significance of medicinal trees that are crucially helpful for diseases.

Key words: Medicinal trees, Biodiversity, Communities, Betla national park, Minerals

INTRODUCTION

Human beings have been contributed about authentic observation for medicinal trees and they are being contributed authentic observation using treatments of several ailments from immemorial time to modern time. Almost 17000 species of flowering plants have been reported in the country out of them 1000 species are medicinal plants, comprising 540 herbs, 100 shrubs, 200 trees and 160 climber species.[1] Palamu district is recognized as a rich biodiversity, major production of minerals and rich resources of medicinal trees as well as the identification of district has been done due to Betla National Park (BNP), Palamu fort, Bheem chulha and addressing the rich resources of medicinal trees. Palamu district is one of the famous districts of Jharkhand state that is situated in the western part of the state. The district is surrounded in the north by Bihar, in the south by Latehar, in the east by Chatra and in the west by Garhwa district. The state has 24 districts whereas Palamu is one out of the twenty-four districts of the state Jharkhand.[2] According to the census of 2011 the total population of Jharkhand is counted 32.9 million (http://jharkhand.gov.in)[3] which contributes 2.72% population of country with total 79,710 Sq.Km geographical area. The state is reported tribal communities like Santhal (35%), Oraon (18.14%), Munda (14.56%), Ho (9.23%), Khariya (2.14%), Asur and Baiga.[4] State has been covered 28.82% forest areas as well as country 21.05% of total areas of forest that shared 3.32% total forest areas of country. The average annual rainfall of the district is 1160mm and average annual rainfall of state is 140cm with tropical deciduous forest. The state is located at 651m average elevation of above the sea level.[5] Medicinal trees are being observed from immemorial to modern time by people of the world in the treatments of various diseases. Biodiversity

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is a very crucial portion of our day to day lives and liveliness that is comprised rich resources thereby families, community, nation and future generations depend.[6] The state Jharkhand is famous as the name of Chotanagpur plateau that is built by the north-eastern part of India.[7] The state lies between 22°00' and 24°01'N latitude and between 83°15' and 87°01E longitude while Palamu district is lies between 23°50' and 24°8'N latitude and between 83°55' and 84°30'E longitude.[8] The district is contained 13 blocks namely Pandwa, Patan, Pandu, Bishrampur, Satbarwa, Lesliganj, Chainpur, Chhatarpur, Hariharganj, Daltonganj, Manatu, Panki and Hussainabad.

MATERIAL & METHODS



Fig. 1- Map representing Pandwa block of Palamu district

Study Area

Pandwa is one of the blocks of district Palamu that is widely bounded by the several number of medicinal trees. It has been famous for Rajhara coal mines and Kathautiya coal mines including Sadabah and Gurgawati river. Pandwa block has 34 villages, 8 panchayat that is covered with the large number of medicinal trees. According to the census 2011 the total population of block is 46,957 out of which 24,330 are males whereas 22,627 are females. The literacy rate is about 65.56% with the male literacy rate is 64.9% and female rate is 44.17%. Pandwa block containing several medicinal trees with major remedies of various diseases thereby medicinal trees have been assumed invaluable by vaidyas, tribal men and people of this block.



Fig. 2- Map representing Jharkhand State

Data Collection

This study was conducted during 2021-2022 from the Pandwa block that is located in the western region of Jharkhand. The data were collected from the various villages of Pandwa block namely Kajri, Murma, Rajhara Colliery, Pandwa and Garigaon with the help of vaidyas, tribal men, local people, informants, taxonomical experts, knowledgeable persons and forest dwellers men through personal interview, personal discussion and questionnaires.

Description of Trees

***Aegle mermelos* Corr. (Bel)**

Significance of these trees are in the remedy of Dysentery: Wet bark (5-30g) is extracted from stem, prepared juice by cloth filteration and provided once time in the morning with a spoonful[9], Sunstroke, Diarrhoea and Constipation: From rippen fruits are removed upper layer and seed, extract furthermore fruits pulp containing (10-15ml) water and conferred to patients with five spoonful daily in the morning.

***Ailanthus excelsa* Roxb. (Ghorkarn)**

It is used in the treatments of Fever, Menstruation and Skin diseases: Bark or Root (5-25g) is crushed into powder and given to patients orally with some water, Wound: Dried seed (5-10g) is removed upper scale and powdered into fine powder and imposed on the wound.

***Azadirachta indica* A. (Neem)**

It is used in the treatments of Jaundis and Fever: Leaves (5-10g) are ground, extracted juice and taken two times daily in a day until well, Heat rash & Pimple: Wet bark (10-20g) is boiled with some water obtained juice by

Table 1- List of medicinal trees observed in Pandwa block of Palamu District, Jharkhand, India.

S.No	Botanical Names	Common Names	Family	Part uses
1	*Acacia arabica* Wild.	Babul	Mimosaceae	Bark, Seed, Gum, Leaves
2	*Acacia catechu* Wild.	Khair	Fabacea	Bark, Resin, Leaves
3	*Acacia nilotica* Wild.	Babul	Fabacea	Stem, Leaves pod
4	*Aegle marmelos* Corr.	Bel	Rutaceae	Fruit, Bark
5	*Ailanthus excelsa* Roxb	Ghorkarn	Simaroubaceae	Bark, Root, Leaves
6	*Albizzia lebbek* Benth.	Siris	Fabaceae	Bark
7	*Alstonia scholaris* Linn.	Devil tree	Apocynaceae	Bark, Leaves, Root
8	*Annona squamosa* Linn.	Sarifa	Annonaceae	Bark, Leaves, fruits, Root, Seed
9	*Artocarpus heterophyllus* Lam.	Kathal	Moraceae	Fruit, Seed
10	*Azadirachta indica* A.	Neem	Meliaceae	Root, Stem, Leaves, Seed
11	*Butea monosperma* Linn.	Palas	Fabaceae	Whole part
12	*Borassus flabelliformis* Roxb.	Tad	Arecaceae	Stem, Fruit
13	*Bombax ceiba* Linn.	Semal	Malvaceae	Bark
14	*Cassia fistula* Linn.	Amaltaash	Leguminosae	Bark, Leaves, Root
15	*Cordia dichotoma* G.Forst.	Lasoda	Boraginaceae	Fruits, Bark, Leaves
16	*Dalbergia sissoo* Roxb.	Seesam	Fabaceae	Stem, Leaves pod
17	*Eucalyptus globulus* Labill.	Eucalyptus	Myrtaceae	Leaves
18	*Ficus benghalensis* Linn.	Bargad	Moraceae	Whole part
19	*Ficus glomerata* Roxb.	Gular	Moraceae	Whole part
20	*Ficus religiosa* Linn.	Peepal	Moraceae	Leaves, bark
21	*Gmelina arborea* Linn.	Gamhar	Lamiaceae	Stem, Leaves
22	*Holoptelea integrifolia planch.*	Chilbil	Ulmaceae	Whole part
23	*Madhuca longifolia* J.F macbr.	Mahua	Sapotaceae	Fruit, Bark, Seed, Leaves
24	*Mangifera indica* Linn.	Aam	Anarcardiaceae	Bark, Fruit
25	*Melia azedarach* Linn.	Bakain	Meliaceae	Bark, Leaves
26	*Moringa oleifera* Linn.	Sahjan	Moringaceae	Whole part
27	*Neolamarckia cadamb* Roxb.	Kadam	Rubiaceae	Bark, Fruit, Leaves
28	*Phoenix sylvestris* Roxb.	Khajur	Arecaceae	Stem,Fruit, Root
29	*Phyllanthus emblica* Linn.	Awla	Phyllanthaceae	Fruit, Seed
30	*Pongamia pinnata* Linn.	Karanj	Leguminosae	Bark, Seed
31	*Psidium guajava* Linn.	Amrud	Myrtaceae	Fruit, Leaves, Bark
32	*Punica granatum* Linn.	Anar	Lythraceae	Bark, Leaves, Fruit
33	*Sapindus saponaria* Linn.	Ritha	Sapindaceae	Fruit
34	*Saraca asoca* Roxb.	Ashoka	Fabaceae	Bark
35	*Shorea robusta* Roth.	Saal	Dipterocarpeae	Bark, Fruit, Leaves, Seed, Gum
36	*Spondias mangifera*	Aamra	Anarcardiaceae	Fruit
37	*Syzygium cumini* Linn.	Jamun	Myrtaceae	Seed, Fruit, Bark, Leaves
38	*Tamarindus indica* Linn.	Imli	Fabaceae	Fruit, Bark, Seed
39	*Tectona grandis* L.f	Teak	Lamiaceae	Bark, Leaves, Flower, Seed
40	*Terminalia arjuna* Roxb.	Arjun	Combretaceae	Bark
41	*Terminalia bellirica* Roxb.	Bahera	Combretaceae	Fruit, Seed
42	*Terminalia chebula* Retz.	Hrar	Combretaceae	Leaves, Fruit, Seed
43	*Terminalia elliptica* Wild.	Asan	Combretaceae	Bark
44	*Wrightia tinctoria* Roxb.	Koraiya	Apocynaceae	Bark, Seed
45	*Ziziphus mauritima* Lam.	Bair	Rhamnaceae	Fruit

cloth filteration and administered orally daily in the morning along with half glass of water, Wound: Dried fruits or seeds (30-50g) are crushed, extracted oil and interpolate on the wound. waste particles are used in the manufacturing of Neem soap, Toothach: Stem is used in the form of tooth brush.[10].

***Bombax ceiba* L. (Semal)**

It is utilized in the remedies of Stomach ache & Dhat syndrome: Dried bark (10-20g) is ground in the form of powder, added 5 ml water and provided to women twice daily in a day with a table spoon, Dysentery: Stem scale (10-20g) is removed from dwarf small semal's tree and

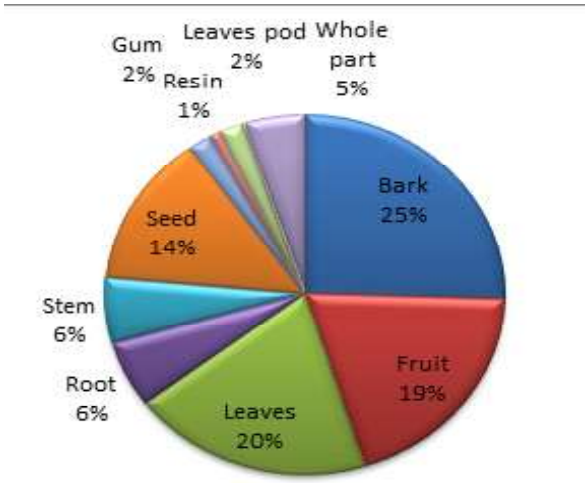


Fig. 3- Showing percentage wise analysis of tree's part

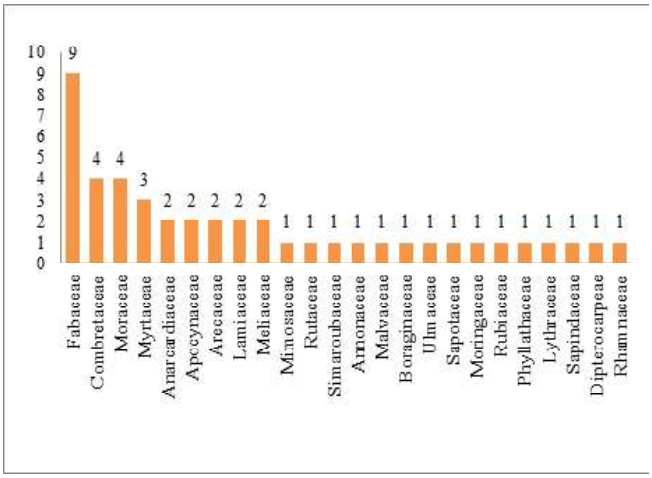


Fig. 4- Graph representing number of genera on the basis of their respective families.

get stem that is ground it into a paste and conferred to patients daily along with a glass of water until well.

***Butea monosperma* L. (Palas)**

It is used in the treatment of Dysentery: Dried seed (10-30g) is crushed, powdered and taken with a glass of water by adding a little amount of powder in the morning, Snakebite: Dried root is powdered and given with a few by adding some water.[11] Toothache: Stem is used in the form of toothbrush.

***Eucalyptus globulus* Labill. (Eucalyptus)**

It is used in the treatments of cough: Raw leaves (5-10g) intermingled with a 1 or 2L water further boil and inhaled vapour to patients, Head pain: Green leaves (10-20g) are added a little water and extracted paste by crushing, put on the head in form of massage.

***Mangifera indica* L. (Aam)**

Using of these trees are crucial role in the treatments of several diseases such as Jaundice: Raw bark (10-30g) is crushed into paste and provided to patients with a little water, Anemia: One ripen fruit is taken daily along with a glass of milk till well, Sunstrock & Stomach heat: From the raw fruits (100-200g) seeds are removed, extracted scale afterward dried and ground into fine powder subsequently imparting orally with some water in the morning, Scorpion & Snake bite: Dried inflorescence (10-20g) is ground and obtained powder , adding some water given to patient.[12]

***Moringa oleifera* L. (Sahjan)**

It is used in the remedies of Diabetes & Blood pressure: Dried bark or seed (10-20g) is ground into fine particles given to patients once times daily along with a glass of water until well, Boost immunity: Fruits & Leaves are taken in the form of vegetable, Hypotension: Ground wet leaves (10-25g) and filterated by cloth filteration method as well as given to patient daily twice in a day along with one glass of water.[13]

***Phyllanthus emblica* L. (Awla)**

It is used in the treatments of Anemia: Juice is extracted from (10-20g) of fruits and taken empty stomach till 7 days in the morning as well as its constituents are used as "Triphala in Ayurveda".[14] Gastric & Acidity: A fruit is taken daily in the morning. *Phyllanthus emblica* L. is being used in the form of great medicinal potential, Head cold: Oil is extracted from the dried seed (10-50g) and confered in the form of massage.

***Pongamia pinnata* L. (Karanch)**

Usage of these trees are several remedies of Skin diseases & Itching: Juice extracted from bark (5-10g) and applied on infected part daily until well. Wound: Oil is obtained by crushing of (5-50g) dried seed afterward put on the wound,[15] Tooth pain & Tooth decay: Wet stem is taken as a tooth brush.

***Psidium guajava* L. (Amrud)**

It is imparted in the treatments of Dysentery: New raw leaves (5-20g) are crushed and extracted juice by cloth filteration method and provided to the patients daily twice in a day, Jaundice & Stomach ache: One fruit is taken each day in the morning.

***Shorea robusta* Gaerin. (Sakhua)**

It is used in the remedies of Stomach ache: Dried bark (10-30g) is powdered, added with a little water and

made round pellets that given daily two pills to patients in the morning, Loose motion: Raw fruits (2-20g) is crushed into the paste and taken to patients along with a glass of water till well, Diarrhoea: Dried gum (5-10g) is ground into fine powder and mixed with some pure curd that imparted to patients daily in the morning.

***Syzygium cumini* L (Jamun)**

It is used in the ailments of Diabetes: Dried seed (5-10g) is ground in the grinding machine into powder and conferred to patients orally in a little quantity along with some water, Gastric: Rippen fruits are taken once time in daily morning, Dysentery: Juice is extracted from (10-20g) of raw bark and imparted two times daily with a spoonful in the morning, Diabetes and Dysentery: Powder is prepared with the help of (5-10g) dried bark by grinding machine and provided to patients daily along with one glass of water.[16]

***Terminalia bellirica* Roxb. (Bahera)**

Using of these trees are in the treatments of Respiratory problem, Cough & Sore throat: Extracts powder with (10-100g) upper scale of dried fruits by crushing and intermingled a little amount of water to provide of patients daily in the morning with five tablespoon until well, Immune system:obtained oil from (10-20g) of dried seed and put on in the form of massage.

***Mauritima ziziphus* Lam. (Bair)**

It is used in the form the treatment of stomach ache & Jaundice: Removed seed from (5-50g) ripen fruits, dried uppermost part and ground in to fine particles further added (5ml) water that gets round-tablet to imparted to patients daily in the morning.



***Aegle marmelos* Cor.,**



***Psidium guajava* L.,**



***Syzygium cumini* L.,**



***Mangifera indica* L.,**



***Ficus racemosa* L.**

RESULT & DISCUSSION

Medicinal trees are enhanced curiosity of those people that were being used trees parts for various purposes as far as almost 45 trees species belonging 24 families that are propounded in the table.1 through botanical name, common name, families and part used. The formulation and exploration of trees have expressed in favour of treatment of several diseases by indigenous people, forest dwellers and vaidh. Those are being used underlying own traditional knowledge in order to treatment of numerous diseases with the help of trees parts viz. bark, leaves, flower and seed that are making paste, juice, powder and pellets as well as granted to the patients for treatments of several diseases until cure.

CONCLUSION

The observation of medicinal trees have explored about multidimensional properties, however medicinal trees have great signification thereby indigenous people are being applied of trees parts in the treatment of various diseases namely malaria, diabetes, cough, anemia, dysentery and skin diseases.The families represents Fabaceae (9 genera), Combretaceae (4 genera), Moraceae (4 genera), Myrtaceae (3 genera), Anarcardiaceae (2 genera) and more were shown in above Fig. 4. Koraiya (*Wrightia tinctoria* roxb.) was rare species which belongs to Apocynaceae family as well as Palas (*Butea monosperma* L.) was dominating species which belongs to Fabaceae family that is widely used with humongous medicinal values. People were very conscious regarging medicinal trees which are used ancient to modern time for the treatment of diseases.

ACKNOWLEDGEMENT

The authors are grateful to Dr. Shashibala Akhaury Former Dean, Faculty of Science, Ranchi University &

Retired Professor, Department of Botany Ranchi University, Ranchi, India and The authors are also grateful to the villagers of Pandwa block who were helpful and supported during survey of medicinal trees.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 213-220

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Green fabrication and characterization of silver nanoparticles from leaf and bark extracts of *Alstonia* sp.

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Received : 30th December, 2022 ; Revised :30th January, 2023

Abstract- In this study, leaf extracts of *Alstonia scholaris* (L.) R.Br (E3)., *Alstonia macrophylla* Wall (E1), and *Alstonia venenata* R.Br.(E2) were used to compare by synthesis and characterization of silver nanoparticles from them. Characterization of synthesized nanoparticles were primarily done by significant colour changes and silver nanoparticle characteristic peak of 450 nm in UV-Visible Spectroscopy. Further confirmation were done using FTIR, particles size analysis and zeta potential. Morphology of the synthesized nanoparticles were imaged using Field emission scanning electron microscopy (FE-SEM). Active phytochemical constituents of the leaf extracts of *Alstonia scholaris* and *Alstonia macrophylla* were characterized using Gas Chromatography-Mass Spectrometry (GC-MS) method. The silver nanoparticles synthesized from three species of *Alstonia* showed similar pattern in FTIR spectroscopy with prominent peaks at 3291.33 cm^{-1} (E1), 3304.77 cm^{-1} (E3), and 3303.89 cm^{-1} (E2), 1637.26 cm^{-1} (E1), 1637.15 cm^{-1}(E3), 1637.17 cm^{-1} (E2) 1065.84 cm^{-1} (E1), 1065.86 (E3), 1065.95 (E2) and 591.32 cm^{-1} (E1) 598.44 cm^{-1}, 585.14 cm^{-1}. Zeta potential and particle size analyzer showed Particle size range of 150-201nm and their zeta potential ranged from -19 to -22 mV. The GC-MS profiles of *Alstonia scholaris* and *Alstonia macrophylla* revealed several common phytoconstituents, proving that identical capping and reducing agents resulted in the emergence of similar peaks among the nanoparticles produced from their plant extract. Some phytoconstituents that were observed in both the plant extracts were lupeol, stigmasterol, campesterol, 2(4H)-Benzofuranone 5, 6, 7, 7a-tetrahydro-4, 4, pentane,3-methyl, 7a trimethyl, (-)-Loliolide, ethane,1- chloro-1-fluoro Neophytadiene, Hexahydrofarnesyl acetone.

Key words: Green synthesis, silver nanoparticles, SEM, DLS, Fourier-transform infrared spectroscopy

INTRODUCTION

Nanobiotechnology, a subfield of nanoscience, has played a pivotal role in the 21st century in understanding a variety of challenges, especially in the domains of agriculture, medicine, and electronics. Nanoscience presents a fundamental scientific challenge since it needs control over atomic interactions. All physicochemical characteristics methods of synthesis of nanomaterials have intrinsic constraints that provide a significant barrier to the development of this field of study. The prospect of using biological materials for nanoparticle synthesis has emerged as the most cost-effective and environmentally friendly method.[1]

Physical, chemical, and biological properties of nanomaterials differ significantly from those of their bulk equivalents.[2] Although physical and chemical approaches are more prevalent for nanoparticle manufacturing,[3] the usage of hazardous substances restricts their uses.[4] In fact, plants, algae, fungi, bacteria, and viruses have been employed to produce metallic nanoparticles in recent years.[5] Eco-friendly and non-toxic, green synthesis of metallic nanoparticles from plants is an intriguing aspect.[6]

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When compared to the raw materials, NPs reveal remarkable features. In terms of environmental protection, this is considered one of the most major motivations for the reliance on the usage of NPs over raw materials.[7] The surface to volume ratio of NPs is greater; as the particle's surface area increases, the particle's radius drops. This is one of the favourable qualities of NPs for drug delivery in the medical industry.[8] The ratio of surface to volume, which determines the surface chemistry, has led in more applications for AgNPs than for other metallic NPs.[9] When the NPs' stability is high, they can be employed as an effective antibacterial agent.[10] This stability is a result of the zeta potential, which offers NPs electrostatic mobility; this enables significant applications in pollution management or removal of any environmental pollutant.

Various kinds and types of metallic NPs have been utilised in a variety of fields. The group of noble metals includes gold (Au), silver (Ag), and platinum (Pt), while the group of metal oxides includes silicon oxide (SiO_2) and zinc oxide (ZnO).[11] Among the metal NPs, silver nanoparticles (AgNPs) have the broadest range of applications, and numerous research groups are now investigating additional uses for AgNPs. Ag is regarded one of the earth's fundamental elements because it exists naturally, and Ag is harder than Au. The thermal and electrical conductivity of AgNPs is superior than that of Au. Metallic Ag is currently utilised in numerous surgical operations, as well as fungicides and preservatives. Consequently, AgNPs have garnered greater interest than other metallic NPs.

Alstonia scholaris (L.) R.Br. (Apocynaceae) is a tropical evergreen tree indigenous to the Indian subcontinent and South East Asia with greyish, rough bark and milky, alkaloid-rich sap. The bark, also known as dita bark, is utilised by various ethnic groups of Northeast India and other areas of the world as an antibacterial agent fighting fungal infections, malarial fever, toothache, rheumatism, snake bite, diarrhoea, etc. The latex is used to cure coughs, oral sores, and fever.[12,13] Only the *scholaris* species of the *Alstonia* genus has been investigated for antibacterial activity.[14] Silver has long been known to suppress the growth of certain bacterial strains and germs prevalent in medical and industrial settings.[15] Numerous attempts to employ silver nanoparticles as an anticancer agent have all shown favourable results.[16,17]

Biota with the single most powerful biocidal properties include bacteria, viruses, fungi, algae, and higher nematodes and helminths. In addition, it is non-toxic to animal cells and compatible with human cell lines, providing numerous applications for biological products.[18] There are reports of silver nanoparticles synthesized from leaf and bark extracts of *Alstonia scholaris* (L.) R.Br. that have showed substantial antibacterial activity against various pathogenic microbes. This study aims to further compare the synthesis and characterization of nanoparticles from leaf of *Alstonia scholaris* (L.) R.Br., *Alstonia macrophylla* Wall, and *Alstonia venenata* R.Br.

MATERIALS & METHOD

Preparation of plant extract

All of the chemicals were supplied by Merck. The leaves of *Alstonia macrophylla* Wall (E1), Alstonia *venenata* R.Br. (E2) and *Alstonia scholaris* (L.) R.Br. (E3) were collected from Rambagh, Bengaluru. The leaves were shade dried and powdered using electric grinder. Extract of each sample were prepared by mixing 0.5 gm of sample with 50 ml of distilled water. This mixture was boiled for 15 minutes before being chilled. Mixture was left for 25 hours shaking at 200 rpm. Whatman No. 1 filter paper was used to filter the incubated solution.

Biosynthesis of Silver nanoparticles

To synthesize AgNPs, 2ml of plant extract was added to a 98ml (1mM) aqueous solution of $AgNO_3$ and the mixture was heated to 80°C. The reduction of metal ions was initially established by observing a shift in hue from light brown to dark brown, followed by spectrophotometric absorption between 200 and 800 nm. Several reaction parameters, including the amount of plant extract, the concentration of the silver nitrate solution, and the reaction time, were altered to increase the nanoparticle yield. During the reaction time, spectrophotometric absorption and DLS measurements were taken for each trial. The reaction medium containing AgNPs was centrifuged for 15 minutes at 10000rpm to eliminate unreacted extract from the final solution.

Characterization of Synthesized Nanoparticles

UV–Vis spectroscopy (UV-1900, 190-110 nm) was used to evaluate the reduction of Ag ions. For the morphology, SEM (JEOL JSM-7500F SEM) with EDX was used. The biomolecules were identified using FTIR spectroscopy (Bruker Alpha II) and the XRD was used to identify the crystalline structure of the AgNPs.

[A] UV-Vis absorption spectra

Characteristic spectra were acquired using a double beam spectrophotometer 1800 (Shimadzu, Tokyo, Japan) in the wavelength range of 300-600 nm with a resolution of 2 nm.

[B] Dynamic light scattering (DLS)

Analyzer for measuring the particle size and zeta potential of manufactured nanoparticles. The aqueous suspension of the produced nanoparticles was filtered via a 0.22-lm syringe-driven filter unit, and the size and dispersion of the nanoparticles were determined using the dynamic light scattering method (Nanopartical, HORIBA, SZ-100)

[C] Fourier Transform Infrared (FTIR) Spectroscopy

A Fourier transform infrared (FTIR) spectrum was obtained using an FTIR-8400 spectrophotometer (Shimadzu, Tokyo, Japan) with a resolution of 4 cm^{-1} over the range of 500 to 4000 cm^{-1}. The materials were formed into pellets by combining potassium bromide (KBr) as a binding agent.

[D] Field Emission-Scanning Electron Microscopy (FE-SEM)

The surface morphology was evaluated using field emission-scanning electron microscopy (FE-SEM) (Hitachi S-4800, Tokyo, Japan) and energy dispersive X-ray spectroscopy (EDAX) for the elemental analysis. Before viewed under a microscope, the samples were dispersed in ethanol and placed on specimen tubs with a gold coating.

Characterization of the Extract

Extract characterization was done using Gas Chromatography-Mass Spectrometry (GC-MS) technique. Sample was diluted in Methanol and injected into a GC-MS QP2010 model (Shimadzu®), Column, GC, SH-I-5 Sil MS Capillary, 30m x 0.25mm x 0.01um, with Splitless injection mode. The GC–MS operating conditions for the analysis were as follows: 45°C for 2 minutes, then 140°C at a rate of 5°C per minute, then 280°C and kept isothermally for 10 minutes. The sample injection volume was 2 L, and the carrier gas was 1 mL/min of helium. The components of the sample were ionised with 70 eV. The duration of the GC ranged from 9.10 minutes to 52.0 minutes. The NIST14.L library (2020) was then queried to compare the compound structures to the NIST database. The compounds were subsequently identified by comparing their retention periods and mass spectra to those of previously identified compounds in the NIST library (C:DatabaseNIST14.L).

RESULT & DISCUSSION

Surface plasmon oscillations in aqueous media give silver nanoparticles (AgNPs) a yellowish-brown appearance. The solution's color altered from pale light to yellowish brown to reddish brown to colloidal brown when various leaf extracts were added to the aqueous silver nitrate solution, all of which are indicative of AgNP production.[19] Previous research has also validated the completeness of the interaction between leaf extract and by seeing similar shifts in color.[20-24] Figure 1(A-E) displays the colour change observed after 15 min, 30 min, 45 min, 60 min, and 24 hours following the start of the process. Figure 1 (F-H) displays the UV-vis spectra acquired for nanoparticles synthesized from leaves of *Alstonia macrophylla* (E1), *Alstonia venenata* (E2) and *Alstonia scholaris* (E3). Due to their surface plasmon resonance, AgNPs generated in reaction media have absorption maxima between 425 and 475 nm in their absorption spectra.

UV visible spectroscopy was employed to understand the formation of nanoparticles through the change of colour of AgNO solution from colourless to dark brown. Excitation of surface Plasmon resonance and reduction of Ag+ to Ag0 are the two major causes for the change of colour of solution (shown in figure 1(A-E)). Temperature, time and stirring accelerated the reaction. Many studies reported the characteristic peak of AgNPs in the range of 400- 450 nm which was present in all the three synthesized nanoparticles E1, E2 and E3, with a peak at 440 nm observed in UV spectrum as shown in figure 1(F-H). The UV-vis spectra also showed that AgNPs formed fast within the first 15 minutes and were stable in solution for up to 24 hours after the reaction had finished.

Characterization using DLS

The polydispersity index, particle size, and size distribution profile of the colloidal solution of ASPL nanoparticles were all measured using dynamic light scattering (DLS). The potential stability of molecules in colloidal suspension was shown by negatively charged plant nanoparticles with negative zeta potential values. Figure 2A-C (A1, A2, B1, B2, C1 and C2) shows particle size distribution and Zeta potential for silver nanoparticles synthesized from leaf extracts of *Alstonia macrophylla* (leaf), *Alstonia scholaris* (leaf) and *Alstonia venenata* (leaf). Particle size range is between150-201nm and their zeta potential ranged from -19 to -22 mV among 3 NPs.

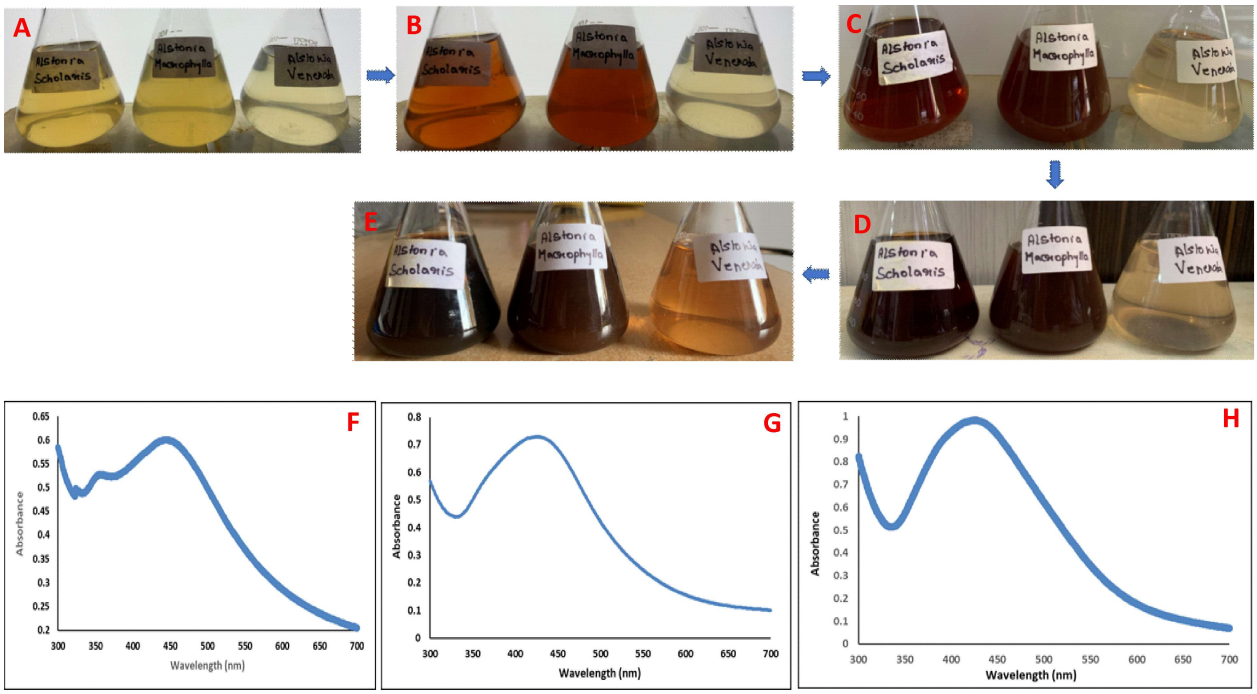


Fig 1: Synthesis of AgNPs using leaves of the plant *Alstonia* and its various species. Colour change during Nanoparticles synthesis at [A]15 min [B] 30 min [C] 45 min [D] 60 min [E] 24 hours. UV–vis absorption spectrum of silver nanoparticles using green synthesis method. [F]E1-*Alstonia macrophylla* (leaf) [G]E3-*Alstonia scholaris* (leaf) and [H] E2- *Alstonia venenata* (leaf)

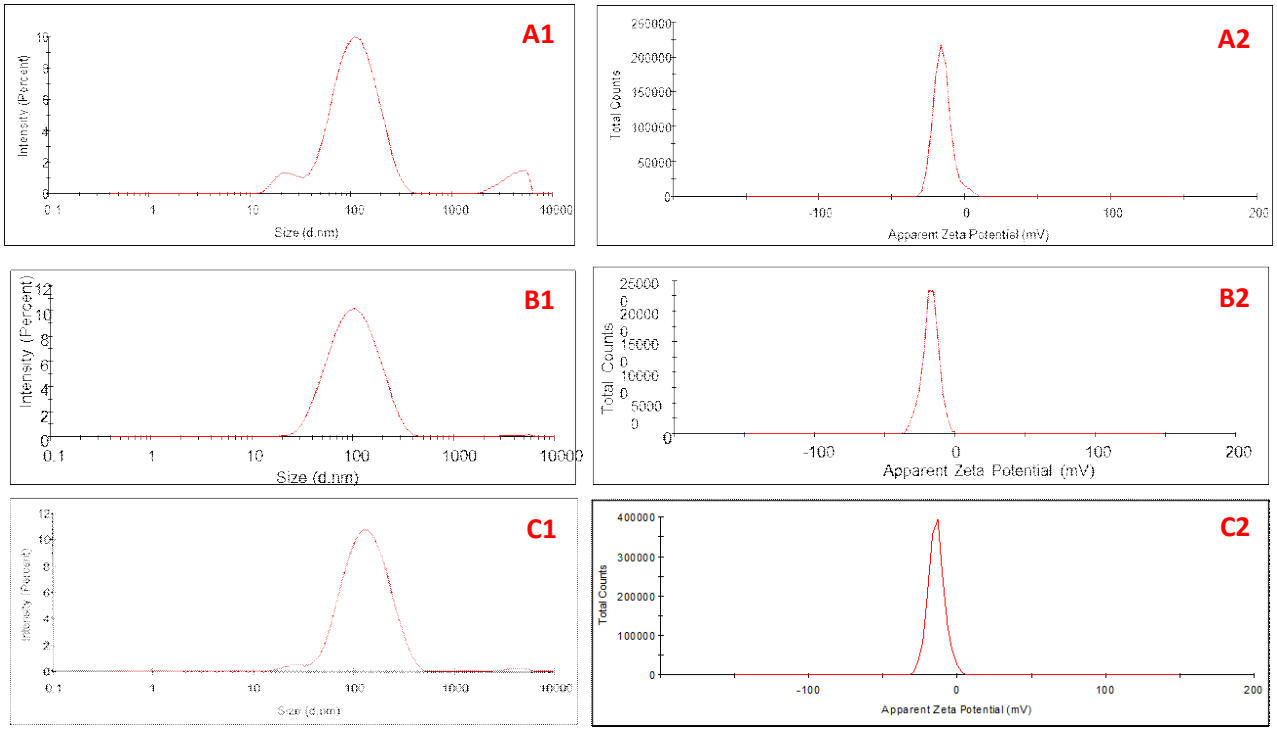
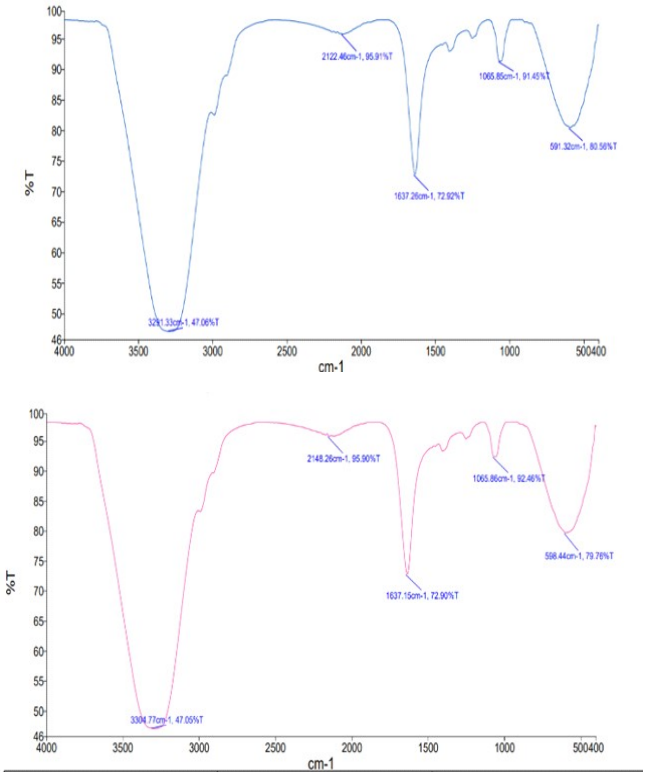


Fig. 2: Graph showing particle size distribution and Zeta potential for silver nanoparticles AgNPs synthesized from leaf extracts of [A1- A2] E1-*Alstonia macrophylla* (leaf) [B1-B2]E3-*Alstonia scholaris* (leaf) and [C1-C2] E2-*Alstonia venenata* (leaf) with labelled prominent peaks

Characterization using FTIR

FTIR spectroscopy was used to identify the various functional groups contributing to the production of the AgNPs seen in Figure 3. Bonding structures generated between metal particles and biomolecules may be analyzed via spectrum of FTIR.[25] This FTIR analysis has been used to examine how several phytochemicals contribute to the reduction of inorganic silver nitrate to silver.[26] Nanoparticles from all the three plants E1, E2 and E3 showed similar pattern having significant bands of absorbance at around 3291.33, 2122.46, 1637.26, 1065.84, 591.32 in E1, 3304.77, 2148.26, 1637.15, 1065.86, 598.44 for E3 and 3303.89, 2144.59, 1637.17, 1065.95, 585.14 for E2. The AgNPs formed by using the *Alstonia* leaf extract have a peak at 3291.33 cm^{-1} (E1), 3304.77 cm^{-1} (E3), and 3303.89 cm^{-1} (E2) that indicates the O-H stretch of the phenolic group, 1637.26 cm^{-1} (E1), 1637.15 cm^{-1} (E3), 1637.17 cm^{-1} (E2) corresponds to the C=O stretch of the ketone group, C-O stretch was assigned at 1065.84 cm^{-1} (E1), 1065.86 (E3), 1065.95 (E2) and C-Cl alkyl halide group has a peak at 591.32 cm^{-1} (E1) 598.44 cm^{-1}, 585.14 cm^{-1}. As a result, silver nanoparticles' surfaces often include phenolic and carbonyl chemicals. Flavonoids, alkaloids, and polyphenols, all of which are water-soluble, provided these chemicals. These bands agree with earlier studies.[27-29]



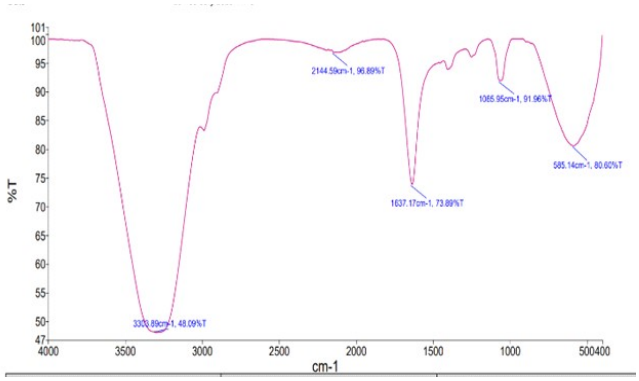


Fig 3: FTIR spectrums of AgNPs synthesized from leaf extracts of [A] E1-*Alstonia macrophylla* (leaf) [B]E3-*Alstonia scholaris* (leaf) and [C] E2- *Alstonia venenata* (leaf) with labelled prominent peaks

Characterization using FE-SEM

The form and size features of the produced nanoparticles have been further elucidated with the use of FESEM as shown in figure 4 [A-C]. It is found that the silver nanoparticles synthesized were mostly spherical in shape and the size is found as 100 nm for *Alstonia macrophylla* (leaf) and *Alstonia scholaris* and 1µm for *Alstonia venenata* (leaf).[25,30] Because of their high surface tension and energy, nanoparticles were also found to exhibit agglomeration characteristics. It was observed that nanoparticles synthesized from three different species of *Alstonia* did not have high difference in morphology evident from the images shown in fig (4A-C). The phytochemicals possessed by three extracts were found to be similar as observed in GCMS profiles (fig 5), thus providing similar quantity and nature of reducing agents for synthesis of nanoparticles. Capping agents were very similar in three cases as the FTIR study, which showed a shift and a variation in peak area, corroborated this.

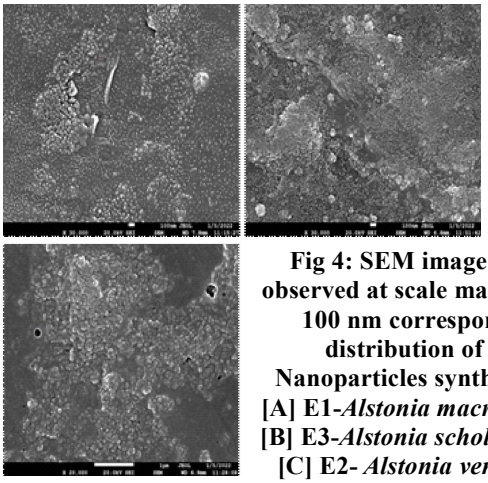
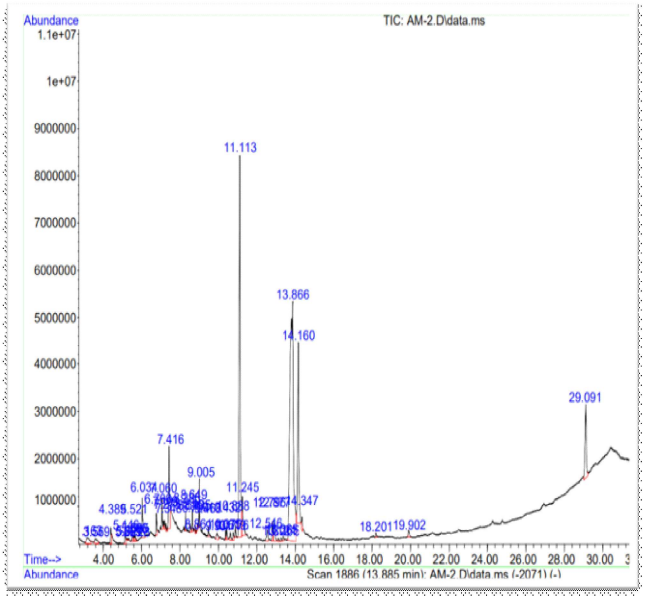


Fig 4: SEM images of AgNPs observed at scale magnification of 100 nm corresponding size distribution of AgNPs. Nanoparticles synthesized from [A] E1-*Alstonia macrophylla* (leaf) [B] E3-*Alstonia scholaris* (leaf) and [C] E2- *Alstonia venenata* (leaf)

Characterization of Extract using GC-MS

Extraction and characterization of phytoconstituents in leaf extracts of *Alstonia macrophylla* and *Alstonia scholaris* were evaluated using GC-MS technique. The total ion chromatogram (TIC) of the both extracts are shown in the figure 5(A-B). *Alstonia scholaris* extract showed presence of oxime-methoxy-phenyl, 2-methoxy 4-vinyl phenol, pentane,2-methyl-, 2(4H)-Benzofuranone 5, 6, 7, 7a-tetrahydro-4, 4, pentane,3-methyl-, 7a trimethyl, (-)-Loliolide, ethane,1- chloro-1-fluoro Neophytadiene, Hexahydrofarnesyl acetone, Phytol, 4, 8, 12, 16-Tetramethylheptadecan-4-olide, 1,3-propanediol, 2-(hydroxymethyl)-2-nitro-, 9,12-Octadecadienoic acid (Z,Z)-, γ-sitosterol, 24-methylenecycloartanol, lupeol, stigmasterol. The antiarthritic properties of *Alstonia scholaris* may be linked to its analgesic, anti-inflammatory, immunosuppressant, and anti-oxidant actions.[31] Previous studies have also reported the antiepileptic and sedative potential properties in the ethanolic extracts of the plant.[32] In another study it was revealed that the antivenom property of leaf extract of the plant and therefore highlighting their significance in snakebite treatment.[33] The extensive medicinal role of *A.scholaris* have been further explained by their *in-vitro* cytotoxic and antiproliferative properties.[34] His findings were also supported by Malick *et al.* in his study.[35]



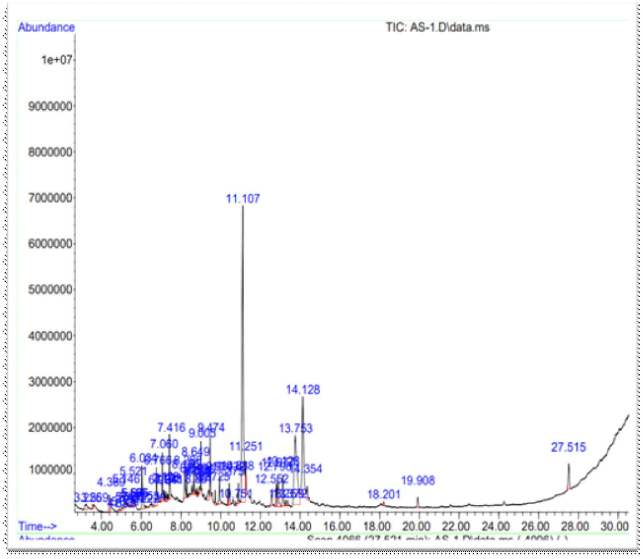


Fig 5: GC-MS Profile of leaf extracts showing Total ion chromatogram (TIC) of [A] *Alstonia macrophylla* (leaf) [B] E3-*Alstonia scholaris*

Whereas the extract of *Alostonia macrophylla* showed presence of 2,5-dimethyl-4-hydroxy-3(2*H*)-furanone, 2-methoxy-4-vinylphenol, 2,6-dimethoxyphenol, 2,6-dimethoxy-4-(2-propenyl)-phenol, 3,5-dimethoxy-4-hydroxyphenylacetic acid, burnamicine, γ-sitosterol, 24-methylenecycloartanol, lupeol, stigmasterol, campesterol, 2(4H)-Benzofuranone 5, 6, 7, 7a-tetrahydro-4, 4, pentane,3-methyl-, 7a trimethyl, (-)-Loliolide, ethane,1-chloro-1-fluoro Neophytadiene, Hexahydrofarnesyl acetone. There were similarity in pattern of metabolites observed among the two extracts. Some metabolites were common while some were solely present in each of them. Medicinal properties of aqueous and alcoholic extracts of *Alostonia scholaris* are very well studied and reported in comparison to *Alostonia macrophylla*. This study highlights several active constituents from GC-MS profile that can play integral role in drug development and medication.

CONCLUSION

In this work, we provide a biomimetic, single-step, and eco-friendly method for manufacturing silver nanoparticles from extracts of *Alstonia* leaves containing compounds with therapeutic potential. This method exemplifies a safe, green, and efficient strategy for biosynthesizing silver nanoparticles. This research demonstrates that $AgNO_3$ may be efficiently reduced to silver nanoparticles with an average diameter in the range

of 15.14 nm using leaf extracts of the *Alstonia macrophylla, Alstonia scholaris* and *Alstonia venenata.* Other benefits of these nanoparticles are also their stability for a long period of time. The absence of UV-Visible alterations in synthesized silver nanoparticles, even after long term storage, may be owing to the presence of proteins, enzymes, and sugars that may function as a capping agent. Synthesized silver nanoparticles can be further tested for their potential antibacterial activity against pathogenic strains of bacteria. This work provides further evidence that different *Alstonia* species have high content of active constituents as observed from results of GCMS thus it may open up vast opportunities for the development of antibiotics and medicines against a wide range of harmful bacterial types.

ACKNOWLEDGEMENT

The authors of this paper are thankful to CytoGene Research & Development, Lucknow for helping and providing necessary facilities for this research work.

CONFLICT OF INTEREST

There is no conflict with other interest in the manuscript content

FUNDING ACKNOWLEDGMENT

No Funding Sources

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 221-224

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Impact of Brick Kiln emission on physico-chemical properties of soil in agriculture field of Madhepura

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Received : 31st December, 2022 ; Revised : 31st January, 2023

Abstract- Physico-chemical parameter of soil near brick kiln of Mathahi village of Madhepura district was examined and compared with physico-chemical parameter of soil half km away from brick kiln. It was observed that the pH value was lowest in all directions at 100mt away from brick kiln followed by 50mt away and 150mt away. Electrical conductivity doubled at each spot. Nitrogen content, Phosphorus content and Potash content also decreased near brick kiln. The pH was lowest (5.8) at 100mts away from brick kiln in its east side, pH 5.4 at North and west side and 5.5 in South side. The lowest value of percentage organic carbon in soil was at 100mt away in east side (0.35). Nitrogen was also lower near all spots of brick kiln. The lowest value was observed in east side at 100mt away from brick kiln (42.35kg/Ha). Phosphorus was also lowest at this spot (4.3kg/Ha). The lowest value of Potash was observed at 100mt away from brick kiln in west side (100.6).

Key words: Physico-chemical parameters, Brick kiln, pH, Electrical conductivity

INTRODUCTION

In the manufacturing of Bricks, generally top soil's up to 1.5mt depth is used. Deeper soil removal resulted in drastic loss of soil fertility and the land becoming agriculturally unproductive.[1] This layer of soil consists of organic matter, nutrients and microbes essential for crop production. 50-100 meter distance from Brick kilns soil is affected and is not suitable for cultivation.[2] It was reported that brick kiln alters the soil physico-chemical properties and soil nutrients of its surrounding areas.[3] Deterioration of soil quality, nutrient content and microbial content indicate that Brick kiln is not suitable near agriculture field. The emission of Brick kiln contains heavy metal like Pb, Cr and Ni which severely affect the biodiversity.[4] Accumulation of heavy metals in the soil adversely affect soil texture. Adrees *et al.* (2016)[5] found that concentration of Ni, Pb and Co causes decrease in organic matter content and lower the concentration of Ca, Mg and Na. The electrical conductivity near Brick kiln fields becomes almost double and pH of soil decreases. Saha *et al.* (2021)[6] concluded that the Brick kiln surrounding areas have very low pH and organic matter content. The organic matter content near the soil of Brick kiln significantly lowers than that of soil away from Brick kiln. The organic carbon content surrounding Brick kiln area also decreases.

Brick kilns are considered harmful for the environment. It is largest source of greenhouse gas emission including black carbon, carbon dioxide, sulphur oxides, nitrogen oxides, carbon monoxide and suspended particulate matters as well as fluoride and small amount of carcinogenic substance. Skinder *et al.* (2014)[7] reported that the average emission of gases per 1000 bricks were 6.35

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to 12.3kg of CO, 0.52 to 5.9 kg of SO_2 and 0.64 to 1.4 kg of particulate matter. Brick kilns are major contributor of black carbon. Thygerson & S.J. (2016)[8] reported that Brick kilns contribute about 31% total suspended particles in environmental air pollution. Hossain *et al.* (2019)[9] reported that acid deposit from sulphur oxides and nitrogen oxides emitted from Brick kilns negatively affect agricultural productivity.

MATERIAL & METHOD

For the present study, a brick kiln near Mathahi village of Madhepura district was selected. The brick kiln is nearby state road which passes from Madhepura to Saharsa. The road it situated at 50mts away from the brick kiln in its south side. Soil samples were collected at a distance of 50mts., 100mts. and 150 mts. in four directions (North, South, East, West) around the Brick Kiln at a depth of 0-30cm in Rainy season, summer season and winter season. soil samples were brought to the laboratory for physico-chemical analysis.

Physico-chemical analysis of soil samples:

(a) pH- pH was determined by the method of Jackson (1967). 10gm soil was dissolved in 20ml distilled water and stirred well for 10 minutes. It was kept undisturbed for some time and stirred again. pH was measured by cystronic pH meter.

(b) Electrical conductivity - Electrical conductivity was measured by the method described by Jackson (1967). 10gm soil was dissolved in 20ml distilled water in a beaker and stirred vigorously. The soil water mixture allowed to stand undisturbed till soil settled completely. The E.C was measured using cystronic conductivity meter.

(c) Estimation of Total organic carbon - For the estimation of Total organic carbon, method of Walkley & Black (1934) was applied.

Required Reagents:

i) 1M potassium dichromate
ii) 0.5N ferrous ammonium sulphate
iii) Conc. Sulphuric acid
iv) Ortho-phosphoric acid
v) Sodium fluoride
vi) Di-phenyl amine

Procedure: For the measurement of total carbon, soil samples were dried. 1gm dry soil sample was transferred in a conical flask and 10ml potassium dichromate solution was added in it. Two blanks were also prepared. 20ml conc. sulphuric acid was added in the flask in samples and blanks. It was swirled well and allowed to stand for 30 minutes. 250ml water was added both in sample and blanks. 10ml ortho-phosphoric acid was added and allowed to cool. 1ml di-phenyl amine as indicator was added. It was titrated against ferrous ammonium sulphate solution. Appearance of violet-blue color shows end point. The amount of total organic carbon was calculated by the following formula:

$$\%Carbon = M \times \frac{V1 - V2}{S} \times 0.39$$

Where,

M = Molarity of ferrous ammonium sulphate solution
V1 = Volume of ferrous ammonium sulphate solution for blank
V2 = Volume of ferrous ammonium sulphate solution for sample
S = Weight of soil sample

(d) Estimation of Nitrogen: Nitrogen was estimated by Kjeldahl method.

Required Reagent:

i) Sulphuric acid
ii) Selenium powder
iii) Hydrogen peroxide
iv) Sodium hydroxide
v) Methyl red
vi) Bromocresol green
vii) Boric acid solution
viii) 0.01M Hydrochloric acid

Procedure: 1gm of sieved dry soil sample was added in the digestion tube. 2.5ml digestion mixture was added and 1ml hydrogen peroxide was added in it. The digestion tubes were placed on heater and heated up to 200°C temperature for 1 hour and then temperature was raised to 330°C for about 2 hours. The digestion tube was removed from heater and allowed to cool. A beaker containing 20ml Boric acid indicator solution was placed beneath the condenser tip. 20ml NaOH was added to the digestion tube. The beaker was removed from its place. Condenser tip was rinsed. Distillate was titrated against 0.01M HCl. Appearance of pink color was the end point. The amount of % Nitrogen was calculated by the formula:

$$\%N = \frac{a - b}{s} \times M \times 1.4$$

Where,

N = Nitrogen

a = volume of 0.01M HCl required for titration of sample
b = volume of 0.01M HCl required for titration of blank
M = Molarity of HCl

(e) Estimation of Phosphorus: Phosphorus was estimated by Olsen method. For this purpose, Extracting solution was prepared which contains following chemicals:

Extracting solution:

i) 0.5M Sodium Bicarbonate solution
ii) 4M Sulphuric acid
iii) 4% ammonium molybdate solution
iv) Potassium antimony tartrate solution

Ascorbic acid was used as reagent and standard solution of phosphate is prepared (100mg/L).

Procedure:

i) 5gm fine dry soil sample was added in 100ml distilled water and stirred well.
ii) 100ml extracting solution was added in it.
iii) Mixture was shaked for 30 minutes and filtered through Whatman filter paper No. 42.
iv) 3ml blank, 3ml sample extract and 3ml standard phosphate solution were taken in 3 test tubes.
v) 3ml reagent was added in each test tube and allowed to stand for 1 hour.
vi) A Blue color appears.
vii) OD was measured on Spectrophotometer at 720nm.
viii) Phosphate concentration was measured by the formula:

$$P\left({}^{mg}/_{gm\ of\ soil}\right) = \left(\frac{a-b}{c-b}\right) X \frac{100}{S}$$

Where,

a= OD of sample
b= OD of blank
c= OD of standard solution
s= Sample weight in gm

(f) Estimation of Potassium: Potassium was estimated by the flame photometer. 10ml normal ammonium acetate was added in 1gm dry soil and shaken for 5 minutes. The result is obtained in ppm.

RESULT

Soil samples were collected near Brick kiln at a distance of 50mts, 100mts and 150mts in all directions (North, South, East and West) and half km away from the Brick Kiln. Physico chemical parameter of soil near Brick kiln and half km away from brick kiln were compared. It was observed that the soil pH became acidic near Brick kiln while it was neutral at half km away from brick kiln. The pH was lowest (5.8) at 100mts away from brick kiln in its east side, pH 5.4 at North and west side and 5.5 in South side. Electrical conductivity became almost double at each spot, organic carbon was lower in comparison to the organic carbon in soil at half km away from the brick kiln. The lowest value of percentage organic carbon in soil was at 100mt away in east side (0.35). Nitrogen was also lower near all spots of brick kiln. The lowest value was observed in east side at 100mt away from brick kiln (42.35kg/Ha). Phosphorus was also lowest at this spot (4.3kg/Ha). The lowest value of Potash was observed at 100mt away from brick kiln in west side (100.6). The result is mentioned in Table No. 01 and the result of all soil parameter at half km away from brick kiln is mentioned in

Table 1- Physico-chemical parameter of soil near Brick Kiln

Direction	Distance	pH	E.C. (dsm^{-1})	Organic carbon (%)	N (kg/Ha)	P (kg/Ha)	K (kg/Ha)
East	50mt	6.7	0.52	0.40	44.82	4.8	170.4
	100mt	5.8	0.68	0.35	42.35	4.3	110.6
	150mt	6.3	0.66	0.45	45.62	5.0	140.3
West	50mt	6.2	0.64	0.43	62.68	6.6	140.4
	100mt	5.4	0.50	0.40	58.93	6.2	100.6
	150mt	6.1	0.54	0.47	60.72	6.8	150.5
North	50mt	5.8	0.70	0.45	68.64	6.4	150.3
	100mt	5.4	0.60	0.42	63.32	6.1	120.5
	150mt	5.6	0.62	0.45	69.62	6.5	170.6
South	50mt	Road					
	100mt	5.5	0.61	0.41	64.31	5.8	154.3
	150mt	6.5	0.63	0.45	68.30	6.1	174.1

Table No. 02. The comparison of pH at different spot near brick kiln and half km away from brick kiln is represented in fig. 1.

Table 2- Physico-chemical parameter of soil half km away from Brick kiln

Season	pH	E.C. (dsm^{-1})	Organic carbon (%)	N (kg/Ha)	P (kg/Ha)	K (kg/Ha)
East	7.5	0.31	0.62	55.26	6.3	220.9
West	7.5	0.30	0.57	50.25	7.2	218.3
North	7.1	0.24	0.61	61.28	8.4	235.4
South	7.2	0.28	0.62	59.65	8.3	225.6

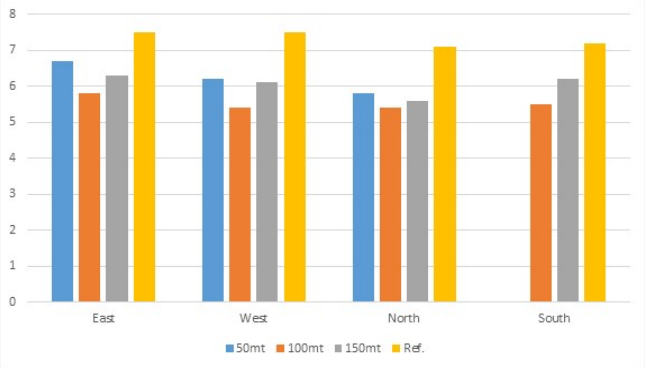


Fig. 1- Comparison of pH at different site near brick kiln and half km away from brick kiln.

CONCLUSION

The physico-chemical parameter of soil is adversely affected by brick kiln emission. In the present study, a brick kiln near Mathahi village of Madhepura district and physico-chemical parameter of soil at a distance of 50mt, 100mt and 150mt were examined. It was compared with physico-chemical parameter of soil half km away from brick kiln. It was observed that the pH near brick kiln decreased and Electrical conductivity became double. All essential nutrients like Nitrogen, Phosphorus and Potash also decreased near brick kiln. The pH was lowest (5.8) at 100mts away from brick kiln in its east side, pH 5.4 at North and west side and 5.5 in South side. Electrical conductivity became almost double at each spot, organic carbon was lower in comparison to the organic carbon in soil at half km away from the brick kiln. The lowest value of percentage organic carbon in soil was at 100mt away in east side (0.35). Nitrogen was also lower near all spots of brick kiln. The lowest value was observed in east side at 100mt away from brick kiln (42.35kg/Ha). Phosphorus was also lowest at this spot (4.3kg/Ha). The lowest value of Potash was observed at 100mt away from brick kiln in west side (100.6).

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 225-230

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Phyto-chemical studies of methanol extracts of *Tinospora cordifolia* growing with different supporting trees by GC-MS

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Received : 30th December, 2022 ; Revised :24th January, 2023

Abstract- Plants are the basis of life on the planet earth as plants and their products directly or indirectly support the life processes. Specially, wild plants constitute an integral part of customs and cultures of different ethnic communities. Plants are the almost exclusive source of drugs for majority of the world population. The phyto components present in *Tinospora cordifolia*, locally known as Gilloy, is used in the treatment of various ailments in the long-established medicinal system in the state of Jharkhand, India and is also an anti-oxidant and immuno-modulator. In the present investigation *Tinospora cordifolia*, independent climber and climbers on different supporting trees has been identified for phytochemistry by screening and quantitative estimation. In the proposed work three categories of experimental plant has been identified: (i) *Tinospora cordifolia* independent climber, (TC); (ii) Climber on Aam or *Mangifera indica*, TC (A) and (iii) Climber on Neem or *Azadirachta indica*, TC (N). Present study was carried out to investigate the phytoconstituents of the *Tinospora cordifolia* which contain terpenoids, steroids, glycosides, flavonoids and Phlobatannins etc. are confirmed by preliminary Phyto-chemical studies. GC-MS analysis of methanol extract of the plant showed the presence of a variety of bioactive compounds. Some of them are of great interest as they are either anti-cancer or anti-oxidant or both

Key words: *Tinospora cordifolia*, Phyto-constituent, GC-MS analysis, Anti-cancer, Bioactive compounds.

INTRODUCTION

Tinospora cordifolia belongs to family Menispermaceae is a large deciduous climbing shrub found throughout tropical Indian subcontinent. It consists of about 70 genera and 450 species. This plant is known for its variety of uses in Ayurveda. Some of the active components present in the plant are alkaloids, steroids, diterpenoid lactones, aliphatics and glycosides.[1] This plant is of great interest to researchers across the globe because of its wide-ranging medicinal properties like anti-diabetic, anti-periodic, anti-spasmodic, anti-inflammatory, anti-arthritic, anti-oxidant, anti-allergic, anti-stress, anti-leprotic, anti-malarial, hepatoprotective, immunomodulatory and anti-neoplastic activities.[3-4] Therefore, the plant *T. cordifolia* can be chosen as a source for the development of industrial products for treatment of various diseases.[5] In the present study *Tinospora cordifolia* is subjected to phytochemical screening and GC-MS analysis in order to screen and confirm the presence of various pharmaceuticals compounds.

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MATERIALS & METHODS

Collection of samples: Plant material *Tinospora cordifolia* (independent climber) and climbers on Aam and Neem trees was collected from the campus of Arogya Bhawan, Ranchi. Jharkhand, India.

Preparation of sample for phytochemical screening:

The plant parts (leaf and stem) were cleaned, dried and powdered with the help of mixer grinder separately. Methanol extracts were prepared and concentrated using rotary evaporator and stored at 4ºC in air tight containers.

Preparation of extract for Gas Chromatography Mass Spectroscopy (GC-MS) Analysis:

15 grams of dried stem powder of *Tinospora cordifolia* was taken and soaked in 150 methanol and it was kept in room temperature for 72 hours with constant shaking. After incubation, solutions were filtered with Whatman filter paper no. 1 and filtrate were kept at room temperature for drying. After drying, the weight of extract was measured and according to weight solvent was added and maintained the concentration of extract as 25 mg/ml. It was sent to Advanced Instrumentation Research Facility, Jawaharlal Nehru University, New Delhi, India.

Methodology for phytochemical screening:

Chemical tests were carried out on the extract and on the powdered specimens using standard procedures based on the protocols of Edeoga *et al.* (2005)[6], Harborne (1973)[7] and Sofowara (1993)[8] to identify the various constituents present.

RESULT & DISCUSSION

Phytochemical Screening:

In the present study, preliminary phytochemical screening of the extract of leaves and stem of experimental plant *T. cordifolia*, TCE (A) and TCE (N) revealed the presence of various bioactive components such as alkaloids, saponins, tannins, Phlobatannins, cardiac glycosides, steroids, flavonoids, terpenoids etc. were identified in appreciable, moderate or in trace amount. Terpenoids, steroids, alkaloids and Phlobatannins were found in moderate amount in *T. cordifolia* stem extract whereas other bioactive compounds such as saponins, tannins, flavonoids and cardiac glycosides were trace in amount. Phlobatannins was absent in case of TCE leaf, TCE (A) stem and leaf. In case of TCE (N), both stem and leaf contained flavonoids in appreciable amount, however it was in moderate amount in TCE leaf, TCE (A) stem and leaf. Other components were in trace amount (Table 1).

Chemistry and medicinal properties of *Tinospora cordifolia* were also reported by many workers.[6,9] These phytochemicals have been attributed to the medicinal properties of the experimental plant.[10] Screening active compounds from plants has led to the discovery of new medicinal drugs which have efficient protection and treatment roles against various diseases, including cancer.[11]

Table 1- Phytochemical screening of secondary metabolites of different parts of *Tinospora cordifolia* grown with different supporting trees

Sl. No.	Plant Part	Saponin	Tannins	Steroids	Terpenoides	Phlobatanins	Flavanoids	Cardiac Glycosides	Alkaloids
1.	TC (stem)	+	+	++	++	++	+	+	++
2.	TC (Leaf)	+	+	+	+	-	++	+	+
3.	TC (A) stem	+	+	+	+	-	++	+	+
4.	TC (A) leaf	+	+	+	+	-	++	++	+
5.	TC (N) stem	+	+	+	+	++	+++	+	++
6.	TC (N) leaf	+	+	+	+	+	+++	+	+

TC : *Tinospora cordifolia* **TC (A)** : *Tinospora cordifolia* on Aam (*Mangifera indica*)
TC (N) : *Tinospora cordifolia* on Neem (*Azadirachta indica*) **(+++)** : Appreciable amount
(++) : Moderate amount
(+) : Trace **(-)** : Absent

Quantitative estimation- Gas Chromatography-Mass Spectroscopy (GC-MS) Analysis: For GC-MS analysis, methanolic extract of the plant sample (stem) was sent to Advance Instrumentation Research Facility (AIRF), Jawaharlal Nehru University, New Delhi. Knowledge of chemical constituents of plants is important and desirable because such information will be important for synthesis of chemical substances. It could be well qualified for application in pharmaceutical industry.[12]

Analysis of bioactive compounds by GC-MS has been reported by different workers.[13-15] The results of GC-MS analysis of *Tinospora cordifolia* extracts were very interesting. GC-MS of the *T. cordifolia* extract revealed the presence of 55 to 75 components having different pharmacological importance along with antioxidant and/or anticancer potential.

n-Hexadecanoic acid or Palmitic acid, Heptadecanoic acid or margaric acid, Stigmasterol, STIGMAST-5-EN-3-

OL, (3.BETA.)- and 1-Heptacosanol were present in TCE (A) and TCE (N) and anticancer in nature. Oleic Acid and Calcitriol was found to be present in TCE (A) and TCE (N) and absent in *T. cordifolia* extract (independent climber). Tetradecanoic acid or Myristic acid was antioxidant and anticancer and Pentadecanoic acid was found to be antioxidant in nature. These two bioactive compounds were present only in TCE (A) and TCE (N). However, there were some compounds which were present only in TCE and anticancer in nature such as pentanoic acid, propyl ester or valproic acid, phytol or 2-hexadecen-1-ol and 9,12-octadecadienoic acid (Z, Z)-. Other antioxidant compounds found were benzenepropanoic acid, 2,5-dimethoxy-, hexadecanoic acid, methyl Ester, pntadecanoic acid, octadecanoic acid, n-hexadecanoic acid or palmitic acid (Table 2). Among all these anticancer compounds, n-hexadecanoic acid or palmitic acid was having highest per cent area (23.12%) in the extract of *Tinospora cordifolia* normal or independent climber as compared to TCE (A) and TCE (N) in which area per cent were 9.67% and 7.15% respectively.

In our investigation, GC-MS analysis of methanolic extract of the stem of experimental plant showed presence of bioactive compounds which are of pharmacological importances. There are 16-17 compounds showing antioxidant and/or anticancer activity of the experimental plant (Table 2).

Table 2- Anticancer and antioxidant compounds obtained from GC-MS analysis of *Tinospora cordifolia*, TCE, TCE (A) and TCE (N)

Name of compound	Common name	Chemical formula	Mol. wt.	TCE (A)	TCE (N)	TCE	Activity
Pentanoic acid, Propyl ester	Valproic acid	$C_8H_{16}O_2$	144	–	–	+	Anticancer[16]
Calcitriol		$C_{27}H_{44}O_3$	416	+	+	–	Anti-proliferative, Pro-apoplotic, Anticarcinogenic and tumor inhibitory[17]
n-Hexadecanoic acid	Palmitic acid	$C_{16}H_{32}O_2$	256	+	+	+	Antioxidant and anticancer[18]
Heptadecanoic acid	Margaric acid	$C_{17}H_{34}O_2$	270	+	+	+	Anticancer[19]
Stigmasterol		$C_{29}H_{48}O$	412	+	+	+	Anticancer inhibit tumor promotion[20]
STIGMAST-5-EN-3-OL, (3. BETA)	Beta sitosterol	$C_{29}H_{50}O$	414	+	+	+	Anticancer and antioxidant[15]
1-Heptacosanol		$C_{27}H_{56}O$	396	+	+	+	Anticancer and antioxidant[20]
Oleic acid		$C_{18}H_{34}O_2$	282	+	+	–	Cancer preventive[19]
Octadecanoic acid	Stearic acid	$C_{18}H_{36}O_2$	284	+	+	+	Antioxidant and cancer preventive[21]
Benzene propanoic acid, 2,5-dimethoxy-		$C_{11}H_{14}O_4$	210	+	–	–	Antioxidant[20]
Phytol	2-Hexadecen-1-ol	$C_{20}H_{40}O_6$	296	–	–	+	Anticancer[20]
9,12-Octadecadienoic acid (Z,Z)-				–	–		Cancer preventive[19]
Hexadecanoic acid, methyl ester	Palmitic acid methyl ester	$C_{17}H_{34}O_2$	270	+	+	+	Antioxidant[19]
Tetradecanoic acid	Myristic acid	$C_{14}H_{28}O_2$	228	+	–	–	Antioxidant and cancer preventive[19]
Pentadecanoic acid		$C_{15}H_{30}O_2$	242	+	–	–	Antioxidant[18]
Icosanoic acid						+	Antitumor[18]
Cholest-4-en-3-one	Cholesterol			+	+	+	Antioxidant[21]

TCE : Extract of *Tinospora cordifolia* independent climber,
TCE (A) : Extract of *Tinospora cordifolia* climbing on Aam (*Mangifera indica*),
TCE (N) : Extract of *Tinospora cordifolia* climbing on Neem (*Azadirachta indica*)

SUMMARY & CONCLUSION

Knowledge of chemical constituents of a plant is important and desirable because such information will be important for synthesis of chemical substances. It could be well qualified for application in pharmaceutical industry. The GC-MS analysis of methanolic extract of experimental plant showed the presence of pharmacologically active compounds such as antioxidant and anticancerous. Phytochemical screening showed the presence of Alkaloids, Flavonoids, Phlobatannins, Saponins, Steroids and Tannins. A comparative analysis of phyto-chemicals and GC-MS of *Tinospora cordifolia* which grows independently and

growing on two different supporting trees *Azadirachta indica* and *Mangifera indica*, established the superiority of the one which grows independently. However, both the antioxidant potential and alkaloid content was higher in TCE (N). Presence of different phytochemicals in the experimental plant suggests its profuse therapeutic efficacy. Quality of this plant can be improved through secondary metabolites production and thus it can be used as a source for developing new drugs and commercialization. Further investigations on preclinical and clinical trials of these extracts could become a part of standard drug designing and treatment protocols for cancer and hence a promising and powerful weapon for cancer treatment.

ACKNOWLEDGEMENT

Authors are grateful and thankful to Head of the Department, Prof. Dr. Kunul Kandir for her valuable suggestions, constant support and inspiration throughout this study.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 231-236

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Effect of ethanol and chloroform extracts of selected plants against *Fusarium oxysporum* f. sp. *pisi*

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Received : 30th December, 2022 ; Revised :24th January, 2023

Abstract- Effect of ethanol and chloroform extract of leaf, stem, bark and seed of selected plants *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. was studied against *Fusarium oxysporum* f. sp. *pisi* (ITCC No.-4814), the causal organism of *Fusarium* wilt in pea. Almost all the extracts inhibited mycellial growth of test pathogen. Among them the maximum inhibition was found in 10 % concentration of chloroform extract of *Millettia pinnata* L. seed with 70 mm inhibition zone and *Cassia tora* L. seed or bark with 65.7 mm against *Fusarium oxysporum* f. sp. *pisi.* Significant inhibition was also occurred in ethanol extract of *Millettia pinnata* L. seed with 65 mm inhibition zone. Ethanol and chloroform extracts were highly significant on inhibition of mycelial growth in different concentration.

Key words: Plant extracts, ethanol, chloroform, *Fusarium oxysporum* f. sp. *pisi*

INTRODUCTION

Pea (*Pisum sativum* L.) is the most important and popular legume crop. It is grown in garden or field of temperate regions of the world. It is the most important Rabi pulse crop in India. It is consumed as pulse and vegetable. It is rich source of protein, also good source of carbohydrates, vitamins A and vitamins C, phosphorus and calcium, and less amount of iron.[1] Cultivation of pea is hindered by several diseases caused by plant pathogens. *Fusarium* wilt of pea caused by *Fusarium oxysporum* f. sp. *pisi* is the destructive disease and losses to farmers. Chemical fungicides are used for controlling the wilt disease which may cause adverse effects on environment and health of human being. Now a day management of *Fusarium* wilt disease has been done through plant extracts.[2-10] To search plant extracts as an alternative source for control of plant diseases is challenge of time. The antifungal activity of some medicinal plant extracts has been recognized and reported by some researchers.[11-13] Antimicrobial activity of plant extracts of *Terminalia chebula, Mangifera indica* and *Eucalyptus citriodora* have been reported against plant pathogens. The chloroform, aqueous or other extracts of some plants like *Piper betle* L. was found to be effective in controlling *Fusarium* spp.[14]

In present study, effects of extracts of selected plants such as *Millettia pinnata* L, *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. were studied.

Millettia pinnata L.is common medicinal plant, belongs to Fabaceae family. The common name of plant is "Karanja". It is grown all over the India especially near the coast and extending from the central to Eastern Himalayas to Ceylon. All parts of *Millettia pinnata* L. are

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used in Ayurveda as remedies for different diseases such as fever, piles, ulcer, bronchitis, whooping cough, skin diseases, rheumatic arthritis, liver pain and leprosy. It contains flavonoid constituents, alkaloids, phenolic compounds, phytosterol, saponins and terpenoids.

Cassia tora L. belongs to Fabaceae family. It is found in different parts of India. All parts of plant such as leaf, root, flower, fruit and seed possess medicinal properties and used in medicine. *Cassia tora* L. are used in simple cough, ringworm, leprosy, dyspepsia, constipation, bronchitis and cardiac disorders. *Cassia tora* L. contains chemicals constituents such as palmitate, stearate, oleate, Emodin, tricontan-1-0l, stigmasterol, succinic and d-tartaric acids uridine, quercitrin and isoquercitrin.

Cymbopogon citratus Stapf. (Family - Poaceae), is commonly known as lemon grass or oil grass, it is widely found in tropical countries. Various compounds mainly alcohols, ketones, terpenes, aldehyde, phenolic compounds and esters are present in lemon grass. It contains essential oils that contain Citral α, Citral β, Terpinolene, Geranyl acetate, Nerol Geraniol, Citronellal, Myrecene and Terpinol Methylheptenone. It is commonly used in teas, soups and curries. It is also used for poultry, fish and seafood.

Moringa oleifera Lam. belongs to the family Moringaceae, found in the tropical and subtropical regions worldwide. It is commonly known as miracle tree, drumstick or horseradish tree. It contains zeatin, quercetin, kaempferom and other phytochemicals. It is good source of vitamins A, C, B and minerals. It contains niazirin, niazirinin, Benzoic acid, gallic acid, beta benzaldehyde. It regulates the metabolism of thyroid hormone, central nervous system, digestive system, genito-urinary system. It is used in the treatment of gastric ulcers, scurvy and asthma.

In this study, we observed the effectiveness of different concentration of ethanol and chloroform extracts of selected plants in reducing the mycelial growth of *Fusarium oxysporum* f. sp. *pisi*.

MATERIALS & METHODS

Preparation of plant Extracts

Fresh leaves, bark, stem and seeds of selected medicinal plants *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. were collected from the surrounding areas of Ranchi, Jharkhand and washed thoroughly under running tap water and then with sterilized distilled water.

All collected parts of plants kept for shade dry at room temperature for 20 - 25 days. After making fine powder by pestle & mortar and electric grinder, 10 g of each fine powder was used for making plant extract in 100 ml of solvent such as ethanol and chloroform. The solution was filtered through Whatman filter paper.

Test Pathogen

The test pathogen *Fusarium oxysporum* f. sp. *pisi* (ITCC No.-4814) was collected from Department of Mycology and Plant Pathology, ICAR-Indian Agricultural Research Institute, New Delhi.

Antifungal activity of plant extracts

Effect of ethanol and chloroform extracts of selected medicinal plants was evaluated by Poisoned food technique against Fungal Pathogen *Fusarium oxysporum* f. sp. *pisi.*[14] Antifungal activity of plant extracts was recorded at different concentration like 1.0 %. 2.5 %, 5.0% and 10.0% against test pathogen. Each experiment was repeated three times.

Inhibition zone and Inhibition percentage of *Fusarium oxysporum* f. sp. *pisi* was calculated by using formula (Vincent, 1947): I=[C-T)/C] X 100, where, I = Percent growth inhibition, C = Growth in control plates, T = Treated plates growth.

RESULT & OBSERVATION

Effects of extracts of different parts of selected medicinal plants – *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. in different solvents like ethanol and chloroform were recorded in Table 1 and Table 2. Mycelial growth of test pathogen *Fusarium oxysporum* f. sp. *pisi.* was inhibited by plant extracts.

For controlling of *Fusarium* wilt disease of pea crop, the antifungal activity of ethanol and chloroform extracts of selected medicinal plants at different concentration was tested. It was recorded that at 10% concentration of ethanol extracts of selected plants, almost all plant extracts were effective in reducing the mycelial growth of *Fusarium oxysporum* f. sp. *pisi* (Table 1).

Inhibition zone of *Fusarium oxysporum* f. sp. *pisi* in different extracts of plants was recorded in Table. Maximum inhibition zone of test pathogen was observed at 10% concentration of seed extract of *Millettia pinnata* L. with 65 mm, followed by bark of *Millettia pinnata* L. with 61.7 mm. Ethanol extracts of *Cassia tora* L. showed significant inhibition of *Fusarium oxysporum* f. sp. *pisi* with the range

of 55 mm to 64 mm. Selected plant extracts of *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. also inhibited the mycelial growth of test pathogen. Bark extract of *Moringa oleifera* Lam. showed 59.3 mm inhibition zone. Root extract of *Cymbopogon citratus* was least effective in compare to other plant extracts. However, leaf of *Cymbopogon citratus* Stapf. showed 57.7 mm inhibition zone.

Effects of chloroform extracts of *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. on inhibition of plant pathogenic fungi *Fusarium oxysporum* f. sp. *pisi.* were recorded in Table 2. It was observed that seed extract of *Millettia pinnata* L. at 10% effectively suppressed the mycelial growth of the *Fusarium oxysporum* f. sp. *pisi* with 70 mm

Table 1- Effect of ethanol extracts of *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. against *Fusarium oxysporum* f. sp. *pisi.*

Plant Extracts	**1%**	**2.5%**	**5%**	**10%**
***Millettia pinnata* L.**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**
a) Seed	51.7	59.3	61.7	65
b) Bark	42.3	49.7	54.7	61.7
c) Leaf	25.7	39.3	50	58
d) Stem	25.7	31.7	41.7	55.3
***Cassia tora* L.**				
a) Seed	49.3	51.3	57.7	60.7
b) Bark	44.3	48.7	60	64
c) Leaf	33	39.7	56	60.3
d) Stem	22.7	35.3	51	55.7
Cymbopogon citrates				
a) Leaf	31	37.7	49.3	57.7
b) Root	27	32.3	36.3	38.3
***Moringa oleifera* Lam.**				
a) Bark	47.7	50	55.3	59.3
b) Leaf	36	42.7	51.3	56
c) Stem	24.3	40.7	47.7	51.3
Check	0	0	0	0

Table 2- Effect of chloroform extracts of *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. against *Fusarium oxysporum* f. sp. *pisi.*

Plant Extracts	**1%**	**2.5%**	**5%**	**10%**
***Millettia pinnata* L.**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**	**Inhibition Zone (mm)**
a) Seed	38.7	47.7	57	70
b) Bark	34.3	45.3	51.3	64.3
c) Leaf	31.7	41.7	46	61
d) Stem	24.3	34.7	40.3	49
***Cassia tora* L.**				
a) Seed	33.7	40.3	55	65.7
b) Bark	26.7	37.3	46	65.7
c) Leaf	24	28.7	40	62.7
d) Stem	22.7	20	35	51.7
Cymbopogon citrates				
a) Leaf	20	29	38.7	43.3
b) Root	16.3	29.7	34.3	38
***Moringa oleifera* Lam.**				
a) Bark	41	48	52.7	64
b) Leaf	31.7	40.3	52.7	59.3
c) Stem	23	32	35.3	47.7
Check	0	0	0	0

inhibition zone. The inhibitory effect of seed or bark extracts of *Cassia tora* L. was effective in 10% concentration using chloroform with 65.7 mm inhibition zone. Antifungal activity of *Millettia pinnata* L. bark and leaf with different concentrations was found to be effective even at 1% concentration against *Fusarium oxysporum* f. sp. *pisi.* Inhibition zone was observed at 10% concentration by bark or leaf extract of *Millettia pinnata* L. with 64.3 mm and 61mm respectively. Bark extract of *Moringa oleifera* Lam. exhibited 64 mm inhibition zone. Other extracts were also inhibited the mycelial growth of *Fusarium oxysporum* f. sp. *pisi.*

Aqueous extracts of 48 plants belonging to six diûerent major groups of the plant kingdom, two commercially available botanicals and different fungicides were screened for antifungal activity against *Drechslera bicolor* causing leaf blight of bell pepper. Aqueous extracts of 48 plants belonging to six different major groups of the plant kingdom, two commercially available botanicals and diûerent fungicides were screened for antifungal activity against *Drechslera bicolor* causing leaf blight of bell pepper. Aqueous extracts of 48 plants belonging to six different major groups of the plant kingdom, two commercially available botanicals and diûerent fungicides were screened for antifungal activity against *Drechslera bicolor* causing leaf blight of bell pepper.

CONCLUSION

Antifungal activity of ethanol and chloroform extracts of selected medicinal plants are due to the presence of minerals, vitamins, essential phytochemicals, alkaloids and secondary metabolites such as flavonoids, phenolic compounds, phytosterol, saponins, polysaccharides, terpenoids and resins. Seed extract of *Millettia pinnata* L. showed maximum antifungal activity using chloroform in compare to other treatments. Ethanol and chloroform extracts of *Millettia pinnata* L., *Cassia tora* L., *Cymbopogon citratus* Stapf. and *Moringa oleifera* Lam. can be used as an alternative source for diseases management of pea crop.

ACKNOWLEDGEMENTS

The author is thankful to Dr. Ajit Kumar Jha, Senior Scientist, ICAR- RCER-FSRCHPR, Ranchi, for his continuous support and encouragement.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 237-240

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Comparative moisture content of *Cuscuta reflexa* Roxb. grown on *Bougainvillae, Vitex negundo* & *Nerium olender* in Ranchi district.

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Received : 18th January, 2023 ; Revised :18th February, 2023

Abstract- Medicinal plant plays a crucial role for the advancement of new drug. Plant material remains a significant resource to combat severe disease in the world. The present study deal with the moisture content of *C. reflexa* grown on three different host. *Cuscuta reflexa* Roxb. belong to family Convolvulaceae. It is a parasitic plant commonly known as Amarbel, Amarlatta, etc. it is used as herbal medicine to cure number of disorders. Several studies have been confirmed that the plant possess potential anti-microbial, anti-oxidant, anti-inflammatory, anti-convulsant, anti-arthritics & hair growth promoting activities.

Key words: *Cuscuta reflexa* Roxb, Parasitic plant, Moisture content, Herbal medicine, Host plant.

INTRODUCTION

The nature's garden of medicine is being exposed since ancient civilization era. Still more than three fourth (3/4) of the world population depends on medicinal plants or their extracts for life sustaining process.[1] *Cuscuta reflexa* Roxb. is also known as "Akashvel" in traditional system of medicine. It is parasitic plant. The tendrils and seeds are used as various disorders.[2] *Cuscuta reflexa* is a leafless parasitic angiosperm belonging to the family Convolvulaceae and it is directly attaches to the host plants through the haustorium.[3] The parasitic plant *Cuscuta reflexa* Roxb is locally named as Amarlatta, Amarbell in hindi, Alagjaria in khartha and Bandh by santhali.[4] It has a only glabrous stem. It is used in various ethnopharma cological uses. The moisture content is the amount of moisture in the sample given as a percentage of the sample of an original weight. Dry content is the amount of solid which are left after drying given as a percentage of the original weight.

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MATERIAL & METHODS

In the study of moisture content, *Cuscuta reflexa* (Fig.1) stem collected from three host plant i.e., *Bougainville spectabilis*, *Nerium oleander* & *Vitex negundo* in the month of January from the different area of Ranchi District. Identification of plant was done by university prof. Dr. Kunul Kandir, head of university department of botany, Ranchi University, Ranchi & with the help of flora book "The Botany of Bihar & Orissa" Vol. I-IV.[5]

The stems were separated from collected plant & washed thrice in running tap water to remove all dirt, soil or any surface contaminants. After being washed all parts were kept in cool area for drying, before being initially weighted in weighing machine. Then taking initial weight, then kept in hot air oven at 55°C-60°C temperature till it dried properly. Then the dried stems of plant were again being weighted in weighing machine. The difference

between initial weight of fresh stem plant and final weight after drying was the moisture content.[6]

The percentage of moisture content were calculated with the given formula.[6]

$$\text{Moisture content} = \frac{\text{initial weight of fresh parts of plant} - \text{final weight of dried parts of plant}}{\text{initial weight of fresh parts of plant}} \; x\; 100$$

RESULT & DISCUSSION

The comparative study on moisture content of *Cuscuta reflexa* stem grown on 3 different hosts was done in the present study. It was observed that stem of *Cuscuta reflexa* collected from *Bougainville spectabilis* has higher moisture value i.e., 79.45% (Table 1). Respectively, *Vitex negundo* has 77.10% (Table 2) & *Nerium oleander* has lowest moisture content value i.e., 76.81% (Table 3).

The investigated plant was having highly medicinal properties. Traditionally the parasitic plant parts are used to cure a number of ailments. Whole plant is used to cure mental disorders conjunctivitis, respiratory disorder, piles, ulcer, stomach problem.[4,7] The whole plant infused with oil & used as massage oil for polio & arthritis.



Fig.1: Photograph of *Cuscuta reflexa* Roxb. on different host *Bougainville spectabilis, Nerium oleander, Vitex negundo*

Table 1- Moisture content of *Cuscuta reflexa* on host *Bougainville spectabilis*

Material	Number of observations	Weight of fresh stem (W1)	Weight of dried stem (W2)	Difference (W1-W2)	Moisture content %
Cuscuta reflexa Roxb on host *Bougainville spectabilis*	1	10 gm	1.813	8.187	81.87
	2	10 gm	2.510	7.490	74.90
	3	10 gm	1.912	8.088	80.88
	4	10 gm	2.075	7.925	79.25
	5	10 gm	1.805	8.195	81.95
	6	10 gm	2.033	7.967	79.67
	7	10 gm	2.431	7.569	75.69
	8	10 gm	1.926	8.074	80.74
	9	10 gm	2.130	7.870	78.40
	10	10 gm	1.908	8.091	80.91
Total					79.456%

Average moisture content = 79.456%

Table 2- Moisture content of *Cuscuta reflexa* on host *Vitex negundo*

Material	Number of observations	Weight of fresh stem (W1)	Weight of dried stem (W2)	Difference (W1-W2)	Moisture content %
Cuscuta reflexa Roxb on host *Vitex negundo*	1	10 gm	2.018	7.982	79.82
	2	10 gm	2.568	7.432	74.32
	3	10 gm	2.13	7.87	78.7
	4	10 gm	2.524	7.476	74.76
	5	10 gm	2.526	7.474	74.74
	6	10 gm	2.01	7.99	79.9
	7	10 gm	2.406	7.594	75.94
	8	10 gm	2.028	7.972	79.72
	9	10 gm	2.46	7.54	75.4
	10	10 gm	2.024	7.776	77.76
Total					771.06%

Average moisture content = 77.106%

Table 3- Moisture content of *Cuscuta reflexa* on host *Nerium oleader*

Material	Number of observations	Weight of fresh stem (W1)	Weight of dried stem (W2)	Difference (W1-W2)	Moisture content %
Cuscuta reflexa Roxb on host *Nerium oleander*	1	10 gm	2.464	7.536	75.36
	2	10 gm	2.368	7.632	76.32
	3	10 gm	2.272	7.728	77.28
	4	10 gm	2.224	7.776	77.76
	5	10 gm	2.349	7.651	76.51
	6	10 gm	2.228	7.772	77.72
	7	10 gm	2.46	7.54	75.4
	8	10 gm	2.364	7.636	76.36
	9	10 gm	2.192	7.808	78.08
	10	10 gm	2.268	7.732	77.32
Total					76.811%

Average moisture content = 76.811%

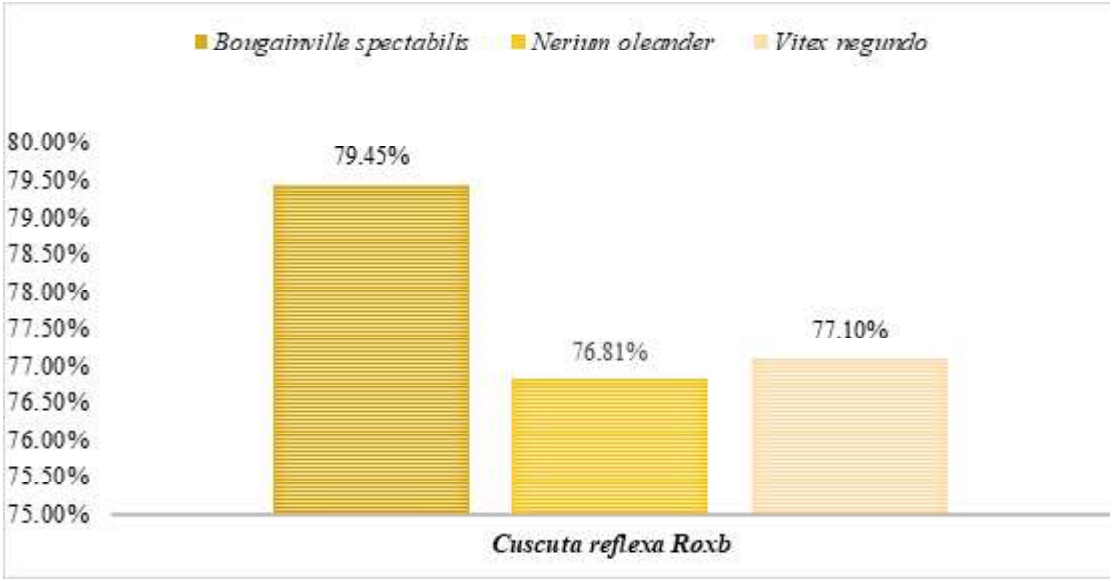


Fig.2: Comparative study on moisture content of *Cuscuta reflexa* on different host.

ACKNOWLEDGEMENT

The first author is thankful to the Head, University Department of Botany, University Professor, Dr. Kunul Kandir & her supervisor Associate Professor, Dr. Anil Kumar for their suggestions. Also, she is very grateful to her friend Ms. Riya, for her valuable knowledge, encouragement and help in completing this work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 241-250

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Assessment of vegetation cover of Palghar taluka using NDVI technique

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Received : 01st March, 2023 ; Revised :29th March, 2023

Abstract- Forest is a part of an ecosystem to maintain the ecological balance, regularize the climate, conserve the soil, etc. Natural vegetation is getting degraded over a period of time due to anthropogenic activities such as expansion of agriculture, urbanization, developmental projects etc. Over exploitation of forest resources leads to shrinking of the forest land which pose a great threat to biodiversity and environment. Thus, there is an urgent need of action to be taken on these vital issues and rapid action for its conservation. The present investigation aims to study the presence of vegetation cover in Palghar Tehsil with the help of remote sensing data and geoinformatics. The investigation applies NDVI (Normalized Difference Vegetation Index) technique to analyze density of vegetation cover using Landsat 8 RESOURCESAT data. The study depicts NDVI differences according to the density of vegetation cover in the last decade. Further the investigation assured the change in vegetation density with the implementation of change detection technique. Majority of area in Palghar Tehsil comes under Less dense vegetation in the decade. Maximum change detection is observed from moderate density vegetation area to less density vegetation area. The present study will help in the implementation and conservation of vegetation conservation strategies in Palghar region. Remote sensing technology and Geoinformatics softwares for data analysis propagate very precise real time information very accurately and are useful for vegetation assessment, management and conservation of flora and fauna.

Key words: vegetation density, FCC, satellite image and interpretation, NDVI, change detection

INTRODUCTION

Vegetation cover is the crown of vegetation to ground surface expressed as fraction or percent. It may be defined as the green vegetation area which is directly detected by the remote sensing sensor.[1] Reduction of natural vegetation cover in tropical countries over the decades of the 21st century is one of the biggest challenges leading to many environmental problems. The spatio-temporal change in vegetation cover is mainly caused by extension of land utilization due to human activities. Over exploitation has a negative impact upon the natural environment especially the vegetation cover. It is a difficult task to assess the real time spatio-temporal change in vegetation by conventional methods. Observing the earth from space is gaining importance to study the effect of human activities on the natural resources over space and time.[2,3] A rapid urbanization, industrialization and expansion of agriculture as well as developmental projects like dams and transport

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caused the disturbance of natural vegetation and habitat of a region of Palghar tehsil. The vegetation plays a vital role in sustenance and regularization of natural environment for the human survival. Thus, there is urgent need to study the vegetation characteristics and distribution patterns. The present investigation is an effort to understand the characteristics, distribution and change in vegetation cover in Palghar tehsil in last decade with the support of remote sensing data. Mapping of earth surface, managing natural resources and studying environmental change has become easier, precise and real-time analysis with the advent of Geospatial Technology especially the third eye like remote sensing. In the recent times, Geospatial Technologies can significantly assist in data collection and analysis of natural resources at low cost and timely manner.[2,3] The research is helpful to the researchers, forest officers and government to identify the problems related to natural environment and plan and implement the programs and policies for the improvement and recovery of natural forest resources. The present study is limited to Palghar tehsil which is influenced by industrialization and urban expansion as well as change in agricultural patterns having impact on natural environment due to which the investigator felt a need of investigation related to the change in vegetation cover.

REVIEW OF LITERATURE

The review of related literature gives an insight to understand the spatiotemporal assessment of vegetation cover using NDVI in Vasai tehsil of Palghar district in research article by Dongre *et al.* (2019)[4]. Further, NDVI and vegetation change detection using remote sensing and GIS - A case study of Vellore district is research article published by Gandhi *et al.* (2015)[5]. Change detection of vegetation cover by NDVI technique on catchment area of the Panchet Hill Dam, India recommends the use of remote sensing for vegetation cover is published by Sumanta *et al.* (2016)[6]. Assessment of spatiotemporal changes in vegetation cover using NDVI in SAGA and QGIS - tutorial credits to Dr. Shital Shukla, Mr. Yogesh Pawar, Ms. Jolly Desai, Ms. Manisha Patil and Mr. S. R. Patil (https://dst-iget.in/) recommends NDVI products and methods for analysis of vegetation existence and density in the area.[7]

AIMS & OBJECTIVES

Assessment of Vegetation Cover of Palghar Taluka Using NDVI Technique:

- To identify the areas of vegetation cover from remote sensing data.
- To assess the vegetation density by using NDVI technique in Palghar Tehsil.
- To evaluate the change in density of vegetation cover in Palghar Tehsil.

MATERIALS & METHODS

Landsat 8 multi spectral remote sensing data with spatial resolution 30 m was used to assess the spatiotemporal change of vegetation cover in the Palghar Tehsil downloaded from earthexplorer.usgs.gov.[8] The downloaded images with Band 3 (Visible green) - 0.525 to 0.600 µm; Band 4 (visible red) - 0.630 to 0.680 µm; Band 5 (Near-infrared) - 0.845 to 0.885 µm and Band 6 (Short wavelength infrared) - 1.56 to 1.66 µm were mosaicked, subset or merged and clipped in QGIS and SAGA softwares.[9-11] (Fig. 1).

FCC and true colour composites were prepared to recognize the vegetation areas in Palghar Tehsil. Colour

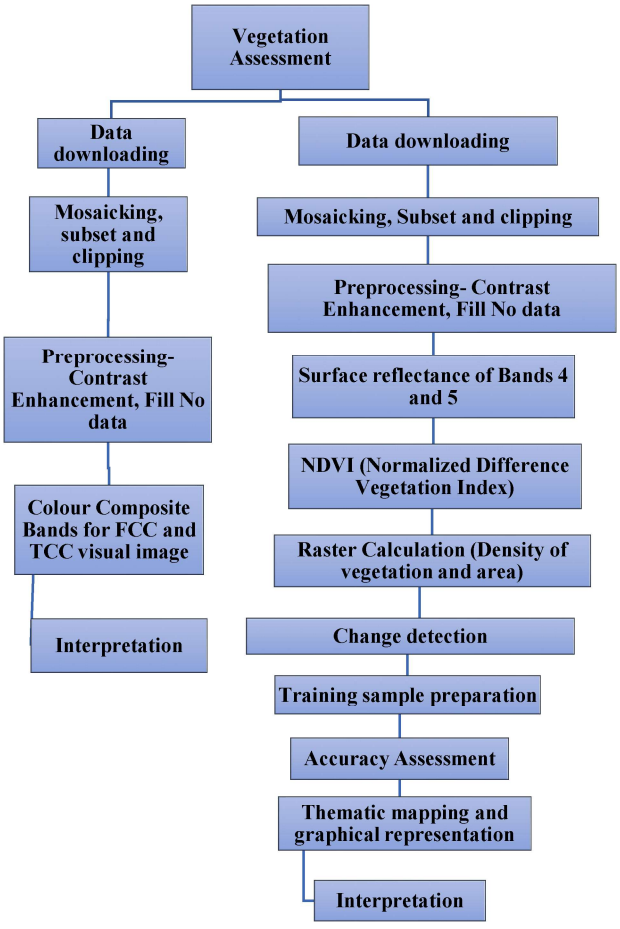


Fig. 1- Flowchart for Vegetation assessment

Infrared band combination (5, 4, 3) is also known as the near-infrared (NIR) composite. It utilizes near-infrared (5), red (4), and green (3). Because chlorophyll reflects near-infrared light, this band composition is useful for analyzing vegetation. In particular, areas in red have better vegetation health. Dark areas depict water and urban areas indicate bluish-white in standard false colour composite image.[9] In true colour composite image, the vegetation is visualized in green colour.

One of the most important biophysical indicators derived from the Satellite images is Vegetation cover. Vegetation indices are of immense help in explaining the distribution of vegetation and soil based on the characteristic of energy reflectance or absorption patterns of green vegetation.[12] The electromagnetic responses of green, healthy vegetation are within the green visible and near-infrared wavelengths. Healthy leaves of plants absorb 70-90% of incident visible radiation, particularly in the blue and red wavelengths, however, the green light is reflected back by the chlorophyll pigment within the chloroplasts in the palisade cells, hence, leaves appear green to the human eye. Blue and red light are absorbed and used in the process of photosynthesis by the chlorophyll pigment. However, wavelengths in the NIR region are mostly reflected and transmitted through the leaves; the cell interfaces in the mesophyll tissue being responsible for their scattering.[13] (Fig. 2).

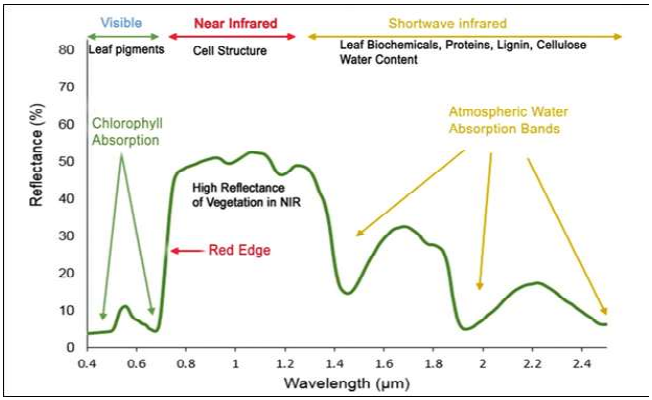


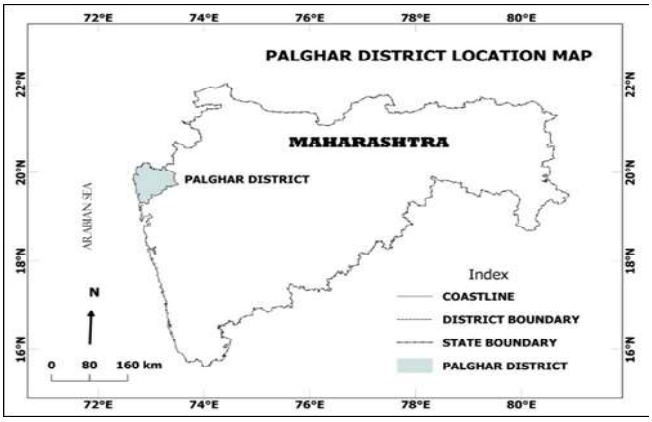
Fig 2- The spectral reflectance curve of vegetation
Source: https://www.researchgate.net/publication/315770910_Landscape_archaeology_on_the_northern_frontier_of_the_Roman_Empire_at_Poroliss vm_An_interdisciplinary_approach/link/5aa8f0000f7e9b0ea3084182/download

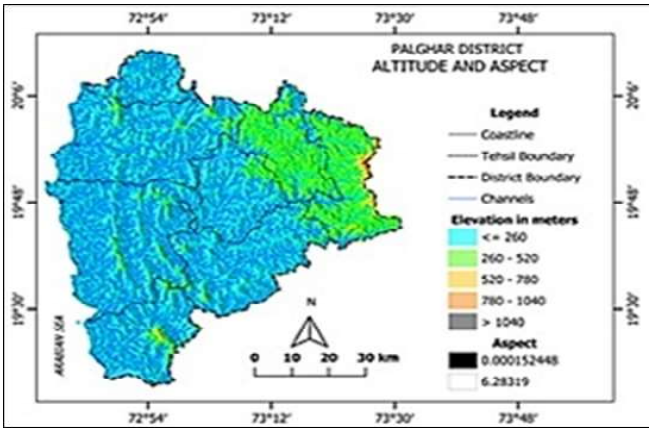
The NDVI is a simple numerical indicator to assess whether the target or object being observed contains live green vegetation or not.[14,15] NDVI is calculated by using SAGA software. It is an index of plant greenness and density existing in a particular area, calculated by formula: NDVI = (Band 5 - Band 4) / (Band 5 + Band 4).[6,16] The NDVI technique is used for extracting the vegetation characteristics presented in the Band 4 and Band 5 (Landsat 8) satellite image. Barren areas or rock, sand, or snow are represented by very low value of NDVI (0.1 and below). Moderate values indicate shrub and grassland (0.2 to 0.3), while high value corresponds to temperate and tropical rainforests (0.6 to 0.8). NDVI values closest to 0 indicate bare soil and those with negative values represent water bodies.[6,17] The degree of greenness in plants is equal to the chlorophyll concentration. The absorption of red light by chlorophyll pigment and the reflection of infrared radiation by water-filled leaf cells leads to the variation in NDVI values.[5,12,18] All visible ranges are captured by the Satellite camera in form of bands through which features can be extracted after applying the NDVI method for different characteristics. The bands are expressed in terms of wavelengths and only three visible bands are used (near infrared, visible red and visible green) in this work for the feature extraction.[12]

NDVI process needs to separate each and every band, for the detection of the vegetation index from a multi spectral remote sensing image. After the separation of different bands, NDVI method is applied according to its characteristic like vegetation at different NDVI threshold values such as below 0.1, 0.2, 0.3, 0.4 and above 0.4. Various NDVI threshold values are used to extract best result from Satellite image of Palghar taluka.[5,19]

Study Area:

Palghar tehsil is the part of the Palghar District located between Arabian coast and Western Ghats (Map 1). It experiences the seasonal hot and humid and mild climate. The district has total of 8 talukas Jawhar, Mokhada, Talasari, Vasai, Vikramgad, Palghar, Dahanu and Wada in the Konkan division of Maharashtra state in India.[20] It is in the Mumbai Metropolitan Region.[21] Forest occupies 48,827 hectares of area out of 1,07,577 hectares of geographic area of Palghar tehsil. Subtropical semi-evergreen, mangrove and deciduous forests exist in the Palghar district. The plantation farming, fodder cultivation and vegetable gardening is also popular agricultural practices in Palghar tehsil.[22]



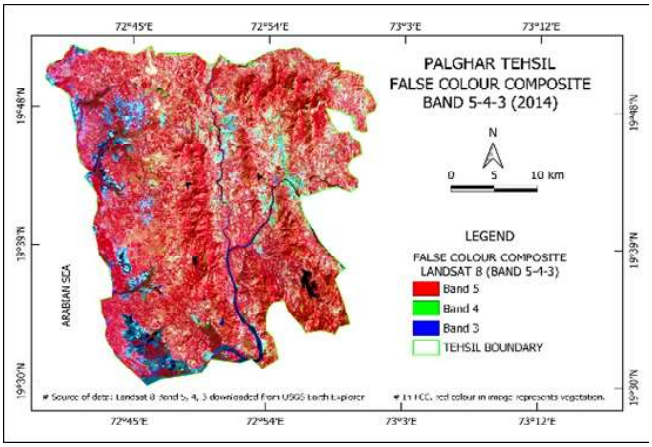


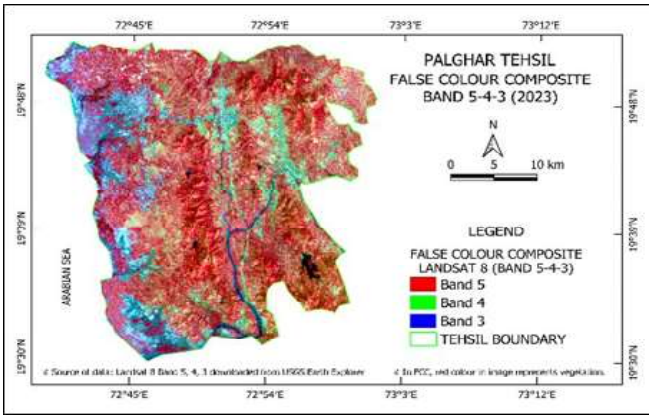
Map 1: Location of palghar district and altitude

RESULTS & DISCUSSIONS

Data Analysis and Findings:

1. False Colour Composite Images: The band combination is also called the near-infrared (NIR) composite. It uses near-infrared (5), red (4), and green (3) as chlorophyll reflects near-infrared light. This band composition is useful for analyzing vegetation as red indicates vegetation, dark blue or black represents water bodies and buildup areas are whitish-blue. The LULC (Land Use Land Cover) identification is possible from the images.



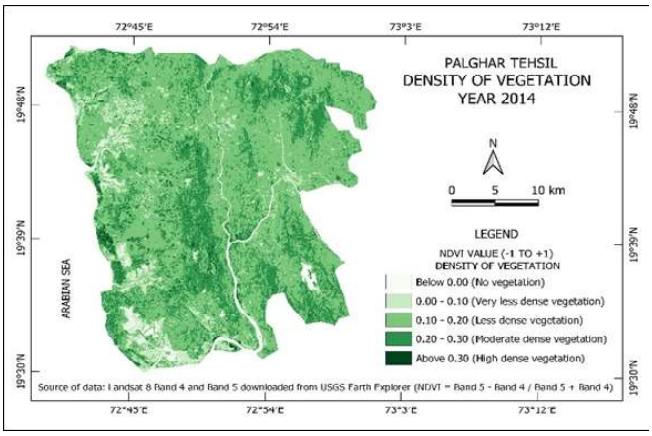


Map 2: FCC images

Visual image interpretation helps to identify the vegetation areas in Palghar tehsil. The FCC image of Palghar tehsil having larger area covered by red compared to other colors means that larger area is under forest cover especially hilly areas and slope land of Asawa fort, Kokaner, Shelwadi, Chahade etc. Coastal plains and river bank areas are under whitish-blue means either buildup urban or rural areas along with transport network especially Palghar town and areas along the western railway and western express highway as well as Surya River. Water bodies in dark blue or black are also present in Palghar tehsil as a river, creeks and dams or lakes. Few patches of barren land or waste land is also observed in Palghar tehsil. Coastal wetlands and low-lying areas are also observed at Kharekuran and Satpati in Palghar tehsil. The comparative observation of FCC images of the year 2014 and 2023 shows that it is very much changed over a period of time. The red colour areas are reduced and whitish-blue colour increased along transport route and surroundings of Palghar town indicating that area under forest or vegetation is reduced due to expansion of urbanization or increasing the size of rural settlements (Map 2).

2. Normalized Difference Vegetation Index (NDVI): The vegetation index is the properties of the red (which vegetation absorbs) and near-infrared bands (which vegetation strongly reflects) to monitor vegetation health and vigor. NDVI always ranges from -1 to +1. Negative values are indicative of water and moisture. Low NDVI values represent barren land or buildup areas and high NDVI values suggest a dense vegetation canopy. The Density of vegetation is relative to low, moderate and high depend on the scale of highest to lowest greenery in the study area. In the present study, the areas having greater than 0.3 NDVI values are considered as high dense

vegetation areas, 0.2 to 0.3 NDVI values indicate moderate dense and below 0.2 NDVI values are considered as a less dense vegetation areas.



Map 3: Palghar Tehsil: Density of vegetation - Year 2014

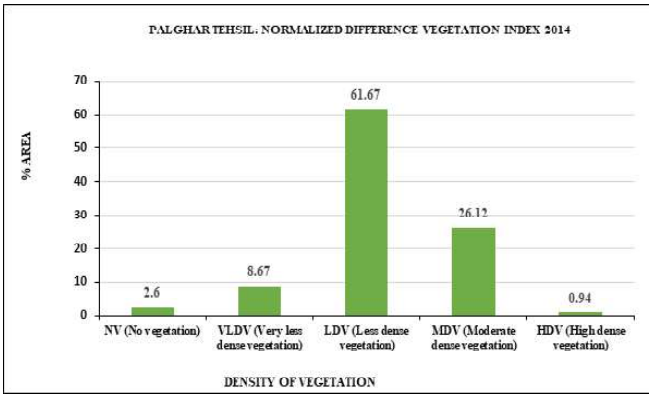
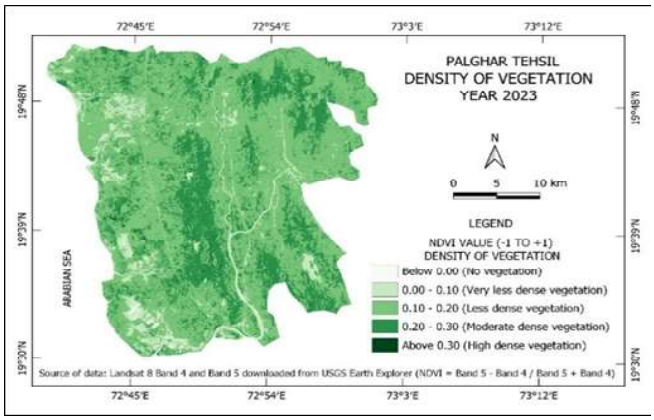


Fig 3- Palghar Tehsil: Normalized Difference Vegetation Index 2014

Table 1- Palghar Tehsil: Normalized Difference Vegetation Index 2014

Sl. No	Class	NDVI Min	NDVI Max	Area in Sq. Km.	Percentage Area
1	NV (No vegetation)	-0.17	0.00	18.53	2.60
2	VLDV (Very less dense vegetation)	0.00	0.10	61.88	8.67
3	LDV (Less dense vegetation)	0.10	0.20	440.26	61.67
4	MDV (Moderate dense vegetation)	0.20	0.30	186.47	26.12
5	HDV (High dense vegetation)	0.30	1.00	6.71	0.94

The area under high dense vegetation is 6.71 sq. km which is around 0.94 % of total area in 2014. In map no. 3, we could observe 8 to 10 patches of high dense vegetation, most probably in the hilly and protected or reserved forest areas of villages in the year 2014. The area under moderate dense vegetation cover is 186.47 sq. km. which is around 26.12 %. The areas with moderate dense vegetation are more or less located at the foothill area which is moderate slope area, represented by NDVI value 0.2 to 0.3. The area under less dense vegetation is 440.26 sq. km which is approximately 61.67 % and that of under very less dense vegetation cover is 61.88 sq. km. which is approximately 8.67 %. The areas of very less dense vegetation cover are found along the coastal plains due to the settlements, industries, transport and agriculture, NDVI value represented between 0 to 0.1. The area under no vegetation is 18.53 sq. km which is approximately 2.60 % (Map 3, Fig. 3, Table 1).



Map 4: Palghar Tehsil: Density of vegetation - Year 2023

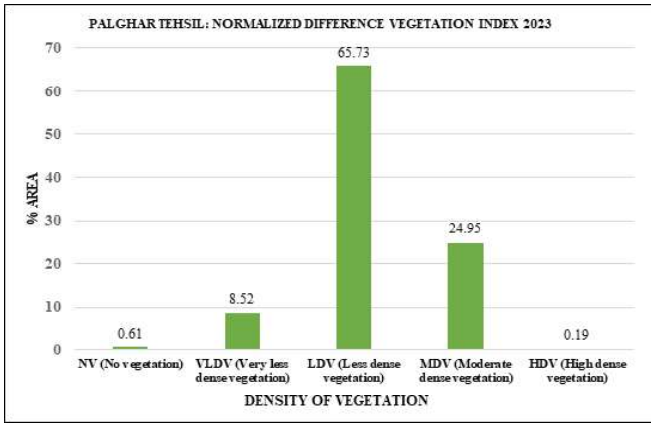


Fig. 4- Palghar Tehsil: Normalized Difference Vegetation Index 2023

Table 2- Palghar Tehsil: Normalized Difference Vegetation Index 2023

Sl. No	Class	NDVI Min	NDVI Max	Area in Sq. Km.	Percentage Area
1	NV (No vegetation)	-0.10	0.00	4.35	0.61
2	VLDV (Very less dense vegetation)	0.00	0.10	60.80	8.52
3	LDV (Less dense vegetation)	0.10	0.20	469.25	65.73
4	MDV (Moderate dense vegetation)	0.20	0.30	178.09	24.95
5	HDV (High dense vegetation)	0.30	1.00	1.36	0.19

The area under very high dense vegetation is 1.36 sq. km which is around 0.19 % of total area in 2023. We could observe 6 to 8 patches of high dense vegetation, most probably in the hilly and protected or reserved forest areas of villages like Asawa, Kokaner, Shelwadi, Chahade etc. which are represented by NDVI value greater than 0.3. The area under moderate dense vegetation cover is 178.09 sq. km. which is around 24.95 %. The areas with moderate dense vegetation are more or less located at the foothill area which is moderate slope area, represented by NDVI value from 0.2 to 0.3. The area under less dense vegetation cover is 469.25 sq. km which is approximately 65.73 % and that of very less dense vegetation cover is 60.80 sq. km. which is approximately 8.52 %. The areas of very less dense vegetation cover are found along the coastal plains due to the settlements, industries, transport and agriculture; represented by NDVI value between 0 to 0.1. The area under no vegetation cover is 4.35 sq. km which is around 0.61 % (Map 4, Fig. 4, Table 2).

Very negligible change in very less dense vegetation area is found from the year 2014 to 2023. Less dense vegetation area increased from 61.67 % to 65.73 %. Moderate dense vegetation area decreased at a considerable extent from 26.12 % to 24.95 %. HDV also decreased from 0.94 % to 0.19 %. Only hilly area patches are left with high dense vegetation (Table 3, Fig. 5).

Table 3- Palghar Tehsil: Normalized Difference Vegetation Index 2014 and 2023

Type of vegetation	% Area in 2014	% Area in 2023
NV (No vegetation)	2.60	0.61
VLDV (Very less dense vegetation)	8.67	8.52
LDV (Less dense vegetation)	61.67	65.73
MDV (Moderate dense vegetation)	26.12	24.95
HDV (High dense vegetation)	0.94	0.19

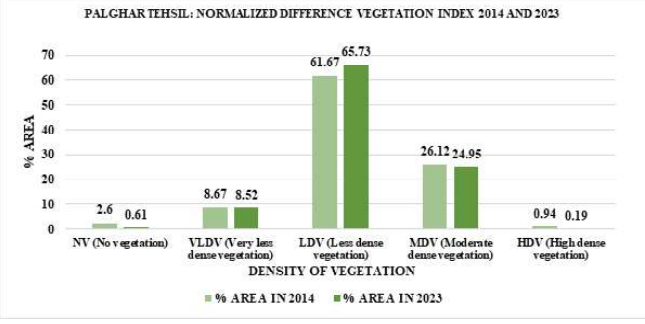


Fig. 5- Palghar Tehsil: Normalized Difference Vegetation Index 2014 and 2023

High dense vegetation areas decreased due to deforestation which is caused by growth of population, rise in built up areas, agricultural expansion and developmental projects like roads, railways, dams, etc. The investigation shows that the vegetation cover gets reduced over a period of time very effectively assessed by using remote sensing data and compilations and analysis with the support of geospatial technology.

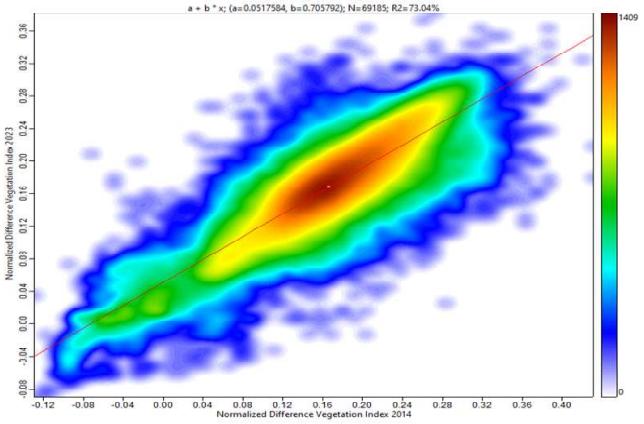
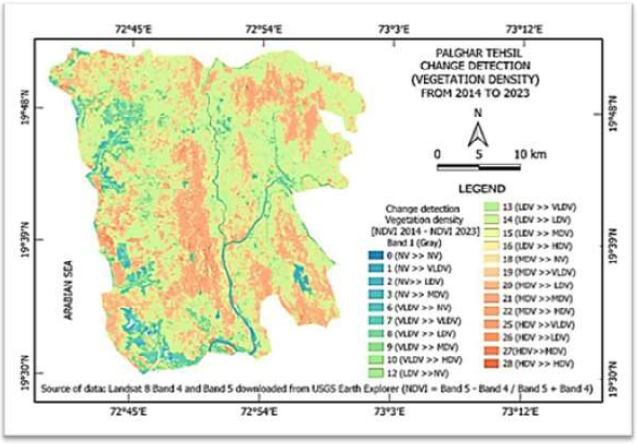


Fig. 6- Palghar Tehsil: Scatterplot (Raster image of NDVI)

The scatterplot is a technique to find the linear relationship of two raster images indicating correlation of DN values of image pixels describing feature (vegetation) characteristics. The scatterplot represents that 73.04 % DN values of NDVI image pixels are correlating with the NDVI classes and the remaining do not correlate. It means only 26.96 % area of NDVI classes changed from high dense to low dense vegetation areas or vice versa (Fig. 6).

3. Change Detection: Change detection is an innovative application of remote sensing image interpretation. It aids to compare and analyze two (or more) remote sensing images taken at different times of the same region and obtain spatio-temporal change information about the ground object especially the LULC.[23] The vegetation change is an important parameter and helpful to the researcher and planners for the conservation of green areas and biodiversity. We can derive change detection using the classified NDVI values of different time period images.

54.25 % of less dense vegetation area, 18.33 % of moderate dense vegetation area and 4.92 % of very less dense vegetation area remained same from 2014 to 2023. 7.67 % of moderate dense vegetation area was converted to less dense vegetation area. 5.92% and 1.45% of less dense vegetation area was converted to moderate dense vegetation and very less dense vegetation areas respectively.



Map 5: Palghar tehsil- Change Detection (Vegetation Density) from 2014 to 2023

Table 4- Palghar Tehsil- Change Detection (Vegetation Density) From 2014 To 2023

Change category	% Change
NV>>NV	0.5040
NV>>VLDV	2.0583
NV>> LDV	0.0287
NV>>MDV	0.0007
VLDV>>NV	0.0964
VLDV>>VLDV	4.9163
VLDV>>LDV	3.5907
VLDV>>MDV	0.0732
VLDV>>HDV	0.0025
LDV>>NV	0.0054
LDV>>VLDV	1.4533
LDV>>LDV	54.2491
LDV>>MDV	5.9213
LDV>>HDV	0.0271
MDV>>NV	0.0014
MDV>>VLDV	0.0737
MDV>>LDV	7.6704
MDV>>MDV	18.3274
MDV>>HDV	0.0610
HDV>>VLDV	0.0022
HDV>>LDV	0.1967
HDV>>MDV	0.6401
HDV>>HDV	0.0999

3.59 % very less dense vegetation area was converted to less dense vegetation area while, 2.06 % of no vegetation area was converted to very less dense vegetation area (Map 5, Fig. 7, Table 4).

Accuracy Assessment:

The accuracy assessment of raster data of NDVI was carried out with the help of training sample areas by observing the topographical maps, Google images and Google Earth Pro, FCC images and NDVI Bhuvan map as well as field survey for high dense, moderate dense and less dense vegetation areas by using the semi-automatic

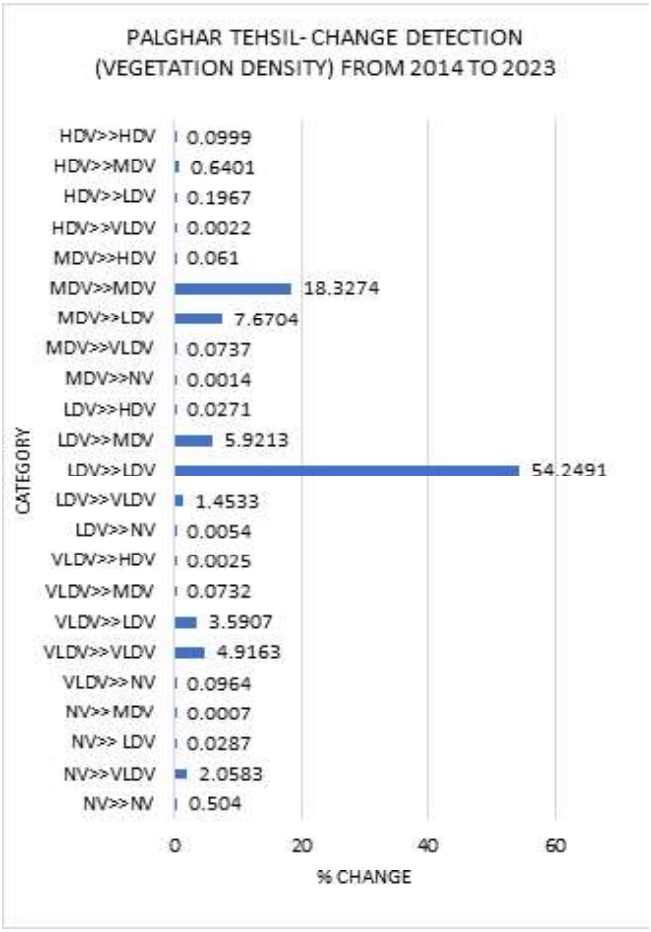


Fig. 7- Palghar Tehsil- Change Detection (Vegetation Density) From 2014 To 2023

classification plugin in QGIS software. The value of accuracy assessment is 82.83 which means approximately 83 % area of actual assessment from images and training samples correlated. It indicates high probability of correct assessment. Also, the accuracy assessment carried out during change detection (Confusion Matrix) as Kappa coefficient which is 0.78 and Overall Accuracy is 0.85 in SAGA software and that too depicts the high probability of correct assessment.

FINDINGS & CONCLUSIONS

- FCC images are useful to identify the LULC and detect the vegetation areas which are represented by red colour in standard FCC image. Few patches of hilly areas and fruit farming areas of coastal plains are covered by dense vegetation. The area around the Palghar town and along the rivers and transport lines is reduced in the vegetation cover and increased in the buildup area.

- The areas under high dense vegetation situated most probably in the hilly and protected or reserved forest areas are represented by NDVI value of greater than 0.3 in 2014 and 2023. The areas with moderate dense vegetation are more or less located at the foothill area which is moderate slope area and represented by NDVI value 0.2 to 0.3. The areas of very less dense vegetation cover are found along the coastal plains due to the settlements, industries, transport and agriculture which are represented by NDVI value between 0 to 0.1. Maximum area is covered by less dense vegetation and minimum or negligible area is under very dense vegetation in Palghar tehsil.
- Area under very less dense vegetation decreased due to the growth of settlements, road, railways, etc. from 2014 to 2023. However, area under less dense vegetation increased while moderate dense vegetation area decreased at a considerable extent. As agricultural and industrial activities increased intensively, high dense vegetation area converts into moderate dense vegetation and less dense vegetation areas. Hence, high dense vegetation area has decreased. Only hilly area patches are left with high dense vegetation area which decreased due to deforestation which is caused by growth of population, rise in built up areas, agricultural expansion and developmental projects like roads, railways, dams, etc.
- Very less dense vegetation areas, less dense vegetation areas and moderate dense vegetation areas remained same from 2014 to 2023. Maximum change detection was observed from moderate dense vegetation area to less dense vegetation area in the same period.
- Industrialization, urban growth, expansion of agricultural land and change in agricultural practices as well as development of transport, dams and other assets are responsible for reduction of forest land in Palghar tehsil. Vegetable Gardening, Plantation Agriculture and Fodder Cultivation are observed as positive signs to increase the area under vegetation in coastal plains. Even tree plantation activities of Government and NGOs are also responsible to increase in vegetation in limited pockets of Palghar tehsil. 8000 trees are planted around 16 km. along the road under the social forestry programme.[22] Agroforestry and conservation of Vanrai-Devrai (Sacred groves) are responsible to conserve the forest in Palghar Tehsil.[24]
- Geospatial Technology platforms helped to detect change in vegetation characteristics of the Palghar tehsil very precisely.

Suggestions for future research:

Afforestation, efficient use of resources, implementation of environmental laws and public awareness on biodiversity conservation should be undertaken in order to conserve existing natural vegetation. Local level research should be undertaken for better understanding of density of vegetation.

ACKNOWLEDGEMENT

The authors express their sincere thanks to Principal, Parle Tilak Vidyalaya Association's Sathaye College (Autonomous), Mumbai for providing the facilities to carry out this work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 251-256

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Morphological characterization and fertility of pollens of two economically important seed species from Apiaceae

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Received : 12th February, 2023 ; Revised :14th March, 2023

Abstract- Pollen morphology is a disciplinary area with the primary objective to describe the pollen grains by their characteristics that also defines different plant groups. On the other hand, pollen fertility indicates the pollen viability. This present work deals with study of pollen morphology and pollen fertility on two varieties of each, of *Foeniculum vulgare* Mill. and *Anethum graveolens* L.A. sowa., two very important economical crops belong to the family Apiaceae. The investigation was carried out under the light microscope with the aid of oculo-micrometre and stage micrometre.

Key words: Pollen grains, Pollen morphology, Pollen fertility, Apiaceae, *Anethum, Foeniculum*.

INTRODUCTION

The family Apiaceae is the 16th largest family of the flowering plants, comprising more than 3600-3780 species under more than 455 genera worldwide.[1] Several ethno-medicinally important horticultural crops belong to this aromatic family, are raised for seed spices, leaf spices and condiments. Fennel (*Foeniculum vulgare* Mill.) and Indian Dill (*Anethum graveolens* L.A. sowa) are two such economically important seed spices from this family cultivated in our country, makes us the largest producer, consumer and exporter in the world.[2] They are mostly grown in rabi season in arid and semi-arid part of India.[3,4] Both the plants are herbaceous, featured with compound umbel inflorescence, presents small yellow, radially symmetrical, hermaphrodite flowers arranged in umbels.[5,6] Our Indian Dill (*Anethum graveolens* L.A. sowa) popularly known as sowa due to the presence of dillapiole and is the most valuable medicinal herb of Apiaceae.[7] Rather than a commercial seed spice, it is used as a antispasmodic and carminative in pharmaceutical industry.[8] Where Fennel (*Foeniculum vulgare* Mill.) is considered one of the oldest medicinal plant in the world.[9] It is valuable for the different ether-essential oils and useful in disease related to chest, kidney, spleen and digestive system. It is also found that fennel is even used to stimulate lactation[10].

A major role of the pollens of these two plant species found in melissopalynology for the production of multiflora honeys, word widely, including India. Now, microspore or pollen grain is the first cell of male gametophyte of spermatophytes.[11] Pollens release after their maturation from anthers. Taxonomists use pollen grain as one of the important tools to generate different phylogenetic

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information about the species. Not only, here, in other disciplines like paleobotany, forensic, cytogenetics, medicine, knowledge of pollen plays crucial role.[12] The pollen grains show case a great variation of morphological characteristics, established by all genetic heredity and generally do not vary upon the environmental changes or events. Thus, they are very defined by their own way in each genera to species of different plant groups. Only the production can be increased or reduced depending on the environmental conditions.

OBJECTIVE OF RESEARCH

Study of pollen morphological characteristics of two varieties of Fennel (*Foeniculum vulgare*)-Ajmer Fennel-(AF-1), Ajmer Fennel-2 (AF-2) and two varieties of Indian Dill (*Anethum graveolens* L.A. sowa) Ajmer Dill-1 (AD-1), Ajmer Dill-2 (AD-2). Along with, the estimation of pollen fertility by stainability test of concerned four materials.

MATERIALS & METHODS

Plant Material: -The seeds of plants were brought from ICAR-National Research Centre on Seed Spices, Ajmer, Rajasthan, India. They were sown in pots and grown in Purulia, West Bengal, India.

Morphological Analysis and Viability Test: - A regular method to study the morphological characteristics and assess the pollen viability is by staining and direct count.[13] For this purpose, anthers were collected from open flowers at full flowering stage from the plants.[14] A cytological dye aceto-orcein was used to stain the fresh pollen grains.[15] Firstly, the anthers were taken, on a watch glass and warmed in 2% aceto-orcein for 2-4 minutes. Then on a clean slide, anthers were placed, teased with needle to uniform distribution of pollen grains and discard the debris. A drop of 45% acetic acid was added to the slide, warmed, covered with coverslip and squashed.

Slides were examined under light microscope, equipped with DSLR camera and photographs were taken at 40X power. Morphological observation was done and measurements of 10 pollens of each variety were calculated with the help of ocular and stage micrometre. To determine pollen fertility at least 200 pollen grains of each variety were analysed. Pollens which are fully expanded, with intact exine and strongly coloured in red considered as fertile or viable, where those were abnormal in shape and remain unstained or very slightly stained were sterile or non-viable.[16,17]

RESULTS

Pollen morphology: - The various pollen morphological characters and measurements of different parameters of four materials have been tabulated in Table-1. All the pollen grains of AF-1, AF-2, AD-1 and AD-2 were exclusively isopolar and radially symmetrical. They were all monads, medium in size[18], 3-zonocolporate[19], prolate[20] and erectus. The polar caps were obtuse type. In case of shape, pollens of AF-1 and AF-2 were elliptical to slightly bone shaped, where pollens of AD-1 were clearly elliptical and AD-2 pollens were bone-shaped.

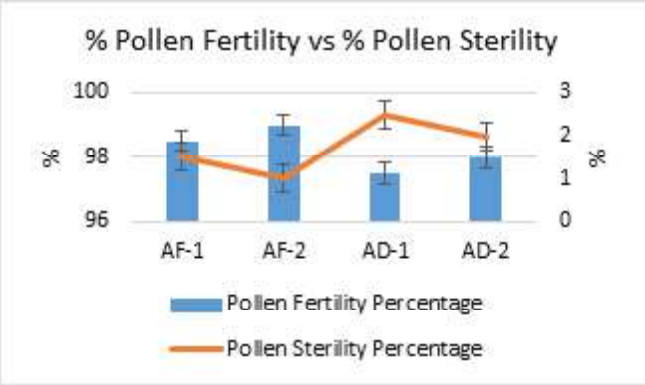
The polar diameter ranges from 29.25±0.6 µm to 35.37±0.67 µm. The longest pollen was observed in AF-2 while the smallest in AD-1. The equatorial diameter ranges from 18.33±0.56 µm to 20.12±0.39 µm. The highest equatorial diameter was found in AD-2 where the lowest was in AF-1. The P/E ratio of all pollen grains were showing ranges from 1.6±0.06 µm to 1.885±0.05 µm. That was below 2.0. At the polar view of the pollen of AD-2 was almost triangular with apertures inside.

Table 1- Pollen morphological measurements and characteristics of two varieties of Fennel–Ajmer Fennel-1 (AF-1), Ajmer Fennel-2 (AF-2) and two varieties of Dill – Ajmer Dill-1 (AD-1), Ajmer Dill-2(AD-2).

Species	Variety	Polar Diameter (P) (µm)	Equatorial Diameter (E) (µm)	P/E	Pollen Shape		Size	Unit	Pollen Class	Polarity	Symmetry	Outline	Polar Cap
					Equatorial View	Polar View							
***Foeniculum vulgare* Mill.**	AF-1	33.68 ±0.64	18.33 ±0.56	1.85 ±0.05 Erect	Prolate	–	Medium	Monad	3-Zonocolporate	Isopolar	Radially symmetrical	Elliptical to slightly bone shaped	Obtuse
	AF-2	35.37 ±0.67	19.42 ±0.55	1.83 ±0.06 Erect	Prolate	–	Medium	Monad	3-Zonocolporate	Isopolar	Radially symmetrical	Elliptical to slightly bone shaped	Obtuse
***Anethum graveolens* L.A. sowa**	AD-1	29.25 ±0.61	18.55 ±0.88	1.6 ±0.06 Erect	Prolate	–	Medium	Monad	3-Zonocolporate	Isopolar	Radially symmetrical	Elliptical	Obtuse
	AD-2	34.40 ±0.48	20.12 ±0.39	1.7 ±0.03 Erect	Prolate	Triangular with apertures inside	Medium	Monad	3-Zonocolporate	Isopolar	Radially symmetrical	Bone shaped	Obtuse

Table 2- Percentage of pollen fertility and sterility of two varieties of Fennel-Ajmer Fennel-1 (AF-1), Ajmer Fennel-2 (AF-2) and two varieties of Dill - Ajmer Dill-1 (AD-1), Ajmer Dill-2(AD-2).

Sl. No.	Species	Variety	Total No. of Pollen studied	Fertile Pollens	Pollen Fertility Percentage	Sterile Pollen	Pollen Sterility Percentage
1	*Foeniculum vulgare* Mill.	AF-1	200	197	98.5	3	1.5
		AF-2	200	198	99	2	1
2	*Anethum graveolens* L.A. sowa	AD-1	200	195	97.5	5	2.5
		AD-2	200	196	98	4	2



Graph 1- Percentage of pollen fertility and percentage of pollen sterility of two varieties of Fennel-Ajmer Fennel-1 (AF-1), Ajmer Fennel-2 (AF-2) and two varieties of Dill - Ajmer Dill-1 (AD-1), Ajmer Dill-2 (AD-2).

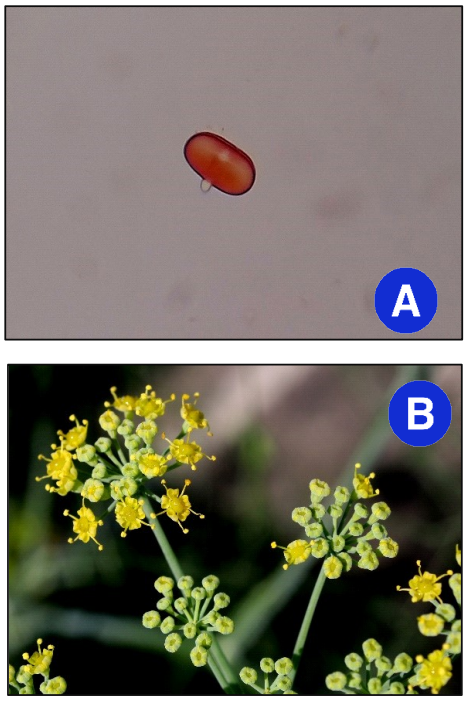


Fig. 1- *Foeniculum vulgare* Mill. Variety: Ajmer Fennel 1 (AF-1), A: Pollen grain in equatorial view (Fully stained, Germinated) B: Flowers on Umbels

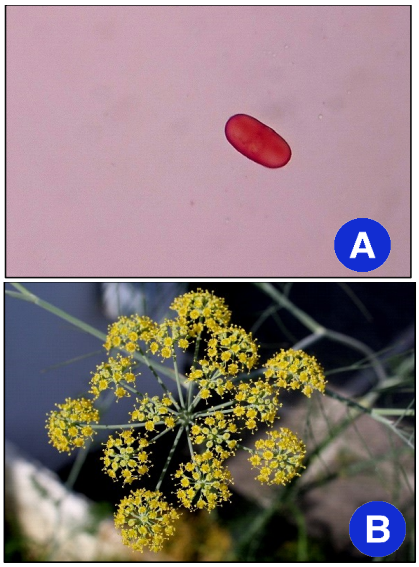


Fig. 2- *Foeniculum vulgare* Mill. Variety: Ajmer Fennel 2 (AF-2), A: Pollen grain in equatorial view (Fully stained), B: Flowers on Umbels

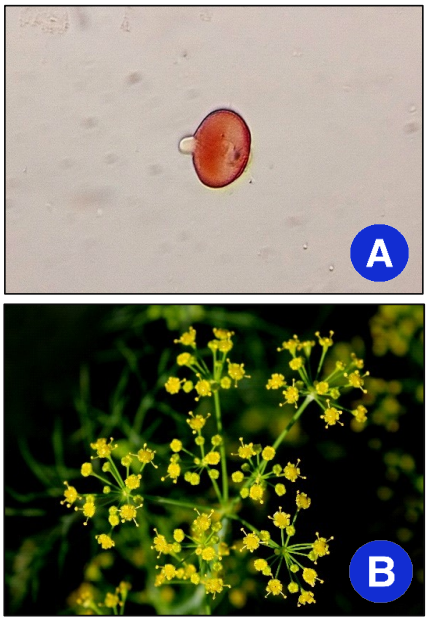


Fig. 3- *Anethum graveolens* L.A. sowa Variety: Ajmer Dill 1 (AD-1), A: Pollen grain in equatorial view (Fully stained, Germinated), B: Flowers on Umbel

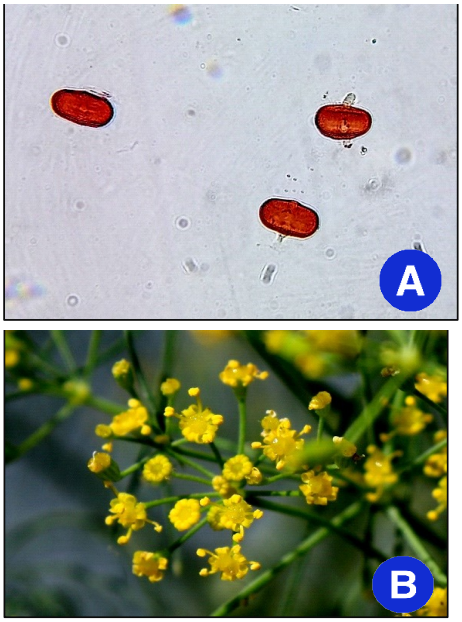


Fig. 4- ***Anethum graveolens*** **L.A. sowa Variety: Ajmer Dill 2 (AD-2), A: Pollen grain in equatorial view (Fully stained, Germinated), B: Flowers on Umbel**



Fig. 5- ***Anethum graveolens*** **L.A. sowa Variety: Ajmer Dill 2 (AD-2), A: Pollen grain in polar view**

Pollen Fertility: -The results of fertility through stainability observed in fresh pollen by aceto-orcein have shown a high pollen fertility in all four varieties. The data has been tabulated in Table -2. The highest pollen fertility percentage was shown in AF-2 i.e., 99%, where the lowest percentage of pollen fertility was found in AD-1 i.e., 97.5%. The pollen fertility- sterility percentage has been showing in Graph-1 to simplify understanding.

DISCUSSION

On the view of palynological contribution to the family Apiaceae, the most extensive and interdisciplinary studies were done by Cerceau-Larrival (1959, 1962, 1963, 1965, 1967a, 1967b, 1971, 1973, 1981). Where the study of pollen by Ertdtman (1943, 1952) and Erdtman et. al. (1961, 1963) has provided us the basic categories with enlisted features to describe a pollen grain very easily and scientifically. A glossary on terminologies of pollen grains and spores by W. Punt et. al. (2006) has become the widely accepted reference guide to the scientists to prepare accurate description for their study materials.[21]

In this investigation, we have found that the pollens of four varieties showed a greater uniformity with very slight difference in their shape and size, presents almost similar external morphology, supported the stenopalynous characteristics of pollen grains of Apiaceae.[14,20] The pollens were having longer polar axis than the equatorial axis, resulted P/E ratio always below 2.0, which reflects the erectus nature of the grains.[22] Now, the natural tendency of an elongated pollen is to lay parallel in a microscopic slide. Therefore, it has become very difficult or even tends to impossible to get the polar view.[23] Where Cerceau (1962) and Punt (1984) mentioned in their literature that polar view of Apiaceae pollens were very important for differentiation. Fortunately, during our experimental studies, we have found the polar view of the pollen of AD-2. The term viability has been defined as "having the capacity to live, grow, germinate or develop". In this experiment we have used aceto-orcein method to test the fertility of pollens. Here, it has considered, that the aceto-carmine or aceto-orcein stainability method is the simplest technique for pollen viability analysis. It is a less sensitive test. It ensures the presence of cytoplasm in the pollen.[24] In our study, even germinating pollens have been also observed confirmed that the pollens were fully viable. It has marked a high fertility percentage over sterility, gives us he idea, why our country has been leading on the production of these crops.

ACKNOWLEDGEMENT

The authors are grateful to the Director Dr. Vinay Bhardwaj and Mr. Siya Ram Meena (Assistant Chief Technical Officer) of ICAR-NRCSS Ajmer, Rajasthan, India for providing us Truthful Labelled seed materials on our asking for research purpose. Also thankful to Prof. Dr. Kunul Kandir, Head, University department of botany, Ranchi university, Ranchi for providing the laboratory facilities.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 257-264

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

In-vitro propagation of *Catharanthus roseus* and their pharmaceutical value

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Received : 30th January, 2023 ; Revised :24th February, 2023

Abstract- *In-vitro* propagation is a method using culturing explant in appropriate culture nutrient medium under sterile condition. Experimental plant *Catharanthus roseus* commonly known as Madagascar belong to the family Apocynaceae. It tropical and has been spread throughout the tropical and subtropical by human activities through its primary traditional use were for people with diabetes, and anticancer effect. It is medicinally important plant widely used for the production of anticancerous drugs by pharmaceuticals companies due to the presence of vinblastine alkaloid. Due to the over exploitation of this plant for pharmaceuticals use as raw materials. We use axillary bud; shoot tip, node and internode explants of *Catharanthus roseus*. Morphogenic effect of various concentrations of cytokinins and auxins added singly or in combination supplemented with MS medium was studied during experimental work. All the cultures were incubated at 25±26°C under 16 hour's photoperiod with light and 8 hours in dark. Somatic embryogenesis was observed on combination of 2, 4-D (1.0mg/l) + Kinetin (1.0mg/l) supplemented with MS medium from shoot apex. Effect of natural additives (compost of sand + gold sand + soil homogenate and shootex) on axillary bud and callus formation was also studied. In our study, combination of BAP (0.5mg/l) + NAA (1.0mg/l) supplemented with MS medium proved to be optimal for the production of good number of shoots from nodal explant. Different size and different positions (Basal, Middle and Distal) of nodal explants along the stem length of *Catharanthus roseus* were cultured on MS medium supplemented with BAP (3.0mg/l). Present study provides the mass production and conserved of *Catharanthus roseus* in short times, for pharmaceutical raw materials.

Key words: *In-vitro* propagation, Explant, *Catharanthus roseus*, Alkaloids, MS medium, Cytokinins, Auxins, Morphogenic.

INTRODUCTION

In-vitro propagation is a method using culturing explant in appropriate culture nutrient medium under sterile condition. Micro propagation is also used to promote germplasm storage for maintenance of disease-free stock in controlled environmental conditions[1], and long term via cryopreservation[2]. *In-vitro* propagation is a method using culturing explant in appropriate culture nutrient medium under sterile condition. Plant parts will be used in culture technique and referred to as micro-propagation. Plant tissue culture forms the back bone of Plant Biotechnology; this includes micro propagation, induction of some clone's somatic hybridization, cryopreservation and regeneration of transgenic plants Plant cell and tissue culture has already contributed significantly to crop improve and has great potential for the future. It has also a great potential for the propagation of important crops like, *Cassava* spp, *Phaseolus* spp, *Solanum* spp.[3] The micro-propagation of elite or selected plants showed good results which benefit the agriculture, horticulture and forestry.[4] *In-vitro*

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Propagation of ornamental and medicinal plants is widely used to produce active compounds for herbal and pharmaceutical industries.[5] *In-vitro Propagation* will enable rapid multiplication and sustainable use of medicinal plants for future generations. *Catharanthus roseus* (Apocynaceae) is an important medicinal plant (2n). Linnaeus, 1759 published *Catharanthus roseus* as *Vinca rosea* in his – Systema Naturae. Reichenbach, 1928 used the name Lochenra in his -Conspectus Regni Vegetabiles but without giving a description or references. G. Don, 1835 published *Catharanthus* in -General system of Gardening and Botany. *Lochnera* is a synonym of -*Catharanthus*. In this study, tissue culture propagation of *Vinca roseus* using explants from stems and leaves was investigated. Single and combined treatments of different phytohormones were added to Murashige and Skoog (MS) media.[6] Catharanthus is an important herbal medicinal plant, in Ayurveda its numerous diseases including diabetes, malaria and Hodgkin's lymphoma, heigh blood pressure, Vinca alkaloid use for cancer.[7] Antiviral, hypoglycemic, anti-inflammatory, antimicrobial, antiparasitic antimicrobial, tranquilizer and immune modulating activities through tissue culture technology.[8] In recent years, plant tissue culture represents a potential renewable source to obtain genetically pure elite population under *in-vitro* conditions.

The concept of *in vitro* plant tissue culture was first developed by a German scientist Gottlieb Haberlandt (Father of tissue culture) in 1902.

He isolated single fully differentiated individual plant cell from different plant species and culture them in nutrient medium containing glucose, peptone and knop salt solution.[9] However, those cultures did not grow further. But he concluded that every cell of the plant body is totipotent i.e. capable of giving rise to a new plant under proper nature condition. *In-vitro* propagation has now become a well-established technique for culturing and studying the physiological behavior of isolated plant organs, tissue, cells, protoplasts and even cell organelles under precisely controlled physical and chemical conditions.[10] Large scale plant tissue culture is found to be an alternative approach to traditional method of plantation as it offers a control supply of biochemical independent of plant availability.[11]

Micro-propagation has become an important tool to obtain genetically pure elites rather than having genetically different populations.

Micro-propagation is now a well-established technique commercialized globally for the rapid production of a number of commercially important plants. Moreover, the plant multiplication can continue throughout the year irrespective of season and the stocks of germplasm can be maintained for many years.[12]

Catharanthus roseus (L) G. Don (2n = 16) is an important medicinal plant of family Apocynaceae which contains a virtual cornucopia of useful alkaloids used in different diseases. It is a perennial evergreen herbaceous plant growing to 1 m tall. *Catharanthus roseus* has more than 130 know alkaloids known as Vinca alkaloids, which are used in cancer, menstrual problems, asthma, diabetes, high blood pressure, and constipation.[13] Recently, Vinblastine and Vincristine alkaloids of *Catharanthus roseus* have been shown to be effective in the treatment of various dreaded diseases like childhood leukemia, skin cancer, malignant lymphoma, breast cancer, Hodgkin's disease, Wilm's disease, choriocarcinoma, Kaposi sarcoma, neuroblastoma, mycosis, fungoides and cardio vascular maladies.

It was also used as folk medicine in the ancient period. In India, the juice of leaves is used as application to bee sting/wasp sting. As home-made remedies, it was used to ease prolong congestion, inflammation, sore throats and carribean. An extract from the flowers was used to make a solution to treat eye irritation and injections. It has been used as a poultice to stop bleeding. It is used as an astringent and diuretic.

MATERIAL & METHODS

Plant material and culture conditions:

In this Research, healthy and elite axillary buds, shoot apex, nodal, and intermodal explants (3-4cm long) of *Catharanthus roseus* were obtained from campus of Jai Prakash Vishwavidyalaya Chapra (Saran) Bihar, India. Young branches of cut and use as explants were to sterilization procedures washing in running tap water.[14] They were further washed in 1% detergent for 1-2 Minutes and then washed in tap water for 6-8 times. Finally, 2 washing were done using sterilized distilled water and then explants were taken for culture.

Medium and Culture Conditions:

Murashige and Murashige and Skoog (MS) medium containing 0.8% agar and sucrose 30gm were used during the experiment. pH of the medium was adjusted to 5.8.

Growth supplemented with various concentrations. Plants Growth Regulators used were: 2,4-D, Kinetin, NAA, IBP and BAP in different combinations. The experiments were conducted in a completely randomized design. Thirty-five replicates were used and repeated three times. All the treatments were statistically analyzed by Duncan's Multiple Range Test (DMRTS).

RESULTS & DISCUSSION

For the present study, different explants (Nodal, Internodal portion and Shoot tip) of experimental plant *Catharanthus roseus* were taken from young plants. Explants having the meristamatic tissue were preferred for the culture experiments. Explants were cultured in the medium after following surface sterilization procedure and then incubated in growth chamber.

The sterilization procedure included- washing of explants with running tap water, soaking of explants in an aqueous solution containing antibiotic (0.03 % streptomycin), repeated washing with distilled water, treating explant with antifungal reagent (0.2% Bavistin - BASF India limited) containing solution followed by treatment with 0.01% Mercuric chloride for 1-2 minutes. Explants were than washed thoroughly with autoclaved sterilized water to remove traces of all used disinfectants and washed repeatedly (Plate 1 fig. C) *Catharanthus roseus* (L).

Effect of growth hormones on different explants of *Catharanthus roseus* L.

Explant culture: Experiments were conducted for large scale multiplication of plant in-vitro by using various explants e.g., Internodal segment, nodal portion and shoot tip of *Catharanthus roseus*. Effect of cytokinins (BAP and kinetin) and auxins (NAA, IBA, IAA and 2, 4-D) added singly was observed and recorded.

Table 1- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and Kinetin) added singly.

Hormone concentration mg/l Kinetin	Hormone concentration mg/l BAP	Culture Response (%)	No. of Shoots (mean ± SE)	Shoot length (in cm) (mean ± SE)
0.5	-	55-60	1.0± 0.35	1.11. ±0.71
1	-	60-65	1.01± 0.28	1.70±0.54
1.5	-	60-65	2.0±0.37	1.52±0.92
2	-	70-75	1.02±0.22	1.74±0.30
2.5	-	50-55	1.0±0.32	1.64±0.27
3	-	40-45	1.0±0.15	1.68±0.11
-	0.5	60-65	2.57± 0.21	1.21 ±0.24
-	1	80-85	4.67± 1.22	1.78±0.26
-	1.5	70-80	4.78±0.26	2.52±0.63
-	2	55-60	5.36±0.29	1.38±0.76
-	2.5	50-55	2.50±0.20	1.17±0.19
-	3	30-35	2.31±0.30	1.10±0.15
Hormone concentration mg/l	No. of Shoots	Shoot length (cm)	No. of Shoots (SE)	Shoot length (SE)
Kinetin 0.5	1	1.11	0.35	0.71
Kinetin 1.0	1.01	1.7	0.28	0.54
Kinetin 1.5	2	1.52	0.37	0.92
Kinetin 2.0	1.02	1.74	0.22	0.3
Kinetin 2.5	1	1.64	0.32	0.27
Kinetin 3.0	1	1.68	0.15	0.11
BAP 0.5	2.57	1.21	0.21	0.24
BAP 1.0	4.67	1.78	1.22	0.26
BAP 1.5	4.78	2.52	0.26	0.63
BAP 2.0	5.36	1.38	0.29	0.76
BAP 2.5	2.5	1.17	0.2	0.19
BAP 3.0	2.31	1.1	0.3	0.15

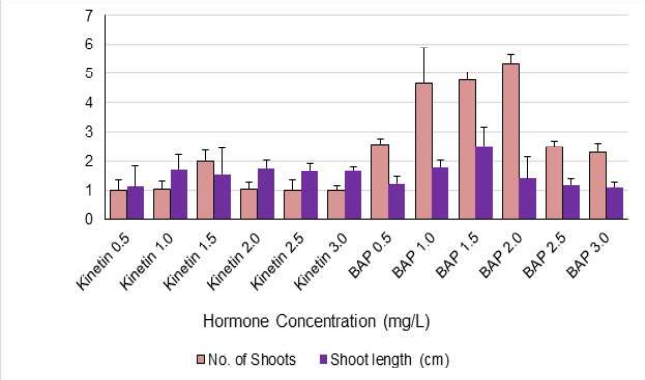
Effect of BAP

In this set of experiment, from shoot explants bud formed in BAP supplemented medium but there was no callusing observed even after 6 weeks. Good number of shoots formed on medium containing 1.0 mg/l BAP cultures. But these shoots did not grow further (Graph 1, and Table-1).

In case of nodal explants, shoot buds or very small shoots were observed on MS medium supplemented with 0.5 mg/l-3.0 mg/l BAP but on these concentrations, callus formation was totally absent. Shoot bud initiation was noticeable after 2 weeks of inoculation. Among all tried concentration of BAP (1.0 mg/l and 3.0 mg/l) proved conducive for shoot proliferation. In this experiment, BAP did not prove stimulatory effect on callusing from both the explants (internodal segment and nodal portion).

Effect of kinetin

On 0.5 mg/l – 3.0 mg/l kinetin containing MS medium cultures, callus induction was noticed from Internodal explants with shoot bud or organized structure (Graph 1, Table 1). From nodal explants, shoot emergence without callusing was noticed on all tried concentration of kinetin (1.0 mg/l – 3.0 mg/l).



Graph 1- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and Kinetin) added singly.

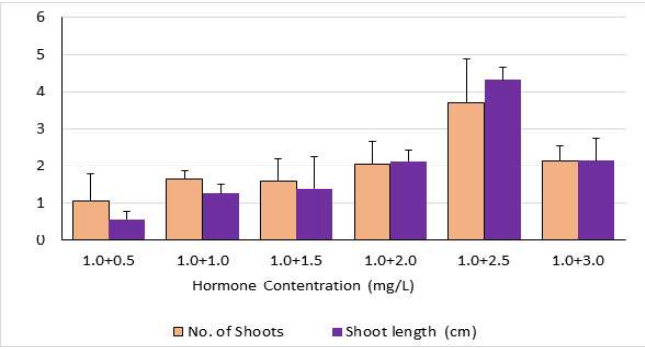
Table 2- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and 2,4-D) added singly.

Hormone concentration				
BAP+2,4-D (mg/l)	Culture Response (%)	No. of Shoots (mean ± SE)	Shoot length (in cm) (mean ±SE)	
1.0 + 0.5	55-60	1.06 ± 0.74	0.56 ± 0.21	
1.0 + 1.0	60-65	1.66 ± 0.23	1.26 ± 0.26	
1.0 + 1.5	55-60	1.60 ± 0.61	1.41 ± 0.84	
1.0 + 2.0	60-65	2.07 ± 0.61	2.14 ± 0.29	
1.0 + 2.5	60-70	3.71 ± 1.18	4.32 ± 0.34	
1.0 + 3.0	60-65	2.14 ± 0.40	2.16 ± 0.60	
BAP+2,4-D (mg/l)	No. of Shoots	Shoot length (cm)	No. of Shoots (SE)	Shoot length (SE)
1.0 + 0.5	1.06	0.56	0.74	0.21
1.0 + 1.0	1.66	1.26	0.23	0.26
1.0 + 1.5	1.60	1.41	0.61	0.84
1.0 + 2.0	2.07	2.14	0.61	0.29
1.0 + 2.5	3.71	4.32	1.18	0.34
1.0 + 3.0	2.14	2.16	0.40	0.60

Effect of 2,4-D

Prolific callusing was observed on higher concentration of 2,4-D (1.0mg/l and 3.0 mg/l) from shoot segment explants but there were no shoot buds or shoot formation noticed on any of the concentration of 2, 4-D even after 6 weeks. On low concentration of 2, 4-D that is 0.5 mg/l, callus induction was noticed but not better than higher strength of 2, 4-D tried in this set of experiment (Graph 2, Table 2). Formed callus was slow going.

From nodal explants, there was no callus formation was recorded but shoot emergence was noticed in few cultures however, they dried later. So, 2, 4-D did not prove favorable for callusing but found conducive for shoot bud formation from nodal explants.



Graph 2- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and 2,4-D) added singly.

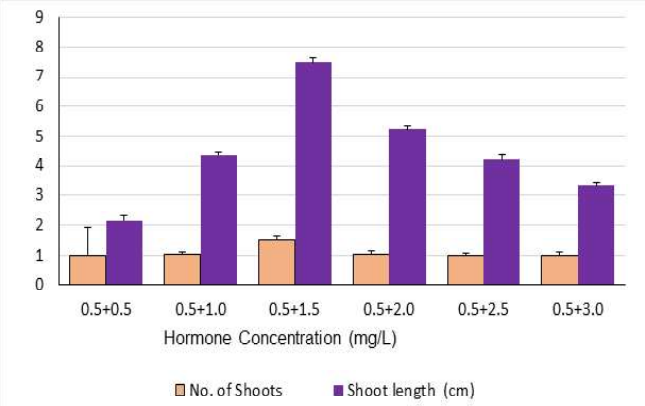
From root tip explant, there was no response observed (neither callus nor shoot formation) on all tried concentration of cytokinins and auxins used singly.

Effect of IBA

All tried concentration of IBA could not initiate callusing and shoot formation even after 2 months of inoculation from nodal explants. Very few shoots were noticed from nodal explants without callus formation (Graph 3, Table 3).

Effect of NAA

On MS medium supplemented with 0.5 mg/l - 3.0 mg/l NAA, only scanty callus induction was noticed from internodal explant with no shoot buds. Formed callus was whitish and slow growing (Graph 4, Table 4)

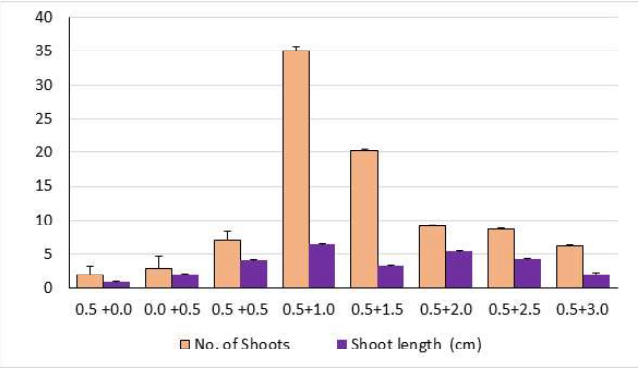


Graph 3- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and IBA) added singly.

Table 3- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium supplemented with various concentrations of cytokinins (BAP and IBA) added singly.

Hormone concentration BAP+IBA (mg/l)	Culture Response (%)	No. of Shoots (mean ± SE)	Shoot length (in cm) (mean±SE)	
0.5+0.5	20-30	1.0± 09	2.15. ±0.16	
0.5+1.0	50-60	1.01± 0.10	4.37±0.12	
0.5+1.5	60-65	1.5± 0.15	7.48±0.14	
0.5+2.0	70-75	1.02±0.11	5.25±0.13	
0.5+2.5	75-80	1.0±0.08	4.20±0.17	
0.5+3.0	80-85	1.0±0.12	3.30±0.15	
BAP+IBA (mg/l)	No. of Shoots	Shoot length (cm)	No. of Shoots (SE)	Shoot length (SE)
0.5+0.5	1	2.15	0.9	0.16
0.5+1.0	1.01	4.37	0.1	0.12
0.5+1.5	1.5	7.48	0.15	0.14
0.5+2.0	1.02	5.25	0.11	0.13
0.5+2.5	1	4.2	0.08	0.17
0.5+3.0	1	3.3	0.12	0.15

In case of nodal explants, few shoots could be noticed on all concentration Effect of 2,4-D: Prolific callusing was observed on higher concentration of 2,4-D (1.0mg/l and 3.0 mg/l) from leaf segment explants but there were no shoot buds or shoot formation noticed on any of the concentration of 2, 4-D even after 6 weeks. On low concentration of 2, 4-D that is 0.5 mg/l, callus induction was noticed but not better than higher strength of 2, 4-D tried in this set of experiment (Plate 2, figure D). Formed callus was slow going of NAA (1.0 mg/l to 3.0 mg/l) but there was no callus formation in any of the cultures even after 6 weeks.



Graph 4- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium Supplemented with various concentrations of cytokinins (BAP and NAA) added singly.

Table 4- Morphogenic Response of nodal explants of *Catharanthus roseus* L. on MS medium Supplemented with various concentrations of cytokinins (BAP and NAA) added singly.

Hormone concentration BAP+NAA (mg/l)	Culture Response (%)	No. of Shoots (mean ± SE)	Shoot length (in cm) (mean±SE)	
0.5 +0.0	10-15	2.0±1.2	1.0±0.06	
0.0 +0.5	20-25	3.0±1.75	2.10±0.07	
0.5 +0.5	60-65	7.11± 1.30	4.13 ±0.08	
0.5+1.0	80-85	35.12± 0.65	6.46±0.09	
0.5+1.5	70-75	20.24±0.24	3.32±0.11	
0.5+2.0	55-60	9.20±0.13	5.34±0.10	
0.5+2.5	50-55	8.74±0.15	4.32±0.12	
0.5+3.0	30-35	6.32±0.14	2.10±0.12	
BAP+NAA (mg/l)	No. of Shoots	Shoot length (cm)	No. of Shoots (SE)	Shoot length (SE)
0.5 +0.0	2	1	1.2	0.06
0.0 +0.5	3	2.1	1.75	0.07
0.5 +0.5	7.11	4.13	1.3	0.08
0.5+1.0	35.12	6.46	0.65	0.09
0.5+1.5	20.24	3.32	0.24	0.11
0.5+2.0	9.2	5.34	0.13	0.1
0.5+2.5	8.74	4.32	0.15	0.12
0.5+3.0	6.32	2.1	0.14	0.12

Experimental steps for *In- vitro* propagation of *Catharanthus roseus* with various concentration of plant growth hormones.



A. Healthy plant **B.** Explant **C.** Surface sterilization in soap **D.** Explant transfer



E. 1 + 2. Explant innoculate in medium **F.** Growth shows **G.** Shoot initiation



H. Shoot apex **I.** Shoot proliferation **J.** Shoot Growth **K.** Shoot and ROOT Initiation



M. Shoot growth **N.** Shoot and Root develop.

Fig:- Steps of *In-vitro* propagation of *Catharanthus roseus in* M. S. medium with various P. R. G.

Acclimatization of *In-vitro* raised plantlets of *Catharanthus roseus*.



Fig. A- Transfer in pot **Fig. B- Putting in shad place** **Fig. C- Planting for sell**

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 265-274

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Phytochemical analysis and therapeutic values of certain plant-based gums and resins found in Ranchi district

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Received : 31st January, 2023 ; Revised :28th February, 2023

Abstract- *Moringa oleifera* Lam., *Shorea robusta* Roxb. ex Gaertn.f., *Butea monosperma* (Lam.) Taub. are some of the documented plants whose excretory products are known for their potential in the indigenous systems of medicine. However, little is known about active compounds found within these plants, which could be responsible for their therapeutic potential. This paper deals with the phytochemical activity of these plants to decipher active chemical compounds present in them which will also provide credence to traditional medicine.

Key words:- Gums, Resins, *Moringa oleifera, Shorea robusta, Butea monosperma,* Phyto-chemical analysis, Therapeutic values

INTRODUCTION

For ages, plants have been a vital source of medicine. Plants are the source of many potent and powerful drugs. Different parts of the plant are traditionally used for the treatment of diverse diseases. The use of different parts of the plant like leaves, stems, roots, flowers, fruits, etc. and their excretory products such as gums, resins and gums-resins as a medicament for treatment of various conditions is well documented in literature.[1]

Gums and resins are low volume, high value product. India is one of the biggest producers of gums and resins in the world. India is a rich centre of plant biodiversity having more than 45,000 plant species including about 120 gum and resin yielding plants. In India, resins and gums are sold to earn a living by the tribes. In the present day, the uses of natural gums are numerous and they are employed by a large number of manufacturing industries including food and pharmaceutical industries. The use of gums and resins in India as medicine can be traced back to the ancient days of Acharya Charak who wrote Charak Samhita Granth.[2]

Several studies have been carried out on various plant secreting gums and resins but majority of the work has been done on *Boswellia serrata*, *Commiphora mukul*, *Gardenia resinifera* and *Shorea robusta* against some plant pathogenic fungi.[3-7]

Moringa oleifera Lam. is a deciduous-to-evergreen shrub or small tree, fast growing perennial tree which can reach a maximum height of 7 to 12 m and a diameter of 20-60 cm at chest height. The stem is normally straight that reaches a height of 1.5 - 2.0m before it begins branching and it can reach upto a height of 3.0m. Its leaves are alternate, twice or thrice pinnate leaves crowded at the end of the branches, long petiole with 8-10 pair of pinnate leaves each bearing two pairs of opposite, elliptic or obovate, rounded or emarginated, entire, dull green on both sides, at first shortly grey, pubescent, glabrous. This tree is widely

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cultivated and naturalized worldwide in the tropics and sub-tropical regions of the world.[8]

Shorea robusta Roxb. ex Gaertn.f. (Sal) belongs to family Dipterocarpacea, which is usually well-known as Shal, Sal in Hindi and Indian Dammer and Sal tree in English. It is a deciduous tree generally found in India, from Himachal to Orissa Eastern districts spreading to the Eastern Ghats of Andhra Pradesh. It is a plant drug having multi-therapeutic potential to cure a variety of human ailments.[9-13]

Butea monosperma (Lam.) Taub. commonly known as flame tree, belongs to the sub-family Caesalpinioideae, family Fabaceae (formerly Leguminosae). This plant is commonly called as Palash tree in India. It is a medium-sized deciduous tree. It grows about 10-15 meters in height. It looks like a small bush when its height is 1-2 meters due to more branching. Its flower is odourless and looks reddish in the flowering season during springs and leaves are trifoliate.[14]

Various screening has been done for antimicrobial activity of plant secreting gums and resins, but most of the studies have primarily been done on gram-positive and gram-negative bacteria.[15-18]

Similarly, various compounds have been isolated from plant secreting gums and resins in which it has been found that maximum of them possessed alkaloids, flavonoids, terpenoids, tannins, saponins and glycosides and many phenolic compounds but concluded that sesquiterpenes is present as a major constituent for the inhibition of bacterial growth.[17,19-22]

However, the studies on phytochemical constituents and medicinal properties in the gums and resins of the documented plants found in Ranchi district are scanty.

Therefore, the present study has been undertaken to deal with the phytochemical screening of the documented excretory parts of the plants such as *Moringa oleifera* Lam., *Butea monosperma* (Lam.) Taub., *Shorea robusta* Roxb. ex Gaertn.f. in different extract and also focus on their therapeutic values.

Study Area: Ranchi District of Jharkhand

Plant samples/ gums were collected from different areas of Ranchi district (Figures:1 and 2).

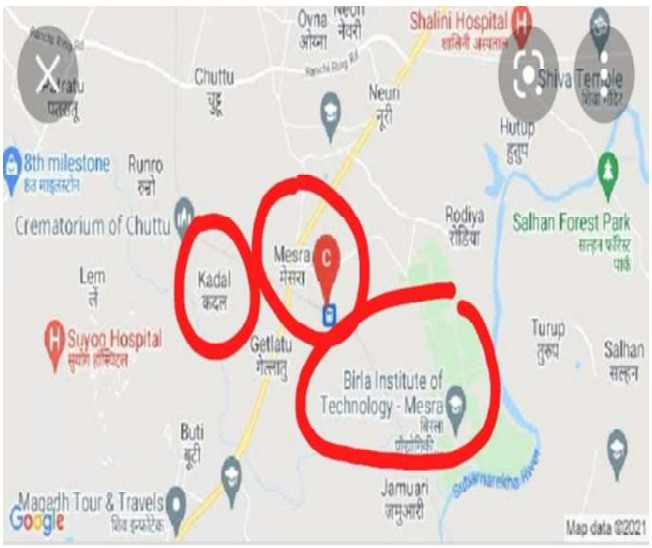




Fig. 1 & 2- Map of the different areas of Ranchi district from where plant samples have been collected.

The list of collected plant materials are as follows (Tables 1 and 2).

Table 1- List of plant-based gums and resins collected from different field of Ranchi district.

Sl. No.	Scientific Name	Common Name	Date of Collection	Place of collection
1.	***Moringa oleifera* Lam.**	Sahjan	15th Sept., 2021	Lalpur, Ranchi
2.	***Shorea robusta* Roxb. ex Gaertn.f.**	Sal, Sakhua, Sarai	18th March, 2022	Namkum, Ranchi
3.	***Butea monosperma* (Lam.) Taub.**	Palash	10th April, 2022	Pithoria, Ranchi

Table 2- Images of plants in natural condition and their collected samples.

Sl. No.	Scientific Name	Plant in Natural Condition	Collected Gums and Resins
1.	*Moringa oleifera* Lam.		
2.	*Shorea robusta* Roxb. ex Gaertn.f.		
3.	*Butea monosperma* (Lam.) Taub.		

MATERIALS & METHODS

Collection of plant materials

Plant-based gums and resins were collected from different areas of Ranchi district in different seasons on the basis of their availability. Plant twigs were also collected and brought to the laboratory for identification by using 'The Botany of Bihar and Orissa', H. H. Haines (1922)[23]. Collected plant twigs were also authenticated by the taxonomists of the University Department of Botany, Ranchi University, Ranchi and herbarium was prepared. Fully dried collected samples were stored in an air tight container.

Steps involved in the preparation of plants extracts

At first, the air dried plant samples/ gums (2 gms in each beaker) were taken and ground with the help of mortar and pestle and then the powdered materials of different samples collected were each soaked in (20 ml) three different respective solvents namely methanol, ethanol and aqueous (taken in 1:10) at 37°C in closed containers for 24 hours. The next day, the plant extracts were filtered with the help of Whatmann filter paper no.1. The filtrate was further diluted using the respective solvent used for the phytochemical screening and the analysis was carried out.

The plant extracts were screened for the presence of biologically active compounds like alkaloids, flavonoids, cardiac glycosides, phenolics, saponins, steroids, tannins, terpenoids, proteins and resins.

RESULTS & DISCUSSION

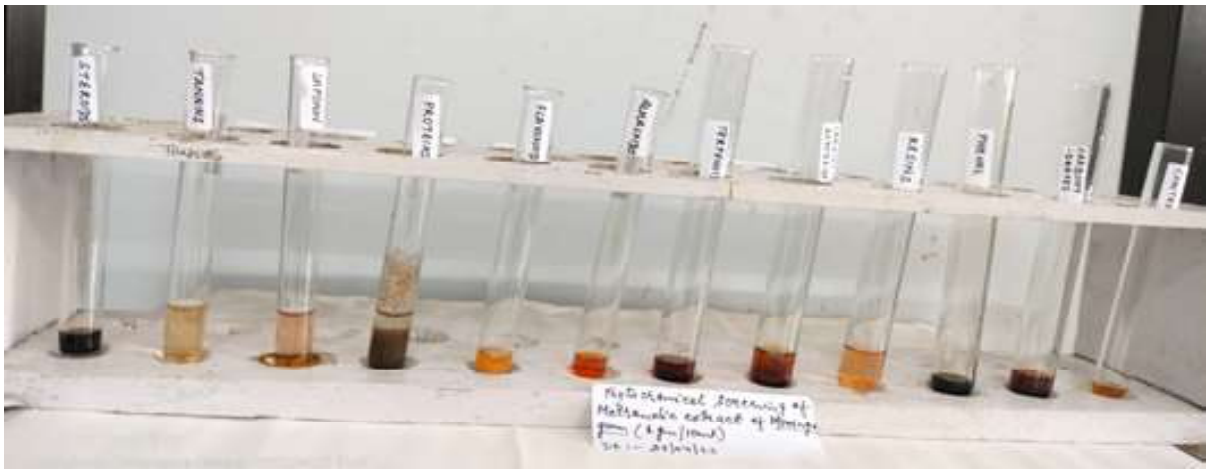
This study seeks to help the indigenous people of Jharkhand to know their therapeutic values of studied plants and also make them aware with the phytochemical analysis such as alkaloids, flavonoids, glycosides, terpenoids, steroids, tannin and proteins for the documented plants secreting gums and resins in different solvents.

Phytochemical screening of methanol, ethanol and aqueous extracts of gums of *Moringa oleifera* Lam. as found in Ranchi district suggests the presence of alkaloids, flavonoids, cardiac glycosides, terpenoids, carbohydrates, proteins, tannins, etc. (Table 3 and photographs a to c).

Table 3- Phytochemical screening of Methanol (M), Ethanol (E), Aqueous (A) extracts of gums of *Moringa oleifera* Lam. in Ranchi district.

Sl. No.	Constituents	Chemical Tests	Methanol Extract (M)	Ethanol Extract (E)	Aqueous Extract (A)
1.	Alkaloids	Dragendorff's Test	+	+	+
2.	Flavonoids	Alkaline Reagent Test	+	+	+
3.	Cardiac Glycosides	Keller Kelliani's Test	+	+	+
4.	Terpenoids	Salkowski's Test	+	+	+
5.	Carbohydrates	Mollisch's Test	+	+	+
6.	Phenol	Ferric Chloride Test	+	+	--
7.	Saponin	Foam Test	--	--	+
8.	Tannins	Bromine Water Test	+	+	+
9.	Resins	HCl Test/ $FeCl_3$ Test	--	--	--
10.	Steroids	Salkowaski Test	+	+	--
11.	Proteins	Biuret Test	+	+	+

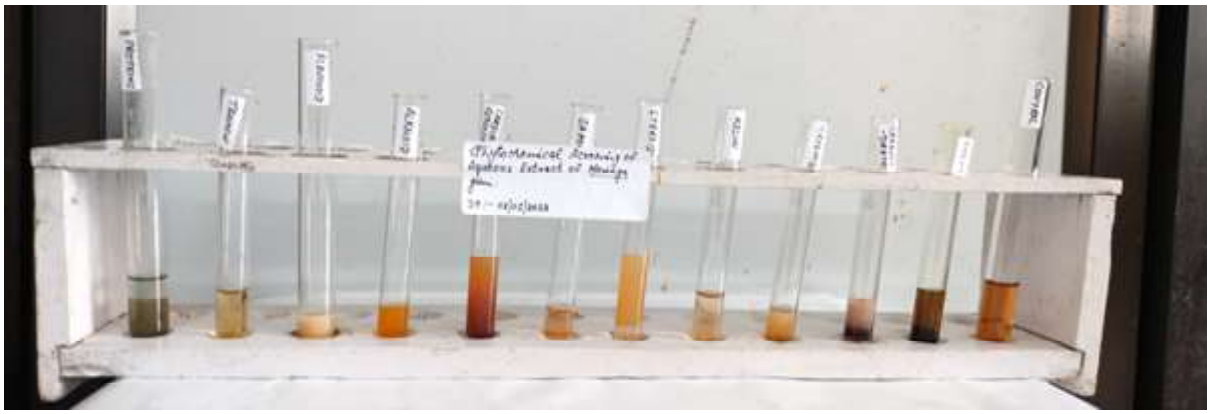
Note: + = Indicates presence of the phytochemical, - = Indicates absence of the phytochemical.



Photograph (a) of Phytochemical Screening of Methanolic Extract of gums of *Moringa oleifera* Lam.



Photograph (b) of Phytochemical Screening of Ethanolic Extract of gums of *Moringa oleifera* Lam.



Photograph (c) of Phytochemical Screening of Aqueous Extract of gums of *Moringa oleifera* Lam.

Phytochemical screening of methanol, ethanol and aqueous extracts of gums of *Butea monosperma* (Lam.) Taub. as found in Ranchi district suggests the presence of alkaloids, flavonoids, cardiac glycosides, terpenoids, carbohydrates, proteins, tannins, etc.(Table 4 and photographs d to f).

Table 4- Phytochemical Screening of Methanol (M), Ethanol (E), Aqueous (A) extracts of gums of *Butea monosperma* (Lam.) Taub. in Ranchi district.

Sl. No.	Constituents	Tests	Methanol Extract (M)	Ethanol Extract (E)	Aqueous Extract (A)
1.	Alkaloids	Dragendorff's Test	+	+	+
2.	Flavonoids	Alkaline Reagent Test	+	+	+
3.	Cardiac Glycosides	Keller Kelliani's Test	+	+	+
4.	Terpenoids	Salkowski's Test	+	+	+
5.	Carbohydrates	Mollisch's Test	+	+	+
6.	Phenol	Ferric Chloride Test	+	+	--
7.	Saponin	Foam Test	--	--	+
8.	Tannins	Bromine Water Test	+	+	+
9.	Resins	HCl / $FeCl_3$ Test	--	--	--
10.	Steroids	Salkowaski Test	+	+	--
11.	Proteins	Biuret Test	+	+	+

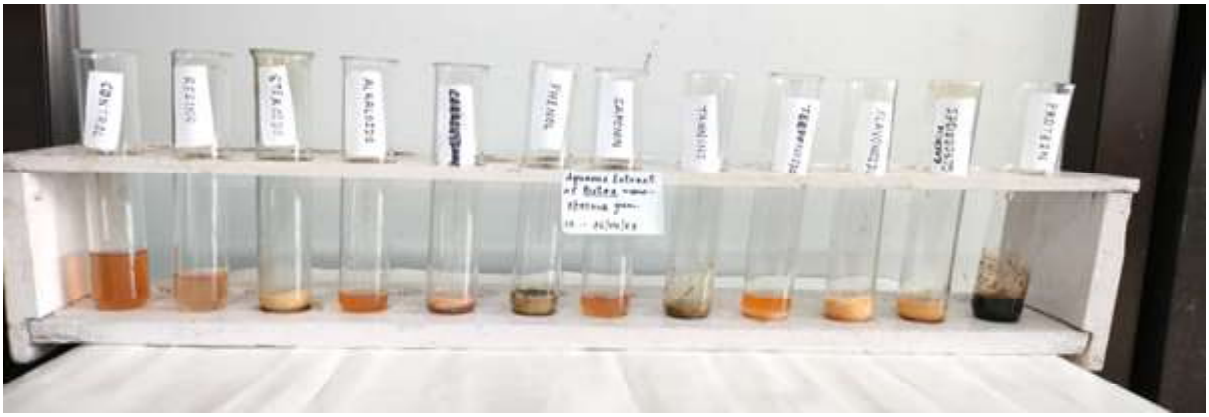
Note: + = Indicates presence of the phytochemical, - = Indicates absence of the phytochemical**.**



Photograph (d) of phytochemical screening of Methanolic Extract of gums of *Butea monosperma*.



Photograph (e) of phytochemical screening of Ethanolic Extract of gums of *Butea monosperma*.



Photograph (f) of phytochemical screening of Aqueous Extract of gums of *Butea monosperma*.

Phytochemical screening of methanol and ethanol extracts of oleoresins of *Shorea robusta* Roxb. ex Gaertn. f. as found in Ranchi district suggests the presence of alkaloids, flavonoids, cardiac glycosides, terpenoids, carbohydrates, saponins, proteins, resins, tannins and steroids (Table 5 and photographs g to h).

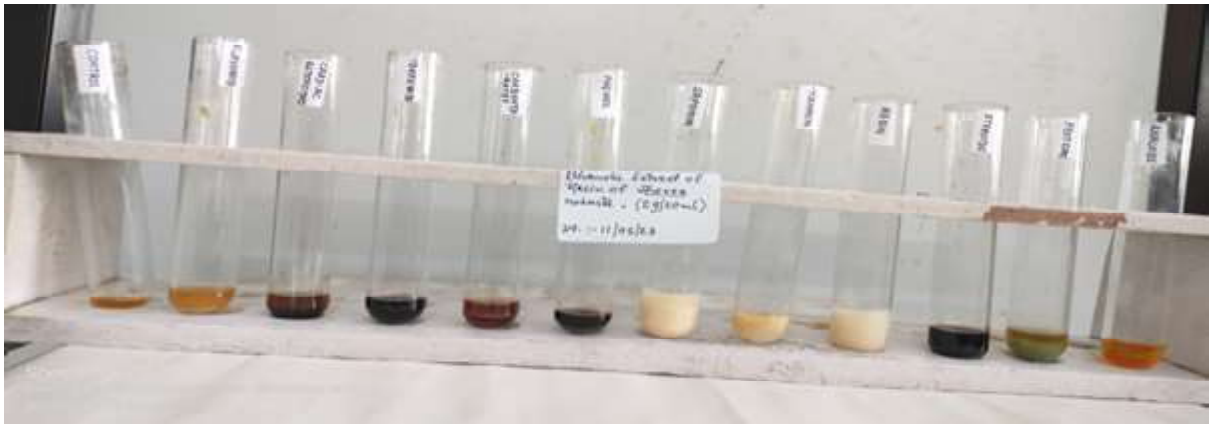
Table 5- Phytochemical Screening of Methanol (M), Ethanol (E) extracts of oleoresins of *Shorea robusta* Roxb. ex Gaertn.f. in Ranchi district.

Sl. No.	Constituents	Tests	Methanol Extract (M)	Ethanol Extract (E)
1.	Alkaloids	Dragendorff's Test	+	+
2.	Flavonoids	Alkaline Reagent Test	+	+
3.	Cardiac Glycosides	Keller Kelliani's Test	+	+
4.	Terpenoids	Salkowski's Test	+	+
5.	Carbohydrates	Mollisch's Test	+	+
6.	Phenol	Ferric Chloride Test	--	+
7.	Saponin	Foam Test	+	+
8.	Tannins	Bromine Water Test	+	+
9.	Resins	HCl/ $FeCl_3$ Test	+	+
10.	Steroids	Salkowaski Test	+	+
11.	Proteins	Biuret Test	+	+

Note: + = Indicates presence of the phytochemical, - = Indicates absence of the phytochemical



Photograph (g) of Methanolic extract of oleoresins of *Shorea robusta* Roxb. ex Gaertn.f.



Photograph (h) of Ethanolic extract of oleoresins of *Shorea robusta* Roxb. ex Gaertn.f.

Medicinal uses or economic importance of the gums and resins of these plants are well documented in literature as follows (Table 6).

Table 6- List of some plant-based gums and resins which exhibit their medicinal values.

Scientific Name	Medicinal uses / Economic importance
***Moringa oleifera* Lam.**	Gelling agent, binder, release retardant in tablet formulations, used in herbal medicine, diuretic, astringent, fever, dysentery, asthma, intestinal cancer. [8,24-30]
***Shorea robusta* Roxb. ex Gaertn.f.**	Used as an ingredient of ointments to heal wounds, burns, pains, skin diseases and to control diarrhoea and dysentery. [12,13,31-34]
***Butea monosperma* (Lam.) Taub.**	Used as medicine, food, fibre and for other miscellaneous purposes such as fish poison, dye, fodder, utensils, *etc*.[21]

CONCLUSION

The phytochemical screening of gums of *Moringa oleifera* Lam. and *Butea monosperma* (Lam.) Taub. indicated the presence of alkaloids, flavonoids, cardiac glycosides, terpenoids, carbohydrates, phenol, tannins, steroids and proteins in the methanolic and ethanolic extracts whereas saponin and resins were not found to be present in the both extracts whereas in aqueous extract, all the active constituents were found to be present except phenol, resins and steroids. In the case of oleoresins of *Shorea robusta* Roxb. ex Gaertn.f., all the active constituents were present in both methanolic and ethanolic extracts but only phenol was not present in methanolic extract.

ACKOWLEDGEMENTS

We extend our profound gratitude to Prof. (Dr.) Kunul Kandir, Head, University Department of Botany, Ranchi University, Ranchi for her constant guidance and support and motivation for this present study.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 275-280

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Antifungal activity of extracts of *Millettia pinnata* L. and *Cassia tora* L. against plant fungal pathogen *Alternaria solani*

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Received : 3rd January, 2023 ; Revised : 2nd February, 2023

Abstract- *Alternaria solani*, plant pathogenic fungus causes Early blight of Potato, main reason of significant economical losses of potato. Present research focus on antifungal activity of seed, bark, leaf and stem extracts of *Millettia pinnata* L. and *Cassia tora* L. against plant fungal pathogen *Alternaria solani* (ITCC No. 4632). Extracts were prepared in organic solvents such as acetone, ethanol, chloroform and petroleum ether. Mycelial growth of *Alternaria solani* was recorded in all prepared extracts. Minimum mycelial growth of *Alternaria solani* was recorded in acetone extract of *Millettia pinnata* L. seed with 2.7 mm. Ethanol extract of Karanj seed also exhibited less growth of mycelium that was 3.3 mm. In compare to Ethanol, Chloroform and Petroleum ether, Acetone solvent is more effective in controlling the Mycelial growth of *Alternaria solani.* All other treatments are effective than check.

Key words: Antifungal activity, *Millettia pinnata* L. (Karanj), *Cassia tora* and *Alternaria solani*

INTRODUCTION

A plant pathogenic fungus *Alternaria solani* causes significant economic losses of potato crop.[1,2] Use of chemical fungicides are harmful for health and also pollute the environment. Plant extracts may be used for the management of different plant diseases. Some scientists are involved to search more environmentally friendly natural compounds for the control of plant diseases using plant extracts from different plants.[3-6] Present research focus on antifungal activity of extracts of *Millettia pinnata* L. and *Cassia tora* L.

Millettia pinnata L. belongs to family Fabaceae, mostly found in tropical and temperate Asia. It is also commonly known as Indian beech, Pongam oil tree, karanj, honge, pungai, karach and naktamala. Various parts of *Millettia pinnata* L. have their medicinal value.[7,8] *Millettia pinnata* L. is used for the treatment of head pains, leucoderma, skin diseases, wounds, piles, liver pain, leprosy and chronic fever. It is used for anti-inflammatory, antidiarrhoeal, anti-ulcer, anti-hyperammonic and antioxidant activity. Different parts of plant contain alkaloids demethoxy-kanugin, gamatay, glabrin, kaempferol, kankone, kanugin, karangin, pinnatin, pongamol, pongapin, quercitin, saponin, β-sitosterol and tannin.

Cassia tora L. belongs to family of Fabaceae, is mostly found in South-East Asia and the South West Pacific. In India it is distributed in tropical parts. It is a dicot legume, commonly known as sickle senna, sickle pod, tora, coffee pod, chakvad and thakara. The whole plant parts have been widely used to control different diseases, especially in Satpura region of Madhya Pradesh,

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India. *Cassia tora* L. is used for anti-inflammatory, antiperiodic, anthelmintic, ophthalmic, liver tonic and cardiotonic. The leaves and seeds of *Cassia tora* contain several anthraquinone glycosides, cinnamaldehyde, gum, tannins, mannitol, coumarins and essential oils which are useful in leprosy, ringworm, colic, dyspepsia, constipation, cough, bronchitis, cardiac disorders.

Antifungal activity of *Millettia pinnata* L. and *Cassia tora* L. against few bacteria was evaluated by some scientist.[9,10] In present research, it was observed that acetone extract of selected plants showed maximum inhibitory activity against plant fungal pathogen *Alternaria solani*, which is due to the presence of some secondary metabolites.

MATERIALS & METHODS

Preparation of the Extract

Collection of plant material

Cassia tora leaves were collected from Chalisgaon, Dist. Leaves, stems, bark and seeds of *Millettia pinnata* L. and *Cassia tora* L. were collecte collected from different locations of Ranchi district of Jharkhand, India. Selected plant parts were cleaned with water followed by shade-dried. The different parts of the selected plants were powdered, stored and used for extraction. 10 g of powder was mixed to 100 mL of different solvents such as acetone, ethanol, petroleum ether and chloroform for 72 hr. The crude extract was filtered through Whatman No. 1 filter paper which was considered as a standard solution.[11]

Test pathogen

Test fungal pathogen *Alternaria solani* (ITCC No.-4632) was obtained from Department of Mycology and Plant Pathology, ICAR-Indian Agricultural Research Institute, New Delhi. The test pathogen was cultured and maintained on Potato dextrose agar medium (PDA) at 28°C for further use.

Antifungal Activity

Poison food technique was used for the examining the antifungal activity of *Millettia pinnata* L. and *Cassia tora* L. against *Alternaria solani*.[12] At different concentration of selected plants extracts like 1.0 %. 2.5 %, 5.0% and 10.0%, diameter of mycelial growth was measured. The experiments were replicated three times and average of diameter of mycelial growth was measured. Percentage of inhibition was calculated by the following formula of Vincent (1947).[13-17]

$$\% \text{ of Mycelial inhibition} = \frac{\text{Radial growth in check} - \text{Radial growth in treatment}}{\text{Radial growth in check}} \times 100$$

RESULT & OBSERVATIONS

Results of antifungal activity of extracts of selected plants – *Millettia pinnata* L. and *Cassia tora* L. using different solvents - acetone, ethanol, petroleum ether and chloroform were given in Table 1, Table 2, Table 3 and Table 4. Selected plants extract significantly inhibited the mycelial growth of test pathogen *Alternaria solani*.

Table 1 showed the antifungal activity of acetone extracts of *Millettia pinnata* L. and *Cassia tora* L. Minimum mycelial growth of *Alternaria solani* was recorded in acetone extract of *Millettia pinnata* L. seed with 2.7 mm followed by Karanj bark or *Cassia tora* seed that was 4.3 mm at 10% concentration. Karanj seed extracts using acetone showed maximum percentage of inhibition against *Alternaria solani* with 97%. Karanj bark or *Cassia tora* seed extracts showed 95.2% of inhibition at 10% concentration. Leaf extract of *Cassia tora* or Karanj was also effective to control *Alternaria*, exhibited 94.4% and 93.7% of inhibition respectively in acetone as a solvent. Other extracts were moderately effective to control *Alternaria solani*.

The ethanol extract of Karanj seed showed minimum mycelial growth of *Alternaria solani* with 3.3 mm. Karanj bark extract showed mycelial growth with 3.7 mm and in *Cassia tora* seed, it was observed growth with 4.3 mm. It was recorded that Karanj leaf or *Cassia tora* bark extract also effective to control test pathogen, in which average diameter of growth was 5.7 mm at 10% concentration (Table 2). Maximum percentage of inhibition of mycelia growth of *Alternaria solani* was recorded in case of Karanj seed that was 96.3%, followed by Karanj bark with 95.9% or *Cassia tora* seed with 95.2%. Leaf extract of Karanj or Bark extract of *Cassia tora* with 93.7 % of inhibition, was also effective to control *Alternaria solani*,

Effect of chloroform extracts are depicted in Table-3. Minimum mycelial growth of *Alternaria solani* was recorded in chloroform extract of Karanj seed with 3.7 mm followed by Karanj bark with 4.0 mm, Karanj leaf with 4.7 mm, *Cassia tora* seed with 5.3 mm and *Cassia tora* bark with 5.7 mm at 10% concentration. Karanj seed extract in chloroform showed maximum percentage of inhibition against *Alternaria solani* which was 95.9%.

Karanj bark showed 95.6% of inhibition, Karanj leaf extract with 94.8%. *Cassia tora* seed and bark exhibited 94.1% and 93.7% of inhibition. Other extracts using chloroform were also effective than check.

Results of antifungal activity of Pertroleum ether extracts were given in Table 4. Minimum mycelial growth of *Alternaria solani* was also recorded in Petroleum ether extract of Karanj seed with 4.3 mm followed by Karanj bark with 4.7 mm, *Cassia tora* seed with 6.0 mm and Karanj leaf or stem with 11.3 mm and 14.7 mm respectively at 10% concentration. Maximum percentage of inhibition 95.2% was recorded in Karanj seed extract followed by Karanj bark with 94.8% or *Cassia tora* seed with 93.3%. Leaf or stem extract of Karanj was moderately effective to control test pathogen.

Table 1- Effect of different concentration of acetone extracts of different parts of *Millettia pinnata* L. and *Cassia tora* L on mycelial growth and Percentage of inhibition of *Alternaria solani*.

Plant Extracts	1%		2.5%		5%		10%	
Millettia pinnata L.	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition
a) Seed	27.7	69.2	22	75.6	9.3	89.7	2.7	97
b) Bark	40.3	55.2	27.7	69.2	12.3	86.3	4.3	95.2
c) Leaf	44.7	50.3	33.3	63	17.3	80.8	5.7	93.7
d) Stem	55.3	38.6	47.3	47.4	20.7	77	15.3	83
Cassia tora L.								
a) Seed	39.3	56.3	31.3	65.2	14.3	84.1	4.3	95.2
b) Bark	45	50	38.3	57.4	19.3	78.6	5.3	94.1
c) Leaf	52	42.2	42	53.3	24.7	72.6	5	94.4
d) Stem	63	30	51.3	43	33	63.3	14.7	83.7
Check	90	0	90	0	90	0	90	0

Table 2- Effect of different concentration of ethanol extracts of different parts of *Millettia pinnata* L. and *Cassia tora* L on mycelial growth and Percentage of inhibition of *Alternaria solani*.

Plant Extracts	1%		2.5%		5%		10%	
Millettia pinnata L.	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition
a) Seed	43.3	51.9	32	64.4	14.7	83.7	3.3	96.3
b) Bark	50.7	43.7	39.3	56.3	20	77.8	3.7	95.9
c) Leaf	59.7	33.7	44	51.1	25	72.2	5.7	93.7
d) Stem	69.3	23	46.7	48.1	31.7	64.8	17.7	80.3
Cassia tora L.								
a) Seed	43.7	51.4	40	55.6	22	75.6	4.3	95.2
b) Bark	55.3	38.6	43.3	51.9	23.7	73.7	5.7	93.7
c) Leaf	65	27.8	51.3	43	32	64.4	7.3	91.9
d) Stem	73	18.9	61.3	31.9	39	56.7	20	77.8
Check	90	0	90	0	90	0	90	0

Table 3- Effect of different concentration of chloroform extracts of different parts of *Millettia pinnata* L. and *Cassia tora* L on mycelial growth and Percentage of inhibition of *Alternaria solani*.

Plant Extracts	1%		2.5%		5%		10%	
Millettia pinnata L.	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition
a) Seed	45.7	49.2	37.3	58.6	21.7	75.9	3.7	95.9
b) Bark	50	44.4	44.7	50.3	32	64.4	4	95.6
c) Leaf	64.3	28.6	47.3	47.4	34.7	61.4	4.7	94.8
d) Stem	69.3	23	52	42.2	40	55.6	16	82.2
Cassia tora L.								
a) Seed	48.3	46.3	41.3	54.1	30.7	65.9	5.3	94.1
b) Bark	55.7	38.1	46.7	48.1	34.7	61.4	5.7	93.7
c) Leaf	65.3	27.4	52.3	41.9	46.3	48.6	6.3	93
d) Stem	64.3	28.6	57	36.7	52.7	41.4	17.3	80.8
Check	90	0	90	0	90	0	90	0

Table 4- Effect of different concentration of Petroleum ether extracts of different parts of *Millettia pinnata* L. and *Cassia tora* L on mycelial growth and Percentage of inhibition of *Alternaria solani*.

Plant Extracts	1%		2.5%		5%		10%	
Millettia pinnata L.	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition	mm	% of inhibition
a) Seed	47.3	47.4	41.3	54.1	26	71.1	4.3	95.2
b) Bark	52	42.2	50	44.4	36	60	4.7	94.8
c) Leaf	70.7	21.4	50	44.4	40	55.6	11.3	87.4
d) Stem	75.3	16.3	52.7	41.4	43.7	51.4	14.7	83.7
Cassia tora L.								
a) Seed	51.7	42.6	45.3	49.7	29.7	67	6	93.3
b) Bark	62.7	30.3	54.3	39.7	40.7	54.8	36	60
c) Leaf	70	22.2	54.3	39.7	46.3	48.6	36.7	59.2
d) Stem	77.3	14.1	59.7	33.7	53.3	40.8	43	52.2
Check	90	0	90	0	90	0	90	0

CONCLUSION

Overall, Extracts of selected plants- *Millettia pinnata* L. and *Cassia tora* L. using different solvents such as acetone, ethanol, petroleum ether and chloroform are significantly the most effective and reduced the mycelial growth of the *Alternaria solani*. Result showed selected plant extract using acetone was most effective against test pathogen. Acetone extract of Karanj seed exhibited maximum inhibition against *Alternaria solani* which was 97%, followed by ethanol extract of Karanj seed which showed inhibition percent of 96.3 at 10 % concentration. Other selected plant extracts in different solvents like chloroform and petroleum ether were also effective to control *Alternaria solani*.

From the present research work, it may be concluded that *Millettia pinnata* L. and *Cassia tora* L. can be utilized for the control of fungal diseases caused by *Alternaria solani*.

ACKNOWLEDGEMENTS

Author is thankful to Dr. Ajit Kumar Jha, Senior Scientist, ICAR- RCER-FSRCHPR, Ranchi, for providing necessary support during research work.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 281-286

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Floristic survey of dicotyledonous angiospermic flora of Kelaghagh Dam Road, Simdega block in Jharkhand

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Received : 11th January, 2023 ; Revised : 12th February, 2023

Abstract- The current floristic survey of dicotyledonous plants was conducted from March to October 2021 along the Kelaghagh Dam Road in Simdega, Jharkhand. In this survey, dicotyledonous plants were evaluated because they are crucial for food, shelter, and biodiversity both in dicot and, eventually, angiosperm flora. Twenty-four extensive field surveys resulted in the identification of 82 dicot plant species across 33 families. All the species were classified into 77 genera.

Key words: Floristic, Dicotyledonous, Angiosperms, Simdega

INTRODUCTION

The variety and variability of plants in a specific area is known as floral diversity. Floristic diversity can be evaluated on any scale, from the global level through toward an ecosystem, community, or even at solitary species level. The largest global, most successful, and most diversified main group that contributes to the earth's predominant vegetation is the angiosperms. The monocotyledons (monocots) and the dicotyledons are the two major classes, within which the angiosperms had conventionally been completely separated (dicots).[1]

Identification of various species of plants and their nomenclature is a crucial task of the taxonomists.

Simdega district is located between latitudes of 220'20" to 220'51" North, and between longitudes of 840' 01" with 850'05" East. Simdega, Kurdeg, Bolba, Thethaitangar, Kolebira, Bano, Jaldega, Pakartanr, Bansjore, and Kersai comprise the ten blocks or circles that make up this district, which has 94 Panchayats. In the summer, the maximum temperature ranged from 38ºC to 42ºC, and in the winter, it was somewhere around 25ºC and 28ºC. The total average annual rainfall is 473 mm.[2] Due to persistent drought-prone conditions in the region and unfavorable geographic conditions, however little known about the diversity of plants. The goal of the current study is to document the diversity of dicot flora along Kelaghagh dam road and to emphasize the diversity of dicot plant resources from a conservation perspective.

Study Area

The study area is Kelaghagh dam road Simdega block in Jharkhand (Map 1).

MATERIAL & METHODS

In 2021, frequent surveys and explorations monitoring the growth during the wet winter and hot summer months were conducted. On the basis of its physical and reproductive characteristics, plant samples were gathered. By reviewing all of the literature that has been accessible, a list of the species from the Kelaghagh Dam Road was constructed. The Flora of Ranchi District[3], A supplement

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Source: https://www.sameti.org/Soil_Inventory/Simdega_Soil_Analysis.pdf

Map 1- Location map of study area

to the botany of Bihar and Orissa[4], and The Botany of Bihar and Orissa[5], all of which are available in the University Department of Botany's Library as well as in the Central Library of Ranchi University, were utilized to confirm the identification of the majority of the specimens. The University Department of Botany at Ranchi University of Ranchi's taxonomists also authenticated the specimens of collected plants, and a herbarium was constructed. International Plant Name Index (IPNI)[6] and e-flora of Botanical Survey of India[7] were also used to check authenticity.

RESULTS

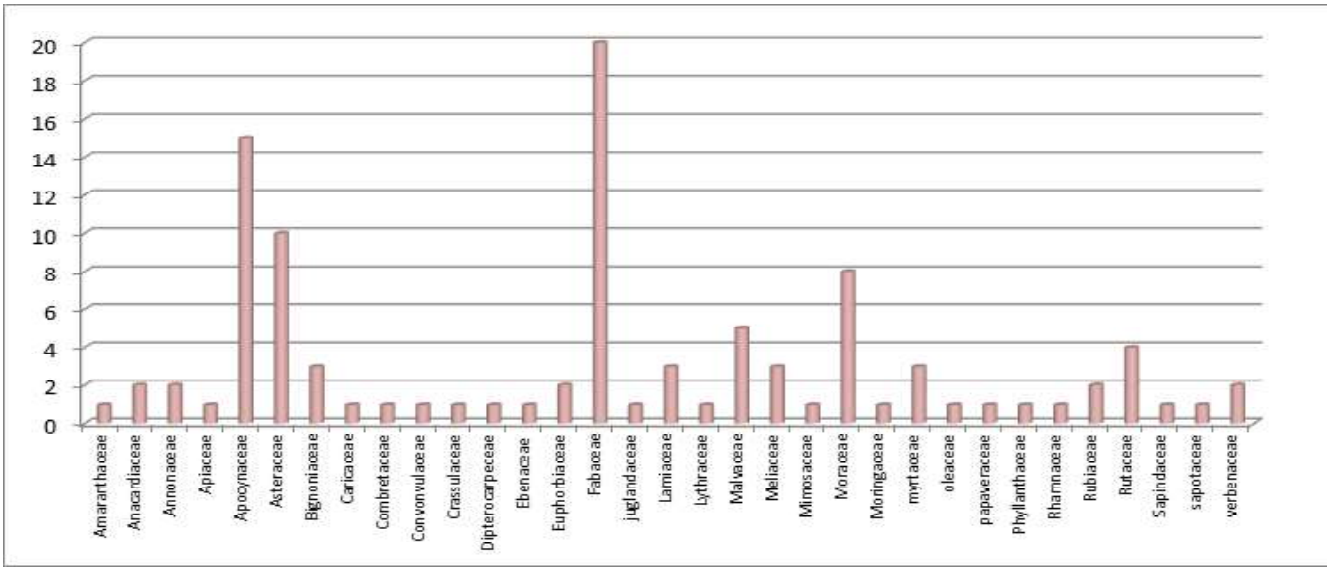
The findings shown in Table no. 1 demonstrate that the dicot plants in the research area were diverse. There seem to be 33 families including 82 species of dicot plants across all. There are 77 genera for all the species.

Fabaceae was the largest family with 16 species, Apocynaceae represented by 9 species with second largest number and it was followed by Asteraceae and Moraceae.

Table 1- Checklist of dicotyledonous angiospermic plants species in Kelaghagh dam road Simdega

Sl. No.	Botanical name	Family	Date of collection	Common Name	Fig. No
1.	*Amaranthus virdis* L.	Amaranthaceae	13.04.2021	Bhajisaag	
2.	*Mangifera indica* L.	Anacardiaceae	03.04.2021	Aam	14
3.	*Buchanania lanzan* Spreng.	Anacardiaceae	01.04.2021	Chironji	
4.	*Annona reticulata* L.	Annonaceae	31.03.2021	Sitafal	
5.	*Annona squamosa* L	Annonaceae	31.03.2021	Sarifa	
6.	*Polyalthia longifolia* (Sonn.) Benth.& Hook.f. ex Thwaites	Annonaceae	03.04.2021	Ashoka	15
7.	*Centella asiatica* (L.) Urb.	Apiaceae	31.03.2021	Brahmi	
8.	*Catharanthus roseus* (L) G. Don	Apocynaceae	13.04.2021	Sadabahar	10
9.	*Calotropis gigantea* (L.) W. T. Aiton	Apocynaceae	31.03.2021	Akwan	19
10.	*Plumeria rubra* L.	Apocynaceae	30.03.2021	Gulaichi	
11.	*Calotropis procera* (Aiton)W.T.Aiton	Apocynaceae	30.03.2021	Aak	
12.	*Plumeria alba* L.	Apocynaceae	03.04.2021	Safed gulaichi	
13.	*Thevetia peruviana* (Pers.) K.Schum.	Apocynaceae	01.04.2021	Peelakaner	8
14.	*Alstonia scholaris* (L.) R.Br.	Apocynaceae	01.04.2021	Chatwan	
15.	*Sphaeranthus indicus* L.	Asteraceae	30.03.2021	Gorakhmundi	7
16.	*Chromolaena odorata* (L.) R.M.King & H.Rob.	Asteraceae	11.08.2021	Tivragandha	
17.	*Parthenium hysterophorus* L.	Asteraceae	13.04.2021	Gajarghash	
18.	*Xanthium stramarium* L.	Asteraceae	15.10.2021	Chotadhatura	
19.	*Cosmos sulphureus* Cav.	Asteraceae	15.10.2021	Yellow cosmos	17
20.	*Pyrostegia venusta* (kerGawl) Miers	Bignoniaceae	30.03.2021	Jallebibali	
21.	*Tecoma stans* (L.) Juss. ex Kunth	Bignoniaceae	15.10.2021	Champa	20
22.	*Cordia myxa* L.	Boraginaceae	30.03.2021	Lasoda	
23.	*Terminalia chebula* Retz.	Combretaceae	07.05.2021	Harra	13
24.	*Carica papaya* L.	Caricaceae	03.04.2021	Papita	
25.	*Cascuta reflexa* Roxb.	Convonvulaceae	13.04.2021	Amarbel	
26.	*Bryophyllum pinnatum* (Lam.) Oken	Crassulaceae	13.04.2021	Patharchoor	

27.	*Coccinia grandis* (L.) Voigt	Cucurbitaceae	03.04.2021	Kundri	
28.	*Shorea robusta* C.F.Gaertn.	Dipterocarpeceae	01.04.2021	Sal	16
29.	*Diospyros malanoxylon* Roxb.	Ebenaceae	01.04.2021	Kusum	
30.	*Ricinus communis* L.	Euphorbiaceae	13.04.2021	Arandi	
31.	*Euphorbia hirta* L.	Euphorbiaceae	23.07.2021	Bada dudhi	
32.	*Senna alata* (L.) Roxb.	Fabaceae	15.10.2021	Dadmurdan	
33.	*Albizia lebbeck* (L.) Benth.	Fabaceae	30.03.2021	Safed siris	
34.	*Crotalaria pallida* Aiton.	Fabaceae	10.11.2021	Jungli tag	4
35.	*Dalbergia sisso* Roxb. ex DC.	Fabaceae	15.10.2021	Sheesham	
36.	*Leucaena leucocephala* (Lam.) de Wit.	Fabaceae	30.03.2021	Subabool	1
37.	*Bauhinia variegate* L.	Fabaceae	15.10.2021	Koinar	
38.	*Peltoforum pterocarpum* (DC.) K.Heyne	Fabaceae	30.03.2021	Peela gulmohar	
39.	*Pongamia pinnata* (L.) Pierre	Fabaceae	31.03.2021	Karanj	5
40.	*Senna tora* L.	Fabaceae	15.10.2021	Chakod	
41.	*Cajanus cajan* (L.) Huth	Fabaceae	03.04.2021	Arhar	
42.	*Tephrosia purpurea* (L.) Pers.	Fabaceae	11.08.2021	Sharpunkha	11
43.	*Lablab purpureous* (L.) Sweet	Fabaceae	03.04.2021	Sem	
44.	*Cassia fistula* L.	Fabaceae	01.04.2021	Amaltash	
45.	*Tamarindus indica* L.	Fabaceae	15.10.2021	Imli	3
46.	*Butea monosperma* (Lam.) Kuntze	Fabaceae	15.10.2021	Palash	2
47.	*Delonix regia* (bojer) Raf	Fabaceae	15.10.2021	Gulmohar	
48.	*Juglans regia* L	Juglandaceae	31.03.2021	Akhrot	
49.	*Gmelina arborea* Roxb. ex Sm.	Lamiaceae	01.04.2021	Gamhar	
50.	*Tectona grandis* L.	Lamiaceae	30.03.2021	Sagwan	
51.	*Leucas cephalotes* (Roth) Spreng.	Lamiaceae	12.08.2021	Goma	
52.	*Punica granatum* L.	Lythraceae	13.04.2021	Annar	
53.	*Bombax cieba* L.	Malvaceae	13.04.2021	Green Semal	
54.	*Hibiscus sabdariffa* L.	Malvaceae	10.11.2021	Kudrum	18
55.	*Melia azedarach* L.	Meliaceae	30.03.2021	Bakain	
56.	*Azadirachta indica* A.Juss.	Meliaceae	03.04.2021	Neem	
57.	*Samanea saman* (Jacq.) Merr.	Mimosaceae	13.04.2021	Gulabi siris	
58.	*Ficus benghalensis* L.	Moraceae	30.03.2021	Banyan tree	12
59.	*Ficus recemosa* L.	Moraceae	03.04.2021	Gullar	
60.	*Ficus religiosa* L.	Moraceae	03.04.2021	Peepal	
61.	*Artocarpus heterophyllus* Lam.	Moraceae	03.04.2021	Kathal	
62.	*Ficus geniculata* Kurz.	Moraceae	13.04.2021	Putkal	
63.	*Artocarpus lacucha* Buch.-Ham.	Moraceae	13.04.2021	Barhar	
64.	*Morus nigra* L.	Moraceae	13.04.2021	Shahtut	
65.	*Moringa oleifera* Lam.	Moringaceae	30.03.2021	Munga	
66.	*Syzygium cumini* (L.) Skeels	Myrtaceae	01.04.2021	Jamun	
67.	*Eucalyptus dalrympleana* Maiden	Myrtaceae	01.04.2021	Eucalyptus	
68.	*Psidium guajava* L.	Myrtaceae	30.03.2021	Amrood	
69.	*Jasminum sambac* (L.) Aiton	Oleaceae	13.04.2021	Bela	
70.	*Argemone mexicana* L.	Papaveraceae	30.03.2021	Satyanshi	
71.	*Phyllanthus acidus* (L.) Skeels.	Phyllanthaceae	30.03.2021	Harpharauri	
72.	*Ziziphus jujuba* mill.	Rhamnaceae	30.03.2021	Ber	
73.	*Adina cordifolia* (Roxb.) Brandis	Rubiaceae	01.04.2021	Karam	
74.	*Mitracarpus hirtus* (L) DC.	Rubiaceae	11.08.2021	Gothaigobi	
75.	*Murraya koenigii* J.Koenig ex L.	Rutaceae	30.03.2021	Curry leaf	
76.	*Aegle marmelous* (L.) Correa	Rutaceae	30.03.2021	Bel	
77.	*Citrus limon* (L.) Osbeck	Rutaceae	31.03.2021	Nimbu	
78.	*Citrus maxima* (Burm.) Merr.	Rutaceae	13.04.2021	Dambha	6
79.	*Litchi chinensis* Sonn.	Sapindaceae	31.03.2021	Lychee	
80.	*Madhuca indica* (J.Koenig ex L.) J.F.Macbr	Sapotaceae	01.04.2021	Mahua	9
81.	*Lantana camara L.*	Verbenaceae	30.03.2021	Putush	
82.	*Vitex negundo* L.	Verbenaceae	30.03.2021	Sindwar	



Graph 1- Family-wise distribution of dicot species in Kelaghagh Dam Road of Simdega district.



Fig. 1
Leucaena leucocephala
(Lam.) de Wit.



Fig. 2
Butea monosperma
(Lam.) Kuntze



Fig. 3
***Tamarindus indica* L.**



Fig. 4
***Crotalaria pallida* Aiton.**



Fig. 5
Pongamia pinnata
(L.) Pierre



Fig. 6
Citrus maxima
(Burm.) Merr.



Fig. 7
Sphaeranthus indicus
L.



Fig. 8
Thevetia peruviana
(Pers.)K.Schum.



Fig. 9
***Madhuca indica* (J.Koenig**
ex L.) J.F.Macbr



Fig. 10
Catharanthus roseus
(L) G. Don



Fig. 11
Tephrosia purpurea
(L.) Pers.



Fig. 12
***Ficus benghalensis* L.**
(Moraceae)

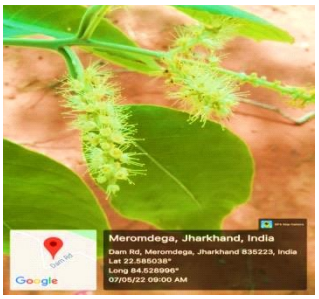


Fig. 13
Terminalia chebula Retz.



Fig. 14
Mangifera indica L.



Fig. 15
Polyalthia longifolia (Sonn.) Benth. & Hook.f. ex



Fig. 16
Shorea robusta C.F.Gaertn.

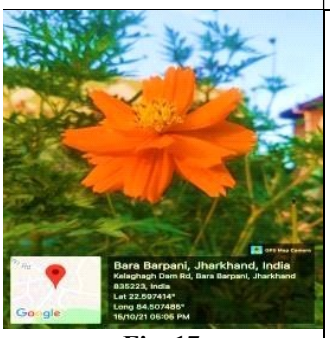


Fig. 17
Cosmos sulphureus Cav.



Fig. 18
Hibiscus sabdariffa L.

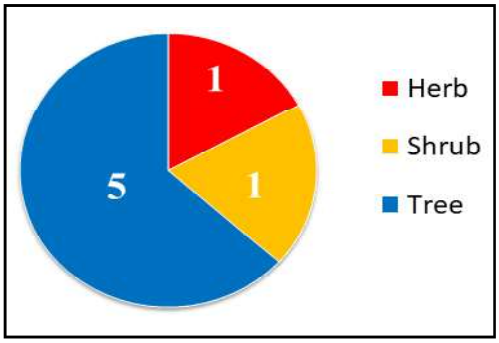


Fig. 19
Calotropis gigantea



Fig. 20
Tecoma stans

Habit		
Herb	Shrub	Tree
14	16	52



Pie chart 1- Diagram showing habit of Dicotyledonous angiospermic Plants

DISCUSSION

To enhance the conservation of rare, threatened, vulnerable and endangered plants species, dicotyledonous angiospermic plants species should indeed be documented, and a floristic survey must be performed to acquire more concerning the local flora. This research study is based on observations of floristic diversity. Although as indigenous societies become more industrialized and many species are currently on the threat of extinction. As a consequence, there is an urgent need to explore and document dicotyledenous angiospermic plant species. As many plant species are rare, threatened, vulnerable, endangered, or extinct, efforts must be made to preserve and conserve the forest area.

CONCLUSION

Present study deals with collecting evidences of certain extinct species of the Kelaghagh Dam Road of Simdega, and this survey includes floristic research of the region. The preparation of the herbarium is essential for the conservation of the endangered species. It also provides significant scientific support to the Department of Forest and Environment in the Simdega district as well as the Department of Botany at Ranchi University, Ranchi. Simdega has a tremendously diverse range of flora, but as of now, we have only encountered 33 families, 77 genera, and 82 species, with the Fabaceae family dominating all of these plant species and *Buchanania lanzan* is the dominant species from this region and the seed of *Buchanania lanzan* used as commercialization purpose by tribal people of that region. Kendu (*Diospyros melanoxylon*) is used to be a minor forest product in this region, commercialization of its leaves for making bidi, a traditional cigarette can turn out to be a boon for local tribal people of that region.

Seeds yield fatty oils of *Pongamia pinnata* is used in tanning industry. It is also used in the preparation of washing soaps and candles. Traditionally, wood of *Pongamia pinnata,* with a calorific value of 4600 Kcal/ Kg is used as a fuel in rural areas. *Tamarindus indica* has exported to different state for making sauces, chutneys, drinks and desserts. Seeds of *Schleichera oleosa* are used for production of biodiesel. *Butea monosperma* tree is used as a host of lac insects for the production of Rangeeni lac, which is an important source of income for tribal people of this region. Dicotyledonous angiospermic plants have not yet been the focus of any research in this region, as far as we were aware. The main goal of this research seems to be to identify new species of dicotyledonous angiospermic plants and to explore the flora along the Kelaghagh Dam Road in Simdega district.

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Biospectra : ISSN: 0973-7057, Vol. 18(1), March, 2023, pp. 287-290

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

Effect of *Azadirachta indica* and *Citrus sinensis* on *Sitophilus zeamais* (L.) (Coleoptera: Curculionidae) insect of stored maize grains.

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Received : 08th January, 2023 ; Revised : 10th February, 2023

Abstract- Powders of Neem (*Azadirachta indica*) leaf and Orange (*C. sinensis*) peels were tested for their insecticidal actions against adult *Sitophilus zeamais*, (Coleoptera: Curculionidae) under controlled conditions at 75±5 percent relative humidity and 27±2°C temperature during the Winter season. Experiments consisted of exposing adult *S. zeamais* to both the Neem leaf and orange peels powders. The maize seeds were treated with Neem leaf powder and Citrus peel powder in different dosage from 10g, 20g and 30gm during the study period. These biopesticides were thoroughly mixed with required quantity of seeds and kept in 1 kg capacity glass jars by manual operation. The result indicated that pulverized plant powder had no significant reproductive effect on adult *S. zeamais*. The maximum number of eggs was found in Neem powder (22.34%) rather than Citrus peel powder in different dosages in this study. The incubation period did not differ significantly among various plant protectant treatments. The maximum incubation period was in Neem leaf powder (12.6 days) while it was minimum in citrus peel powder. The hatchability percentage of the pest was minimum (27.24 %) in the grains treated with Neem powder and this treatment was followed by Citrus peel powder in different concentrations in this research.

Key words: Neem Leaf, Orange Peel, *S. zeamais*, Maize grains

INTRODUCTION

The Maize is another important cereal grain belongs to the family of Gramineae. It is an important staple food for human beings and animals. Maize can flourish in diverse climates condition and is grown-up by small holder families in many countries of the world. Maize is the gift of the new world to old world. Maize has few insect-pests under storage situation, but it is vulnerable to insect attack after harvest. This pest is an internal feeder and causes significant loss to cereals affecting the quantity, as well as quality of stored cereal grains through storage. Insect pests cause heavy economic losses to stored grains. Maize weevil (*S. zeamais*) is a major pest that attack stored maize grains in the tropic and temperate regions of the world. The control of stored-grain insect pests by chemical insecticides and fumigants are tools for preventing or minimizing insect damage to stored grain. Research work done over the years has revealed that, two plant extracts possess pesticidal properties and could therefore replace the hazardous synthetic products currently being used. Such plant materials include powders from parts of the Neem tree (*Azadirachta indica*). Mixing dried neem leaves with grain in storage is classic example of natural produce use that has been practiced by farmers in many countries for many years. The main active insecticidal ingredients are present mainly in the seeds, leaves, and other parts of the neem.[1] A particular grain variety showed different responses by similar mode of infestation on insect pests. The cereal food

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grain differs each other in their nutritional and physical factors and varied responses, which greatly influences the grain resistance. It is principally an energy giving food because of its high starch content. The grain in also used to feed monogastric such as pigs and poultry as the main basis of energy whereas the plant also serves as fodder for ruminants.[2] The starch is used in creation ice cream, alcohol, beverages, cosmetics and adhesive, cooking oil, salad oil, glycerine, soap and pharmaceuticals can as well be made from maize.[3]

Post harvest losses occur between harvest and the moment of human use. They include on-farm losses, such as when grain is threshed, winnowed, and dried, as well as losses next to the chain through carrying, storage and processing. Harvesting is done by defiance the ear from the stalk using the hand with both cutlass and sickle. This is done through natural draying using Sun draying or non-natural draying like photovoltaic solar energy.[4]

The pests attack the grains, cause losses to the crops in many ways such as, reduce the yield of crops and grains, Lower the quality of crops and grains and increase the cost of production. Therefore, measures should be taken to prevent damage to various crops in the field and during storage. For the purpose of saving the grains from insect attacks during storage it needs systematic research approach to determine the source of physico-biochemical resistant techniques, so that the resistant factors can be utilized in breeding programmes to save the grains from insect infestation. Various factors responsible for the grain resistance have been suggested by many workers. Similarly, the chemical factors responsible for the resistance have also been established in a few cases.[5]

The protection of grains from the stored grain pests is one of the most important problems before us. The discovery of selective plant insecticides meant a great advancement in our fight against insect enemies because they were very toxic to insects and comparatively harmless to humans. They could, therefore, be used freely as domestic insecticides. The discovery of chlorinated hydrocarbons and organophosphate insecticides overshadowed these plant insecticides which were considered to be non-toxic and comparatively safe for humans and domestic animals. The recent studies, however, showed that the synthetic insecticides badly suffer from the presence of poisonous residual toxicity and development of insect resistance. This has led to the revival of interest in plant insecticides. In addition, the chemical pesticides cause serious health hazards by transmitting pathogens and viruses to human beings and livestock's. The control of the stored grain pest by the chemical and other method is being explored to effect further improvement in stored pest control technique. Though chemicals have proved to give effective control of various pests infesting grains, but simultaneously they also create several problems such as appearance of the resistant strain of pest and toxic hazards due to chemical residue. The plant parts have been habitually used to control domestic insect pests in stored grains, crop, in human and cattle medicine. Hence, the objective of this laboratory experiment was to determine the effect of ground Neem leaf and Orange peels against maize weevil and storage Neem leaf and Orange peels on its insecticides movement. Many studies reported its effective use in administration insect pest. Plant based necessary oil like Citrus and Neem have stepped in market place and is best suited for organic food invention in developing and industrial Countries. Azadirachtin, extracted from Neem leaf, act on target organism as antifidant growth regulator. For example, Plant powders from Neem and Citrus had strong insecticidal activity on *Sitophilus oryzae and S. granaries,* and *S. zeamais.* The study proves the insecticidal activity of plant powder of Neem leaf and Orange peels against *S. zeamais* in addition; the current study was designed to investigate the suitability of these plant powders to protect stored maize grains. Recently voluminous work has been done on Neem and its chemical components to control the pests in the field and in storage. Considering the foresaid facts in view, the following studies have been planned against *Sitophilus zeamais* so as to develop effective and feasible management of this serious pest of maize grains in storage.

MATERIAL & METHODS

Experimental Site: -

This research was carried out in the Department of Botany laboratory of Jai Prakash University Chapra. Chapra is located between latitude 180^0 (90^0 North and 90^0 South) and 360^0 of longitude ($+180^0$ eastward and -180^0 westward). This distance between each degree of latitude is about 67 miles 9110 Kilometers).

Insect culture: - The adult *S. zeamais* used for the experiment were cultured in a plastic container under ambient laboratory temperature of 27 ± 2^oC and $75\pm5\%$ humidity during the Winter season.

Collection and Preparation of Plant Materials

The fresh leaves of Neem were collected from a farm at Mashrak in Shastritola while the Orange Peels were collected from Orange sellers in Chapra market. The collection of adults for mass rearing was made local *S. zeamais* of Mashrak. The plant materials were air dried for one week. The dried leaves were ground separately into powder and sieved to obtain fine powders. The powders were stored in a cool dried place until when needed.

Experimental Condition

The Maize seeds were treated with Neem leaf powder and Citrus peel powder in different dosage from 5 gm to 30 gm during the study period. These bio-pesticides were thoroughly mixed with required quantity of seeds and kept in 1 kg capacity glass jars by manual operation. Each experiment was divided into 3 replicates having 25g seed each in separate glass vials measuring 15 x 6 cm. In individual treatment of each replication, 10 pairs of matured adults of beetles were released into every vial obtained from the pure culture raised in the laboratory. All vials were kept under room temperature of 27±2°C. and relative humidity at 75±5%. The mouth of each vial was covered with muslin cloth and tied with rubber bands. For taking the observations on fecundity, adult emergence, grain damage and mortality of the insects, four sets of experiments were analyzed separately for each study.

It was clear from the data shown in Table 1 that all grain protectants showed their superiority over control check (88.04%). The females laid minimum number of eggs (22.34%) on grains treated with Neem leaf powder which proved highly protected in comparison to Citrus peel powder in this study.

RESULTS & DISCUSSION

As regards the incubation period of the eggs, there were significant differences in various treatments (Table 1). The maximum incubation period was observed in grains treated with Neem Seed rather than citrus peel powder and similar results was obtained for hatchability of maize weevils in this investigation.

All the grain protectants were found to be effective in reducing the number of eggs laid by female maize weevils. This is in consonance with the reports of Don-Pedro (1989); Kumar & Okonkwo (1991).[6,7] The maximum number of eggs was found in Neem powder (22.34%) rather than Citrus peel powder in different dosages in this study. The incubation period did not differ significantly among various plant protectant treatments. The maximum incubation period was in Neem leaf powder (12.6 days) while it was minimum in citrus peel powder. The hatchability percentage of the pest was minimum (27.24 %) in the grains treated with Neem powder and this treatment was followed by Citrus peel powder in different concentrations in this research. Plant natural products that constitute effective safer alternatives to synthetic insecticides producing adverse effects on the ecosystems have been tested in the management of stored product pests.[8-10]

These findings were in accordance with the results given by Radha (2014)[11] who found that Neem reduced the oviposition rate of *R. dominica* in maize grains. The present findings were also found in conformity with the results of Srivastava (1994)[12] who also reported that *A. indica* adversely affects the fecundity and hatchability of rice moth.

Table 1- Effect of biopesticides upon Maize weevil reproductive performance

Sl. No.	Protectant	Dosages	Fecundity % female	Incubation (days)	Hatchability (Percentage)
1.	Neem powder	10g	22.34 (5.40)	12.06	27.24 (32.62)
2.	Neem powder	20g	20.13 (4.60)	10.23	24.58 (27.31)
3.	Neem powder	30g	18.52 (4.14)	08.56	20.18 (23.42)
4.	Citrus peel powder	10g	24.42 (5.82)	14.13	30.36 (38.62)
5.	Citrus peel powder	20g	22.18 (4.46)	13.03	28.86 (36.83)
6.	Citrus peel powder	30g	21.46 (3.98)	12.28	26.91 (34.43)

CONCLUSION

The study shows that neem leaf powder was most effective in suppressing weevils population. The two plant powder indicates that higher dosage is more efficient in management of pests. The effects of neem leaf and citrus peels powders in different dosage have been tested on growth and development of maize weevils. The neem plant is very common in India, therefore making its leaf radily available for the control of insect pests of stored grains. The insecticidal action of neem leaf and orange peels

powder when mixed was not very effective in the control of insect pest of stored maize grains S. zeamais. The consist use of neem leaf and orange peels in pest management is recommended as it will help reduce heaps of waste capable of polluting the environment.

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26th January 1953 - 20th April, 2021

- Former Professor & Head, University Department of Zoology, Ranchi University, Ranchi
- Former Dean, Faculty of Science, Ranchi University, Ranchi
- Founder President MSET-ICCB, Ranchi
- Former Editor-in-Chief of Biospectra
- M.Sc. and Ph.D. in Zoology (Entomology)- From Patna University, Patna, Bihar - 1973, 1984
- Date of Joining Ranchi University Service through BPSC Recommendation - 31-08-1977
- Started organising national/international level conferences from 2008 and so far successfully organised following conferences :-
  1. ICTBR, Namkum, Ranchi, Jharkhand - 2008
  2. ICBMSL, Lalgutwa, Ranchi, Jharkhand - 2009
  3. ICBRD & NRMCU, KGVK, Ranchi, Jharkhand - 2010
  4. ICLRAD, CPRS, ICAR, Patna, Bihar - 2011
  5. RAMBSWERD, KSSRDI, Bengaluru - 2013
  6. ICBNEHRI, IINRG, Namkum, Ranchi - 2014
  7. GOSLANRUF, METAS, Ranchi - 2017
- Established five Life Scientist Memorial chairs in Zoology and four in Botany under MSET-ICCB from 2010 through which several PG Scholarships, young, senior, emerging scientist awards as well as life time achievement awards have been conferred to outstanding researchers in the field of animal and plant science research.

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